
GTK4 Tutorial for Beginners

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GTK 4 Tutorial for beginners

Now that GTK 4 is widely used, “GTK” usually means GTK 4. Therefore, this tutorial simply says “GTK” except where the version needs to be explicitly mentioned.

This tutorial illustrates how to write C programs with the GTK library. It is aimed at beginners, so the contents are limited to the basics. The table of contents is at the end of this README.md.

- Section 3 to 23 describe the basics, with the example of a simple editor `tfe` (Text File Editor).
- Section 24 to 27 describe `GtkDrawingArea`.
- Section 28 describes Drag and Drop.
- Section 29 to 33 describe the list model and the list view (`GtkListView`, `GtkGridView`, and `GtkColumnView`). It also describes `GtkExpression`.

The latest version of the tutorial is located at the [GTK4-tutorial GitHub repository](#). You can read it directly without downloading.

There’s a [GitHub Page](#) which is the HTML version of the tutorial.

GTK Documentation

Please refer to the [GTK API Reference](#) and the [GNOME Developer Documentation Website](#) for further information. These websites were opened in August 2021.

If you want to know about GObject and the type system, please refer to the [GObject tutorial](#). GObject is the base of GTK, so it is important for developers to understand it as well as GTK.

Contribution

This tutorial is still under development and unstable. Even though the example codes have been tested on GTK (version 4.10.1), bugs may still exist. If you find any bugs, errors, or mistakes in the tutorial and C examples, please let me know. You can post them to [GitHub issues](#). You can also post updated files to [pull request](#). When submitting a pull request, please ensure that you only modify the files in the `src` directory. Don’t modify the files under `gfm` or `html` directories. After modifying some source files, run `bundle exec rake` to create GFM (GitHub Flavoured Markdown) files and run `bundle exec rake html` to create HTML files. Please refer to the [Author Guide](#) for further information.

If you have a question, feel free to post it to `issue` of this repository. Your questions and feedback are welcome and invaluable for improving this tutorial.

How to Get Gtk 4 Tutorial in HTML or PDF Format

If you want to get the tutorial in HTML or PDF format, build them with `rake` command, which is a ruby version of make. Type `bundle exec rake html` for HTML. Type `bundle exec rake pdf` for PDF. An appendix [Author Guide](#) describes how to write source files.

License

The license of this repository is written in [Section1](#).

In short:

- GFDL1.3 for documents
- GPL3 for programs

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Chapter 1

Introduction

1.1 Prerequisite and License

1.1.1 Prerequisite

1.1.1.1 GTK on a Linux OS

This tutorial describes GTK version 4 libraries. Although GTK is primarily designed for Linux using the C compiler, it is now widely used on other platforms, including Windows and macOS, through various language bindings such as Vala, Python, Rust, and Ruby. However, this tutorial describes only **C programs on Linux**.

If you want to try the examples in the tutorial, you need:

- PC with a Linux distribution like Ubuntu or Debian (including WSL - Windows Subsystem for Linux).
- GCC.
- GTK 4 (version 4.10.1 or later).

The stable version of GTK is 4.20.3 at present (December 28, 2025). The examples in this tutorial have been tested with GTK version 4.10. Version 4.10 added some new classes and functions and deprecated some old ones. Some example programs in this tutorial don't work on older versions.

1.1.1.2 Note for WSL2 (WSLg) Users

When running GTK4 applications on WSL2, you may encounter several warnings in your terminal. These are typically due to interactions between GTK4's rendering system and the WSLg Wayland compositor, rather than bugs in your code. These warnings are often harmless, but in some cases they may affect rendering performance or popup behavior.

For professional-grade performance testing or pixel-perfect UI behavior, a native Linux installation is recommended.

1.1.1.3 Ruby and Rake for Making the Document

This repository includes Ruby programs. They are used to make GFM (GitHub Flavored Markdown), HTML, and PDF files.

You need:

- Linux OS (including WSL)
- Ruby 3.0 or later (tested on 4.0.1)
- Rake

There are two ways to install Ruby.

- Install your distribution's package.
- Use a Ruby version manager such as `rbenv` or `mise`.

Note: If you choose to use a Ruby version other than 4.0.1, you must update the version settings in `Gemfile`. Additionally, depending on your version manager, you may need to modify `.ruby-version` or `mise.toml` as well.

1.1.2 License

Copyright (C) 2020, 2023 ToshioCP (Toshio Sekiya)

The GTK4-tutorial repository contains tutorial documents and programs such as converters, generators and controllers. All of them comprise the 'GTK4-tutorial' package. This package is simply called 'GTK4-tutorial' in the following description.

GTK4-tutorial is free; you can redistribute it and/or modify it under the terms of the following licenses.

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The licenses above are effective as of April 15, 2023. Before that, GPL covered all the contents of the GTK4-tutorial. But GFDL1.3 is more appropriate for documents so the license was changed.

1.2 Preparation

1.2.1 Installing GTK on Linux Distributions

This section describes how to install GTK on Linux distributions.

There are two ways to install GTK:

- Install from distribution packages.
- Build from source.

1.2.1.1 Installation from Distribution Packages

This is the easiest way to install. The following command installs the GTK (version 4) development files on Ubuntu.

```
$ sudo apt install libgtk-4-dev
```

It is important to install the development package (`libgtk-4-dev`). Otherwise, you will not be able to compile any GTK-based programs.

Fedora, Debian, Arch, Gentoo, and OpenSUSE also have GTK packages. See the website of your distribution for further information.

Package information for Arch, Debian/Ubuntu, and Fedora is described on the GTK website: [Installing GTK from packages](#).

1.2.1.2 Installation from Source

If you want to install the development version of GTK, you need to build it from source. See the [Compiling the GTK Libraries](#) section of the GTK API reference.

1.2.2 How to Download This Repository

There are two ways: zip and git. Downloading a zip file is the easiest way. However, if you use git and clone this repository, you can easily update your local repository with the `git pull` command.

1.2.2.1 Download a zip File

- Open your browser and navigate to [this repository](#).
- Click on the green <> Code button. A dropdown menu will appear; click Download ZIP.
- The repository will be downloaded as a zip file to your computer.
- Unzip the file.

1.2.2.2 Clone the Repository

- Click on the green <> Code button. A dropdown menu will appear. The first section is Clone with three tabs. Click the HTTPS tab and click the copy icon to the right of `https://github.com/ToshioCP/Gtk4-tutorial.git`.
- Open a terminal and type `git clone`, then press Ctrl+Shift+V. The command line will look like `git clone https://github.com/ToshioCP/Gtk4-tutorial.git`. Press Enter.
- The directory `Gtk4-tutorial` will be created. This directory is a local copy of the repository.

1.2.3 Examples in the Tutorial

Examples are under the `src` directory. For example, the first example of the tutorial is `pr1.c` and its path is `/src/misc/pr1.c`. Therefore, you do not need to type the examples manually.

Chapter 2

GTK Basics

2.1 GtkApplication and GtkApplicationWindow

2.1.1 GtkApplication

2.1.1.1 GtkApplication and g_application_run

People write programming codes to make an application. What are applications? Applications are software that runs using libraries, which are the OS, frameworks and so on. In GTK 4 programming, the `GtkApplication` is a program (or executable) that runs using `Gtk` libraries.

The basic way to write a `GtkApplication` is as follows.

- Create a `GtkApplication` instance.
- Run the application.

That's all. Very simple. The following is the C code representing the scenario above.

```
1  #include <gtk/gtk.h>
2
3  int
4  main (int argc, char **argv) {
5      GtkApplication *app;
6      int stat;
7
8      app = gtk_application_new ("com.github.ToshioCP.pr1", G_APPLICATION_DEFAULT_FLAGS);
9      stat = g_application_run (G_APPLICATION (app), argc, argv);
10     g_object_unref (app);
11     return stat;
12 }
```

The first line says that this program includes the header files of the `Gtk` libraries. The function `main` is a startup function in C language. The variable `app` is defined as a pointer to a `GtkApplication` instance. The function `gtk_application_new` creates a `GtkApplication` instance and returns a pointer to the instance. The `GtkApplication` instance is a C structure data in which the information of the application is stored. The arguments will be explained later. The function `g_application_run` runs an application that the instance defined. (We often say that the function runs `app`. Actually, `app` is not an application but a pointer to the instance of the application. However, it is simple and short, and probably no confusion occurs.)

Here I used the word `instance`. Instance, class and object are terminologies in Object Oriented Programming. I use these words in the same way. But, I will often use “object” instead of “instance” in this tutorial. That means “object” and “instance” are the same. Object is a bit ambiguous word. In a broad

sense, object has wider meaning than instance. So, readers should be careful of the contexts to find the meaning of “object”. In many cases, object and instance are interchangeable.

The function `gtk_application_new` has two parameters.

- Application ID (`com.github.ToshioCP.pr1`). It is used to distinguish applications by the system. The format is reverse-DNS. See [GNOME Developer Documentation – Application ID](#) for further information.
- Application flag (`G_APPLICATION_DEFAULT_FLAGS`). If the application runs without any arguments, the flag is `G_APPLICATION_DEFAULT_FLAGS`. Otherwise, you need other flags. See [GIO API reference](#) for further information.

Notice: If your GLib-2.0 version is older than 2.74, use `G_APPLICATION_FLAGS_NONE` instead of `G_APPLICATION_DEFAULT_FLAGS`. It is an old flag replaced by `G_APPLICATION_DEFAULT_FLAGS` and deprecated since version 2.74.

To compile this, run the following command. The string `pr1.c` is the filename of the C source code above.

If you’ve downloaded this repository, you don’t need to create the file. There’s the same file at [/src/misc/pr1.c](#) in your local repository. All the example codes are under the [/src](#) directory as well.

```
$ gcc `pkg-config --cflags gtk4` pr1.c `pkg-config --libs gtk4`
```

The C compiler `gcc` generates an executable file, `a.out`. Let’s run it.

```
$ ./a.out
```

```
(a.out:5084): GLib-GIO-WARNING **: 09:52:04.236: Your application does not implement
g_application_activate() and has no handlers connected to the 'activate' signal.
```

```
It should do one of these.
```

```
$
```

Oh, it just produces an error message. This error message shows that the `GtkApplication` object ran, without a doubt. Now, let’s think about what this message means.

2.1.1.2 Signals

The message tells us that:

1. The application doesn’t implement `g_application_activate()`,
2. It has no handlers connected to the “activate” signal, and
3. You will need to solve at least one of these.

These two problems are related to signals. So, I will explain that first.

A signal is emitted when something happens. For example, a window is created, a window is destroyed and so on. The signal “activate” is emitted when the application is activated. (Activated is a bit different from started, but you can think of them both as almost the same so far.) If the signal is connected to a function, which is called a signal handler or simply handler, then the function is invoked when the signal is emitted.

The flow is like this:

1. Something happens.
2. If it’s related to a certain signal, then the signal is emitted.
3. If the signal has been connected to a handler in advance, then the handler is invoked.

Signals are defined in objects. For example, the “activate” signal belongs to the `GApplication` object, which is a parent object of `GtkApplication` object.

The `GApplication` object is a child object of the `GObject` object. `GObject` is the top object in the hierarchy of all the objects.

```
GObject -- GApplication -- GtkApplication
<---parent          --->child
```

A child object inherits signals, functions, properties and so on from its parent object. So, `GtkApplication` also has the “activate” signal.

Now we can solve the problem in `pr1.c`. We need to connect the “activate” signal to a handler. We use a function `g_signal_connect` which connects a signal to a handler.

```
1  #include <gtk/gtk.h>
2
3  static void
4  app_activate (GApplication *app, gpointer *user_data) {
5      g_print ("GtkApplication is activated.\n");
6  }
7
8  int
9  main (int argc, char **argv) {
10     GtkApplication *app;
11     int stat;
12
13     app = gtk_application_new ("com.github.ToshioCP.pr2", G_APPLICATION_DEFAULT_FLAGS);
14     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
15     stat = g_application_run (G_APPLICATION (app), argc, argv);
16     g_object_unref (app);
17     return stat;
18 }
```

First, we define the handler `app_activate` which simply displays a message. The function `g_print` is defined in GLib and it’s like a `printf` in the C standard library. In the function `main`, we add `g_signal_connect` before `g_application_run`. The function `g_signal_connect` has four arguments.

1. An instance to which the signal belongs.
2. The name of the signal.
3. A handler function (also called callback), which needs to be cast by `G_CALLBACK`.
4. Data to pass to the handler. If no data is necessary, `NULL` is given.

It is described in the [GObject API Reference](#). Correctly, `g_signal_connect` is a macro (not a C function).

```
#define g_signal_connect (
    instance,
    detailed_signal,
    c_handler,
    data
)
```

You can find the description of each signal in the API reference manual. For example, “activate” signal is in the [GApplication section](#) in the GIO API Reference.

```
void
activate (
    GApplication *self,
    gpointer user_data
)
```

This is a declaration of the “activate” signal handler. You can use any name instead of “activate” in the declaration above. The parameters are:

- `self` is an instance to which the signal belongs.

- `user_data` is a data defined in the fourth argument of the `g_signal_connect` function. If it is `NULL`, then you can ignore and leave out the second parameter.

API reference manual is very important. You should see and understand it.

Let's compile the source file above (`pr2.c`) and run it.

```
$ gcc `pkg-config --cflags gtk4` pr2.c `pkg-config --libs gtk4`
$ ./a.out
GtkApplication is activated.
$
```

OK, well done. However, you may have noticed that it's painful to type such a long line to compile. It is a good idea to use shell script to solve this problem. Make a text file which contains the following line.

```
gcc `pkg-config --cflags gtk4` $1.c `pkg-config --libs gtk4`
```

Then, save it under the directory `$HOME/bin`, which is usually `/home/(username)/bin`. (If your user name is James, then the directory is `/home/james/bin`). And turn on the execute bit of the file. If the filename is `comp`, do like this:

```
$ chmod 755 $HOME/bin/comp
$ ls -log $HOME/bin
... ..
-rwxr-xr-x 1 62 May 23 08:21 comp
... ..
```

If this is the first time that you make a `$HOME/bin` directory and save a file in it, then you need to logout and login again.

```
$ comp pr2
$ ./a.out
GtkApplication is activated.
$
```

2.1.2 GtkWindow and GtkApplicationWindow

2.1.2.1 GtkWindow

A message "GtkApplication is activated." was printed out in the previous subsection. It was good in terms of a test of `GtkApplication`. However, it is insufficient because `Gtk` is a framework for graphical user interface (GUI). Now we go ahead with adding a window into this program. What we need to do is:

1. Create a `GtkWindow`.
2. Connect it to the `GtkApplication`.
3. Show the window.

Now rewrite the function `app_activate`.

2.1.2.1.1 Create a GtkWindow

```
1 static void
2 app_activate (GApplication *app, gpointer user_data) {
3     GtkWidget *win;
4
5     win = gtk_window_new ();
6     gtk_window_set_application (GTK_WINDOW (win), GTK_APPLICATION (app));
7     gtk_window_present (GTK_WINDOW (win));
8 }
```

Widget is an abstract concept that includes all the GUI interfaces such as windows, labels, buttons, multi-line text, boxes and so on. And GtkWidget is a base object from which all the GUI objects derive.

```
parent <-----> child
GtkWidget -- GtkWidget
```

GtkWindow includes GtkWidget at the top of its object.

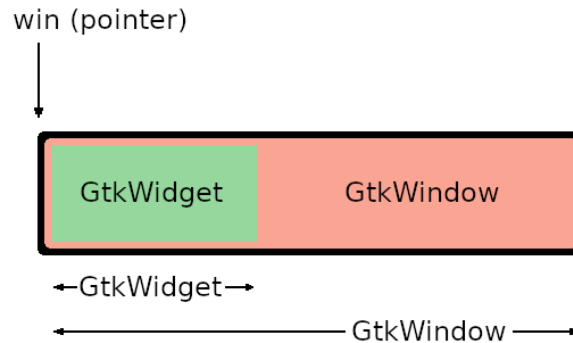


Figure 2.1: GtkWindow and GtkWidget

The function `gtk_window_new` is defined as follows.

```
GtkWidget *
gtk_window_new (void);
```

By this definition, it returns a pointer to GtkWidget, not GtkWindow. It actually creates a new GtkWindow instance (not GtkWidget) but returns a pointer to GtkWidget. However, the pointer points to the GtkWidget and at the same time it also points to the GtkWindow that contains the GtkWidget in it.

If you want to use `win` as a pointer to a GtkWindow type instance, you need to cast it.

```
(GtkWindow *) win
```

It works, but isn't usually used. Instead, `GTK_WINDOW` macro is used.

```
GTK_WINDOW (win)
```

The macro is recommended because it not only casts the pointer but also checks the type.

2.1.2.1.2 Connect it to the GtkApplication. The function `gtk_window_set_application` is used to connect GtkWindow to GtkApplication.

```
gtk_window_set_application (GTK_WINDOW (win), GTK_APPLICATION (app));
```

You need to cast `win` to GtkWindow and `app` to GtkApplication with `GTK_WINDOW` and `GTK_APPLICATION` macro.

GtkApplication continues to run until the related window is destroyed. If you didn't connect GtkWindow and GtkApplication, GtkApplication destroys itself immediately. Because no window is connected to GtkApplication, GtkApplication doesn't need to wait anything. As it destroys itself, the GtkWindow is also destroyed.

2.1.2.1.3 Show the Window. The function `gtk_window_present` presents the window to a user (shows it to the user).

GTK 4 changes the default widget visibility to on, so every widget doesn't need to change it to on. But, there's an exception. Top level window (this term will be explained later) isn't visible when it is created. So you need to use the function above to show the window.

You can use `gtk_widget_set_visible (win, true)` instead of `gtk_window_present`. But the behavior of these two is different. Suppose there are two windows `win1` and `win2` on the screen and `win1` is behind `win2`. Both windows are visible. The function `gtk_widget_set_visible (win1, true)` does nothing because `win1` is already visible. So, `win1` is still behind `win2`. The other function `gtk_window_present (win1)` moves `win1` to the top of the stack of the windows. Therefore, if you want to present the window, you should use `gtk_window_present`.

Two functions `gtk_widget_show` and `gtk_widget_hide` are deprecated since GTK 4.10. You should use `gtk_widget_set_visible` instead.

Save the program as `pr3.c`, then compile and run it.

```
$ comp pr3
$ ./a.out
```

A small window appears.

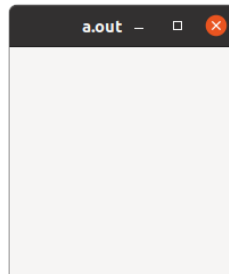


Figure 2.2: Screenshot of the window

Click on the close button then the window disappears and the program finishes.

2.1.2.2 GtkApplicationWindow

`GtkApplicationWindow` is a child object of `GtkWindow`. It has some extra feature for better integration with `GtkApplication`. It is recommended to use it as the top-level window of the application instead of `GtkWindow`.

Now rewrite the program and use `GtkApplicationWindow`.

```
1 static void
2 app_activate (GApplication *app, gpointer user_data) {
3     GtkWidget *win;
4
5     win = gtk_application_window_new (GTK_APPLICATION (app));
6     gtk_window_set_title (GTK_WINDOW (win), "pr4");
7     gtk_window_set_default_size (GTK_WINDOW (win), 400, 300);
8     gtk_window_present (GTK_WINDOW (win));
9 }
```

When you create `GtkApplicationWindow`, you need to give `GtkApplication` instance as an argument. Then it automatically connect these two instances. So you don't need to call `gtk_window_set_application` any more.

The program sets the title and the default size of the window. Compile it and run `a.out`, then you will see a bigger window with the title "pr4".

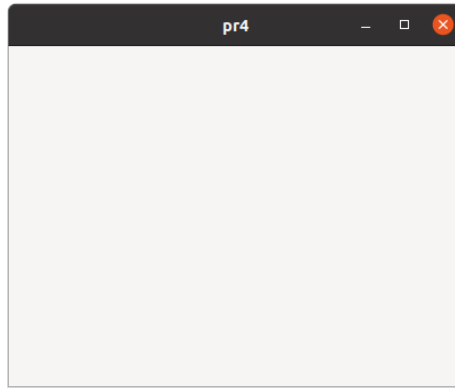


Figure 2.3: Screenshot of the window

2.2 Widgets (1)

2.2.1 GtkLabel, GtkButton and GtkBox

2.2.1.1 GtkLabel

We made a window and showed it on the screen in the previous section. Now we go on to the next topic: widgets. The simplest widget is GtkLabel. It is a widget with text in it.

```

1  #include <gtk/gtk.h>
2
3  static void
4  app_activate (GApplication *app) {
5      GtkWidget *win;
6      GtkWidget *lab;
7
8      win = gtk_application_window_new (GTK_APPLICATION (app));
9      gtk_window_set_title (GTK_WINDOW (win), "lb1");
10     gtk_window_set_default_size (GTK_WINDOW (win), 400, 300);
11
12     lab = gtk_label_new ("Hello.");
13     gtk_window_set_child (GTK_WINDOW (win), lab);
14
15     gtk_window_present (GTK_WINDOW (win));
16 }
17
18 int
19 main (int argc, char **argv) {
20     GApplication *app;
21     int stat;
22
23     app = gtk_application_new ("com.github.ToshioCP.lb1", G_APPLICATION_DEFAULT_FLAGS);
24     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
25     stat = g_application_run (G_APPLICATION (app), argc, argv);
26     g_object_unref (app);
27     return stat;
28 }

```

Save this program to a file `lb1.c`. (You can use </src/misc/lb1.c> if you've downloaded this repository.) Then compile and run it.


```
$ cd src/misc
$ comp lb1
$ ./a.out
```

A window with a message “Hello.” appears.

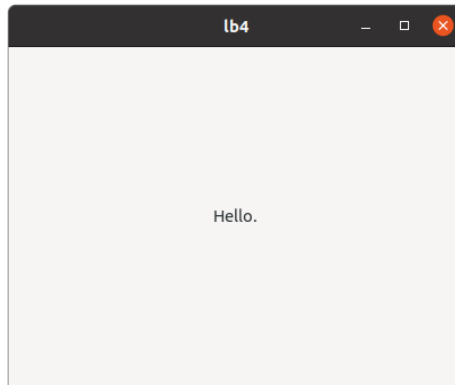


Figure 2.4: Screenshot of the label

There are only a few changes between `pr4.c` and `lb1.c`. A program `diff` is useful to know the difference.

```
$ cd misc; diff pr4.c lb1.c
4c4
< app_activate (GApplication *app, gpointer user_data) {
---
> app_activate (GApplication *app) {
5a6
> GtkWidget *lab;
8c9
< gtk_window_set_title (GTK_WINDOW (win), "pr4");
---
> gtk_window_set_title (GTK_WINDOW (win), "lb1");
9a11,14
>
> lab = gtk_label_new ("Hello.");
> gtk_window_set_child (GTK_WINDOW (win), lab);
>
18c23
< app = gtk_application_new ("com.github.ToshioCP.pr4", G_APPLICATION_DEFAULT_FLAGS);
---
> app = gtk_application_new ("com.github.ToshioCP.lb1", G_APPLICATION_DEFAULT_FLAGS);
$
```

This tells us:

- A signal handler `app_activate` doesn't have `user_data` parameter. If the fourth argument of `g_signal_connect` is `NULL`, you can leave out `user_data`.
- The definition of a new variable `lab` is added.
- The title of the window is changed.
- A label is created and connected to the window as a child.

The function `gtk_window_set_child (GTK_WINDOW (win), lab)` makes the label `lab` a child widget of the window `win`. Be careful. A child widget is different from a child object. Objects have parent-child relationships and widgets also have parent-child relationships. But these two relationships are totally different. Don't be confused. In the program `lb1.c`, `lab` is a child widget of `win`. Child widgets are

always located in their parent widget on the screen. See how the window has appeared on the screen. The application window includes the label.

The window `win` doesn't have any parents. We call such a window top-level window. An application can have more than one top-level window.

2.2.1.2 GtkButton

The next widget is `GtkButton`. It displays a button on the screen with a label or icon on it. In this subsection, we will make a button with a label. When the button is clicked, it emits a “clicked” signal. The following program shows how to catch the signal and do something.

```

1  #include <gtk/gtk.h>
2
3  static void
4  click_cb (GtkButton *btn) {
5      g_print ("Clicked.\n");
6  }
7
8  static void
9  app_activate (GApplication *app) {
10     GtkWidget *win;
11     GtkWidget *btn;
12
13     win = gtk_application_window_new (GTK_APPLICATION (app));
14     gtk_window_set_title (GTK_WINDOW (win), "lb2");
15     gtk_window_set_default_size (GTK_WINDOW (win), 400, 300);
16
17     btn = gtk_button_new_with_label ("Click me");
18     gtk_window_set_child (GTK_WINDOW (win), btn);
19     g_signal_connect (btn, "clicked", G_CALLBACK (click_cb), NULL);
20
21     gtk_window_present (GTK_WINDOW (win));
22 }
23
24 int
25 main (int argc, char **argv) {
26     GtkApplication *app;
27     int stat;
28
29     app = gtk_application_new ("com.github.ToshioCP.lb2", G_APPLICATION_DEFAULT_FLAGS);
30     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
31     stat = g_application_run (G_APPLICATION (app), argc, argv);
32     g_object_unref (app);
33     return stat;
34 }
```

Look at the line 17 to 19. First, it creates a `GtkButton` instance `btn` with a label “Click me”. Then, adds the button to the window `win` as a child. Finally, connects the “clicked” signal of the button to the handler `click_cb`. So, if `btn` is clicked, the function `click_cb` is invoked. The suffix “cb” means “call back”.

Name the program `lb2.c` and save it. Now compile and run it.

A window with the button appears. Click the button (it is a large button, you can click everywhere in the window), then a string “Clicked.” appears on the terminal. It shows the handler was invoked by clicking

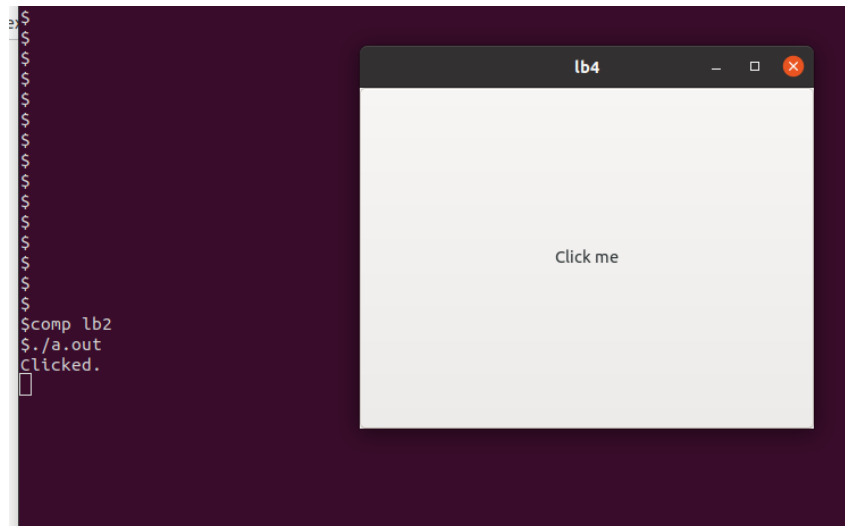


Figure 2.5: Screenshot of the label

the button.

It's good that we've made sure that the clicked signal was caught and the handler was invoked by using `g_print`. However, using `g_print` doesn't fit well with GTK, which is a GUI library. So, we will change the handler. The following code is extracted from `lb3.c`.

```

1  static void
2  click_cb (GtkButton *btn, GtkWidget *win) {
3      gtk_window_destroy (win);
4  }
5
6  static void
7  app_activate (GApplication *app) {
8      GtkWidget *win;
9      GtkWidget *btn;
10
11     win = gtk_application_window_new (GTK_APPLICATION (app));
12     gtk_window_set_title (GTK_WINDOW (win), "lb3");
13     gtk_window_set_default_size (GTK_WINDOW (win), 400, 300);
14
15     btn = gtk_button_new_with_label ("Close");
16     gtk_window_set_child (GTK_WINDOW (win), btn);
17     g_signal_connect (btn, "clicked", G_CALLBACK (click_cb), win);
18
19     gtk_window_present (GTK_WINDOW (win));
20 }

```

And the difference between `lb2.c` and `lb3.c` is as follows.

```

$ cd misc; diff lb2.c lb3.c
4,5c4,5
< click_cb (GtkButton *btn) {
<     g_print ("Clicked.\n");
---
> click_cb (GtkButton *btn, GtkWidget *win) {
>     gtk_window_destroy (win);
14c14

```

```

<  gtk_window_set_title (GTK_WINDOW (win), "lb2");
---
>  gtk_window_set_title (GTK_WINDOW (win), "lb3");
17c17
<  btn = gtk_button_new_with_label ("Click me");
---
>  btn = gtk_button_new_with_label ("Close");
19c19
<  g_signal_connect (btn, "clicked", G_CALLBACK (click_cb), NULL);
---
>  g_signal_connect (btn, "clicked", G_CALLBACK (click_cb), win);
29c29
<  app = gtk_application_new ("com.github.ToshioCP.lb2", G_APPLICATION_DEFAULT_FLAGS);
---
>  app = gtk_application_new ("com.github.ToshioCP.lb3", G_APPLICATION_DEFAULT_FLAGS);
35d34
<
$

```

The changes are:

- The function `g_print` in `lb2.c` was deleted and two lines are inserted.
 - `click_cb` has the second parameter, which comes from the fourth argument of the `g_signal_connect` at line 17. One thing to be careful is the types are different between the second parameter of `click_cb` and the fourth argument of `g_signal_connect`. The former is `GtkWindow *` and the latter is `GtkWidget *`. The compiler doesn't complain because `g_signal_connect` uses `gpointer` (general type of pointer). In this program the instance pointed to by `win` is a `GtkApplicationWindow` object. It is a descendant of `GtkWindow` and `GtkWidget` class, so both `GtkWindow *` and `GtkWidget *` are correct types of the instance.
 - `gtk_window_destroy (win)` destroys the top-level window. Then the application quits.
- The label of `btn` is changed from "Click me" to "Close".
- The fourth argument of `g_signal_connect` is changed from `NULL` to `win`.

The most important change is the fourth argument of the `g_signal_connect`. This argument is described as "data to pass to the handler" in the definition of [g_signal_connect](#).

2.2.1.3 GtkBox

`GtkWindow` and `GtkApplicationWindow` can have only one child. If you want to add two or more widgets in a window, you need a container widget. `GtkBox` is one of the containers. It arranges two or more child widgets into a single row or column. The following procedure shows the way to add two buttons in a window.

- Create a `GtkApplicationWindow` instance.
- Create a `GtkBox` instance and add it to the `GtkApplicationWindow` as a child.
- Create a `GtkButton` instance and append it to the `GtkBox`.
- Create another `GtkButton` instance and append it to the `GtkBox`.

After this, the widgets are connected as shown in the following diagram.

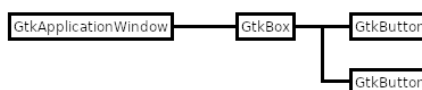


Figure 2.6: Parent-child relationship

The program lb4.c is as follows.

```

1  #include <gtk/gtk.h>
2
3  static void
4  click1_cb (GtkButton *btn) {
5      const char *s;
6
7      s = gtk_button_get_label (btn);
8      if (g_strcmp0 (s, "Hello.") == 0)
9          gtk_button_set_label (btn, "Good-bye.");
10     else
11         gtk_button_set_label (btn, "Hello.");
12 }
13
14 static void
15 click2_cb (GtkButton *btn, GtkWidget *win) {
16     gtk_window_destroy (win);
17 }
18
19 static void
20 app_activate (GApplication *app) {
21     GtkWidget *win;
22     GtkWidget *box;
23     GtkWidget *btn1;
24     GtkWidget *btn2;
25
26     win = gtk_application_window_new (GTK_APPLICATION (app));
27     gtk_window_set_title (GTK_WINDOW (win), "lb4");
28     gtk_window_set_default_size (GTK_WINDOW (win), 400, 300);
29
30     box = gtk_box_new (GTK_ORIENTATION_VERTICAL, 5);
31     gtk_box_set_homogeneous (GTK_BOX (box), TRUE);
32     gtk_window_set_child (GTK_WINDOW (win), box);
33
34     btn1 = gtk_button_new_with_label ("Hello.");
35     g_signal_connect (btn1, "clicked", G_CALLBACK (click1_cb), NULL);
36
37     btn2 = gtk_button_new_with_label ("Close");
38     g_signal_connect (btn2, "clicked", G_CALLBACK (click2_cb), win);
39
40     gtk_box_append (GTK_BOX (box), btn1);
41     gtk_box_append (GTK_BOX (box), btn2);
42
43     gtk_window_present (GTK_WINDOW (win));
44 }
45
46 int
47 main (int argc, char **argv) {
48     GApplication *app;
49     int stat;
50
51     app = gtk_application_new ("com.github.ToshioCP.lb4", G_APPLICATION_DEFAULT_FLAGS);
52     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);

```

```

53     stat = g_application_run (G_APPLICATION (app), argc, argv);
54     g_object_unref (app);
55     return stat;
56 }

```

Look at the function `app_activate`.

After the creation of a `GtkApplicationWindow` instance, a `GtkBox` instance is created.

```

box = gtk_box_new(GTK_ORIENTATION_VERTICAL, 5);
gtk_box_set_homogeneous (GTK_BOX (box), TRUE);

```

The first argument arranges the children of the box vertically. The orientation constants are defined like this:

- `GTK_ORIENTATION_VERTICAL`: the children widgets are arranged vertically
- `GTK_ORIENTATION_HORIZONTAL`: the children widgets are arranged horizontally

The second argument is the size of the space between the children. The unit of the length is pixel.

The next function `gtk_box_set_homogeneous` fills the box with the children, giving them the same space.

After that, two buttons `btn1` and `btn2` are created and the signal handlers are set. Then, these two buttons are appended to the box.

```

1  static void
2  click1_cb (GtkButton *btn) {
3      const char *s;
4
5      s = gtk_button_get_label (btn);
6      if (g_strcmp0 (s, "Hello.") == 0)
7          gtk_button_set_label (btn, "Good-bye.");
8      else
9          gtk_button_set_label (btn, "Hello.");
10 }

```

The function `gtk_button_get_label` returns the text from the label. The string is owned by the button and you can't modify or free it. The `const` qualifier is necessary for the string `s`. If you change the string, your compiler will give you a warning.

You always need to be careful with the `const` qualifier when you see the GTK 4 API reference.

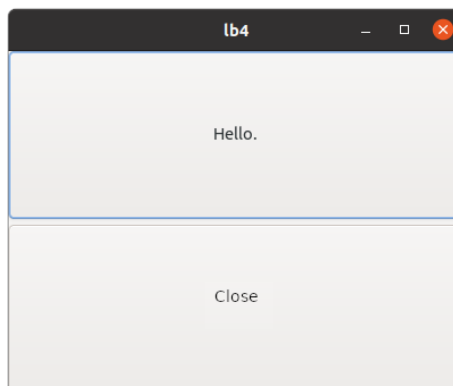


Figure 2.7: Screenshot of the box

The handler corresponding to `btn1` toggles its label. The handler corresponding to `btn2` destroys the top-level window and the application quits.

2.3 Widgets (2)

2.3.1 GtkTextView, GtkTextBuffer and GtkScrolledWindow

2.3.1.1 GtkTextView and GtkTextBuffer

GtkTextView is a widget for multi-line text editing. GtkTextBuffer is a text buffer which is connected to GtkTextView. See the sample program `tfv1.c` below.

```

1  #include <gtk/gtk.h>
2
3  static void
4  app_activate (GApplication *app) {
5      GtkWidget *win;
6      GtkWidget *tv;
7      GtkTextBuffer *tb;
8      gchar *text;
9
10     text =
11         "Once upon a time, there lived an old man known as Taketori-no-Okina. "
12         "His name, meaning \"The Bamboo Cutter,\" reflected his life's work of crafting bamboo baskets
13         "One day, while deep in the hills, he came across a bamboo stalk that shimmered with a mysteri
14         "\"How strange and wondrous!\" he exclaimed. "
15         "When he carefully cut it open, he found a tiny, beautiful baby girl nestled inside, glowing w
16         "Believing she was a gift from the heavens, he brought her home.\n"
17         "Though his wife was astonished by his tale, they were both filled with joy, for they had never
18     ;
19     win = gtk_application_window_new (GTK_APPLICATION (app));
20     gtk_window_set_title (GTK_WINDOW (win), "Taketori");
21     gtk_window_set_default_size (GTK_WINDOW (win), 800, 600);
22
23     tv = gtk_text_view_new ();
24     tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
25     gtk_text_buffer_set_text (tb, text, -1);
26     gtk_text_view_set_wrap_mode (GTK_TEXT_VIEW (tv), GTK_WRAP_WORD_CHAR);
27
28     gtk_window_set_child (GTK_WINDOW (win), tv);
29
30     gtk_window_present (GTK_WINDOW (win));
31 }
32
33 int
34 main (int argc, char **argv) {
35     GApplication *app;
36     int stat;
37
38     app = gtk_application_new ("com.github.ToshioCP.tfv1", G_APPLICATION_DEFAULT_FLAGS);
39     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
40     stat = g_application_run (G_APPLICATION (app), argc, argv);
41     g_object_unref (app);
42     return stat;
43 }

```

Look at line 25. A GtkTextView instance is created and its pointer is assigned to `tv`. When the GtkTextView instance is created, a GtkTextBuffer instance is also created and connected to the GtkTextView

automatically. “GtkTextBuffer instance” will be referred to simply as “GtkTextBuffer” or “buffer”. In the next line, the pointer to the buffer is assigned to `tb`. Then, the text from line 10 to 20 is assigned to the buffer. If the third argument of `gtk_text_buffer_set_text` is a positive integer, it is the length of the text. If it is -1, the string terminates with NULL.

GtkTextView has a wrap mode. When it is set to `GTK_WRAP_WORD_CHAR`, text wraps in between words, or if that is not enough, also between graphemes.

Wrap mode is written in [Gtk_WrapMode](#) in the GTK 4 API document.

In line 30, `tv` is added to `win` as a child.

Now compile and run it. If you’ve downloaded this repository, its pathname is `src/tfv/tfv1.c`.

```
$ cd src/tfv
$ comp tfv1
$ ./a.out
```

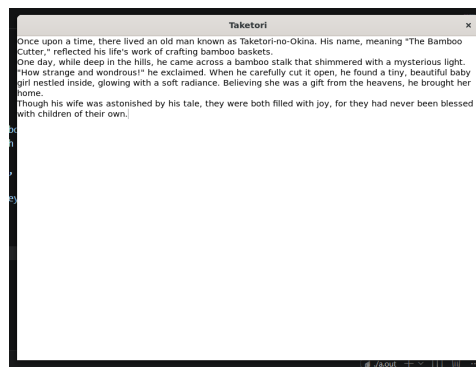


Figure 2.8: GtkTextView

There’s an I-beam pointer in the window. You can add or delete any characters on the GtkTextView, and your changes are kept in the GtkTextBuffer. If you add more characters beyond the limit of the window, the height increases and the window extends. If the height gets bigger than the height of the screen, you won’t be able to control the size of the window or change it back to the original size. This is essentially a bug. This can be solved by adding a GtkScrolledWindow between the GtkApplicationWindow and GtkTextView.

2.3.1.2 GtkScrolledWindow

What we need to do is:

- Create a GtkScrolledWindow and insert it as a child of the GtkApplicationWindow
- Insert the GtkTextView widget to the GtkScrolledWindow as a child.

Modify `tfv1.c` and save it as `tfv2.c`. There are only a few differences between these two files.

```
$ cd tfv; diff tfv1.c tfv2.c
5a6
> GtkWidget *scr;
22a24,26
> scr = gtk_scrolled_window_new ();
> gtk_window_set_child (GTK_WINDOW (win), scr);
>
28c32
< gtk_window_set_child (GTK_WINDOW (win), tv);
---
> gtk_scrolled_window_set_child (GTK_SCROLLED_WINDOW (scr), tv);
```



```

38c42
< app = gtk_application_new ("com.github.ToshioCP.tfv1", G_APPLICATION_DEFAULT_FLAGS);
---
> app = gtk_application_new ("com.github.ToshioCP.tfv2", G_APPLICATION_DEFAULT_FLAGS);
$

```

The whole code of tfv2.c is as follows.

```

1  #include <gtk/gtk.h>
2
3  static void
4  app_activate (GApplication *app) {
5      GtkWidget *win;
6      GtkWidget *scr;
7      GtkWidget *tv;
8      GtkTextBuffer *tb;
9      gchar *text;
10
11     text =
12         "Once upon a time, there lived an old man known as Taketori-no-Okina. "
13         "His name, meaning \"The Bamboo Cutter,\" reflected his life's work of crafting bamboo baskets
14         "One day, while deep in the hills, he came across a bamboo stalk that shimmered with a mysteri
15         "\"How strange and wondrous!\" he exclaimed. "
16         "When he carefully cut it open, he found a tiny, beautiful baby girl nestled inside, glowing w
17         "Believing she was a gift from the heavens, he brought her home.\n"
18         "Though his wife was astonished by his tale, they were both filled with joy, for they had never
19     ;
20     win = gtk_application_window_new (GTK_APPLICATION (app));
21     gtk_window_set_title (GTK_WINDOW (win), "Taketori");
22     gtk_window_set_default_size (GTK_WINDOW (win), 800, 600);
23
24     scr = gtk_scrolled_window_new ();
25     gtk_window_set_child (GTK_WINDOW (win), scr);
26
27     tv = gtk_text_view_new ();
28     tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
29     gtk_text_buffer_set_text (tb, text, -1);
30     gtk_text_view_set_wrap_mode (GTK_TEXT_VIEW (tv), GTK_WRAP_WORD_CHAR);
31
32     gtk_scrolled_window_set_child (GTK_SCROLLED_WINDOW (scr), tv);
33
34     gtk_window_present (GTK_WINDOW (win));
35 }
36
37 int
38 main (int argc, char **argv) {
39     GApplication *app;
40     int stat;
41
42     app = gtk_application_new ("com.github.ToshioCP.tfv2", G_APPLICATION_DEFAULT_FLAGS);
43     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
44     stat = g_application_run (G_APPLICATION (app), argc, argv);
45     g_object_unref (app);
46     return stat;

```

47 }

Compile and run it.

Now, the window doesn't extend even if you type a lot of characters, it just scrolls.

2.4 Strings and Memory Management

GtkTextView and GtkTextBuffer have functions that have string parameters or return a string. The knowledge of strings and memory management is useful to understand how to use these functions.

2.4.1 Strings and Memory

A String is an array of characters that is terminated with `'\0'`. C has no built-in string type. Strings are represented using character arrays (or pointers to characters). It behaves like a string in other languages. So, in practice, a pointer to the first element of such a character array is often loosely called a “string”.

The following is a sample program.

```
char a[10], *b;

a[0] = 'H';
a[1] = 'e';
a[2] = 'l';
a[3] = 'l';
a[4] = 'o';
a[5] = '\0';

b = a;
/* *b is 'H' */
/* *(++b) is 'e' */
```

An array `a` is defined as a `char` type array and its size is ten. The first five elements are `'H'`, `'e'`, `'l'`, `'l'`, `'o'`. They are character codes. For example, `'H'` is the same as `0x48` or `72`. The sixth element is `'\0'`, which is the same as zero, and indicates that the sequence of the data ends there. The array represents the string “Hello”.

The size of the array is 10, so four bytes aren't used. But it's OK. They are just ignored. (If the variable `a` is defined out of functions or its storage class is static, the four bytes are assigned with zero. Otherwise, that is to say, the storage class is auto, they are undefined.)

The variable `b` is a pointer to a character. It is assigned to `a`, so `b` points to the first element of `a` (character `'H'`). The array `a` itself cannot be assigned to. So `a=a+1` causes syntax error.

On the other hand, `b` is a pointer type variable, which is mutable. So, `++b`, which increases `b` by one, is allowed.

If a pointer is `NULL`, it points to nothing. So, the pointer is not a string. It is different from empty strings. An empty string is a pointer to a memory location where the first character is `'\0'`

There are four cases:

- The string is read only
- The string is in static memory area
- The string is in stack
- The string is in memory allocated from the heap area

2.4.2 Read Only Strings

A string literal is enclosed in double quotes like this:

```
char *s;
s = "Hello"
```

“Hello” is a string literal, and is read only. So, the following program is illegal.

```
*(s+1) = 'a';
```

The result is undefined. Probably something bad will happen, such as a segmentation fault.

NOTE: The memory of the literal string is allocated when the program is compiled. Literal strings can be seen using the `strings` command.

```
$ strings src/tvf/a.out
/lib64/ld-linux-x86-64.so.2
cN<5
```

```
... ..
... ..
```

Once upon a time, there was an old man who was called Taketori-no-Okina. It is a japanese word that means a bamboo. One day, he went into a hill and found a shining bamboo. "What a mysterious bamboo it is!," he said. His wife was surprized at his story. They were very happy because they had no children.

```
... ..
... ..
```

This shows that literal strings are embedded in program binary codes.

2.4.3 Strings Defined as Arrays

If a string is defined as an array, it's stored in static memory area or stack. It depends on the storage class of the array. If the array's storage class is `static`, then it's placed in static memory area. The allocated memory lives for the life of the program. This area is writable.

If the array's storage class is `auto`, it's placed in stack. If the array is defined inside a function, its default storage class is `auto`. The stack area will disappear when the function returns to the caller. Arrays defined on the stack are writable.

```
static char a[] = {'H', 'e', 'l', 'l', 'o', '\0'};

void
print_strings (void) {
    char b[] = "Hello";

    a[1] = 'a'; /* Writable because 'a' is a character array (in static storage). */
    b[1] = 'a'; /* Writable because 'b' is a character array (on the stack). */

    printf ("%s\n", a); /* Hallo */
    printf ("%s\n", b); /* Hallo */
}
```

The array `a` is defined out of functions. It is placed in the static memory area even if the `static` is left out. The compiler calculates the number of the elements (six) and allocates six bytes in the static memory area. Then, it copies “Hello” literal string data to the memory.

The array `b` is defined inside the function, so its storage class is `auto`. The compiler calculates the number of the elements in the string literal. It is six, including the null terminator. The compiler allocates six bytes in the stack and copies “Hello” literal string to the stack memory.

Both `a` and `b` are writable.

The memory is allocated and freed by the program automatically so you don't need to allocate or free. The array `a` is alive during the program's life time. The array `b` is alive when the function is called until the function returns to the caller.

2.4.4 Strings in the Heap Area

You can get, use and release memory from the heap area. The standard C library provides `malloc` to get memory and `free` to release memory. GLib provides the functions `g_new` and `g_free`. They are similar to `malloc` and `free`.

```
g_new (struct_type, n_struct)
```

`g_new` is a macro to allocate memory for an array.

- `struct_type` is the type of the element of the array.
- `n_struct` is the size of the array.
- The return value is a pointer to the array. Its type is a pointer to `struct_type`.

For example,

```
char *s;
s = g_new (char, 10);
/* s points to an array of char. The size of the array is 10. */
```

```
struct tuple {int x, y;} *t;
t = g_new (struct tuple, 5);
/* t points to an array of struct tuple. */
/* The size of the array is 5. */
```

`g_free` frees memory.

```
void
g_free (gpointer mem);
```

If `mem` is `NULL`, `g_free` does nothing. `gpointer` is a type of general pointer. It is the same as `void *`. This pointer can be cast to any pointer type. Conversely, any pointer type can be cast to `gpointer`.

```
g_free (s);
/* Frees the memory allocated to s. */
```

```
g_free (t);
/* Frees the memory allocated to t. */
```

If the argument does not point to allocated memory, the behavior is undefined and may result in a segmentation fault.

Some GLib functions allocate memory. For example, `g_strdup` allocates memory and copies a string given as an argument.

```
char *s;
s = g_strdup ("Hello");
g_free (s);
```

The string literal "Hello" has 6 bytes because the string has '\0' at the end. `g_strdup` gets 6 bytes from the heap area and copies the string to the memory. `s` is assigned the start address of the memory. `g_free` returns the memory to the heap area.

`g_strdup` is described in [GLib API Reference](#). The following is extracted from the reference.

The returned string should be freed with `g_free()` when no longer needed.

If you forget to free the allocated memory it will remain until the program ends. Repeated allocation and no freeing cause memory leak. It is a bug and may bring a serious problem.

2.4.5 Const Qualifier

A const-qualified variable must be initialized at definition time, and its value cannot be modified through that variable. However, this does not prevent the memory it refers to from being freed.

```
const int x = 10; /* initialization is OK. */
```

```
x = 20; /* This is illegal because x is qualified with const */
```

If a function returns `const char *` type, the string can't be changed or freed. If a function has a `const char *` type parameter, it ensures that the parameter is not changed in the function.

```
// You never change or free the returned string.
```

```
const char *
gtk_label_get_text (
    GtkWidget *self
)
```

```
// The string 'str' remains valid while the function is executing.
```

```
void
gtk_label_set_text (
    GtkWidget *self,
    const char *str
)
```

2.5 Widgets (3)

2.5.1 Open Signal

2.5.1.1 G_APPLICATION_HANDLES_OPEN flag

We made a very simple editor in the previous section with `GtkTextView`, `GtkTextBuffer` and `GtkScrolledWindow`. We will add file-read ability to the program and improve it to a file viewer.

The simplest way to specify a filename is to use a command line argument.

```
$ ./a.out filename
```

The program will open the file and insert its contents into a `GtkTextBuffer`.

To do this, we need to know how `GtkApplication` (or `GApplication`) recognizes arguments. This is described in the [GIO API Reference – Application](#).

When `GtkApplication` is created, a flag (`GApplicationFlags`) is given as an argument.

```
GtkApplication *
gtk_application_new (const gchar *application_id, GApplicationFlags flags);
```

This tutorial explains only two flags, `G_APPLICATION_DEFAULT_FLAGS` and `G_APPLICATION_HANDLES_OPEN`.

`G_APPLICATION_FLAGS_NONE` was used instead of `G_APPLICATION_DEFAULT_FLAGS` before `GIO` version 2.73.3 (`GLib` 2.73.3 5/Aug/2022). Now it is deprecated and `G_APPLICATION_DEFAULT_FLAGS` is recommended.

For further information, see [GIO API Reference – ApplicationFlags](#) and [GIO API Reference – g_application_run](#).

We’ve already used `G_APPLICATION_DEFAULT_FLAGS`, as it is the simplest option, and no command line arguments are allowed. If you give arguments in the command line, an error will occur.

The flag `G_APPLICATION_HANDLES_OPEN` is the second simplest option. It allows command line arguments, specifically filenames.

```
app = gtk_application_new ("com.github.ToshioCP.tfv3", G_APPLICATION_HANDLES_OPEN);
```

2.5.1.2 Open Signal

When `G_APPLICATION_HANDLES_OPEN` flag is given to the application, two signals are available.

- activate signal: This signal is emitted when there’s no argument.
- open signal: This signal is emitted when there is at least one argument.

The handler of the “open” signal is defined as follows.

```
void
open (
    GApplication* self,
    gpointer files,
    gint n_files,
    gchar* hint,
    gpointer user_data
)
```

The parameters are:

- self: the application instance (usually `GtkApplication`)
- files: an array of `GFiles`. [array length=`n_files`] [element-type `GFile`]
- n_files: the number of the elements of `files`
- hint: an optional context string provided by the caller (often unused)
- user_data: user data that is set when the signal handler was connected.

2.5.2 File Viewer

2.5.2.1 What is a File Viewer?

A file viewer is a program that displays text files. Our file viewer is as follows.

- When arguments are given, it recognizes the first argument as a filename and opens it.
- The second argument and later are ignored.
- If there is no argument, it shows an error message and quits.
- If it successfully opens the file, it reads the content of the file, inserts it to `GtkTextBuffer` and shows the window.
- If it fails to open the file, it shows an error message and quit.

The program is shown below.

```
1  #include <gtk/gtk.h>
2
3  static void
4  app_activate (GApplication *app) {
5      g_printerr ("You need a filename argument.\n");
6  }
7
8  static void
9  app_open (GApplication *app, GFile ** files, int n_files, char *hint) {
10     GtkWidget *win;
11     GtkWidget *scr;
```

```

12  GtkWidget *tv;
13  GtkTextBuffer *tb;
14  char *contents;
15  gsize length;
16  char *filename_bin, *filename_utf8;
17  GError *err = NULL;
18
19  win = gtk_application_window_new (GTK_APPLICATION (app));
20  gtk_window_set_default_size (GTK_WINDOW (win), 800, 600);
21
22  scr = gtk_scrolled_window_new ();
23  gtk_window_set_child (GTK_WINDOW (win), scr);
24
25  tv = gtk_text_view_new ();
26  tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
27  gtk_text_view_set_wrap_mode (GTK_TEXT_VIEW (tv), GTK_WRAP_WORD_CHAR);
28  gtk_text_view_set_editable (GTK_TEXT_VIEW (tv), FALSE);
29  gtk_scrolled_window_set_child (GTK_SCROLLED_WINDOW (scr), tv);
30
31  if (g_file_load_contents (files[0], NULL, &contents, &length, NULL, &err)) {
32      gtk_text_buffer_set_text (tb, contents, length);
33      g_free (contents);
34      if ((filename_bin = g_file_get_basename (files[0])) != NULL) {
35          filename_utf8 = g_filename_display_name (filename_bin);
36          gtk_window_set_title (GTK_WINDOW (win), filename_utf8);
37          g_free (filename_bin);
38          g_free (filename_utf8);
39      }
40      gtk_window_present (GTK_WINDOW (win));
41  } else {
42      g_printerr ("%s.\n", err->message);
43      g_error_free (err);
44      gtk_window_destroy (GTK_WINDOW (win));
45  }
46  }
47
48  int
49  main (int argc, char **argv) {
50      GtkApplication *app;
51      int stat;
52
53      app = gtk_application_new ("com.github.ToshioCP.tfv3", G_APPLICATION_HANDLES_OPEN);
54      g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
55      g_signal_connect (app, "open", G_CALLBACK (app_open), NULL);
56      stat = g_application_run (G_APPLICATION (app), argc, argv);
57      g_object_unref (app);
58      return stat;
59  }

```

The source code is located at [/src/tfv/tfv3.c](#) in the repository. Compile and run it.

```

$ comp tfv3
$ ./a.out tfv3.c

```

The function main has only two changes from the previous version.

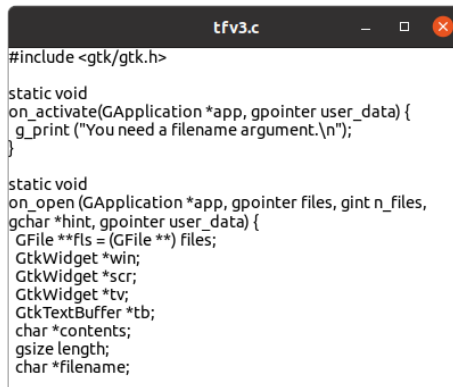


Figure 2.9: File viewer

- `G_APPLICATION_DEFAULT_FLAGS` is replaced by `G_APPLICATION_HANDLES_OPEN`
- `g_signal_connect (app, "open", G_CALLBACK (app_open), NULL)` is added.

When the flag `G_APPLICATION_HANDLES_OPEN` is given to `gtk_application_new` function, the application behaves like this:

- If the application is run without command line arguments, it emits “activate” signal when it is activated.
- If the application is run with command line arguments, it emits “open” signal when it is activated.

The handler `app_activate` becomes very simple. It just outputs a message and returns to the caller. Then the application quits immediately because no window is created.

The main work is done in the handler `app_open`.

- Creates `GtkApplicationWindow`, `GtkScrolledWindow`, `GtkTextView` and `GtkTextBuffer` and connects them together
- Sets wrap mode to `GTK_WRAP_WORD_CHAR` in `GtkTextView`
- Sets `GtkTextView` to non-editable because the program isn’t an editor but only a viewer
- Reads the file and inserts the text into `GtkTextBuffer` (this will be explained later). The window title is set with the filename. In general, filenames are sequences of bytes. To display them on a screen, they must be converted into encoded strings. The function `g_filename_display_name` converts a filename into a valid UTF-8 string.
- If the file is not opened, outputs an error message and destroys the window. This causes the application to exit.

The following is the file reading part of the program.

```
if (g_file_load_contents (files[0], NULL, &contents, &length, NULL, &err)) {
    gtk_text_buffer_set_text (tb, contents, length);
    g_free (contents);
    if ((filename_bin = g_file_get_basename (files[0])) != NULL) {
        filename_utf8 = g_filename_display_name (filename_bin);
        gtk_window_set_title (GTK_WINDOW (win), filename_utf8);
        g_free (filename_bin);
        g_free (filename_utf8);
    }
    gtk_window_present (GTK_WINDOW (win));
} else {
    g_printerr ("%s.\n", err->message);
    g_error_free (err);
    gtk_window_destroy (GTK_WINDOW (win));
}
```



```
}
```

The function `g_file_load_contents` loads the file contents into a temporary buffer, allocated automatically, and sets `contents` to point to the buffer. The length of the buffer is assigned to `length`. It returns `TRUE` if the file's contents are successfully loaded. The caller takes the ownership of the buffer and is responsible for freeing it. If an error occurs, it returns `FALSE` and sets the variable `err` to point to a newly created `GError` structure. The caller takes the ownership of the `GError` structure and is responsible for freeing it. If you want to know the details about `g_file_load_contents`, see [g file load contents](#).

If the file is read successfully, the contents are inserted into `GtkTextBuffer`. The temporary buffer `contents` is then freed. The function `g_file_get_basename` returns the basename of the pathname. Generally, a pathname of a filename is a sequence of byte data. To display the filename, it must be converted to a valid encoded string. The function `g_filename_display_name` converts a filename into a valid UTF-8 string. After the conversion, the window title is set with the filename, the memories for filenames are freed, and the window is shown.

If it fails, `g_file_load_contents` sets `err` to point to a newly created `GError` structure. The structure is:

```
struct GError {
    GQuark domain;
    int code;
    char* message;
}
```

The `message` member is the most commonly used. It points to an error message. A function `g_error_free` is used to free the memory of the structure. See [GError](#).

The program above outputs an error message, frees `err` and destroys the window and finally make the program quit.

2.5.3 GtkNotebook

`GtkNotebook` is a container widget that contains multiple widgets with tabs. It shows only one child at a time. Another child will be shown when its tab is clicked.

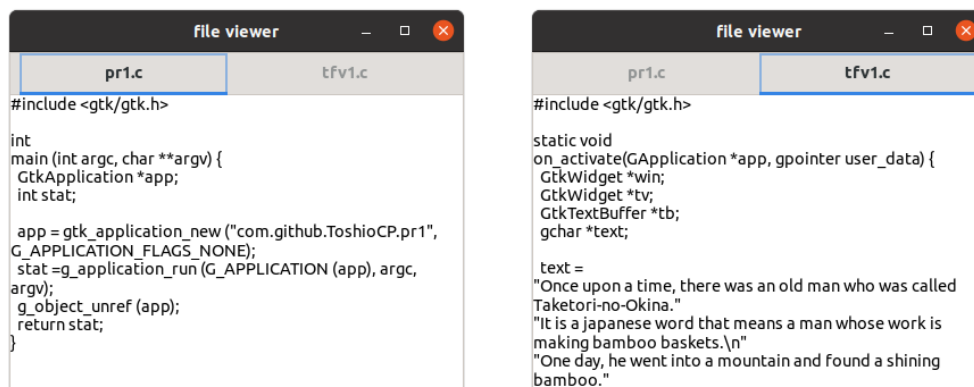


Figure 2.10: GtkNotebook

The left image is the window at the startup. The file `pr1.c` is shown and its filename is in the left tab. After clicking on the right tab, the contents of the file `tfv1.c` is shown (the right image).

The next source code is `tfv4.c`, including a `GtkNoteBook` widget. The widget is inserted as a child of a `GtkApplicationWindow` and contains multiple `GtkScrolledWindow` widgets.

```
1 #include <gtk/gtk.h>
2
```

```

3  static void
4  app_activate (GApplication *app) {
5      g_printerr ("You need filename arguments.\n");
6  }
7
8  static void
9  app_open (GApplication *app, GFile ** files, gint n_files, gchar *hint) {
10     GtkWidget *win;
11     GtkWidget *nb;
12     GtkWidget *lab;
13     GtkNotebookPage *nbp;
14     GtkWidget *scr;
15     GtkWidget *tv;
16     GtkTextBuffer *tb;
17     char *contents;
18     gsize length;
19     char *filename_bin, *filename_utf8;
20     int i;
21     GError *err = NULL;
22
23     win = gtk_application_window_new (GTK_APPLICATION (app));
24     gtk_window_set_title (GTK_WINDOW (win), "file viewer");
25     gtk_window_set_default_size (GTK_WINDOW (win), 800, 600);
26     nb = gtk_notebook_new ();
27     gtk_window_set_child (GTK_WINDOW (win), nb);
28
29     for (i = 0; i < n_files; i++) {
30         if (g_file_load_contents (files[i], NULL, &contents, &length, NULL, &err)) {
31             scr = gtk_scrolled_window_new ();
32             tv = gtk_text_view_new ();
33             tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
34             gtk_text_view_set_wrap_mode (GTK_TEXT_VIEW (tv), GTK_WRAP_WORD_CHAR);
35             gtk_text_view_set_editable (GTK_TEXT_VIEW (tv), FALSE);
36             gtk_scrolled_window_set_child (GTK_SCROLLED_WINDOW (scr), tv);
37
38             gtk_text_buffer_set_text (tb, contents, length);
39             g_free (contents);
40             filename_bin = g_file_get_basename (files[i]);
41             filename_utf8 = g_filename_display_name (filename_bin);
42             lab = gtk_label_new (filename_utf8);
43             g_free (filename_bin);
44             g_free (filename_utf8);
45             gtk_notebook_append_page (GTK_NOTEBOOK (nb), scr, lab);
46             nbp = gtk_notebook_get_page (GTK_NOTEBOOK (nb), scr);
47             g_object_set (nbp, "tab-expand", TRUE, NULL);
48         } else {
49             g_printerr ("%s.\n", err->message);
50             g_clear_error (&err);
51         }
52     }
53     if (gtk_notebook_get_n_pages (GTK_NOTEBOOK (nb)) > 0)
54         gtk_window_present (GTK_WINDOW (win));
55     else

```

```

56     gtk_window_destroy (GTK_WINDOW (win));
57 }
58
59 int
60 main (int argc, char **argv) {
61     GtkApplication *app;
62     int stat;
63
64     app = gtk_application_new ("com.github.ToshioCP.tfv4", G_APPLICATION_HANDLES_OPEN);
65     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
66     g_signal_connect (app, "open", G_CALLBACK (app_open), NULL);
67     stat = g_application_run (G_APPLICATION (app), argc, argv);
68     g_object_unref (app);
69     return stat;
70 }

```

The changes are primarily located in the `app_open` function. The numbers on the left below correspond to the line numbers in the source code.

- 11-13: Variables `nb`, `lab` and `nbp` are defined. They point to a `GtkNotebook`, `GtkLabel` and `GtkNotebookPage` respectively.
- 24: The window's title is set to "file viewer".
- 25: The default size of the window is 800x600.
- 26-27: `GtkNotebook` is created and set as a child of the `GtkApplicationWindow`.
- 29-52: For-loop. The variable `files[i]` points to *i*-th `GFile`, which is created by the `GtkApplication` from the *i*-th command line argument.
- 31-36: A `GtkScrolledWindow` and a `GtkTextView` are created. A `GtkTextBuffer` is obtained from the `GtkTextView`. The `GtkTextView` is connected to the `GtkScrolledWindow` as a child.
- 38-39: inserts the contents of the file into the `GtkTextBuffer` and frees the memory referenced by `contents`.
- 40-44: Retrieves the filename from the `GFile` using `g_file_get_basename`. The filename is a byte-sequence without a specific encoding. It needs to be converted into a UTF-8 string with `g_filename_display_name` before setting the label text. The strings `filename_bin` and `filename_utf8` are freed.
- 45-46: Appends a `GtkScrolledWindow` to the `GtkNotebook` as a child and the `GtkLabel` is set as the child's tab. At the same time, a `GtkNoteBookPage` is created automatically. The function `gtk_notebook_get_page` returns the `GtkNotebookPage` of the child (`GtkScrolledWindow`).
- 47: `GtkNotebookPage` has "tab-expand" property. If it is set to `TRUE` then the tab expands horizontally as long as possible. If it is `FALSE`, then the width of the tab is determined by the size of the label. `g_object_set` is a general function to set properties of objects. See [GObject API Reference – g_object_set](#).
- 48-50: If it fails to read the file, the error message is shown. The function `g_clear_error (&err)` works like `g_error_free (err); err = NULL;`.
- 53-56: If at least one page exists, the window is shown. Otherwise, the window is destroyed and the application quits.

2.6 Defining a Final Class

2.6.1 A Very Simple Editor

We made a very simple file viewer in the previous section. Now we go on to rewrite it and turn it into a very simple editor. Its source file is `tfe1.c` (text file editor 1) under `tfe` directory.

`GtkTextView` is a multi-line editor. So, we don't need to write the editor from scratch. We just add two things to the file viewer:

- Pointers to GFile instances.
- A text-save function.

There are a couple of ways to store the pointers.

- Use global variables
- Make a child class of GtkTextView and its each instance holds a pointer to the GFile instance.

Using global variables is easy to implement. Define a sufficient size pointer array to GFile. For example,

```
GFile *f[20];
```

The variable `f[i]` corresponds to the file associated with the *i*-th GtkNotebookPage.

However, there are two problems. First, the array has a fixed size. If a user provides too many arguments (more than 20 in the example above), it becomes impossible to store all the GFile instances. Second, global variables make maintenance difficult. Our program is small so far, but as a program grows larger, maintaining global variables becomes increasingly challenging. When you need to check a global variable, you must examine every part of the code that references it.

Making a child class is a good idea in terms of maintenance. We prefer this approach over using global variables.

Note that we are discussing “child classes,” not “child widgets.” These two concepts are entirely different: a “class” is a term within the GObject system, whereas a “widget” refers to the UI hierarchy. If you are not familiar with GObject, please see:

- [GObject API reference](#)
- [GObject tutorial for beginners](#)

A child class inherits everything from the parent and, in addition, extends its performance. We will define TfeTextView as a child class of GtkTextView. It has everything that GtkTextView has and adds a pointer to a GFile.

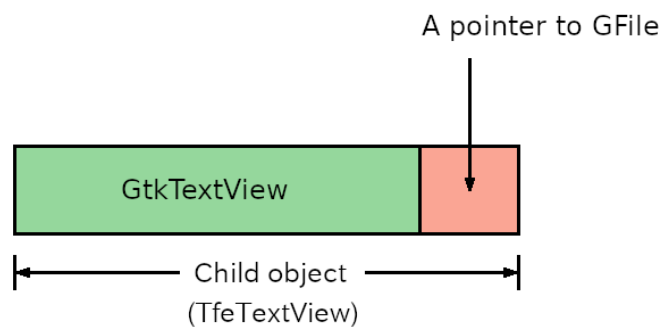


Figure 2.11: Child object of GtkTextView

2.6.2 How to Define a Child Class of GtkTextView

You need to know GObject system convention. First, look at the program below.

```
#define TFE_TYPE_TEXT_VIEW tfe_text_view_get_type ()
G_DECLARE_FINAL_TYPE (TfeTextView, tfe_text_view, TFE, TEXT_VIEW, GtkTextView)

struct _TfeTextView
{
    GtkTextView parent;
    GFile *file;
};
```

```

G_DEFINE_FINAL_TYPE (TfeTextView, tfe_text_view, GTK_TYPE_TEXT_VIEW);

static void
tfe_text_view_init (TfeTextView *tv) {
}

static void
tfe_text_view_class_init (TfeTextViewClass *class) {
}

/* --- Here are some functions --- */

GtkWidget *
tfe_text_view_new (void) {
    return GTK_WIDGET (g_object_new (TFE_TYPE_TEXT_VIEW, NULL));
}

```

- The name “TfeTextView” is composed of two parts: “Tfe” and “TextView.” “Tfe” serves as the prefix (or namespace), while “TextView” represents the object name.
- There are three different identifier patterns. TfeTextView (camel case), tfe_text_view (this is used for functions) and TFE_TYPE_TEXT_VIEW (This is used to cast an object to TfeTextView).
- First, define the TFE_TYPE_TEXT_VIEW macro as tfe_text_view_get_type(). Macro names follow the pattern (prefix)_TYPE_(object) and are written in uppercase. The replacement text is always (prefix)_(object)_get_type() and are written in lower case. This definition must be put before G_DECLARE_FINAL_TYPE macro.
- The arguments of G_DECLARE_FINAL_TYPE macro are the child class name in camel case, lower case with underscore, prefix (upper case), object (upper case with underscore) and parent class name (camel case). The following two C structures are declared in the expansion of the macro.
 - typedef struct _TfeTextView TfeTextView
 - typedef struct {GtkTextViewClass parent_class; } TfeTextViewClass;
- These declarations tell us that TfeTextView and TfeTextViewClass are C structures. “TfeTextView” has two meanings, class name and C structure name. The C structure TfeTextView is called an instance. Similarly, TfeTextViewClass is called a class.
- Declare the structure _TfeTextView. The underscore is necessary. The first member is the parent object (C structure). Notice this is not a pointer but the object itself. The second member and later are members of the child object. TfeTextView structure has a pointer to a GFile instance as a member.
- G_DEFINE_FINAL_TYPE macro. The arguments are the child object name in camel case, lower case with underscore and parent object type (prefix)_TYPE_(module). This macro is mainly used to register the new class to the type system. The type system is a base system of GObject. Every class has its own type. The types of GObject, GtkWidget and TfeTextView are G_TYPE_OBJECT, GTK_TYPE_WIDGET and TFE_TYPE_TEXT_VIEW respectively. For example, TFE_TYPE_TEXT_VIEW is a macro and it is expanded to a function tfe_text_view_get_type(). It returns a unique GType identifier representing the type within the GObject.
- The instance init function tfe_text_view_init is called when the instance is created. It is the same as a constructor in other object oriented languages.
- The class init function tfe_text_view_class_init is called when the class is created.
- Some public functions can be located between tfe_text_view_class_init and tfe_text_view_new functions. Public functions are open to other instances and they can be called anywhere in the program. They are the same as public method in other object oriented languages. The variable tv contains a pointer to the TfeTextView object (C structure). It has a member file and it is pointed to by tv->file.
- TfeTextView instance creation function is tfe_text_view_new. Its name is (prefix)_(object)_new.

It uses `g_object_new` function to create the instance. The arguments are (prefix)_TYPE_(object), a list to initialize properties and NULL. NULL is the end mark of the property list. No property is initialized here. And the return value is cast to `GtkWidget`.

This program shows the overview of how to define a child class.

2.6.3 Close-request Signal

Imagine that you are using this editor. First, you run the editor with arguments. The arguments are filenames. The editor reads the files and shows the window with the text of files in it. Then you edit the text. After you finish editing, you click on the close button of the window and quit the editor. The editor updates files just before the window closes.

`GtkWindow` emits the “close-request” signal when the close button is clicked. We will connect the signal and the handler `before_close`. (A handler is a C function which is connected to a signal.) The function `before_close` is called when the signal “close-request” is emitted.

```
g_signal_connect (win, "close-request", G_CALLBACK (before_close), NULL);
```

The signal “close-request” is defined on the `GtkWindow` class and it is inherited by the descendant classes. The first argument `win` is a `GtkApplicationWindow`, which is the child class of the `GtkWindow` class. The macro `g_signal_connect` connects the “close-request” signal to the `before_close` handler. The handler needs `G_CALLBACK` cast. The `before_close` function is as follows.

```
1  static gboolean
2  before_close (GtkWindow *win, GtkWidget *nb) {
3      GtkWidget *scr;
4      GtkWidget *tv;
5      GFile *file;
6      char *pathname;
7      GtkTextBuffer *tb;
8      GtkTextIter start_iter;
9      GtkTextIter end_iter;
10     char *contents;
11     unsigned int n;
12     unsigned int i;
13     GError *err = NULL;
14
15     n = gtk_notebook_get_n_pages (GTK_NOTEBOOK (nb));
16     for (i = 0; i < n; ++i) {
17         scr = gtk_notebook_get_nth_page (GTK_NOTEBOOK (nb), i);
18         tv = gtk_scrolled_window_get_child (GTK_SCROLLED_WINDOW (scr));
19         file = tfe_text_view_get_file (TFE_TEXT_VIEW (tv));
20         tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
21         gtk_text_buffer_get_bounds (tb, &start_iter, &end_iter);
22         contents = gtk_text_buffer_get_text (tb, &start_iter, &end_iter, FALSE);
23         if (file && ! g_file_replace_contents (file, contents, strlen (contents), NULL, TRUE, G_FILE_CRE
24             g_printerr ("%s.\n", err->message);
25             g_clear_error (&err);
26     }
27     g_free (contents);
28     tfe_text_view_set_file (TFE_TEXT_VIEW (tv), NULL);
29 }
30 return FALSE;
31 }
```

The numbers on the left are line numbers.

- 15: The number of note book pages is assigned to `n`.
- 16-29: For loop with regard to the index to each page.
- 17-19: `scr`, `tv` and `file` hold the pointers to the `GtkScrolledWindow`, `TfeTextView` and `GFile`. The function `tfe_text_view_get_file` returns the pointer to the `GFile` instance that `tv` holds. The `GFile` instance is owned by `tv`. The function will be shown later.
- 20-22: `tb` holds a pointer to the `GtkTextBuffer` of the `TfeTextView`. The function `gtk_text_buffer_get_bounds` assigns the start and end iterators of the buffer to `start_iter` and `end_iter` respectively. An iterator represents a specific position within the buffer. Then, `gtk_text_buffer_get_text` returns the text between `start_iter` and `end_iter`, which encompasses the entire content of the buffer.
- 23-26: The text is saved to the file. If it fails, error messages are displayed. The `GError` instance must be freed and the pointer `err` needs to be `NULL` for the next run in the loop.
- 27: `contents` are freed.
- 28: `GFile` is no longer needed. The function `tfe_tet_view_set_file` decrements the reference count of the `GFile` within `tv` and replaces it with a new `GFile` or `NULL`. Reference counts will be explained in a later section ([section 11](#)). When the reference count reaches zero, the `GFile` instance is automatically destroyed.

2.6.4 Source code of `tfe1.c`

The following is the whole source code of `tfe1.c`.

```

1  #include <gtk/gtk.h>
2
3  /* Define TfeTextView Widget which is the child class of GtkTextView */
4
5  #define TFE_TYPE_TEXT_VIEW tfe_text_view_get_type ()
6  G_DECLARE_FINAL_TYPE (TfeTextView, tfe_text_view, TFE, TEXT_VIEW, GtkTextView)
7
8  struct _TfeTextView
9  {
10     GtkTextView parent;
11     GFile *file;
12 };
13
14 G_DEFINE_FINAL_TYPE (TfeTextView, tfe_text_view, GTK_TYPE_TEXT_VIEW);
15
16 static void
17 tfe_text_view_init (TfeTextView *tv) {
18     tv->file = NULL;
19 }
20
21 static void
22 tfe_text_view_class_init (TfeTextViewClass *class) {
23 }
24
25 void
26 tfe_text_view_set_file (TfeTextView *tv, GFile *f) {
27     if (tv->file) {
28         g_object_unref (tv->file);
29     }
30     if (f)
31         g_object_ref (f); /* get the ownership of the GFile */
32     tv->file = f;

```

```

33 }
34
35 GFile *
36 tfe_text_view_get_file (TfeTextView *tv) {
37     return tv->file;
38 }
39
40 GtkWidget *
41 tfe_text_view_new (void) {
42     return GTK_WIDGET (g_object_new (TFE_TYPE_TEXT_VIEW, NULL));
43 }
44
45 /* ----- end of the definition of TfeTextView ----- */
46
47 static gboolean
48 before_close (GtkWindow *win, GtkWidget *nb) {
49     GtkWidget *scr;
50     GtkWidget *tv;
51     GFile *file;
52     char *pathname;
53     GtkTextBuffer *tb;
54     GtkTextIter start_iter;
55     GtkTextIter end_iter;
56     char *contents;
57     unsigned int n;
58     unsigned int i;
59     GError *err = NULL;
60
61     n = gtk_notebook_get_n_pages (GTK_NOTEBOOK (nb));
62     for (i = 0; i < n; ++i) {
63         scr = gtk_notebook_get_nth_page (GTK_NOTEBOOK (nb), i);
64         tv = gtk_scrolled_window_get_child (GTK_SCROLLED_WINDOW (scr));
65         file = tfe_text_view_get_file (TFE_TEXT_VIEW (tv));
66         tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
67         gtk_text_buffer_get_bounds (tb, &start_iter, &end_iter);
68         contents = gtk_text_buffer_get_text (tb, &start_iter, &end_iter, FALSE);
69         if (file && ! g_file_replace_contents (file, contents, strlen (contents), NULL, TRUE, G_FILE_CRE
70             g_printerr ("%s.\n", err->message);
71             g_clear_error (&err);
72         }
73         g_free (contents);
74         tfe_text_view_set_file (TFE_TEXT_VIEW (tv), NULL);
75     }
76     return FALSE;
77 }
78
79 static void
80 app_activate (GApplication *app) {
81     g_print ("You need to give filenames as arguments.\n");
82 }
83
84 static void
85 app_open (GApplication *app, GFile ** files, gint n_files, gchar *hint) {

```



```

86  GtkWidget *win;
87  GtkWidget *nb;
88  GtkWidget *lab;
89  GtkNotebookPage *nbp;
90  GtkWidget *scr;
91  GtkWidget *tv;
92  GtkTextBuffer *tb;
93  char *contents;
94  gsize length;
95  char *filename_bin, *filename_utf;
96  int i;
97  GError *err = NULL;
98
99  win = gtk_application_window_new (GTK_APPLICATION (app));
100  gtk_window_set_title (GTK_WINDOW (win), "file editor");
101  gtk_window_set_default_size (GTK_WINDOW (win), 600, 400);
102
103  nb = gtk_notebook_new ();
104  gtk_window_set_child (GTK_WINDOW (win), nb);
105
106  for (i = 0; i < n_files; i++) {
107      if (g_file_load_contents (files[i], NULL, &contents, &length, NULL, &err)) {
108          scr = gtk_scrolled_window_new ();
109          tv = tfe_text_view_new ();
110          tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
111          gtk_text_view_set_wrap_mode (GTK_TEXT_VIEW (tv), GTK_WRAP_WORD_CHAR);
112          gtk_scrolled_window_set_child (GTK_SCROLLED_WINDOW (scr), tv);
113
114          tfe_text_view_set_file (TFE_TEXT_VIEW (tv), files[i]);
115          gtk_text_buffer_set_text (tb, contents, length);
116          g_free (contents);
117          filename_bin = g_file_get_basename (files[i]);
118          filename_utf = g_filename_display_name (filename_bin);
119          lab = gtk_label_new (filename_utf);
120          g_free (filename_bin);
121          g_free (filename_utf);
122          gtk_notebook_append_page (GTK_NOTEBOOK (nb), scr, lab);
123          nbp = gtk_notebook_get_page (GTK_NOTEBOOK (nb), scr);
124          g_object_set (nbp, "tab-expand", TRUE, NULL);
125      } else {
126          g_printerr ("%s.\n", err->message);
127          g_clear_error (&err);
128      }
129  }
130  if (gtk_notebook_get_n_pages (GTK_NOTEBOOK (nb)) > 0) {
131      g_signal_connect (win, "close-request", G_CALLBACK (before_close), nb);
132      gtk_window_present (GTK_WINDOW (win));
133  } else
134      gtk_window_destroy (GTK_WINDOW (win));
135  }
136
137  int
138  main (int argc, char **argv) {

```

```

139  GtkApplication *app;
140  int stat;
141
142  app = gtk_application_new ("com.github.ToshioCP.tfe1", G_APPLICATION_HANDLES_OPEN);
143  g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
144  g_signal_connect (app, "open", G_CALLBACK (app_open), NULL);
145  stat = g_application_run (G_APPLICATION (app), argc, argv);
146  g_object_unref (app);
147  return stat;
148 }

```

- 25-33: The function `tfe_text_view_set_file` decreases the reference count of `tv->file` by one (`g_object_unref`). This means `tv` releases the `GFile` instance ownership. Then, the function replaces `tv->file` with the new `GFile f` or `NULL`. If it is a `GFile`, the function increases the reference count of `f` by one and `tv` gets the ownership of `f`. For reference count, see [Section 11](#) for further information.
- 131: The “close-request” signal is connected to `before_close` handler. The fourth argument is called “user data” and it will be the second argument of the signal handler. So, `nb` is given to `before_close` as the second argument.

Now it’s time to compile and run.

```

$ cd src/tfe
$ comp tfe1
$ ./a.out taketori.txt`.

```

Modify the contents and close the window. Make sure that the file is modified.

Now we got a very simple editor. It’s not smart. We need more features like open, save, saveas, change font and so on. We will add them in the following sections.

2.7 GtkBuilder and UI Files

We developed a very simple editor in the previous section. While it functions by reading and writing files at the start and end of the program, there is room for improvement. It would be more user-friendly to have “New”, “Open”, “Save”, and “Close” buttons. This section describes how to incorporate these buttons into the window. We will focus on building the widgets and omit the file I/O functions for now.

2.7.1 New, Open, Save and Close Buttons

The screenshot above shows the layout. The source code `tfe2.c` is as follows.

```

1  #include <gtk/gtk.h>
2
3  static void
4  app_activate (GApplication *app) {
5      GtkWidget *win, *nb, *scr, *tv, *label;
6      GtkWidget *boxv, *boxh;
7      GtkWidget *dmy1, *dmy2, *dmy3;
8      GtkWidget *bttn; /* button for new */
9      GtkWidget *btno; /* button for open */
10     GtkWidget *btns; /* button for save */
11     GtkWidget *btnc; /* button for close */
12     int i;
13     GtkNotebookPage *nbp;
14     char *files[] = {"file-1", "file-2"};

```

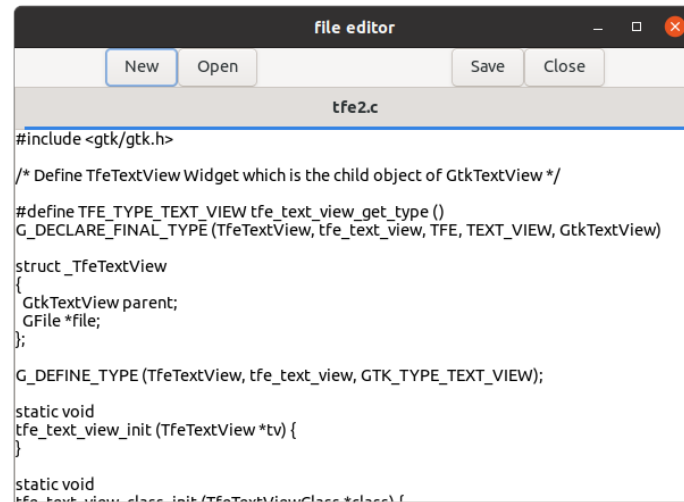


Figure 2.12: Screenshot of the file editor

```

15
16 win = gtk_application_window_new (GTK_APPLICATION (app));
17 gtk_window_set_title (GTK_WINDOW (win), "file editor");
18 gtk_window_set_default_size (GTK_WINDOW (win), 600, 400);
19
20 boxv = gtk_box_new (GTK_ORIENTATION_VERTICAL, 0);
21 gtk_window_set_child (GTK_WINDOW (win), boxv);
22
23 boxh = gtk_box_new (GTK_ORIENTATION_HORIZONTAL, 0);
24 gtk_box_append (GTK_BOX (boxv), boxh);
25
26 dmy1 = gtk_label_new(NULL); /* dummy label for left space */
27 gtk_label_set_width_chars (GTK_LABEL (dmy1), 10);
28 dmy2 = gtk_label_new(NULL); /* dummy label for center space */
29 gtk_widget_set_hexpand (dmy2, TRUE);
30 dmy3 = gtk_label_new(NULL); /* dummy label for right space */
31 gtk_label_set_width_chars (GTK_LABEL (dmy3), 10);
32 btnn = gtk_button_new_with_label ("New");
33 btno = gtk_button_new_with_label ("Open");
34 btns = gtk_button_new_with_label ("Save");
35 btnc = gtk_button_new_with_label ("Close");
36
37 gtk_box_append (GTK_BOX (boxh), dmy1);
38 gtk_box_append (GTK_BOX (boxh), btnn);
39 gtk_box_append (GTK_BOX (boxh), btno);
40 gtk_box_append (GTK_BOX (boxh), dmy2);
41 gtk_box_append (GTK_BOX (boxh), btns);
42 gtk_box_append (GTK_BOX (boxh), btnc);
43 gtk_box_append (GTK_BOX (boxh), dmy3);
44
45 nb = gtk_notebook_new ();
46 gtk_widget_set_hexpand (nb, TRUE);
47 gtk_widget_set_vexpand (nb, TRUE);
48 gtk_box_append (GTK_BOX (boxv), nb);
49

```

```

50  for (i = 0; i < 2; i++) {
51      scr = gtk_scrolled_window_new ();
52      tv = gtk_text_view_new ();
53      label = gtk_label_new (files[i]);
54      gtk_text_view_set_wrap_mode (GTK_TEXT_VIEW (tv), GTK_WRAP_WORD_CHAR);
55      gtk_scrolled_window_set_child (GTK_SCROLLED_WINDOW (scr), tv);
56      gtk_notebook_append_page (GTK_NOTEBOOK (nb), scr, label);
57      nbp = gtk_notebook_get_page (GTK_NOTEBOOK (nb), scr);
58      g_object_set (nbp, "tab-expand", TRUE, NULL);
59  }
60  gtk_window_present (GTK_WINDOW (win));
61  }
62
63  int
64  main (int argc, char **argv) {
65      GtkApplication *app;
66      int stat;
67
68      app = gtk_application_new ("com.github.ToshioCP.tfe2", G_APPLICATION_DEFAULT_FLAGS);
69      g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
70      stat = g_application_run (G_APPLICATION (app), argc, argv);
71      g_object_unref (app);
72      return stat;
73  }

```

This program just builds the widgets. Other features will be implemented in the later sections.

The function `app_activate` builds the widgets in the main application window.

- 16-18: Creates a `GtkApplicationWindow` instance and sets the title and default size.
- 20-21: Creates a vertical `GtkBox` instance `boxv` as a child of `GtkApplicationWindow`.
- 23-24: Creates a horizontal `GtkBox` instance `boxh` and appends it to `boxv` as the first child.
- 26-31: Creates three dummy labels. The `dmy1` and `dmy3` labels are ten characters wide, while the `dmy2` label has its `hexpand` property set to `TRUE` to fill the available space.
- 32-35: Creates four buttons `New`, `Open`, `Save`, and `Close`.
- 37-43: Appends these label and button instances to `boxh`.
- 45-48: Creates a `GtkNotebook` instance with `hexpand` and `vexpand` properties set to `TRUE`, allowing it to expand in both directions. It is then appended to `boxv` as the second child.
- 50-59: Creates, `GtkScrolledWindow`, `GtkTextView`, and `GtkLabel` instances in the for-loop. The scrolled window `scr` takes `tv` as a child. The function `gtk_notebook_append_page` appends `scr` as a page and `label` as a page tab to the note book. The function `gtk_notebook_get_page` retrieves the `GtkNotebookPage` instance that corresponds to `scr`, a child widget of `nb`. `GtkNotebookPage` class does not have a function to set the property “tab-expand”, so the general property setting function `g_object_set` is used here. The property is set to `TRUE` to expand the tab horizontally.

Building widgets takes 44 lines (= 59-16+1). We also needed many variables (`boxv`, `boxh`, `dmy1`, ...), and most of them are used only for building the widgets. Is there a better way to reduce this overhead?

Gtk provides `GtkBuilder`. It reads user interface (UI) data and builds a window. It simplifies this tedious process.

2.7.2 UI Files

Look at the UI file `tfe3.ui` that defines the widget structure.

```

1  <?xml version="1.0" encoding="UTF-8"?>
2  <interface>

```

```

3  <object class="GtkApplicationWindow" id="win">
4    <property name="title">file editor</property>
5    <property name="default-width">600</property>
6    <property name="default-height">400</property>
7    <child>
8      <object class="GtkBox">
9        <property name="orientation">GTK_ORIENTATION_VERTICAL</property>
10       <child>
11         <object class="GtkBox">
12           <property name="orientation">GTK_ORIENTATION_HORIZONTAL</property>
13           <child>
14             <object class="GtkLabel">
15               <property name="width-chars">10</property>
16             </object>
17           </child>
18           <child>
19             <object class="GtkButton">
20               <property name="label">New</property>
21             </object>
22           </child>
23           <child>
24             <object class="GtkButton">
25               <property name="label">Open</property>
26             </object>
27           </child>
28           <child>
29             <object class="GtkLabel">
30               <property name="hexpand">TRUE</property>
31             </object>
32           </child>
33           <child>
34             <object class="GtkButton">
35               <property name="label">Save</property>
36             </object>
37           </child>
38           <child>
39             <object class="GtkButton">
40               <property name="label">Close</property>
41             </object>
42           </child>
43           <child>
44             <object class="GtkLabel">
45               <property name="width-chars">10</property>
46             </object>
47           </child>
48         </object>
49       </child>
50     <child>
51       <object class="GtkNotebook" id="nb">
52         <property name="hexpand">TRUE</property>
53         <property name="vexpand">TRUE</property>
54       <child>
55         <object class="GtkScrolledWindow">

```

```

56         <property name="hscrollbar-policy">automatic</property>
57         <property name="vscrollbar-policy">automatic</property>
58         <child>
59             <object class="GtkTextView">
60                 </object>
61         </child>
62     </object>
63 </child>
64 <child type="tab">
65     <object class="GtkLabel">
66         <property name="label">file-1</property>
67     </object>
68 </child>
69 <child>
70     <object class="GtkScrolledWindow">
71         <property name="hscrollbar-policy">automatic</property>
72         <property name="vscrollbar-policy">automatic</property>
73         <child>
74             <object class="GtkTextView">
75                 </object>
76         </child>
77     </object>
78 </child>
79 <child type="tab">
80     <object class="GtkLabel">
81         <property name="label">file-2</property>
82     </object>
83 </child>
84 </object>
85 </child>
86 </object>
87 </child>
88 </object>
89 </interface>

```

This is an XML file. The extension “ui” means User Interface.

Tags are enclosed in angle brackets, beginning with < and ending with >. They generally come in pairs: a start tag and an end tag. For instance, <interface> opens the element, while </interface> closes it.

The end tag is identified by a forward slash immediately following the opening bracket, and its name must match the corresponding start tag. Additionally, certain tags (such as object tags) can include class and id attributes within their start tag to provide extra metadata.

- 1: XML declaration. This is a special tag and doesn’t follow the “start and end tag” rule. It specifies that the XML version is 1.0 and the encoding is UTF-8.
- 2, 89: The UI file begins and ends with interface tags.
- 3: The object tag defines an instance with the class of `GtkApplicationWindow` and the id of `win`. The class attribute specifies the type of the object to create, while the id attribute provides a unique name used to retrieve the object within your C code.
- 4-6: Property tags define the values of the properties on the object that is defined with an object tag just before the property definitions. These three lines define the properties of the top-level window `win`.
 - title: “file editor”
 - default-width: 600

- default-height: 400
- 7: Child tag means that the following object is a child widget. For example, line 7 defines the GtkBox as a child widget of the GtkApplicationWindow `win`.
- 8-49: Defines a horizontal box comprising labels and buttons.
- 50-85: Defines the GtkNotebook as a child of the vertical box. It contains two pairs of children, where each pair consists of a content area (a scrolled window) and a tab (a label). While a standard `<child>` tag defines the content, a `<child>` tag with `type="tab"` defines the tab itself. Note that `GtkNotebookPage` cannot be defined directly in UI files because it is not a widget; it is an auxiliary object derived directly from `GObject`.

Compare this UI file with the `app_activate` function of `tfe2.c`. While both methods produce the same window and widget hierarchy, the UI file approach offers superior readability and easier maintenance.

2.7.2.1 Gtk4-Builder-Tool

You can verify and optimize your UI files using the `gtk4-builder-tool` utility.

- **Validation:** `gtk4-builder-tool validate <ui_file_name>` checks the file for errors. If the UI file contains syntax errors or invalid object definitions, the tool will point them out.
- **Simplification:** `gtk4-builder-tool simplify <ui_file_name>` cleans up the UI file and prints the result to the terminal.
 - The `--replace` option: Using this option replaces the original file with the simplified version.
 - Removing Defaults: If a property is set to its default value, the tool removes it to save space. For example, since the default orientation for `GtkBox` widgets is horizontal, that line would be deleted (ex. line 12 in `tfe3.ui`).
 - Value Conversion: Some values are shortened; for instance, “TRUE” and “FALSE” are converted to “1” and “0”.

Example:

```
$ gtk-builder-tool validate tfe/tfe3.ui
$
```

Since `tfe3.ui` contains no errors, `gtk-builder-tool` produced no output.

A Note on Maintenance

While the simplify tool makes the file smaller, it can sometimes make it harder to read. For example, it converts “TRUE” and “FALSE” into “1” and “0” respectively. Keeping “TRUE” or “FALSE” is better for maintenance because it is more readable for humans than “1” or “0”.

It is always a good practice to validate your UI file before deploying your application.

2.7.3 GtkBuilder

The file `tfe3.c` builds the same widget as `tfe2.c` using `tfe3.ui`. It is far shorter than `tfe2.c` (39 lines against 73 lines).

```
$ wc tfe/tfe2.c tfe/tfe3.c
 73  274 2557 tfe/tfe2.c
 38  113 1093 tfe/tfe3.c
111  387 3650 total
$
```

The following is the file `tfe3.c`

```
1 #include <gtk/gtk.h>
2
3 static void
4 app_activate (GApplication *app) {
```

```

5   GtkWidget *win;
6   GtkWidget *nb;
7   GtkWidget *scr;
8   GtkNotebookPage *nbp;
9   int n_pages;
10  GtkBuilder *build;
11
12  build = gtk_builder_new_from_file ("tfe3.ui");
13  win = GTK_WIDGET (gtk_builder_get_object (build, "win"));
14  nb = GTK_WIDGET (gtk_builder_get_object (build, "nb"));
15  gtk_window_set_application (GTK_WINDOW (win), GTK_APPLICATION (app));
16  g_object_unref(build);
17
18  n_pages = gtk_notebook_get_n_pages (GTK_NOTEBOOK (nb));
19  for (int i = 0; i < n_pages; i++) {
20      scr = gtk_notebook_get_nth_page (GTK_NOTEBOOK (nb), i);
21      nbp = gtk_notebook_get_page (GTK_NOTEBOOK (nb), scr);
22      g_object_set (nbp, "tab-expand", TRUE, NULL);
23  }
24
25  gtk_window_present (GTK_WINDOW (win));
26 }
27
28 int
29 main (int argc, char **argv) {
30     GtkApplication *app;
31     int stat;
32
33     app = gtk_application_new ("com.github.ToshioCP.tfe3", G_APPLICATION_DEFAULT_FLAGS);
34     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
35     stat = g_application_run (G_APPLICATION (app), argc, argv);
36     g_object_unref (app);
37     return stat;
38 }

```

GtkBuilder builds widgets based on a UI file.

- 12: The function `gtk_builder_new_from_file` reads the file `tfe3.ui`. Then, it builds the widgets and creates a `GtkBuilder` object. All the widgets are connected based on the parent-children relationship described in the UI file.
- 13, 14: We can retrieve objects from the builder object using the `gtk_builder_get_object` function. The top level window, which has an id of “win” in the UI file, is taken and assigned to the variable `win`. Similarly, The `GtkNotbook` instance with the id “nb” is assignd to the variable `nb`.
- 15: The function `gtk_window_set_application` adds `win` to the application’s list of managed windows. You can use `gtk_application_add_window` instead of this function. The result is identical.
- 16: After the window and the application are connected, we no longer need the `GtkBuilder` instance. It is released with the `g_object_unref` function. 18-23: Sets the “tab-expand” properties of the `GtkNotebookPage` instances to `TRUE`. Since `GtkNotebookPage` is not a widget, its properties cannot be set in the UI file; you must set them in your C source code.

2.7.3.1 Using UI String

GtkBuilder can build widgets with a string. Use `gtk_builder_new_from_string` instead of `gtk_builder_new_from_file`.


```

char *uistring;

uistring =
"<interface>"
  "<object class=\"GtkApplicationWindow\" id=\"win\">"
    "<property name=\"title\">file editor</property>"
    "<property name=\"default-width\">600</property>"
    "<property name=\"default-height\">400</property>"
    "<child>"
      "<object class=\"GtkBox\">"
        "<property name=\"orientation\">GTK_ORIENTATION_VERTICAL</property>"
      ... ..
    "</child>"
  "</object>"
"</interface>";

build = gtk_builder_new_from_string (uistring, -1);

```

This method has an advantage and disadvantage. The advantage is that the UI string is written in the source code. So, no UI file is needed at runtime. The disadvantage is that defining the UI as a C string can be tedious, as the xml needs quoting and special characters need escaping. If you want to use this method, you should write a script that converts UI files into C string literals.

- Replace backslashes with two backslashes.
- Add a backslash before each double quote.
- Add double quotes at the left and right of the string in each line.

Or, if you have `jq` installed, you can use `jq -R < tfe3.ui` to do the quoting and escaping for you.

2.7.3.2 GResources

A GResource is similar to a UI string, except that a GResource is a binary data. The `glib-compile-resources` program compiles UI files into GResources. It can compile not only text files but also binary files such as images, sounds and so on. After compilation, it bundles them up into one GResource object.

An xml file is necessary for the resource compiler `glib-compile-resources`. It describes resource files.

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <gresources>
3   <gresource prefix="/com/github/ToshioCP/tfe3">
4     <file>tfe3.ui</file>
5   </gresource>
6 </gresources>

```

- 2: “gresources” tag can include multiple “gresource” tags. However, this xml has only one “gresource” tag.
- 3: A “gresource” tag can have a prefix attribute. This tag defines that the prefix for `tfe3.ui` is `/com/github/ToshioCP/tfe3`. A prefix works like a namespace. Even if you have two same name resources, GTK can distinguish them with their different prefixes.
- 4: The name of the GResource is `tfe3.ui`. The resource will be referred to as “`/com/github/ToshioCP/tfe3/tfe3.ui`” by GtkBuilder. The pattern is “prefix” + “name”. If you want to add more files, insert them between line 4 and 5.

Save this xml text to `tfe3.gresource.xml`. The GResource compiler `glib-compile-resources` shows its usage with the argument `--help`.

```
$ glib-compile-resources --help
```

Usage:

```
glib-compile-resources [OPTION...] FILE
```

Compile a resource specification into a resource file.
 Resource specification files have the extension `.gresource.xml`,
 and the resource file have the extension called `.gresource`.

Help Options:

`-h, --help` Show help options

Application Options:

<code>--version</code>	Show program version and exit
<code>--target=FILE</code>	Name of the output file
<code>--sourcedir=DIRECTORY</code>	The directories to load files referenced in FILE from (default: current directory)
<code>--generate</code>	Generate output in the format selected for by the target filename extension
<code>--generate-header</code>	Generate source header
<code>--generate-source</code>	Generate source code used to link in the resource file into your code
<code>--generate-dependencies</code>	Generate dependency list
<code>--dependency-file=FILE</code>	Name of the dependency file to generate
<code>--generate-phony-targets</code>	Include phony targets in the generated dependency file
<code>--manual-register</code>	Don't automatically create and register resource
<code>--internal</code>	Don't export functions; declare them <code>G_GNUC_INTERNAL</code>
<code>--external-data</code>	Don't embed resource data in the C file; assume it's linked externally
<code>--c-name=IDENTIFIER</code>	C identifier name used for the generated source code
<code>-C, --compiler=COMMAND</code>	The target C compiler (default: the CC environment variable)

\$

Usually, the resources are converted to a C source file. Two options `--target` and `--generate-source` are added to the command line. Move your current directory to `/src/tfe` and run the command below, then a C source file `resources.c` will be generated.

```
$ glib-compile-resources tfe3.gresource.xml --target=resources.c --generate-source
```

Modify `tfe3.c` and save it as `tfe3_r.c`.

```
#include "resources.c"
...
...
build = gtk_builder_new_from_resource ("/com/github/ToshioCP/tfe3/tfe3.ui");
...
...
```

The function `gtk_builder_new_from_resource` builds widgets from a resource.

Then, compile and run it.

```
$ comp tfe3_r
$ ./a.out
```

A window appears and it is the same as the screen shown by `tfe3.c`.

Generally, using GResources is the best approach for C programs. They allow you to use embedded assets in various widgets. For example, a `GtkImage` can be created from a resource path, provided that the original image file has been compiled into the bundle using `glib-compile-resources`.

```
GtkWidget *image;
image = gtk_image_new_from_resource (const char* resource_path)
```

2.8 Build System

2.8.1 Managing Big Source Files

We've compiled a small editor so far. The program is not complicated yet, but if it grows bigger, it will be difficult to maintain. So, we should do the following now.

- We've had only one C source file and put everything in it. We need to divide and organize it.
- There are two compilers, `gcc` and `glib-compile-resources`. We should control them with one build tool.

2.8.2 Divide a C Source File into Two Parts.

When you divide C source file into several parts, each file should contain one thing. For example, our source has two things, the definition of `TfeTextView` and functions related to `GtkApplication` and `GtkApplicationWindow`. It is a good idea to separate them into two files, `tfetextview.c` and `tfe.c`.

- `tfetextview.c` includes the definition and functions of `TfeTextView`.
- `tfe.c` includes functions like `main`, `app_activate`, `app_open` and so on, which relate to `GtkApplication` and `GtkApplicationWindow`.

Now we have three source files, `tfetextview.c`, `tfe.c` and `tfe3.ui`. The 3 of `tfe3.ui` is like a version number. Managing version with filenames is one possible idea but it also has a problem. You need to rename files in each version, and this affects the contents of source files that refer to those filenames. So, we should take 3 away from the filename.

In `tfe.c` the function `tfe_text_view_new` is invoked to create a `TfeTextView` instance. But it is defined in `tfetextview.c`, not `tfe.c`. The absence of a declaration (not definition) of `tfe_text_view_new` will cause a compilation error when `tfe.c` is compiled. The declaration is necessary in `tfe.c`. That public information is usually written in header files. These have a `.h` suffix like `tfetextview.h`. The header files are included by C source files. For example, `tfetextview.h` is included by `tfe.c`.

The source files are shown below.

`tfetextview.h`

```

1  #include <gtk/gtk.h>
2
3  #define TFE_TYPE_TEXT_VIEW tfe_text_view_get_type ()
4  G_DECLARE_FINAL_TYPE (TfeTextView, tfe_text_view, TFE, TEXT_VIEW, GtkTextView)
5
6  void
7  tfe_text_view_set_file (TfeTextView *tv, GFile *f);
8
9  GFile *
10 tfe_text_view_get_file (TfeTextView *tv);
11
12 GtkWidget *
13 tfe_text_view_new (void);

```

`tfetextview.c`

```

1  #include <gtk/gtk.h>
2  #include "tfetextview.h"
3
4  struct _TfeTextView
5  {
6      GtkTextView parent;
7      GFile *file;

```

```

8  };
9
10 G_DEFINE_FINAL_TYPE (TfeTextView, tfe_text_view, GTK_TYPE_TEXT_VIEW);
11
12 static void
13 tfe_text_view_init (TfeTextView *tv) {
14 }
15
16 static void
17 tfe_text_view_class_init (TfeTextViewClass *class) {
18 }
19
20 void
21 tfe_text_view_set_file (TfeTextView *tv, GFile *f) {
22     tv->file = f;
23 }
24
25 GFile *
26 tfe_text_view_get_file (TfeTextView *tv) {
27     return tv->file;
28 }
29
30 GtkWidget *
31 tfe_text_view_new (void) {
32     return GTK_WIDGET (g_object_new (TFE_TYPE_TEXT_VIEW, NULL));
33 }

```

tfe.c

```

1  #include <gtk/gtk.h>
2  #include "tfetextview.h"
3
4  static void
5  app_activate (GApplication *app) {
6      g_print ("You need a filename argument.\n");
7  }
8
9  static void
10 app_open (GApplication *app, GFile ** files, gint n_files, gchar *hint) {
11     GtkWidget *win;
12     GtkWidget *nb;
13     GtkWidget *lab;
14     GtkNotebookPage *nbp;
15     GtkWidget *scr;
16     GtkWidget *tv;
17     GtkTextBuffer *tb;
18     char *contents;
19     gsize length;
20     char *filename;
21     int i;
22     GError *err = NULL;
23     GtkBuilder *build;
24
25     build = gtk_builder_new_from_resource ("/com/github/ToshioCP/tfe/tfe.ui");

```

```

26 win = GTK_WIDGET (gtk_builder_get_object (build, "win"));
27 gtk_window_set_application (GTK_WINDOW (win), GTK_APPLICATION (app));
28 nb = GTK_WIDGET (gtk_builder_get_object (build, "nb"));
29 g_object_unref (build);
30 for (i = 0; i < n_files; i++) {
31     if (g_file_load_contents (files[i], NULL, &contents, &length, NULL, &err)) {
32         scr = gtk_scrolled_window_new ();
33         tv = tfe_text_view_new ();
34         tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
35         gtk_text_view_set_wrap_mode (GTK_TEXT_VIEW (tv), GTK_WRAP_WORD_CHAR);
36         gtk_scrolled_window_set_child (GTK_SCROLLED_WINDOW (scr), tv);
37
38         tfe_text_view_set_file (TFE_TEXT_VIEW (tv), g_file_dup (files[i]));
39         gtk_text_buffer_set_text (tb, contents, length);
40         g_free (contents);
41         filename = g_file_get_basename (files[i]);
42         lab = gtk_label_new (filename);
43         gtk_notebook_append_page (GTK_NOTEBOOK (nb), scr, lab);
44         nbp = gtk_notebook_get_page (GTK_NOTEBOOK (nb), scr);
45         g_object_set (nbp, "tab-expand", TRUE, NULL);
46         g_free (filename);
47     } else {
48         g_printerr ("%s.\n", err->message);
49         g_clear_error (&err);
50     }
51 }
52 if (gtk_notebook_get_n_pages (GTK_NOTEBOOK (nb)) > 0) {
53     gtk_window_present (GTK_WINDOW (win));
54 } else
55     gtk_window_destroy (GTK_WINDOW (win));
56 }
57
58 int
59 main (int argc, char **argv) {
60     GtkApplication *app;
61     int stat;
62
63     app = gtk_application_new ("com.github.ToshioCP.tfe", G_APPLICATION_HANDLES_OPEN);
64     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
65     g_signal_connect (app, "open", G_CALLBACK (app_open), NULL);
66     stat = g_application_run (G_APPLICATION (app), argc, argv);
67     g_object_unref (app);
68     return stat;
69 }

```

The UI file `tfe.ui` is the same as `tfe3.ui` in the previous section.

`tfe.gresource.xml`

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <gresources>
3     <gresource prefix="/com/github/ToshioCP/tfe">
4         <file>tfe.ui</file>
5     </gresource>
6 </gresources>

```

Splitting the source code into multiple files makes maintenance easier. But now we face a new problem. The number of build steps has increased.

- Compiling the UI file `tfe.ui` into `resources.c`, using `tfe.gresource.xml`.
- Compiling `tfe.c` into `tfe.o` (object file).
- Compiling `tfetextview.c` into `tfetextview.o`.
- Compiling `resources.c` into `resources.o`.
- Linking all the object files into application `tfe`.

Build tools manage the steps.

2.8.3 Meson and Ninja

I will explain the Meson and Ninja build systems.

Other possible tools are Make and Autotools. They are traditional tools but slower than Ninja. Many developers have recently migrated to Meson and Ninja. For example, GTK 4 uses them.

You need to create `meson.build` file first.

```

1 project('tfe', 'c')
2
3 gtkdep = dependency('gtk4')
4
5 gnome=import('gnome')
6 resources = gnome.compile_resources('resources','tfe.gresource.xml')
7
8 sourcefiles=files('tfe.c', 'tfetextview.c')
9
10 executable('tfe', sourcefiles, resources, dependencies: gtkdep, install: false)
```

- 1: The function `project` defines things about the project. The first argument is the name of the project and the second is the programming language.
- 3: The function `dependency` defines a dependency that is taken by `pkg-config`. We put `gtk4` as an argument.
- 5: The function `import` imports a module. In line 5, the `gnome` module is imported and assigned to the variable `gnome`. The `gnome` module provides helper tools to build GTK programs.
- 6: The method `.compile_resources` is from the `gnome` module. It compiles files to resources, as instructed by the xml file. In line 6, the resource filename is `resources`, which means `resources.c` and `resources.h`, and the xml file is `tfe.gresource.xml`. This method generates C source file by default.
- 8: Defines source files.
- 10: The `executable` function generates a target file by compiling source files. The first argument is the filename of the target. The following arguments are source files. The last two arguments have keys and values. For example, the fourth argument has a key `dependencies`, a delimiter (`:`) and a value `gtkdep`. This type of parameter is called *keyword parameter* or *kwargs*. The value `gtkdep` is defined in line 3. The last argument tells Meson and Ninja that this project shouldn't install the executable file. it is only built within the build directory.

Now run meson and ninja.

```
$ meson setup _build
$ ninja -C _build
```

The meson command takes two arguments.

- `setup`: The first argument is a command of meson. Setup is the default, so you can leave it out. But it is recommended to write it explicitly since version 0.64.0.
- The second argument is the name of the build directory.

Then, the executable file `tfe` is generated under the directory `_build`.

```
$ _build/tfe tfe.c tfetextview.c
```

A window appears. It includes a notebook with two pages. One is `tfe.c` and the other is `tfetextview.c`.

For further information, see [The Meson Build system](#).

2.9 Instance Initialization and Destruction

A new version of the text file editor (`tfe`) will be made in this section and the following four sections. It is `tfe5`. There are many changes from the prior version. They are located in two directories, `/src/tfe5` and `/src/tfetextview`.

2.9.1 Encapsulation

We've divided C source file into two parts. But it is not enough in terms of encapsulation.

- `tfe.c` includes everything other than `TfeTextView`. It should be divided into at least two parts, `tfeapplication.c` and `tfenotebook.c`.
- Header files also need to be organized.

However, first of all, I'd like to focus on the object `TfeTextView`. It is a child object of `GtkTextView` and has a new member `file` in it. The important thing is how we manage the `GFile` object referenced by file.

- What is necessary to `GFile` when creating (or initializing) `TfeTextView`?
- What is necessary to `GFile` when destructing `TfeTextView`?
- Should `TfeTextView` read/write a file by itself or not?
- How it communicates with objects outside?

You need to know at least class, instance and signals before thinking about them. I will explain them in this section and the next section. After that I will explain:

- Organizing functions.
- How to use `GtkFileDialog`. It is a new class made in the version 4.10 and replaces `GtkFileChooserDialog`.

2.9.2 GObject and its Children

`GObject` and its children are objects, which have both class and object C structures. First, think about instances. An instance is a block of memory allocated for the object structure. The following is the structure of `TfeTextView`.

```
/* This typedef statement is automatically generated by the macro G_DECLARE_FINAL_TYPE */
typedef struct _TfeTextView TfeTextView;

struct _TfeTextView {
    GtkTextView parent;
    GFile *file;
};
```

The members of the structure are:

- The member `parent` is a `GtkTextView` C structure. It is declared in `gtktextview.h`. `GtkTextView` is the parent of `TfeTextView`.
- The member `file` is a pointer to a `GFile`. It can be `NULL` if the `TfeTextView` instance has no file. The most common case is that the instance is newly created.

You can find the declaration of the structures of the ancestors in the source files in GTK or GLib. The following is extracted from the source files (not exactly the same).

```

typedef struct _GObject GObject;
typedef struct _GObject GInitiallyUnowned;
struct _GObject
{
    GTypeInstance g_type_instance;
    volatile guint ref_count;
    GData *qdata;
};

typedef struct _GtkWidget GtkWidget;
struct _GtkWidget
{
    GInitiallyUnowned parent_instance;
    GtkWidgetPrivate *priv;
};

typedef struct _GtkTextView GtkTextView;
struct _GtkTextView
{
    GtkWidget parent_instance;
    GtkTextViewPrivate *priv;
};

```

In each structure, its parent is declared at the top of the members. So, all the ancestors are included in the child object. The structure of `TfeTextView` is like the following diagram.



Figure 2.13: The structure of the instance `TfeTextView`

Derivable classes (ancestor classes) have their own private data area which are not included by the structure above. For example, `GtkWidget` has `GtkWidgetPrivate` (C structure) for its private data.

Notice declarations are not definitions. So, no memories are allocated when C structures are declared. Memories are allocated to them from the heap area when the `tfe_text_view_new` function is called. At the same time, the ancestors' private area allocated for the `TfeTextView`. They are hidden from `TfeTextView` and it can't access to them directly. The created memory is called instance. When a `TfeTextView` instance is created, it is given three data area.

- The instance (C structure).
- `GtkWidgetPrivate` structure.
- `GtkTextViewPrivate` structure.

`TfeTextView` functions can access to its instance only. The `GtkWidgetPrivate` and `GtkTextViewPrivate` are used by the ancestors' functions. See the following example.

```

GtkWidget *tv = tfe_text_view_new ();
GtkTextBuffer *buffer = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));

```

The parent's function `gtk_text_view_get_buffer` accesses the `GtkTextViewPrivate` data (owned by `tv`). There is a pointer, which refers to the `GtkBuffer`, in the private area and the function returns the pointer. (Actual behavior is a bit more complicated.)

`TfeTextView` instances inherit the ancestors functions like this.

A TfeTextView instance is created every time the `tfe_text_view_new` function is called. Therefore, multiple TfeTextView instances can exist.

2.9.3 Initialization of TfeTextView Instances

The function `tfe_text_view_new` creates a new TfeTextView instance.

```

1 GtkWidget *
2 tfe_text_view_new (void) {
3     return GTK_WIDGET (g_object_new (TFE_TYPE_TEXT_VIEW, "wrap-mode", GTK_WRAP_WORD_CHAR, NULL));
4 }

```

When this function is invoked, a TfeTextView instance is created and initialized. The initialization process is as follows.

1. When the instance is created, GtkWidgetPrivate and GtkTextViewPrivate structures are also created
2. Initializes GObject (GInitiallyUnowned) part in the TfeTextView instance.
3. Initializes GtkWidget part (the first `priv`) in the TfeTextView instance and GtkWidgetPrivate structure.
4. Initializes GtkTextView part (the second `priv`) in the TfeTextView instance and GtkTextViewPrivate structure.
5. Initializes TfeTextView part (`file`) in the TfeTextView instance.

The step two through four is done by `g_object_init`, `gtk_widget_init` and `gtk_text_view_init`. They are called by the system automatically and you don't need to care about them. Step five is done by the function `tfe_text_view_init` in `tfetextview.c`.

```

1 static void
2 tfe_text_view_init (TfeTextView *tv) {
3     tv->file = NULL;
4 }

```

This function just initializes `tv->file` to be NULL.

2.9.4 Functions and Classes

In Gtk, all objects derived from GObject have classes and instances (but abstract objects have only classes). Instances are memory of C structure, which are described in the previous two subsections. Each object can have more than one instance. Those instances have the same structure. Instances just have data. Therefore, it doesn't define object's behavior. We need at least two things. One is functions and the other is class methods.

The latest GTK 4 document classifies functions into a constructor, functions and instance methods.

- constructors: Their name are always `gtk_(objectname)_new`. They create the objects.
- functions: Their first parameter (argument) is *NOT* the instance. Usually functions are utility functions for the class.
- instance methods: Their first parameter (argument) is the instance. They do some task for the specific instance.

This tutorial uses **functions** in two ways, broad or narrow sense.

You've already seen many functions. For example,

- `TfeTextView *tfe_text_view_new (void);` is a function (constructor) to create a TfeTextView instance.
- `GtkTextBuffer *gtk_text_view_get_buffer (GtkTextView *textview)` is a function (instance method) to get a GtkTextBuffer from GtkTextView.

Functions are public, which means that they are expected to be used by other objects. They are similar to public methods in object oriented languages.

Class (C structure) mainly consists of pointers to C functions. They are called *class methods* and used by the object itself or its descendant objects. For example, GObject class is declared in `gobject.h` in GLib source files.

```

1  typedef struct _GObjectClass      GObjectClass;
2  typedef struct _GObjectClass      GInitiallyUnownedClass;
3
4  struct _GObjectClass
5  {
6      GTypeClass    g_type_class;
7
8      /*< private >*/
9      GSList        *construct_properties;
10
11     /*< public >*/
12     /* seldom overridden */
13     GObject*      (*constructor)      (GType                type,
14                                         guint                n_construct_properties,
15                                         GObjectConstructParam *construct_properties);
16
17     /* overridable methods */
18     void          (*set_property)      (GObject            *object,
19                                         guint              property_id,
20                                         const GValue      *value,
21                                         GParamSpec        *pspec);
22     void          (*get_property)      (GObject            *object,
23                                         guint              property_id,
24                                         GValue            *value,
25                                         GParamSpec        *pspec);
26     void          (*dispose)           (GObject            *object);
27     void          (*finalize)          (GObject            *object);
28     /* seldom overridden */
29     void          (*dispatch_properties_changed) (GObject      *object,
30                                                   guint          n_pspecs,
31                                                   GParamSpec   **pspecs);
32
33     /* signals */
34     void          (*notify)            (GObject            *object,
35                                         GParamSpec        *pspec);
36
37     /* called when done constructing */
38     void          (*constructed)       (GObject            *object);
39
40     /*< private >*/
41     gsize         flags;
42
43     gsize         n_construct_properties;
44
45     gpointer       pspecs;
46     gsize         n_pspecs;
47
48     /* padding */
49     gpointer       pdummy[3];

```



```

24
25  /* size requests */
26  GtkSizeRequestMode (* get_request_mode)          (GtkWidget      *widget);
27  void                (* measure) (GtkWidget      *widget,
28                                GtkOrientation  orientation,
29                                int              for_size,
30                                int              *minimum,
31                                int              *natural,
32                                int              *minimum_baseline,
33                                int              *natural_baseline);
34
35  /* Mnemonics */
36  gboolean (* mnemonic_activate) (GtkWidget      *widget,
37                                gboolean         group_cycling);
38
39  /* explicit focus */
40  gboolean (* grab_focus)         (GtkWidget      *widget);
41  gboolean (* focus)              (GtkWidget      *widget,
42                                GtkDirectionType direction);
43  void     (* set_focus_child)    (GtkWidget      *widget,
44                                GtkWidget      *child);
45
46  /* keyboard navigation */
47  void     (* move_focus)        (GtkWidget      *widget,
48                                GtkDirectionType direction);
49  gboolean (* keynav_failed)     (GtkWidget      *widget,
50                                GtkDirectionType direction);
51
52  gboolean (* query_tooltip)      (GtkWidget *widget,
53                                int         x,
54                                int         y,
55                                gboolean    keyboard_tooltip,
56                                GtkTooltip *tooltip);
57
58  void     (* compute_expand)     (GtkWidget *widget,
59                                gboolean    *hexpand_p,
60                                gboolean    *vexpand_p);
61
62  void     (* css_changed)        (GtkWidget      *widget,
63                                GtkCssStyleChange *change);
64
65  void     (* system_setting_changed) (GtkWidget      *widget,
66                                GtkSystemSetting  settings);
67
68  void     (* snapshot)          (GtkWidget      *widget,
69                                GtkSnapshot      *snapshot);
70
71  gboolean (* contains)          (GtkWidget *widget,
72                                double      x,
73                                double      y);
74
75  /*< private >*/
76

```

```

77  GtkWidgetClassPrivate *priv;
78
79  gpointer padding[8];
80  };
81
82  struct _GtkTextViewClass
83  {
84      GtkWidgetClass parent_class;
85
86      /*< public >*/
87
88      void (* move_cursor)          (GtkTextView      *text_view,
89                                   GtkMovementStep  step,
90                                   int                count,
91                                   gboolean           extend_selection);
92      void (* set_anchor)           (GtkTextView      *text_view);
93      void (* insert_at_cursor)     (GtkTextView      *text_view,
94                                   const char        *str);
95      void (* delete_from_cursor)   (GtkTextView      *text_view,
96                                   GtkDeleteType     type,
97                                   int                count);
98      void (* backspace)            (GtkTextView      *text_view);
99      void (* cut_clipboard)        (GtkTextView      *text_view);
100     void (* copy_clipboard)        (GtkTextView      *text_view);
101     void (* paste_clipboard)       (GtkTextView      *text_view);
102     void (* toggle_overwrite)      (GtkTextView      *text_view);
103     GtkTextBuffer * (* create_buffer) (GtkTextView      *text_view);
104     void (* snapshot_layer)        (GtkTextView      *text_view,
105                                    GtkTextViewLayer  layer,
106                                    GtkSnapshot        *snapshot);
107     gboolean (* extend_selection)   (GtkTextView      *text_view,
108                                    GtkTextExtendSelection granularity,
109                                    const GtkTextIter  *location,
110                                    GtkTextIter        *start,
111                                    GtkTextIter        *end);
112     void (* insert_emoji)          (GtkTextView      *text_view);
113
114     /*< private >*/
115
116     gpointer padding[8];
117  };
118
119  /* The following definition is generated by the macro G_DECLARE_FINAL_TYPE */
120  typedef struct {
121      GtkTextView parent_class;
122  } TfeTextViewClass;

```

- 120-122: These three lines are generated by the macro `G_DECLARE_FINAL_TYPE`. So, they are not written in either `tfetextview.h` or `tfetextview.c`.
- 3, 84, 121: Each class has its parent class at the first member of its structure. It is the same as instance structures.
- Class members in ancestors are open to the descendant class. So, they can be changed in `tfe_text_view_class_init` function. For example, the `dispose` pointer in `GObjectClass` will be overridden later in `tfe_text_view_class_init`. (Override is an object oriented programming

terminology. Override is rewriting ancestors' class methods in the descendant class.)

- Some class methods are often overridden. `set_property`, `get_property`, `dispose`, `finalize` and `constructed` are such methods.

TfeTextViewClass includes its ancestors' class in it. It is illustrated in the following diagram.

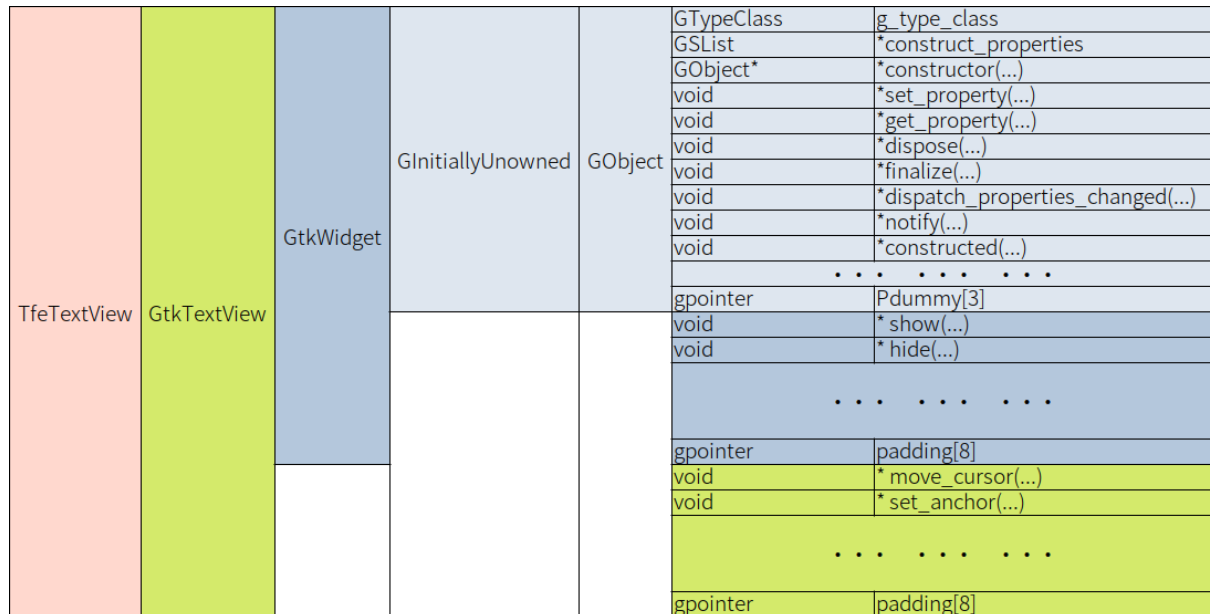


Figure 2.14: The structure of TfeTextView Class

2.9.6 Destruction of TfeTextView

Every Object derived from GObject has a reference count. If an object A refers to an object B, then A keeps a pointer to B in A and at the same time increases the reference count of B by one with the function `g_object_ref (B)`. If A doesn't need B any longer, A discards the pointer to B (usually it is done by assigning NULL to the pointer) and decreases the reference count of B by one with the function `g_object_unref (B)`.

If two objects A and B refer to C, then the reference count of C is two. If A no longer needs C, A discards the pointer to C and decreases the reference count in C by one. Now the reference count of C is one. In the same way, if B no longer needs C, B discards the pointer to C and decreases the reference count in C by one. At this moment, no object refers to C and the reference count of C is zero. This means C is no longer useful. Then C destructs itself and finally the memories allocated to C is freed.

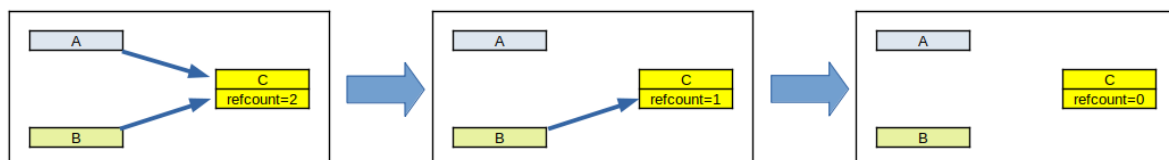


Figure 2.15: Reference count of B

The idea above is based on an assumption that an object referred by nothing has reference count of zero. When the reference count drops to zero, the object starts its destruction process. The destruction process is split into two phases: disposing and finalizing. In the disposing process, the object invokes the function pointed to by `dispose` in its class to release all references to other instances. After that, it invokes the function pointed to by `finalize` in its class to complete the destruction process.

In the destruction process, TfeTextView needs to unref the GFile pointed to by `tv->file`. You must write the dispose handler `tfe_text_view_dispose`.

```

1 static void
2 tfe_text_view_dispose (GObject *gobject) {
3     TfeTextView *tv = TFE_TEXT_VIEW (gobject);
4
5     if (G_IS_FILE (tv->file))
6         g_clear_object (&tv->file);
7
8     G_OBJECT_CLASS (tfe_text_view_parent_class)->dispose (gobject);
9 }

```

- 5,6: If `tv->file` points to a GFile, it decreases the reference count of the GFile instance. The function `g_clear_object` decreases the reference count and assigns NULL to `tv->file`. In dispose handlers, we usually use `g_clear_object` rather than `g_object_unref`.
- 8: invokes parent's dispose handler. (This will be explained later.)

In the disposing process, the object uses the pointer in its class to call the handler. Therefore, `tfe_text_view_dispose` needs to be registered in the class when the TfeTextView class is initialized. The function `tfe_text_view_class_init` is the class initialization function and it is declared in the `G_DEFINE_FINAL_TYPE` macro expansion.

```

static void
tfe_text_view_class_init (TfeTextViewClass *class) {
    GObjectClass *object_class = G_OBJECT_CLASS (class);

    object_class->dispose = tfe_text_view_dispose;
}

```

Each ancestors' class has been created before TfeTextViewClass is created. Therefore, there are four classes and each class has a pointer to each dispose handler. Look at the following diagram. There are four classes – GObjectClass (GInitiallyUnownedClass), GtkWidgetClass, GtkTextViewClass and TfeTextViewClass. Each class has its own dispose handler – `dh1`, `dh2`, `dh3` and `tfe_text_view_dispose`.

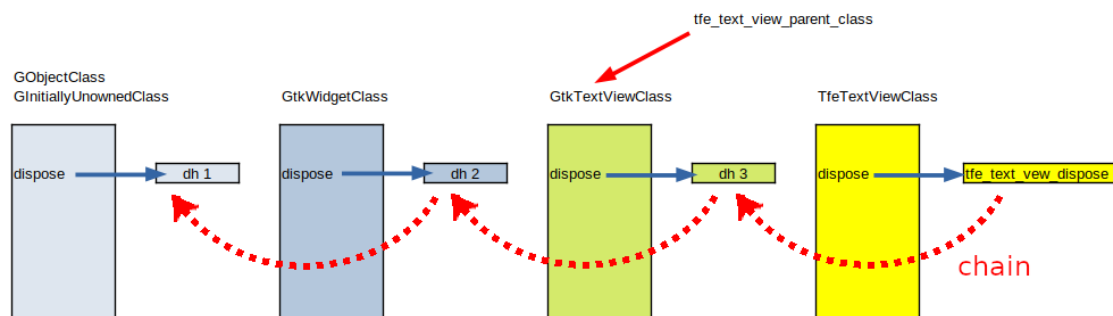


Figure 2.16: dispose handlers

Now, look at the `tfe_text_view_dispose` program above. It first releases the reference to GFile object pointed to by `tv->file`. Then it invokes its parent's dispose handler in line 8.

```
G_OBJECT_CLASS (tfe_text_view_parent_class)->dispose (gobject);
```

A variable `tfe_text_view_parent_class`, which is made by `G_DEFINE_FINAL_TYPE` macro, is a pointer that points to the parent object class. The variable `gobject` is a pointer to TfeTextView instance which is casted as a GObject instance. Therefore, `G_OBJECT_CLASS (tfe_text_view_parent_class)->dispose` points to the handler `dh3` in the diagram above. The statement `G_OBJECT_CLASS (tfe_text_view_parent_class)->dispose (gobject)` is the same as `dh3 (gobject)`, which means it

releases all the reference to the other instances in the `GtkTextViewPrivate` in the `TfeTextView` instance. After that, `dh3` calls `dh2`, and `dh2` calls `dh1`. Finally all the references are released.

2.10 Signals

2.10.1 Signals

Each object is encapsulated in Gtk programming. And it is not recommended to use global variables because they are prone to make the program complicated. So, we need something to communicate between objects. There are two ways to do so.

- Instance methods: Instance methods are functions on instances. For example, `tb = gtk_text_view_get_buffer (tv)` is an instance method on the instance `tv`. The caller requests `tv` for `tb`, which is a `GtkTextBuffer` instance connected to `tv`.
- Signals: For example, `activate` signal on `GApplication` object. When the application is activated, the signal is emitted. Then the handler, which has been connected to the signal, is invoked.

The caller of methods or signals are usually out of the object. One of the differences between these two is that the object is active or passive. In methods, objects passively respond to the caller. In signals, objects actively send signals to handlers.

GObject signals are registered, connected and emitted.

1. Signals are registered in the class. The registration is done usually when the class is initialized. Signals can have a default handler, which is sometimes called “object method handler”. It is not a user handler connected by `g_signal_connect` family functions. A default handler is always called on any instance of the class.
2. Signals are connected to handlers by the macro `g_signal_connect` or its family functions. The connection is usually done out of the object. One important thing is that signals are connected on a specific instance. Suppose there exist two `GtkButton` instances A, B and a function C. Even if you connected the “clicked” signal on A to C, B and C are *not* connected.
3. When Signals are emitted, the connected handlers are invoked. Signals are emitted on the instance of the class.

2.10.2 Signal Registration

In `TfeTextView`, two signals are registered.

- “change-file” signal. This signal is emitted when `tv->file` is changed.
- “open-response” signal. The function `tfe_text_view_open` doesn’t return the status because it can’t get the status from the file chooser dialog. (Instead, the call back function gets the status.) This signal is emitted instead of the return value of the function.

A static variable or array is used to store signal ID.

```
enum {
    CHANGE_FILE,
    OPEN_RESPONSE,
    NUMBER_OF_SIGNALS
};

static guint tfe_text_view_signals[NUMBER_OF_SIGNALS];
```

Signals are registered in the class initialization function.

```
1 static void
2 tfe_text_view_class_init (TfeTextViewClass *class) {
3     GObjectClass *object_class = G_OBJECT_CLASS (class);
```



```

4
5 object_class->dispose = tfe_text_view_dispose;
6 tfe_text_view_signals[CHANGE_FILE] = g_signal_new ("change-file",
7           G_TYPE_FROM_CLASS (class),
8           G_SIGNAL_RUN_LAST | G_SIGNAL_NO_RECURSE | G_SIGNAL_NO_HOOKS,
9           0 /* class offset */,
10          NULL /* accumulator */,
11          NULL /* accumulator data */,
12          NULL /* C marshaller */,
13          G_TYPE_NONE /* return_type */,
14          0 /* n_params */
15          );
16 tfe_text_view_signals[OPEN_RESPONSE] = g_signal_new ("open-response",
17           G_TYPE_FROM_CLASS (class),
18           G_SIGNAL_RUN_LAST | G_SIGNAL_NO_RECURSE | G_SIGNAL_NO_HOOKS,
19           0 /* class offset */,
20          NULL /* accumulator */,
21          NULL /* accumulator data */,
22          NULL /* C marshaller */,
23          G_TYPE_NONE /* return_type */,
24          1 /* n_params */,
25          G_TYPE_INT
26          );
27 }
```

- 6-15: Registers “change-file” signal. `g_signal_new` function is used. The signal “change-file” has no default handler (object method handler) so the offset (the line 9) is set to zero. You usually don’t need a default handler. If you need it, use `g_signal_new_class_handler` function instead of `g_signal_new`. See [GObject API Reference](#) for further information.
- The return value of `g_signal_new` is the signal id. The type of signal id is `guint`, which is an alias for unsigned int. It is used in the function `g_signal_emit`.
- 16-26: Registers “open-response” signal. This signal has a parameter.
- 24: Number of the parameters. “open-response” signal has one parameter.
- 25: The type of the parameter. `G_TYPE_INT` is a type of integer. Such fundamental types are described in [gtype.h](#).

The handlers are declared as follows.

```

/* "change-file" signal handler */
void
user_function (TfeTextView *tv,
              gpointer user_data)

/* "open-response" signal handler */
void
user_function (TfeTextView *tv,
              TfeTextViewOpenResponseType response_id,
              gpointer user_data)
```

- The signal “change-file” doesn’t have any parameters, so the handler’s arguments are a `TfeTextView` instance and a user data.
- The signal “open-response” has one parameter and its arguments are a `TfeTextView` instance, the signal’s parameter and user data.
- The variable `tv` points to the instance on which the signal is emitted.
- The last argument `user_data` comes from the fourth argument of `g_signal_connect`.

- The parameter (`response_id`) comes from the fourth argument of `g_signal_emit`.

The values of the type `TfeTextViewOpenResponseType` are defined in `tfetextview.h`.

```
/* "open-response" signal response */
enum TfeTextViewOpenResponseType
{
    TFE_OPEN_RESPONSE_SUCCESS,
    TFE_OPEN_RESPONSE_CANCEL,
    TFE_OPEN_RESPONSE_ERROR
};
```

- The parameter is set to `TFE_OPEN_RESPONSE_SUCCESS` when `tfe_text_view_open` has successfully opened a file and read it.
- The parameter is set to `TFE_OPEN_RESPONSE_CANCEL` when the user has canceled.
- The parameter is set to `TFE_OPEN_RESPONSE_ERROR` when an error has occurred.

2.10.3 Signal Connection

A signal and a handler are connected by the function macro `g_signal_connect`. There are some similar function macros like `g_signal_connect_after`, `g_signal_connect_swapped` and so on. However, `g_signal_connect` is used most often. The signals “change-file” and “open-response” are connected to their callback functions out of the `TfeTextView` object. Those callback functions are defined by users.

For example, callback functions are defined in `src/tfe6/tfewindow.c` and their names are `file_changed_cb` and `open_response_cb`. They will be explained later.

```
g_signal_connect (TFE_TEXT_VIEW (tv), "change-file", G_CALLBACK (file_changed_cb), nb);

g_signal_connect (TFE_TEXT_VIEW (tv), "open-response", G_CALLBACK (open_response_cb), nb);
```

2.10.4 Signal Emission

A signal is emitted on the instance. A function `g_signal_emit` is used to emit the signal. The following lines are extracted from `src/tfetextview/tfetextview.c`. Each line comes from a different line.

```
g_signal_emit (tv, tfe_text_view_signals[CHANGE_FILE], 0);
g_signal_emit (tv, tfe_text_view_signals[OPEN_RESPONSE], 0, TFE_OPEN_RESPONSE_SUCCESS);
g_signal_emit (tv, tfe_text_view_signals[OPEN_RESPONSE], 0, TFE_OPEN_RESPONSE_CANCEL);
g_signal_emit (tv, tfe_text_view_signals[OPEN_RESPONSE], 0, TFE_OPEN_RESPONSE_ERROR);
```

- The first argument is the instance on which the signal is emitted.
- The second argument is the signal id.
- The third argument is the detail of the signal. The signals “change-file” and “open-response” don’t have details and the arguments are zero.
- The signal “change-file” doesn’t have parameters, so there’s no fourth argument.
- The signal “open-response” has one parameter. The fourth argument is the parameter.

2.11 TfeTextView Class

The `TfeTextView` class will be finally completed in this section. The remaining topic is functions. `TfeTextView` functions, which are constructors and instance methods, are described in this section.

The source files are in the directory `src/tfetextview`. You can get them by downloading the [repository](#).

2.11.1 tfetextview.h

The header file `tfetextview.h` provides:

- The type of TfeTextView, which is TFE_TYPE_TEXT_VIEW.
- The macro G_DECLARE_FINAL_TYPE, the expansion of which includes some useful functions and definitions.
- Constants for the open-response signal.
- Public functions of tfetextview.c. They are constructors and instance methods.

Therefore, Any programs using TfeTextView needs to include tfetextview.h.

```

1  #pragma once
2
3  #include <gtk/gtk.h>
4
5  #define TFE_TYPE_TEXT_VIEW tfe_text_view_get_type ()
6  G_DECLARE_FINAL_TYPE (TfeTextView, tfe_text_view, TFE, TEXT_VIEW, GtkTextView)
7
8  /* "open-response" signal response */
9  enum TfeTextViewOpenResponseType
10 {
11     TFE_OPEN_RESPONSE_SUCCESS,
12     TFE_OPEN_RESPONSE_CANCEL,
13     TFE_OPEN_RESPONSE_ERROR
14 };
15
16 GFile *
17 tfe_text_view_get_file (TfeTextView *tv);
18
19 void
20 tfe_text_view_open (TfeTextView *tv, GtkWindow *win);
21
22 void
23 tfe_text_view_save (TfeTextView *tv);
24
25 void
26 tfe_text_view_saveas (TfeTextView *tv);
27
28 GtkWidget *
29 tfe_text_view_new_with_file (GFile *file);
30
31 GtkWidget *
32 tfe_text_view_new (void);

```

- 1: The preprocessor directive #pragma once makes the header file be included only once. It is non-standard but widely used.
- 3: Includes gtk4 header files. The header file gtk4 also has the same mechanism to avoid being included multiple times.
- 5-6: These two lines define TfeTextView type, its class structure and some useful definitions.
 - TfeTextView and TfeTextViewClass are declared as typedef of C structures.
 - You need to define a structure _TfeTextView later.
 - The class structure _TfeTextViewClass is defined here. You don't need to define it by yourself.
 - Convenience functions TFE_TEXT_VIEW () for casting and TFE_IS_TEXT_VIEW for type check are defined.
- 8-14: A definition of the values of the “open-response” signal parameters.
- 16-32: Declarations of public functions on TfeTextView.

2.11.2 Constructors

A `TfeTextView` instance is created with `tfe_text_view_new` or `tfe_text_view_new_with_file`. These functions are called constructors.

```
GtkWidget *tfe_text_view_new (void);
```

It just creates a new `TfeTextView` instance and returns the pointer to the new instance.

```
GtkWidget *tfe_text_view_new_with_file (GFile *file);
```

It is given a `Gfile` object as an argument and it loads the file into the `GtkTextBuffer` instance, then returns the pointer to the new instance. The argument `file` is owned by the caller and the function doesn't change it. If an error occurs during the creation process, `NULL` will be returned.

Each function is defined as follows.

```

1  GtkWidget *
2  tfe_text_view_new_with_file (GFile *file) {
3      g_return_val_if_fail (G_IS_FILE (file), NULL);
4
5      GtkWidget *tv;
6      GtkTextBuffer *tb;
7      char *contents;
8      gsize length;
9
10     if (! g_file_load_contents (file, NULL, &contents, &length, NULL, NULL)) /* read error */
11         return NULL;
12
13     tv = tfe_text_view_new();
14     tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
15     gtk_text_buffer_set_text (tb, contents, length);
16     TFE_TEXT_VIEW (tv)->file = g_file_dup (file);
17     gtk_text_buffer_set_modified (tb, FALSE);
18     g_free (contents);
19     return tv;
20 }
21
22 GtkWidget *
23 tfe_text_view_new (void) {
24     return GTK_WIDGET (g_object_new (TFE_TYPE_TEXT_VIEW, "wrap-mode", GTK_WRAP_WORD_CHAR, NULL));
25 }
```

- 22-25: `tfe_text_view_new` function. Just returns the value from the function `g_object_new` but casts it to the pointer to `GtkWidget`. The function `g_object_new` creates any instances of its descendant class. The arguments are the type of the class, property list and `NULL`, which is the end mark of the property list. `TfeTextView` “wrap-mode” property has `GTK_WRAP_WORD_CHAR` as the default value.
- 1-20: `tfe_text_view_new_with_file` function.
- 3: `g_return_val_if_fail` is described in [GLib API Reference – g_return_val_if_fail](#). And also [GLib API Reference – Message Logging](#). It tests whether the argument `file` is a pointer to `GFile`. If it's true, the program goes on to the next line. If it's false, it returns `NULL` (the second argument) immediately. And at the same time it logs out the error message (usually the log is outputted to `stderr` or `stdout`). This function is used to check the programmer's error. If an error occurs, the solution is usually to change the (caller) program and fix the bug. You need to distinguish programmer's errors and runtime errors. You shouldn't use this function to find runtime errors.
- 10-11: Reads the file. If an error occurs, `NULL` is returned.

- 13: Calls the function `tfe_text_view_new`. The function creates `TfeTextView` instance and returns the pointer to the instance.
- 14: Gets the pointer to the `GtkTextBuffer` instance corresponds to `tv`. The pointer is assigned to `tb`
- 15: Assigns the contents read from the file to `tb`.
- 16: Duplicates `file` and sets `tv->file` to point to it. `GFile` is *not* thread safe. The duplication makes sure that the `GFile` instance of `tv` keeps the file information even if the original one is changed by other thread.
- 17: The function `gtk_text_buffer_set_modified (tb, FALSE)` sets the modification flag of `tb` to `FALSE`. The modification flag indicates that the contents have been modified. It is used when the contents are saved. If the modification flag is `FALSE`, it doesn't need to save the contents.
- 18: Frees the memories pointed to by `contents`.
- 19: Returns `tv`, which is a pointer to the newly created `TfeTextView` instance. If an error happens, `NULL` is returned.

2.11.3 Save and Saveas Functions

Save and saveas functions write the contents in the `GtkTextBuffer` to a file.

```
void tfe_text_view_save (TfeTextView *tv)
```

The function `tfe_text_view_save` writes the contents in the `GtkTextBuffer` to a file specified by `tv->file`. If `tv->file` is `NULL`, then it shows file chooser dialog and prompts the user to choose a file to save. Then it saves the contents to the file and sets `tv->file` to point to the `GFile` instance for the file.

```
void tfe_text_view_saveas (TfeTextView *tv)
```

The function `saveas` shows a file chooser dialog and prompts the user to select an existing file or specify a new file to save. Then, the function changes `tv->file` and save the contents to the specified file. If an error occurs, it is shown to the user through the alert dialog. The error is managed only in the `TfeTextView` and no information is notified to the caller.

2.11.3.1 save_file Function

```

1  static gboolean
2  save_file (GFile *file, GtkTextBuffer *tb, GtkWidget *win) {
3      GtkTextIter start_iter;
4      GtkTextIter end_iter;
5      char *contents;
6      gboolean stat;
7      GtkAlertDialog *alert_dialog;
8      GError *err = NULL;
9
10     gtk_text_buffer_get_bounds (tb, &start_iter, &end_iter);
11     contents = gtk_text_buffer_get_text (tb, &start_iter, &end_iter, FALSE);
12     stat = g_file_replace_contents (file, contents, strlen (contents), NULL, TRUE, G_FILE_CREATE_NONE,
13     if (stat)
14         gtk_text_buffer_set_modified (tb, FALSE);
15     else {
16         alert_dialog = gtk_alert_dialog_new ("%s", err->message);
17         gtk_alert_dialog_show (alert_dialog, win);
18         g_object_unref (alert_dialog);
19         g_error_free (err);
20     }
21     g_free (contents);
22     return stat;
23 }
```

- The function `save_file` is called from `saveas_dialog_response` and `tfe_text_view_save`. This function saves the contents of the buffer to the file given as an argument. If error happens, it displays an error message. So, a caller of this function don't need to take care of errors. The class of this function is `static`. Therefore, only functions in this file (`tfetextview.c`) call this function. Such static functions usually don't have `g_return_val_if_fail` functions.
- 10-11: Gets the text contents from the buffer.
- 12: The function `g_file_replace_contents` writes the contents to the file and returns the status (true = success/ false = fail). It has many parameters, but some of them are almost always given the same values. See the [GIO documentation](#) for details.
 - `GFile* file`: `GFile` to which the contents are saved.
 - `const char* contents`: contents to be saved. The string is owned by the caller.
 - `gsize length`: the length of the contents
 - `const char* etag`: entity tag. It is usually `NULL`.
 - `gboolean make_backup`: true to make a backup if the file exists. false not to make it. the file will be overwritten.
 - `GFileCreateFlags flags`: usually `G_FILE_CREATE_NONE` is fine.
 - `char** new_etag`: new entity tag. It is usually `NULL`.
 - `Gancellable* cancellable`: If a cancellable instance is set, the other thread can cancel this operation. it is usually `NULL`.
 - `GError** error`: If error happens, `GError` will be set.
- 13,14: If no error happens, set the modified flag to be `FALSE`. This means that the buffer is not modified since it has been saved.
- 16-19: If it fails to save the contents, an error message will be displayed.
- 16: Creates an alert dialog. The parameters are printf-like format string followed by values to insert into the string. `GtkAlertDialog` is available since version 4.10. If your version is older than 4.10, use `GtkMessageDialog` instead. `GtkMessageDialog` is deprecated since version 4.10.
- 17: Show the alert dialog. The parameters are the dialog and the transient parent window. This allows window managers to keep the dialog on top of the parent window, or center the dialog over the parent window. It is possible to give no parent window to the dialog by giving `NULL` as the argument. However, it is encouraged to give parents to dialogs.
- 18: Releases the dialog.
- 19: Frees the `GError` struct pointed to by `err` with `g_error_free` function.
- 21: Frees `contents`.
- 22: Returns the status to the caller.

2.11.3.2 save_dialog_cb Function

```

1  static void
2  save_dialog_cb(GObject *source_object, GAsyncResult *res, gpointer data) {
3      GtkFileDialog *dialog = GTK_FILE_DIALOG (source_object);
4      TfeTextView *tv = TFE_TEXT_VIEW (data);
5      GtkTextBuffer *tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
6      GFile *file;
7      GtkWidget *win = gtk_widget_get_ancestor (GTK_WIDGET (tv), GTK_TYPE_WINDOW);
8      GError *err = NULL;
9      GtkAlertDialog *alert_dialog;
10
11  if (((file = gtk_file_dialog_save_finish (dialog, res, &err)) != NULL) && save_file(file, tb, GTK_
12      // The following is complicated. The comments here will help your understanding
13      // G_IS_FILE(tv->file) && tv->file == file => nothing to do
14      // G_IS_FILE(tv->file) && tv->file != file => unref(tv->file), tv->file=file, emit change_file
15      // tv->file=NULL => tv->file=file, emit change_file
16  if (!(G_IS_FILE (tv->file) && g_file_equal (tv->file, file))) {

```

```

17     if (G_IS_FILE (tv->file))
18         g_object_unref (tv->file);
19     tv->file = file; // The ownership of 'file' moves to TfeTextView.
20     g_signal_emit (tv, tfe_text_view_signals[CHANGE_FILE], 0);
21 }
22 }
23 if (err && err->code != GTK_DIALOG_ERROR_DISMISSED) {
24     alert_dialog = gtk_alert_dialog_new ("%s", err->message);
25     gtk_alert_dialog_show (alert_dialog, GTK_WINDOW (win));
26     g_object_unref (alert_dialog);
27     g_clear_error (&err);
28 }
29 }

```

- The function `save_dialog_cb` is a call back function that is given to the `gtk_file_dialog_save` function as an argument. The `gtk_file_dialog_save` shows a file chooser dialog to the user. The user chooses or types a filename and clicks on the Save button or just clicks on the Cancel button. Then the call back function is called with the result. This is the general way in GIO to manage asynchronous operations. A pair of functions `g_data_input_stream_read_line_async` and `g_data_input_stream_read_line_finish` are one example. These functions are thread-safe. The arguments of `save_dialog_cb` are:
 - `GObject *source_object`: The `GObject` instance that the operation was started with. It is actually the `GtkFileDialog` instance that is shown to the user. However, the call back function is defined as `AsyncReadyCallback`, which is a general call back function for an asynchronous operation. So the type is `GObject` and you need to cast it to `GtkFileDialog` later.
 - `GAsyncResult *res`: The result of the asynchronous operation. It will be given to the `gtk_dialog_save_finish` function.
 - `gpointer data`: A user data set in the `gtk_dialog_save` function.
- 11: Calls `gtk_dialog_save_finish`. It is given the result `res` as an argument and returns a pointer to a `GFile` object the user has chosen. If the user has canceled or an error happens, it returns `NULL`, creates a `GError` object and sets `err` to point to it. If `gtk_dialog_save_finish` returns a `GFile`, the function `save_file` is called.
- 12-21: If the file is successfully saved, these lines are executed. See the comments, line 12-15, for the details.
- 23-28: If an error happens, show the error message through the alert dialog.

2.11.3.3 tfe_text_view_save Function

```

1 void
2 tfe_text_view_save (TfeTextView *tv) {
3     g_return_if_fail (TFE_IS_TEXT_VIEW (tv));
4
5     GtkTextBuffer *tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
6     GtkWidget *win = gtk_widget_get_ancestor (GTK_WIDGET (tv), GTK_TYPE_WINDOW);
7
8     if (! gtk_text_buffer_get_modified (tb))
9         return; /* no need to save it */
10    else if (tv->file == NULL)
11        tfe_text_view_saveas (tv);
12    else
13        save_file (tv->file, tb, GTK_WINDOW (win));
14 }

```

- The function `tfe_text_view_save` writes the contents to the `tv->file` file. It calls `tfe_text_view_saveas` or `save_file`.

- 1-3: The function is public, i.e. it is open to the other objects. So, it doesn't have `static` class. Public functions should check the parameter type with `g_return_if_fail` function. If `tv` is not a pointer to a `TfeTextView` instance, then it logs an error message and immediately returns. This function is similar to `g_return_val_if_fail`, but no value is returned because `tfe_text_view_save` doesn't return a value (void).
- 5-6: `GtkTextBuffer` `tb` and `GtkWidget` (`GtkWindow`) `win` are set. The function `gtk_widget_get_ancestor` (`widget`, `type`) returns the first ancestor of the widget with the type, which is a `GType`. The parent-child relationship here is the one for widgets, not classes. More precisely, the returned widget's type is the `type` or a descendant object type of the `type`. Be careful, the "descendant object" in the previous sentence is *not* "descendant widget". For example, the type of `GtkWindow` is `GTK_TYPE_WINDOW` and the one of `TfeTextView` is `TFE_TYPE_TEXT_VIEW`. The top level window may be a `GtkApplicationWindow`, but it is a descendant of `GtkWindow`. Therefore, `gtk_widget_get_ancestor` (`GTK_WIDGET (tv)`, `GTK_TYPE_WINDOW`) possibly returns `GtkWindow` or `GtkApplicationWindow`.
- 8-9: If the buffer hasn't modified, it doesn't need to be saved.
- 10-11: If `tv->file` is `NULL`, which means no file has been given yet, it calls `tfe_text_view_saveas` to prompt a user to select a file and save the contents.
- 12-13: Otherwise, it calls `save_file` to save the contents to the file `tv->file`.

2.11.3.4 tfe_text_view_saveas Function

```

1 void
2 tfe_text_view_saveas (TfeTextView *tv) {
3     g_return_if_fail (TFE_IS_TEXT_VIEW (tv));
4
5     GtkWidget *win = gtk_widget_get_ancestor (GTK_WIDGET (tv), GTK_TYPE_WINDOW);
6     GtkFileDialog *dialog;
7
8     dialog = gtk_file_dialog_new ();
9     gtk_file_dialog_save (dialog, GTK_WINDOW (win), NULL, save_dialog_cb, tv);
10    g_object_unref (dialog);
11 }

```

The function `tfe_text_view_saveas` shows a file chooser dialog and prompts the user to choose a file and save the contents.

- 1-3: Check the type of `tv` because the function is public.
- 6: `GtkWidget` `win` is set to the window which is an ancestor of `tv`.
- 8: Creates a `GtkFileDialog` instance. `GtkFileDialog` is available since version 4.10. If your `Gtk` version is older than 4.10, use `GtkFileChooserDialog` instead. `GtkFileChooserDialog` is deprecated since version 4.10.
- 9: Calls `gtk_file_dialog_save` function. The arguments are:
 - `dialog`: `GtkFileDialog`.
 - `GTK_WINDOW (win)`: transient parent window.
 - `NULL`: `NULL` means no cancellable object. If you put a cancellable object here, you can cancel the operation by other thread. In many cases, it is `NULL`. See [GCancelable](#) for further information.
 - `save_dialog_cb`: A callback to call when the operation is complete. The type of the pointer to the callback function is [GAsyncReadyCallback](#). If a cancellable object is given and the operation is cancelled, the callback won't be called.
 - `tv`: This is an optional user data which is `gpointer` type. It is used in the callback function.
- 10: Releases the `GtkFileDialog` instance because it is useless anymore.

This function just shows the file chooser dialog. The rest of the operation is done by the callback function.

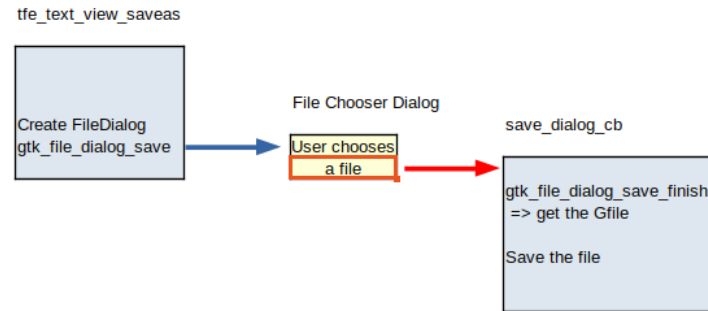


Figure 2.17: Saveas process

2.11.4 Open Related Functions

Open function shows a file chooser dialog to a user and prompts them to choose a file. Then it reads the file and puts the text into GtkTextBuffer.

```
void tfe_text_view_open (TfeTextView *tv, GtkWidget *win);
```

The parameter `win` is the transient window. A file chooser dialog will be shown at the center of the window.

This function may be called just after `tv` has been created. In that case, `tv` has not been incorporated into the widget hierarchy. Therefore it is impossible to get the top-level window from `tv`. That's why the function needs `win` parameter.

This function is usually called when the buffer of `tv` is empty. However, even if the buffer is not empty, `tfe_text_view_open` doesn't treat it as an error. If you want to revert the buffer, calling this function is appropriate.

Open and read process is divided into two phases. One is creating and showing a file chooser dialog and the other is the callback function. The former is `tfe_text_view_open` and the latter is `open_dialog_cb`.

2.11.4.1 open_dialog_cb Function

```

1  static void
2  open_dialog_cb (GObject *source_object, GAsyncResult *res, gpointer data) {
3      GtkFileDialog *dialog = GTK_FILE_DIALOG (source_object);
4      TfeTextView *tv = TFE_TEXT_VIEW (data);
5      GtkTextBuffer *tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
6      GtkWidget *win = gtk_widget_get_ancestor (GTK_WIDGET (tv), GTK_TYPE_WINDOW);
7      GFile *file;
8      char *contents;
9      gsize length;
10     gboolean file_changed;
11     GtkAlertDialog *alert_dialog;
12     GError *err = NULL;
13
14     if ((file = gtk_file_dialog_open_finish (dialog, res, &err)) != NULL
15         && g_file_load_contents (file, NULL, &contents, &length, NULL, &err)) {
16         gtk_text_buffer_set_text (tb, contents, length);
17         g_free (contents);
18         gtk_text_buffer_set_modified (tb, FALSE);
19         // G_IS_FILE(tv->file) && tv->file == file => unref(tv->file), tv->file=file, emit response with
20         // G_IS_FILE(tv->file) && tv->file != file => unref(tv->file), tv->file=file, emit response with
  
```

```

21 // tv->file==NULL => tv->file=file, emit response with
22 // The order is important. If you unref tv->file first, you can't compare tv->file and file anym
23 // And the signals are emitted after new tv->file is set. Or the handler can't catch the new fil
24 file_changed = (G_IS_FILE (tv->file) && g_file_equal (tv->file, file)) ? FALSE : TRUE;
25 if (G_IS_FILE (tv->file))
26     g_object_unref (tv->file);
27 tv->file = file; // The ownership of 'file' moves to TfeTextView
28 if (file_changed)
29     g_signal_emit (tv, tfe_text_view_signals[CHANGE_FILE], 0);
30 g_signal_emit (tv, tfe_text_view_signals[OPEN_RESPONSE], 0, TFE_OPEN_RESPONSE_SUCCESS);
31 } else {
32     if (err->code == GTK_DIALOG_ERROR_DISMISSED) // The user canceled the file chooser dialog
33         g_signal_emit (tv, tfe_text_view_signals[OPEN_RESPONSE], 0, TFE_OPEN_RESPONSE_CANCEL);
34     else {
35         alert_dialog = gtk_alert_dialog_new ("%s", err->message);
36         gtk_alert_dialog_show (alert_dialog, GTK_WINDOW (win));
37         g_object_unref (alert_dialog);
38         g_signal_emit (tv, tfe_text_view_signals[OPEN_RESPONSE], 0, TFE_OPEN_RESPONSE_ERROR);
39     }
40     g_clear_error (&err);
41 }
42 }

```

This function is similar to `save_dialog_cb`. Both are callback functions on a `GtkFileDialog` object.

- 2: It has three parameters like `save_dialog_cb`. They are:
 - `GObject *source_object`: The `GObject` instance that the operation was started with. It is actually the `GtkFileDialog` instance that is shown to the user. It will be cast to `GtkFileDialog` later.
 - `GAsyncResult *res`: The result of the asynchronous operation. It will be given to the `gtk_dialog_open_finish` function.
 - `gpointer data`: A user data set in the `gtk_dialog_open` function. It is actually a `TfeTextView` instance and it will be cast to `TfeTextView` later.
- 14: The function `gtk_file_dialog_open_finish` returns a `GFile` object if the operation has succeeded. Otherwise it returns `NULL`.
- 16-30: If the user selects a file and the file has successfully been read, the codes from 16 to 30 will be executed.
- 16-18: Sets the buffer of `tv` with the text read from the file. And frees `contents`. Then sets the modified status to false.
- 19-30: The codes are a bit complicated. See the comments. If the file (`tv->file`) is changed, “change-file” signal is emitted. The signal “open-response” is emitted with the parameter `TFE_OPEN_RESPONSE_SUCCESS`.
- 31-41: If the operation failed, the codes from 31 to 41 will be executed.
- 32-33: If the error code is `GTK_DIALOG_ERROR_DISMISSED`, it means that the user has clicked on the “Cancel” button or close button on the header bar. Then, “open-response” signal is emitted with the parameter `TFE_OPEN_RESPONSE_CANCEL`. The Dialog error is described [here](#) in the GTK API reference.
- 35-38: If another error occurs, it shows an alert dialog to report the error and emits “open-response” signal with the parameter `TFE_OPEN_RESPONSE_ERROR`.
- 40: Clears the error structure.

2.11.4.2 tfe_text_view_open Function

```

1 void
2 tfe_text_view_open (TfeTextView *tv, GtkWindow *win) {

```

```

3  g_return_if_fail (TFE_IS_TEXT_VIEW (tv));
4  // 'win' is used for a transient window of the GtkFileDialog.
5  // It can be NULL.
6  g_return_if_fail (GTK_IS_WINDOW (win) || win == NULL);
7
8  GtkFileDialog *dialog;
9
10 dialog = gtk_file_dialog_new ();
11 gtk_file_dialog_open (dialog, win, NULL, open_dialog_cb, tv);
12 g_object_unref (dialog);
13 }

```

- 3-6: Check the type of the arguments `tv` and `win`. Public functions always need to check the arguments.
- 10: Creates a `GtkFileDialog` instance.
- 11: Calls `gtk_file_dialog_open`. The arguments are:
 - `dialog`: the `GtkFileDialog` instance
 - `win`: the transient window for the file chooser dialog
 - `NULL`: `NULL` means no cancellable object
 - `open_dialog_cb`: callback function
 - `tv`: user data which is used in the callback function
- 12: Releases the dialog instance because it is useless anymore.

The whole process between the caller and `TfeTextView` is shown in the following diagram. It is really complicated. Because `gtk_file_dialog_open` can't return the status of the operation.

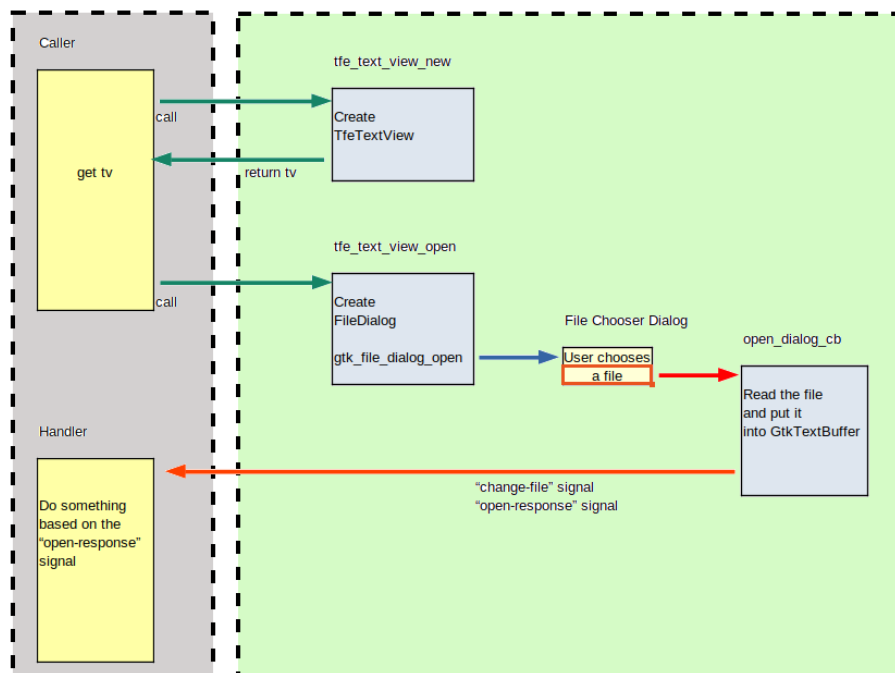


Figure 2.18: Caller and TfeTextView

1. A caller gets a pointer `tv` to a `TfeTextView` instance by calling `tfe_text_view_new`.
2. The caller connects the handler (left bottom in the diagram) and the signal “open-response”.
3. It calls `tfe_text_view_open` to prompt the user to select a file from the file chooser dialog.
4. When the dialog is closed, the callback `open_dialog_cb` is called.
5. The callback function reads the file and inserts the text into `GtkTextBuffer` and emits a signal to inform the status as a response code.

6. The handler of the “open-response” signal is invoked and the operation status is given to it as an argument (signal parameter).

2.11.5 Getting GFile in TfeTextView

You can get the GFile in a TfeTextView instance with `tfe_text_view_get_file`. It is very simple.

```

1  GFile *
2  tfe_text_view_get_file (TfeTextView *tv) {
3      g_return_val_if_fail (TFE_IS_TEXT_VIEW (tv), NULL);
4
5      if (G_IS_FILE (tv->file))
6          return g_file_dup (tv->file);
7      else
8          return NULL;
9  }
```

The important thing is to duplicate `tv->file`. Otherwise, if the caller frees the GFile object, `tv->file` is no more guaranteed to point to the GFile. Another reason to use `g_file_dup` is that GFile isn't thread-safe. If you use GFile in the different thread, the duplication is necessary. See [Gio API Reference – g_file_dup](#).

2.11.6 The API Document and Source File of tfetextview.c

Refer [API document of TfeTextView](#).

You can find all the TfeTextView source codes under `/src/tfetextview` directories.

2.12 Functions in GtkNotebook

GtkNotebook is a very important object in the text file editor `tfe`. It connects the application and TfeTextView objects. A set of public functions are declared in `tfnotebook.h`. The name “tfnotebook” is used only for filenames; no “TfeNotebook” object type is defined. There's no “TfeNotebook” object.

The source files are in the directory `src/tfe5`. You can get them by downloading the [repository](#).

```

1  void
2  notebook_page_save(GtkNotebook *nb);
3
4  void
5  notebook_page_close (GtkNotebook *nb);
6
7  void
8  notebook_page_open (GtkNotebook *nb);
9
10 void
11 notebook_page_new_with_file (GtkNotebook *nb, GFile *file);
12
13 void
14 notebook_page_new (GtkNotebook *nb);
```

This header file describes the public functions in `tfnotebook.c`.

- 1-2: `notebook_page_save` saves the current page to the file with the name specified in the tab.” If the name is `untitled` or `untitled` followed by digits, a file chooser dialog appears and a user can choose or specify a filename.
- 4-5: `notebook_page_close` closes the current page.

- 7-8: `notebook_page_open` shows a file chooser dialog and a user can choose a file. The contents of the file is inserted to a new page.
- 10-11: `notebook_page_new_with_file` creates a new page and a file given as an argument is read and inserted into the page.
- 13-14: `notebook_page_new` creates a new empty page.

You probably find that the functions except `notebook_page_close` are higher level functions of

- `tfe_text_view_save`
- `tfe_text_view_open`
- `tfe_text_view_new_with_file`
- `tfe_text_view_new`

respectively.

There are two layers. One of them is `tfe_text_view` ..., which is the lower level layer. The other is `notebook` ..., which is the higher level layer.

Now let's look at the program of each function.

2.12.1 `notebook_page_new`

```

1  static char *
2  get_untitled () {
3      static int c = -1;
4      if (++c == 0)
5          return g_strdup_printf("Untitled");
6      else
7          return g_strdup_printf ("Untitled%u", c);
8  }
9
10 static void
11 notebook_page_build (GtkNotebook *nb, GtkWidget *tv, const char *filename) {
12     GtkWidget *scr = gtk_scrolled_window_new ();
13     GtkNotebookPage *nbp;
14     GtkWidget *lab;
15     int i;
16
17     gtk_scrolled_window_set_child (GTK_SCROLLED_WINDOW (scr), tv);
18     lab = gtk_label_new (filename);
19     i = gtk_notebook_append_page (nb, scr, lab);
20     nbp = gtk_notebook_get_page (nb, scr);
21     g_object_set (nbp, "tab-expand", TRUE, NULL);
22     gtk_notebook_set_current_page (nb, i);
23     g_signal_connect (tv, "change-file", G_CALLBACK (file_changed_cb), nb);
24 }
25
26 void
27 notebook_page_new (GtkNotebook *nb) {
28     g_return_if_fail(GTK_IS_NOTEBOOK (nb));
29
30     GtkWidget *tv;
31     char *filename;
32
33     tv = tfe_text_view_new ();
34     filename = get_untitled ();

```

```

35  notebook_page_build (nb, tv, filename);
36  g_free (filename);
37  }

```

- 26-37: The function `notebook_page_new`.
- 28: The function `g_return_if_fail` checks the argument. It's necessary because the function is public.
- 33: Creates `TfeTextView` object.
- 34: Creates `filename`, which is "Untitled", "Untitled1",
- 1-8: The function `get_untitled`.
- 3: Static variable `c` is initialized at the first call of this function. After that `c` keeps its value unless it is changed explicitly.
- 4-7: Increases `c` by one and if it is zero, it returns "Untitled". If it is a positive integer, it returns "Untitled<the integer>", for example, "Untitled1", "Untitled2", and so on. The function `g_strdup_printf` creates a string and it should be freed by `g_free` when it becomes useless. The caller of `get_untitled` is in charge of freeing the string.
- 36: Calls `notebook_page_build` to build a new page.
- 37: Frees `filename`.
- 10- 24: The function `notebook_page_build`. A parameter with the `const` qualifier cannot be modified within the function. It means that the argument `filename` is owned by the caller. The caller is responsible for freeing it when it is no longer needed.
- 12: Creates `GtkScrolledWindow`.
- 17: Inserts `tv` to `GtkScrolledWindow` as a child.
- 18-19: Creates `GtkLabel`, then appends `scr` and `lab` to the `GtkNotebook` instance `nb`.
- 20-21: Sets "tab-expand" property to TRUE. The function `g_object_set` sets properties on an object. The object can be any object derived from `GObject`. In many cases, an object has its own function to set its properties, but sometimes does not. In that case, use `g_object_set` to set the property.
- 22: Sets the current page to the newly created page.
- 23: Connects "change-file" signal and the handler `file_changed_cb`.

2.12.2 notebook_page_new_with_file

```

1  void
2  notebook_page_new_with_file (GtkNotebook *nb, GFile *file) {
3      g_return_if_fail(GTK_IS_NOTEBOOK (nb));
4      g_return_if_fail(G_IS_FILE (file));
5
6      GtkWidget *tv;
7      char *filename;
8
9      if ((tv = tfe_text_view_new_with_file (file)) == NULL)
10         return; /* read error */
11     filename = g_file_get_basename (file);
12     notebook_page_build (nb, tv, filename);
13     g_free (filename);
14 }

```

- 9-10: Calls `tfe_text_view_new_with_file`. If the function returns `NULL`, an error has happened. Then, it does nothing and returns.
- 11-13: Gets the filename, builds a new page and frees `filename`.

2.12.3 notebook_page_open

```

1  static void
2  open_response_cb (TfeTextView *tv, int response, GtkNotebook *nb) {
3      GFile *file;
4      char *filename;
5
6      if (response != TFE_OPEN_RESPONSE_SUCCESS) {
7          g_object_ref_sink (tv);
8          g_object_unref (tv);
9      } else {
10         file = tfe_text_view_get_file (tv);
11         filename = g_file_get_basename (file);
12         g_object_unref (file);
13         notebook_page_build (nb, GTK_WIDGET (tv), filename);
14         g_free (filename);
15     }
16 }
17
18 void
19 notebook_page_open (GtkNotebook *nb) {
20     g_return_if_fail(GTK_IS_NOTEBOOK (nb));
21
22     GtkWidget *tv;
23
24     tv = tfe_text_view_new ();
25     g_signal_connect (tv, "open-response", G_CALLBACK (open_response_cb), nb);
26     tfe_text_view_open (TFE_TEXT_VIEW (tv), GTK_WINDOW (gtk_widget_get_ancestor (GTK_WIDGET (nb), GTK_
27 }

```

- 18-27: The function `notebook_page_open`.
- 24: Creates `TfeTextView` object.
- 25: Connects the signal “open-response” and the handler `open_response_cb`.
- 26: Calls `tfe_text_view_open`. The “open-response” signal will be emitted later in this function to inform the result.
- 1-16: The handler `open_response_cb`.
- 6-8: If the response code is not `TFE_OPEN_RESPONSE_SUCCESS`, the instance `tv` will be destroyed. It has floating reference, which will be explained later. A floating reference needs to be converted into an ordinary reference before releasing it. The function `g_object_ref_sink` does that. After that, the function `g_object_unref` releases `tv` and decreases the reference count by one. Finally the reference count becomes zero and `tv` is destroyed.
- 9-15: Otherwise, it builds a new page with `tv`.

2.12.4 Floating Reference

All the widgets are derived from `GInitiallyUnowned`. `GObject` and `GInitiallyUnowned` are almost the same. The difference is like this. When an instance of `GInitiallyUnowned` is created, the instance has a “floating reference”. On the other hand, when an instance of `GObject` (not `GInitiallyUnowned`) is created, it has “normal reference”. Their descendants inherits them, so every widget has a floating reference just after the creation. Non-widget class, for example, `GtkTextBuffer` is a direct sub class of `GObject` and it has normal reference.

The function `g_object_ref_sink` converts the floating reference into a normal reference. If the instance does not have a floating reference, `g_object_ref_sink` simply increases the reference count by one. It is used when an widget is added to another widget as a child.

```
GtkTextView *tv = gtk_text_view_new (); // Floating reference
GtkScrolledWindow *scr = gtk_scrolled_window_new ();
gtk_scrolled_window_set_child (scr, tv); // Scrolled window sinks the tv's floating reference and tv
```

When tv is added to scr as a child, g_object_ref_sink is used.

```
g_object_ref_sink (tv);
```

So, the floating reference is converted into an ordinary reference. That is to say, floating reference is removed, and the normal reference count is one. Thanks to this, the caller does not need to decrease tv's reference count. If an Object_A were not a descendant of GInitiallyUnowned, the program would be like this:

```
Object_A *obj_a = object_a_new (); // reference count is one
GtkScrolledWindow *scr = gtk_scrolled_window_new ();
gtk_scrolled_window_set_child (scr, obj_a); // obj_a's reference count is two
// obj_a is referred by the caller (this program) and scrolled window
g_object_unref (obj_a); // obj_a's reference count is one because the caller no longer refers obj_a.
```

This example tells us that the caller needs to unref obj_a.

If you use g_object_unref to an instance that has a floating reference, you need to convert the floating reference to a normal reference in advance. See [GObject API reference](#) for further information.

2.12.5 notebook_page_close

```
1 void
2 notebook_page_close (GtkNotebook *nb) {
3     g_return_if_fail(GTK_IS_NOTEBOOK (nb));
4
5     GtkWidget *win;
6     int i;
7
8     if (gtk_notebook_get_n_pages (nb) == 1) {
9         win = gtk_widget_get_ancestor (GTK_WIDGET (nb), GTK_TYPE_WINDOW);
10        gtk_window_destroy(GTK_WINDOW (win));
11    } else {
12        i = gtk_notebook_get_current_page (nb);
13        gtk_notebook_remove_page (GTK_NOTEBOOK (nb), i);
14    }
15 }
```

This function closes the current page. If the page is the only page the notebook has, then the function destroys the top-level window and quits the application.

- 8-10: If the page is the only page the notebook has, it calls `gtk_window_destroy` to destroy the top-level window.
- 11-13: Otherwise, removes the current page. The child widget (TfeTextView) is also destroyed.

2.12.6 notebook_page_save

```
1 static TfeTextView *
2 get_current_textview (GtkNotebook *nb) {
3     int i;
4     GtkWidget *scr;
5     GtkWidget *tv;
6
7     i = gtk_notebook_get_current_page (nb);
```



```

8   scr = gtk_notebook_get_nth_page (nb, i);
9   tv = gtk_scrolled_window_get_child (GTK_SCROLLED_WINDOW (scr));
10  return TFE_TEXT_VIEW (tv);
11 }
12
13 void
14 notebook_page_save (GtkNotebook *nb) {
15     g_return_if_fail(GTK_IS_NOTEBOOK (nb));
16
17     TfeTextView *tv;
18
19     tv = get_current_textview (nb);
20     tfe_text_view_save (tv);
21 }

```

- 13-21: `notebook_page_save`.
- 19: Gets the `TfeTextView` instance belongs to the current page. The caller does not have the ownership of `tv` so you don't need to care about the reference count of `tv`.
- 20: Calls `tfe_text_view_save`.
- 1-11: `get_current_textview`. This function gets the `TfeTextView` object belongs to the current page.
- 7: Gets the page number of the current page.
- 8: Gets the child widget `scr`, which is a `GtkScrolledWindow` instance, of the current page. The object `scr` is owned by the notebook `nb`.
- 9-10: Gets the child widget of `scr`, which is a `TfeTextView` instance, and returns it. The returned instance is owned by `scr`.

2.12.7 `file_changed_cb` Handler

The function `file_changed_cb` is a handler connected to “change-file” signal. If a file in a `TfeTextView` instance is changed, the instance emits this signal. This handler changes the label of the `GtkNotebookPage`.

```

1  static void
2  file_changed_cb (TfeTextView *tv, GtkNotebook *nb) {
3      GtkWidget *scr;
4      GtkWidget *label;
5      GFile *file;
6      char *filename;
7
8      file = tfe_text_view_get_file (tv);
9      scr = gtk_widget_get_parent (GTK_WIDGET (tv));
10     if (G_IS_FILE (file)) {
11         filename = g_file_get_basename (file);
12         g_object_unref (file);
13     } else
14         filename = get_untitled ();
15     label = gtk_label_new (filename);
16     g_free (filename);
17     gtk_notebook_set_tab_label (nb, scr, label);
18 }

```

- 8: Gets the `GFile` instance from `tv`.
- 9: Gets the `GkScrolledWindow` instance which is the parent widget of `tv`.
- 10-12: If `file` points to a `GFile` instance, the filename of the `GFile` is assigned to `filename`. Then, unref the `GFile` object `file`.

- 13-14: Otherwise (file is NULL), a string `Untitled(number)` is assigned to `filename`.
- 15-17: Creates a `GtkLabel` instance `label` with the filename and set the label of the `GtkNotebookPage` with `label`.

2.13 Tfe Main Program

The file `tfeapplication.c` is a main program of Tfe. It includes all the code other than `tfetextview.c` and `tfenotebook.c`. It does:

- Application support, mainly handling command line arguments.
- Builds widgets using UI files.
- Connects button signals and their handlers.
- Manages CSS.

2.13.1 The Function main

The function `main` is the first invoked function in C language. It connects the command line given by the user and Gtk application.

```

1  #define APPLICATION_ID "com.github.ToshioCP.tfe"
2
3  int
4  main (int argc, char **argv) {
5      GtkApplication *app;
6      int stat;
7
8      app = gtk_application_new (APPLICATION_ID, G_APPLICATION_HANDLES_OPEN);
9
10     g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
11     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
12     g_signal_connect (app, "open", G_CALLBACK (app_open), NULL);
13
14     stat = g_application_run (G_APPLICATION (app), argc, argv);
15     g_object_unref (app);
16     return stat;
17 }
```

- 1: Defines the application id. Thanks to the `#define` directive, it is easy to find the application id.
- 8: Creates `GtkApplication` object.
- 10-12: Connects “startup”, “activate” and “open” signals to their handlers.
- 14: Runs the application.
- 15-16: Releases the reference to the application and returns the status.

2.13.2 Startup Signal Handler

A startup signal is emitted just after the `GtkApplication` instance is initialized. The handler initializes the whole application which includes not only `GtkApplication` instance but also widgets and some other objects.

- Builds the widgets using a UI file.
- Connects button signals and their handlers.
- Sets CSS.

The handler is as follows.

```

1  static void
2  app_startup (GApplication *application) {
```

```

3   GtkApplication *app = GTK_APPLICATION (application);
4   GtkBuilder *build;
5   GtkApplicationWindow *win;
6   GtkNotebook *nb;
7   GtkButton *btno;
8   GtkButton *bttn;
9   GtkButton *btns;
10  GtkButton *btnc;
11
12  build = gtk_builder_new_from_resource ("/com/github/ToshioCP/tfe/tfe.ui");
13  win = GTK_APPLICATION_WINDOW (gtk_builder_get_object (build, "win"));
14  nb = GTK_NOTEBOOK (gtk_builder_get_object (build, "nb"));
15  gtk_window_set_application (GTK_WINDOW (win), app);
16  btno = GTK_BUTTON (gtk_builder_get_object (build, "btno"));
17  bttn = GTK_BUTTON (gtk_builder_get_object (build, "bttn"));
18  btns = GTK_BUTTON (gtk_builder_get_object (build, "btns"));
19  btnc = GTK_BUTTON (gtk_builder_get_object (build, "btnc"));
20  g_signal_connect_swapped (btno, "clicked", G_CALLBACK (open_cb), nb);
21  g_signal_connect_swapped (bttn, "clicked", G_CALLBACK (new_cb), nb);
22  g_signal_connect_swapped (btns, "clicked", G_CALLBACK (save_cb), nb);
23  g_signal_connect_swapped (btnc, "clicked", G_CALLBACK (close_cb), nb);
24  g_object_unref (build);
25
26  GdkDisplay *display;
27
28  display = gdk_display_get_default ();
29  GtkCssProvider *provider = gtk_css_provider_new ();
30  gtk_css_provider_load_from_string (provider, "textview {padding: 10px; font-family: monospace; font-size: 12px;}");
31  gtk_style_context_add_provider_for_display (display, GTK_STYLE_PROVIDER (provider), GTK_STYLE_PROVIDER_PRIORITY_USER);
32
33  g_signal_connect (win, "destroy", G_CALLBACK (before_destroy), provider);
34  g_object_unref (provider);
35 }

```

- 12-15: Builds widgets using a UI resource. Connects the top-level window and the application with `gtk_window_set_application`.
- 16-23: Gets buttons and connects their signals and handlers. The macro `g_signal_connect_swapped` connects a signal and handler like `g_signal_connect`. The difference is that `g_signal_connect_swapped` swaps the user data for the object. For example, the macro on line 20 swaps `nb` for `btno`. So, the handler expects that the first argument is `nb` instead of `btno`.
- 24: Releases the reference to `GtkBuilder`.
- 26-31: Sets CSS. CSS in Gtk is similar to CSS in HTML. You can set margin, border, padding, color, font and so on with CSS. In this program, CSS is on line 30. It sets padding, font-family and font size of `GtkTextView`. CSS will be explained in the next subsection.
- 26-28: `GdkDisplay` is used to set CSS. The default `GdkDisplay` object can be obtained with the function `gdk_display_get_default`. This function needs to be called after the application is registered.
- 33: Connects “destroy” signal on the main window and `before_destroy` handler. This handler is explained in the next subsection.
- 34: The provider is useless for the startup handler, so it is released. Note: It doesn’t mean the destruction of the provider. It is referred by the display so the reference count is not zero.

2.13.3 CSS in Gtk

CSS is an abbreviation of Cascading Style Sheet. It is originally used with HTML to describe the presentation semantics of a document. You might have found that widgets in Gtk is similar to elements in HTML. It shows that CSS can be applied to Gtk windowing system, too.

2.13.3.1 CSS Nodes and Selectors

The syntax of CSS is as follows.

```
selector { color: yellow; padding-top: 10px; ...}
```

Every widget has CSS node. For example, GtkTextView has `textview` node. If you want to set style to GtkTextView, substitute “textview” for `selector` above.

```
textview {color: yellow; ...}
```

Class, ID and some other things can be applied to the selector like Web CSS. Refer to [GTK 4 API Reference – CSS in Gtk](#) for further information.

The codes of the startup handler has a CSS string on line 30.

```
textview {padding: 10px; font-family: monospace; font-size: 12pt;}
```

- Padding is a space between the border and contents. This space makes the textview easier to read.
- font-family is a name of font. The font name “monospace” is one of the generic family font keywords.
- Font-size is set to 12pt.

2.13.3.2 GtkStyleContext, GtkCssProvider and GdkDisplay

GtkStyleContext is deprecated since version 4.10. But two functions `gtk_style_context_add_provider_for_display` and `gtk_style_context_remove_provider_for_display` are not deprecated. They add or remove a css provider object to the GdkDisplay object.

GtkCssProvider is an object which parses CSS for style widgets.

To apply your CSS to widgets, you need to add GtkStyleProvider (the interface of GtkCssProvider) to the GdkDisplay object. You can get the default display object with the function `gdk_display_get_default`. The returned object is owned by the function and you don’t have its ownership. So, you don’t need to care about releasing it.

Look at the source file of the `startup` handler again.

- 28: The display is obtained by `gdk_display_get_default`.
- 29: Creates a GtkCssProvider instance.
- 30: Puts the CSS into the provider.
- 31: Adds the provider to the display. The last argument of `gtk_style_context_add_provider_for_display` is the priority of the style provider. `GTK_STYLE_PROVIDER_PRIORITY_APPLICATION` is a priority for application-specific style information. Refer to [GTK 4 Reference — Constants](#) for more information. You can find other constants, which have “`STYLE_PROVIDER_PRIORITY_XXXX`” pattern names.

```
1 static void
2 before_destroy (GtkWidget *win, GtkCssProvider *provider) {
3     GdkDisplay *display = gdk_display_get_default ();
4     gtk_style_context_remove_provider_for_display (display, GTK_STYLE_PROVIDER (provider));
5 }
```

When a widget is destroyed, or more precisely during its disposing process, a “destroy” signal is emitted. The “before_destroy” handler connects to the signal on the main window. (See the program list of `app_startup`.) So, it is called when the window is destroyed.

The handler removes the CSS provider from the GdkDisplay.

Note: CSS providers are removed automatically when the application quits. So, even if the handler `before_destroy` is removed, the application works.

2.13.4 Activate and open signal handler

The handlers of “activate” and “open” signals are `app_activate` and `app_open` respectively. They just create a new `GtkNotebookPage`.

```

1  static void
2  app_activate (GApplication *application) {
3      GtkApplication *app = GTK_APPLICATION (application);
4      GtkWidget *win = GTK_WIDGET (gtk_application_get_active_window (app));
5      GtkWidget *boxv = gtk_window_get_child (GTK_WINDOW (win));
6      GtkNotebook *nb = GTK_NOTEBOOK (gtk_widget_get_last_child (boxv));
7
8      notebook_page_new (nb);
9      gtk_window_present (GTK_WINDOW (win));
10 }
11
12 static void
13 app_open (GApplication *application, GFile ** files, gint n_files, const gchar *hint) {
14     GtkApplication *app = GTK_APPLICATION (application);
15     GtkWidget *win = GTK_WIDGET (gtk_application_get_active_window (app));
16     GtkWidget *boxv = gtk_window_get_child (GTK_WINDOW (win));
17     GtkNotebook *nb = GTK_NOTEBOOK (gtk_widget_get_last_child (boxv));
18     int i;
19
20     for (i = 0; i < n_files; i++)
21         notebook_page_new_with_file (nb, files[i]);
22     if (gtk_notebook_get_n_pages (nb) == 0)
23         notebook_page_new (nb);
24     gtk_window_present (GTK_WINDOW (win));
25 }

```

- 1-10: `app_activate`.
- 6: In this program, we traverse the widget hierarchy for the sake of simplicity. However, the most robust design is to extend `GtkApplicationWindow` and include a pointer to the notebook in its instance structure (e.g., `win->nb`).
- 8-10: Creates a new page and shows the window.
- 12-25: `app_open`.
- 20-21: Creates notebook pages with files.
- 22-23: If no page has been created, maybe because of read error, then it creates an empty page.
- 24: Shows the window.

These codes have become really simple thanks to `tfnotebook.c` and `tfetextview.c`.

2.13.5 A primary instance

Only one `GApplication` instance can be run at a time in a session. The session is a bit difficult concept and also platform-dependent, but roughly speaking, it corresponds to a graphical desktop login. When you use your PC, you probably login first, then your desktop appears until you log off. This is the session.

However, Linux is multi process OS and you can run two or more instances of the same application. Isn't it a contradiction?

When first instance is launched, then it registers itself with its application ID (for example, `com.github.ToshioCP.tfe`). Just after the registration, the startup signal is emitted, then activate or open signal is emitted and the instance's main loop runs. I wrote "a startup signal is emitted just after the application instance is initialized" in the prior subsection. More precisely, it is emitted after the registration.

If another instance which has the same application ID is launched, it also tries to register itself. Because this is the second instance, the registration of the ID has already been done, so it fails. Because of the failure, a startup signal isn't emitted. After that, activate or open signal is emitted in the primary instance, not on the second instance. The primary instance receives the signal and its handler is invoked. On the other hand, the second instance doesn't receive the signal and it immediately quits.

Try running two instances in a row.

```
$ ./_build/tfe &
[1] 84453
$ ./build/tfe tfeapplication.c
$
```

First, the primary instance opens a window. Then, after the second instance is run, a new notebook page with the contents of `tfeapplication.c` appears in the primary instance's window. This is because the open signal is emitted in the primary instance. The second instance immediately quits so shell prompt soon appears.

2.13.6 A series of handlers corresponds to the button signals

```
1 static void
2 open_cb (GtkNotebook *nb) {
3     notebook_page_open (nb);
4 }
5
6 static void
7 new_cb (GtkNotebook *nb) {
8     notebook_page_new (nb);
9 }
10
11 static void
12 save_cb (GtkNotebook *nb) {
13     notebook_page_save (nb);
14 }
15
16 static void
17 close_cb (GtkNotebook *nb) {
18     notebook_page_close (GTK_NOTEBOOK (nb));
19 }
```

`open_cb`, `new_cb`, `save_cb` and `close_cb` just call corresponding notebook page functions.

2.13.7 meson.build

```
1 project('tfe', 'c')
2
3 gtkdep = dependency('gtk4')
4
5 gnome=import('gnome')
6 resources = gnome.compile_resources('resources','tfe.gresource.xml')
```

```

7
8 sourcefiles=files('tfeapplication.c', 'tfenotebook.c', '../tfetextview/tfetextview.c')
9
10 executable('tfe', sourcefiles, resources, dependencies: gtkdep)

```

In this file, just the source file names are modified from the prior version.

2.13.8 source files

You can download the files from the [repository](#). There are two options.

- Use git and clone.
- Run your browser and open the [top page](#). Then click on “Code” button and click “Download ZIP” in the popup menu. After that, unzip the archive file.

If you use git, run the terminal and type the following.

```
$ git clone https://github.com/ToshioCP/Gtk4-tutorial.git
```

The source files are under [/src/tfe5](#) directory.

2.14 How to Build tfe (Text File Editor)

2.14.1 How to Compile and Execute the Text Editor ‘tfe’

First, the source files are in the [Gtk4-tutorial repository](#). How to download them is written at the end of the [previous section](#).

The following is the instruction for compilation and execution.

- You need meson and ninja.
- If you have installed gtk4 from source, you need to set environment variables to match your installation.
- Change your current directory to `src/tfe5` directory.
- Type `meson setup _build` for configuration.
- Type `ninja -C _build` for compilation. Then the application `tfe` is built under the `_build` directory.
- Type `_build/tfe` to execute it.

Then the window appears. There are four buttons, **New**, **Open**, **Save** and **Close**.

- Click on **Open** button, then a file chooser dialog appears. Choose a file from the list and click on **Open** button. Then the file is read and a new Notebook Page appears.
- Edit the file and click on **Save** button, then the text is saved to the original file.
- Click **Close**, then the Notebook Page disappears.
- Click **Close** again, then the **Untitled** Notebook Page disappears and at the same time the application quits.

This is a very simple editor. It is a good practice for you to add more features.

2.14.2 Total Number of Lines, Words and Characters

The command below is the output of `wc` executed in the `src` directory.

```

$ LANG=C wc tfe5/meson.build tfe5/tfeapplication.c tfe5/tfe.gresource.xml tfe5/tfenotebook.c tfe5/tf
  10   17   294 tfe5/meson.build
 110  334 3600 tfe5/tfeapplication.c
   6    9   153 tfe5/tfe.gresource.xml
 144  389 3637 tfe5/tfenotebook.c
  15   21   241 tfe5/tfenotebook.h

```

```

235    825   8516 tfetextview/tfetextview.c
    32    54    624 tfetextview/tfetextview.h
    61   100   2073 tfe5/tfe.ui
613   1749  19138 total
$

```

2.15 Menus and Actions

2.15.1 Menus

Users often use menus to give a command to the application. It is like this:

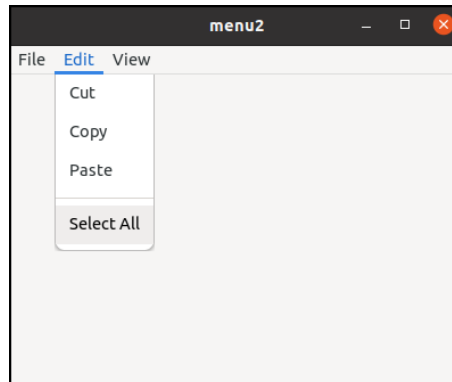


Figure 2.19: Menu

There are two types of objects.

- “File”, “Edit”, “View”, “Cut”, “Copy”, “Paste” and “Select All”. They are called “menu item” or simply “item”. When the user clicks one of these items, then something will happen.
- Menubar, submenu referenced by “Edit” item and two sections. They are called “menu”. Menu is an ordered list of items. They are similar to arrays.

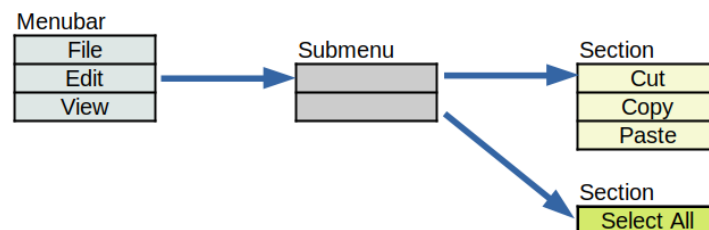


Figure 2.20: Menu structure

- Menubar is a menu which has three items, which are “File”, “Edit” and “View”.
- The menu item labeled “Edit” has a link to the submenu which has two items. These two items don’t have labels. Each item refers to a section.
- The first section is a menu which has three items – “Cut”, “Copy” and “Paste”.
- The second section is a menu which has one item – “Select All”.

Menus can build a complicated structure thanks to the links of menu items.

2.15.2 GMenuModel, GMenu and GMenuItem

GMenuModel is an abstract object which represents a menu. GMenu is a simple implementation of GMenuModel and a child object of GMenuModel.

GObject -- GMenuModel -- GMenu

Because GMenuModel is an abstract object, it isn't instantiatable. Therefore, it doesn't have any functions to create its instance. If you want to create a menu, use `g_menu_new` to create a GMenu instance. GMenu inherits all the functions of GMenuModel.

GMenuItem is an object directly derived from GObject. GMenuItem and GMenu (or GMenuModel) don't have a parent-child relationship.

GObject -- GMenuModel -- GMenu
GObject -- GMenuItem

GMenuItem has attributes. One of the attributes is label. For example, there is a menu item which has "Edit" label in the first diagram. "Cut", "Copy", "Paste" and "Select All" are also the labels of the menu items. Other attributes will be explained later.

Some menu items have a link to another GMenu. There are two types of links, submenu and section.

GMenuItem can be inserted, appended or prepended to GMenu. When it is inserted, all of the attributes and link values are copied and stored in the menu. The GMenuItem itself is not really inserted. Therefore, after the insertion, GMenuItem is useless and it should be freed. The same goes for appending or prepending.

The following code shows how to append GMenuItem to GMenu.

```
GMenu *menu = g_menu_new ();
GMenuItem *menu_item_quit = g_menu_item_new ("Quit", "app.quit");
g_menu_append_item (menu, menu_item_quit);
g_object_unref (menu_item_quit);
```

2.15.3 Menu and Action

One of the attributes of menu items is an action. This attribute points to an action object.

There are two action objects, GSimpleAction and GPropertyAction. GSimpleAction is often used. And it is used with a menu item. Only GSimpleAction is described in this section.

An action that corresponds to a menu item will be activated when the menu item is clicked. Then the action emits an activate signal.

1. menu item is clicked.
2. The corresponding action is activated.
3. The action emits a signal.
4. The connected handler is invoked.

The following code is an example.

```
static void
quit_activated(GSimpleAction *action, GVariant *parameter, gpointer app) { ... .. }

GSimpleAction *act_quit = g_simple_action_new ("quit", NULL);
g_action_map_add_action (G_ACTION_MAP (app), G_ACTION (act_quit));
g_signal_connect (act_quit, "activate", G_CALLBACK (quit_activated), app);
GMenuItem *menu_item_quit = g_menu_item_new ("Quit", "app.quit");
```

- The variable `menu_item_quit` points to a menu item. It is actually a pointer, but we often say that `menu_item_quit` is a menu item. It has a label "Quit" and is connected to an action "app.quit". "app" is a prefix and "quit" is the name of the action. The prefix "app" means that the action belongs to the GtkApplication instance.
- `act_quit` is an action. It has a name "quit". The function `g_simple_action_new` creates a stateless action. So, `act_quit` is stateless. Stateless actions do not maintain any internal state. They are like

a simple push button; when clicked, they perform an operation (like “quit” or “open”) and that’s all. Most menu items, such as “Quit”, “Copy”, or “Paste”, are connected to stateless actions. Most of the actions are stateless and have no parameter. The argument `NULL` means that the action doesn’t have any parameters. Another action is stateful and stateful actions will be explained in the next section.

- The action `act_quit` is added to the `GtkApplication` instance with `g_action_map_add_action`. So, the action’s scope is application. The prefix of `app.quit` indicates the scope.
- “activate” signal of the action is connected to the handler `quit_activated`.

If the menu is clicked, the corresponding action “quit” will be activated and emits an “activate” signal. Then, the handler `quit_activated` is called.

2.15.4 Menu Bar

A menu bar and menus are traditional. Menu buttons are often used instead of a menu bar lately, but the old style is still used widely.

Applications have only one menu bar. If an application has two or more windows which have menu bars, the menu bars are exactly the same. Because every window refers to the same menubar instance in the application.

An application’s menu bar is usually unchanged once it is set. So, it is appropriate to set it in the “startup” handler. Because the handler is called only once in the primary application instance.

I think it is good for readers to clarify how applications behave.

- When an application runs for the first time, the instance is called primary.
- The primary instance registers itself to the system. If it succeeds, it emits “startup” signal.
- When the instance is activated, an “activate” or “open” signal is emitted.
- If the application is run for the second time or later and there exists a primary instance, the instance is called a remote instance.
- A remote instance doesn’t emit “startup” signal.
- If it tries to emit an “activate” or “open” signal, the signals are not emitted on the remote instance but primary instance.
- The remote instance quits.

Therefore, an “activate” or “open” handler can be called twice or more. On the other hand, a “startup” handler is called once. So, the menubar should be set in the “startup” handler.

```
static void
app_startup (GApplication *app) {
    ... ..
    gtk_application_set_menubar (GTK_APPLICATION (app), G_MENU_MODEL (menubar));
    ... ..
}
```

2.15.5 Simple Example

The following is a simple example of menus and actions. The source file `menu1.c` is located at [/src/menu](#) directory.

```
1  #include <gtk/gtk.h>
2
3  static void
4  quit_activated(GSimpleAction *action, GVariant *parameter, GApplication *application) {
5      g_application_quit (application);
6  }
7
```

```

8  static void
9  app_activate (GApplication *application) {
10     GtkApplication *app = GTK_APPLICATION (application);
11     GtkWidget *win = gtk_application_window_new (app);
12     gtk_window_set_title (GTK_WINDOW (win), "menu1");
13     gtk_window_set_default_size (GTK_WINDOW (win), 400, 300);
14
15     gtk_application_window_set_show_menubar (GTK_APPLICATION_WINDOW (win), TRUE);
16     gtk_window_present (GTK_WINDOW (win));
17 }
18
19 static void
20 app_startup (GApplication *application) {
21     GtkApplication *app = GTK_APPLICATION (application);
22
23     GSimpleAction *act_quit = g_simple_action_new ("quit", NULL);
24     g_action_map_add_action (G_ACTION_MAP (app), G_ACTION (act_quit));
25     g_signal_connect (act_quit, "activate", G_CALLBACK (quit_activated), application);
26
27     GMenu *menubar = g_menu_new ();
28     GMenuItem *menu_item_menu = g_menu_item_new ("Menu", NULL);
29     GMenu *menu = g_menu_new ();
30     GMenuItem *menu_item_quit = g_menu_item_new ("Quit", "app.quit");
31     g_menu_append_item (menu, menu_item_quit);
32     g_object_unref (menu_item_quit);
33     g_menu_item_set_submenu (menu_item_menu, G_MENU_MODEL (menu));
34     g_object_unref (menu);
35     g_menu_append_item (menubar, menu_item_menu);
36     g_object_unref (menu_item_menu);
37
38     gtk_application_set_menubar (GTK_APPLICATION (app), G_MENU_MODEL (menubar));
39 }
40
41 #define APPLICATION_ID "com.github.ToshioCP.menu1"
42
43 int
44 main (int argc, char **argv) {
45     GtkApplication *app;
46     int stat;
47
48     app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
49     g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
50     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
51
52     stat = g_application_run (G_APPLICATION (app), argc, argv);
53     g_object_unref (app);
54     return stat;
55 }

```

- 3-6: `quit_activated` is a handler of the “activate” signal on the action `act_quit`. Handlers of the “activate” signal have three parameters.
 1. The action instance on which the signal is emitted.
 2. Parameter. In this example it is `NULL` because the second argument of `g_simple_action_new` (line 23) is `NULL`. You don’t need to care about it.

3. User data. It is the fourth parameter in the `g_signal_connect` (line 25) that connects the action and the handler.
- 5: The function `g_application_quit` immediately quits the application.
- 8-17: `app_activate` is an “activate” signal handler on the application.
- 11-13: Creates a `GtkApplicationWindow` `win`. And sets the title and the default size.
- 15: Sets `GtkApplicationWindow` to show the menubar.
- 16: Shows the window.
- 19-38: `app_startup` is a “startup” signal handler on the application.
- 23: Creates `GSimpleAction` `act_quit`. It is stateless. The first argument of `g_simple_action_new` is a name of the action and the second argument is a parameter. If you don’t need the parameter, pass `NULL`. Therefore, `act_quit` has a name “quit” and no parameter.
- 24: Adds the action to `GtkApplication` `app`. `GtkApplication` implements an interface `GActionMap` and `GActionGroup`. `GtkApplication` (`GActionMap`) can have a group of actions and the actions are added with the function `g_action_map_add_action`. This function is described in [Gio API Reference – g_action_map_add_action](#). Because this action belongs to `GtkApplication`, its scope is “app” and it is referred with “app.quit” if the prefix (scope) is necessary.
- 25: Connects “activate” signal of the action and the handler `quit_activated`.
- 27-30: Creates `GMenu` and `GMenuItem` instances. `menubar` and `menu` are `GMenu`. `menu_item_menu` and `menu_item_quit` are `GMenuItem`. `menu_item_menu` has a label “Menu” and no action. `menu_item_quit` has a label “Quit” and an action “app.quit”.
- 31-32: Appends `menu_item_quit` to `menu`. As I mentioned before, all the attributes and links are copied and used to form a new item in `menu`. Therefore after the addition, `menu_item_quit` is no longer needed. It is freed by `g_object_unref`.
- 33-34: Sets the submenu link in `menu_item_menu` to refer to `menu`. Then, `menu` is no more useful and it is freed.
- 35-36: Appends `menu_item_menu` to `menubar`. Then frees `menu_item_menu`. `GMenu` and `GMenuItem` are built and finally connected to the variable `menubar`. The structure of the menu is shown in the diagram below.
- 38: The menubar is inserted to the application.

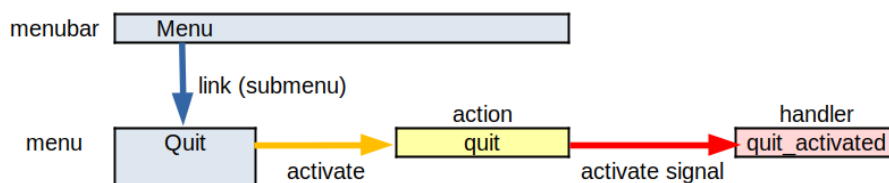


Figure 2.21: menu and action

2.15.6 Compiling

Change your current directory to `src/menu`. Use `comp` to compile `menu1.c`.

```
$ comp menu1
$ ./a.out
```

Then, a window appears. Click on “Menu” on the menubar, then a menu appears. Click on “Quit” menu, then the application quits.

2.15.7 Primary and Remote Application Instances

Let’s try running the application twice. Use `&` in your shell command line, then the application runs concurrently.

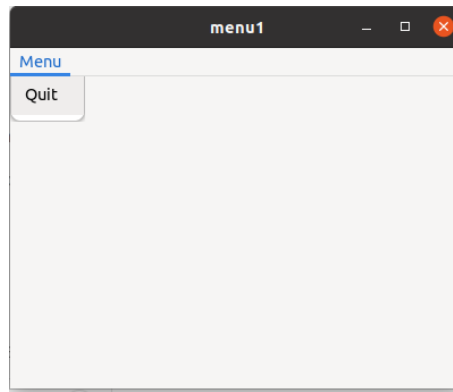


Figure 2.22: Screenshot of menu1

```
$ ./a.out &
[1] 70969
$ ./a.out
$
```

Then, two windows appear.

- The first `./a.out` calls the application and a primary instance is created. It calls “startup” and “activate” handlers and shows a window.
- The second `./a.out` calls the the application again and the created instance is a remote one. It doesn’t emit “startup” signal. And it activates the application but the “activate” signal is emitted on the primary instance. The remote instance quits.
- The primary instance called “activate” handler. The handler creates a new window. It adds a menu bar to the window with `gtk_application_window_set_show_menubar` function.

Both the windows have menu bars. And they are exactly the same. The two windows belong to the primary instance.

If you click on the “Quit” menu, the application (the primary instance) quits.

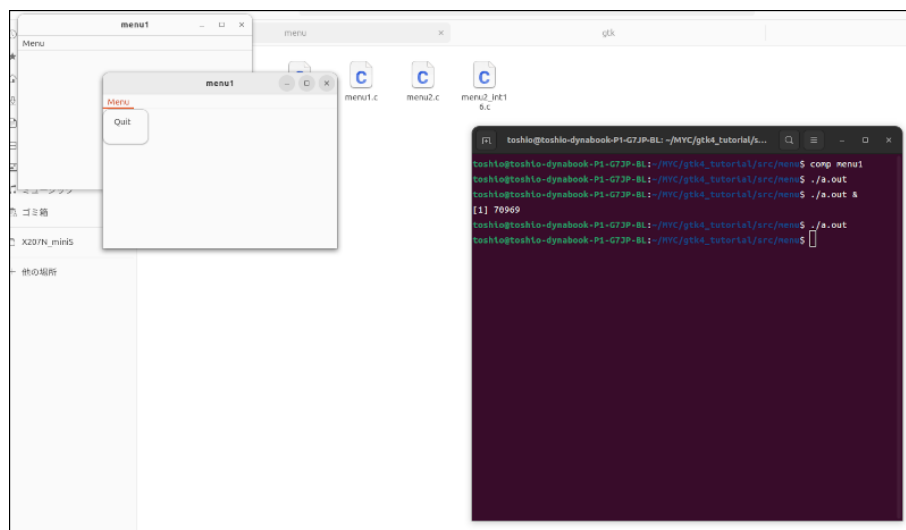


Figure 2.23: menu1 – two windows

The second execution makes a new window. However, it depends on the “activate” handler. If you create your window in the startup handler and the activate handler just presents the window, no new window is created at the second execution. For example, `tfe` (text file editor) doesn’t create a second window. It just

creates a new notebook page. Because its activate handler doesn't create any window but just creates a new notebook page.

Second or more executions often happen on the desktop applications. If you double-click the icon twice or more, the application is run multiple times. Therefore, you need to think about your startup and activate (open) handler carefully.

2.16 Stateful Action

Some actions have states. The typical values of states are boolean or string values. However, other types of states are possible if necessary.

Actions which have states are called stateful.

2.16.1 Stateful Action without a Parameter

Some menus are called toggle menus. For example, the “Maximized” menu has two state values – maximized and unmaximized. The value of the state is changed every time the menu is clicked. An action that corresponds to the “Maximized” menu also has a state. The value of the state is boolean, which is TRUE or FALSE. TRUE corresponds to maximized and FALSE to unmaximized.

The following is an example code to implement a “Maximized” menu. The code of `maximize_state_changed` will be shown later.

```
GSimpleAction *act_maximize = g_simple_action_new_stateful ("maximize",
                                                           NULL, g_variant_new_boolean (FALSE));
g_signal_connect (act_maximize, "change-state", G_CALLBACK (maximize_state_changed), win);
g_action_map_add_action (G_ACTION_MAP (win), G_ACTION (act_maximize));
... ..
GMenuItem *menu_item_maximize = g_menu_item_new ("Maximized", "win.maximize");
```

- `act_maximize` is a `GSimpleAction` instance. It is created with `g_simple_action_new_stateful`. The function has three arguments. The first argument “maximize” is the name of the action. The second argument is a parameter type. NULL means the action doesn't have a parameter. The third argument is the initial state of the action. It is a `GVariant` value. `GVariant` will be explained in the next subsection. The function `g_variant_new_boolean (FALSE)` returns a `GVariant` value of FALSE. If there are two or more top-level windows, each window has its own `act_maximize` action. So, the number of the actions is the same as the number of the windows.
- The `act_maximize` action has a “change-state” signal. The signal is connected to a handler `maximize_state_changed`. If the “Maximized” menu is clicked, then the corresponding action `act_maximize` is activated. But no handler is connected to the “activate” signal. Then, the default behavior for boolean-stated actions with a NULL parameter type like `act_maximize` is to toggle them via the “change-state” signal.
- The action is added to the `GtkWindow`, `win`. Therefore, the scope of the action is “win”.
- `menu_item_maximize` is a `GMenuItem` instance. There are two arguments. The first argument “Maximized” is the label of `menu_item_maximize`. The second argument is an action. The action “win.maximize” has a prefix “win” and an action name “maximize”. The prefix indicates that the action belongs to the window.

```
1 static void
2 maximize_state_changed(GSimpleAction *action, GVariant *value, GtkWindow *win) {
3     if (g_variant_get_boolean (value))
4         gtk_window_maximize (win);
5     else
6         gtk_window_unmaximize (win);
```

```

7   g_simple_action_set_state (action, value);
8 }

```

This is the signal handler that has been connected to the “change-state” signal of the “maximize” action.

- The handler `maximize_state_changed` has three parameters. The first parameter is the action which emits the “change-state” signal. The second parameter is the value of the new state of the action. The third parameter is the user data that was passed to `g_signal_connect`.
- If the value is of boolean type and `TRUE`, then it maximizes the window. Otherwise it unmaximizes it.
- Sets the state of the action with `value`. Note: Before `g_simple_action_set_state` is called, the action still holds its original state. You must call this function to update the action to the new state.

You can use “activate” signal instead of “change-state” signal, or both signals. But the way above is the simplest and the best.

2.16.1.1 GVariant

GVariant is a fundamental type. It isn’t a child of GObject. GVariant can contain boolean, string or other type values. For example, the following program assigns `TRUE` to `value` whose type is GVariant.

```
GVariant *value = g_variant_new_boolean (TRUE);
```

Another example is:

```
GVariant *value2 = g_variant_new_string ("Hello");
```

`value2` is a GVariant and it has a string type value “Hello”. GVariant can contain other types like `int16`, `int32`, `int64`, `double` and so on.

If you want to get the original value, use `g_variant_get` series functions. For example, you can get the boolean value with `g_variant_get_boolean`.

```
gboolean bool = g_variant_get_boolean (value);
```

Since `value` has been created as a boolean type GVariant with `TRUE` value, `bool` equals `TRUE`. In the same way, you can get a string from `value2`

```
const char *str = g_variant_get_string (value2, NULL);
```

The second parameter is a pointer to `gsize` type variable (`gsize` is defined as unsigned long). If it isn’t `NULL`, then the variable is used by the function for the string length. If it is `NULL`, nothing happens. The returned string `str` is owned by the instance and can’t be changed or freed by the caller.

2.16.2 Stateful Action with a Parameter

Another example of stateful actions is an action that corresponds to color select menus. For example, there are three menus and each menu has red, green or blue color respectively. They determine the background color of a GtkLabel widget. One action is connected to the three menus. The action has a state whose value is “red”, “green” or “blue”. The values are string. Those colors are given to the signal handler as a parameter.

```

... ..
GVariantType *vtype = g_variant_type_new("s");
GSimpleAction *act_color
    = g_simple_action_new_stateful ("color", vtype, g_variant_new_string ("red"));
g_variant_type_free (vtype);
GMenuItem *menu_item_red = g_menu_item_new ("Red", "app.color::red");
GMenuItem *menu_item_green = g_menu_item_new ("Green", "app.color::green");
GMenuItem *menu_item_blue = g_menu_item_new ("Blue", "app.color::blue");

```

```
g_signal_connect (act_color, "activate", G_CALLBACK (color_activated), NULL);
... ..
```

- GVariantType is a C structure and it keeps a type of GVariant. It is created with the function `g_variant_type_new`. The argument of the function is a GVariant type string. So, `g_variant_type_new("s")` returns a GVariantType structure containing a string type. The returned value, GVariantType structure, is owned by the caller. So, you need to free it when it is no longer needed.
- The variable `act_color` points to a GSimpleAction instance. It is created with `g_simple_action_new_stateful`. The function has three arguments. The first argument “color” is the name of the action. The second argument is a parameter type which is GVariantType. `g_variant_type_new("s")` creates GVariantType which is a string type (`G_VARIANT_TYPE_STRING`). The third argument is the initial state of the action. It is a GVariant. The function `g_variant_new_string ("red")` returns a GVariant value which has the string value “red”. GVariant has a reference count and `g_variant_new...` series functions return a GVariant value with a floating reference. That means the caller doesn’t own the value at this point. And `g_simple_action_new_stateful` function consumes the floating reference so you don’t need to care about releasing the GVariant instance.
- The GVariantType structure `vtype` is useless after `g_simple_action_new_stateful`. It is released with the function `g_variant_type_free`.
- The variable `menu_item_red` points to a GMenuItem instance. The function `g_menu_item_new` has two arguments. The first argument “Red” is the label of `menu_item_red`. The second argument is a detailed action. Its prefix is “app”, the action name is “color” and the target is “red”. Target is sent to the action as a parameter. The same goes for `menu_item_green` and `menu_item_blue`.
- The function `g_signal_connect` connects the activate signal on the action `act_color` and the handler `color_activated`. If one of the three menus is clicked, then the action `act_color` is activated with the target (parameter) which is given by the menu.

The following is the “activate” signal handler.

```
static void
color_activated(GSimpleAction *action, GVariant *parameter) {
    char *color = g_strdup_printf ("label.lb {background-color: %s;}",
                                   g_variant_get_string (parameter, NULL));
    gtk_css_provider_load_from_string (provider, color);
    g_free (color);
    g_action_change_state (G_ACTION (action), parameter);
}
```

- The handler originally has three parameters. The third parameter is the user data set in the `g_signal_connect` function. But it is left out because the fourth argument of the `g_signal_connect` has been NULL. The first parameter is the action which emits the “activate” signal. The second parameter is the parameter, or target, given to the action. It is a color specified by the menu.
- The variable `color` is a CSS string created by `g_strdup_printf`. The arguments of `g_strdup_printf` are the same as `printf` C standard function. The function `g_variant_get_string` gets the string contained in `parameter`. The string is owned by the instance and you mustn’t change or free it. The string `label.lb` is a selector. It consists of `label`, a node name of `GtkLabel`, and `lb` which is a class name. It selects `GtkLabel` which has `lb` class. For example, menus have `GtkLabel` to display their labels, but they don’t have `lb` class. So, the CSS doesn’t change their background color. The string `{background-color: %s;}` makes the background color `%s` to which the color from `parameter` is assigned.
- Frees the string `color`.
- Changes the state with `g_action_change_state`.

Note: If you haven’t set an “activate” signal handler, the signal is forwarded to “change-state” signal. So, you can use “change-state” signal instead of “activate” signal. See [src/menu/menu2_change_state.c](#).

2.16.2.1 GVariantType

GVariantType gives a type of GVariant. GVariantType is created with a type string.

- “b” means boolean type.
- “s” means string type.

The following program is a simple example. It finally outputs the string “s”.

```

1  #include <glib.h>
2
3  int
4  main (int argc, char **argv) {
5      GVariantType *vtype = g_variant_type_new ("s");
6      const char *type_string = g_variant_type_peek_string (vtype);
7      g_print ("%s\n", type_string);
8      g_variant_type_free (vtype);
9  }
```

- The function `g_variant_type_new` creates a `GVariantType` structure. The argument “s” is a type string. It means string. The returned structure is owned by the caller. When it becomes useless, you need to free it with the function `g_variant_type_free`.
- The function `g_variant_type_peek_string` takes a peek at `vtype`. It is the string “s” given to `vtype` when it was created. The string is owned by the instance and the caller can’t change or free it.
- Prints the string to the terminal. You can’t free `vtype` before `g_print` because the string `type_string` is owned by `vtype`.
- Frees `vtype`.

2.16.3 Example

The following code includes stateful actions above. This program has menus like this:

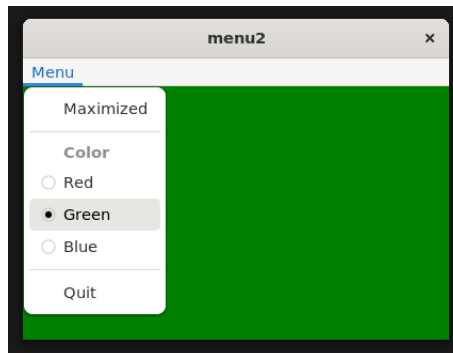


Figure 2.24: menu2

- The “Maximized” menu toggles the size of the window between maximized and unmaximized. If the window is maximized, a check mark is put before the “Maximized” label.
- Red, green and blue menu determines the back ground color of the label in the window. The menus have radio buttons on the left of the menus. And the radio button of the selected menu turns on.
- The “quit” menu quits the application.

The code is as follows.

```

1  #include <gtk/gtk.h>
2
3  /* The provider below provides application wide CSS data. */
4  GtkCssProvider *provider;
```

```

5
6 static void
7 maximize_state_changed(GSimpleAction *action, GVariant *value, GtkWindow *win) {
8     if (g_variant_get_boolean (value))
9         gtk_window_maximize (win);
10    else
11        gtk_window_unmaximize (win);
12    g_simple_action_set_state (action, value);
13 }
14
15 static void
16 color_activated(GSimpleAction *action, GVariant *parameter) {
17     char *color = g_strdup_printf ("label.lb {background-color: %s;}", g_variant_get_string (parameter
18     /* Change the CSS data in the provider. */
19     /* Previous data is thrown away. */
20     gtk_css_provider_load_from_string (provider, color);
21     g_free (color);
22     g_action_change_state (G_ACTION (action), parameter);
23 }
24
25 static void
26 app_shutdown (GApplication *app, GtkCssProvider *provider) {
27     gtk_style_context_remove_provider_for_display (gdk_display_get_default(), GTK_STYLE_PROVIDER (prov
28 }
29
30 static void
31 app_activate (GApplication *app) {
32     GtkWindow *win = GTK_WINDOW (gtk_application_window_new (GTK_APPLICATION (app)));
33     gtk_window_set_title (win, "menu2");
34     gtk_window_set_default_size (win, 400, 300);
35
36     GtkWidget *lb = gtk_label_new (NULL);
37     gtk_widget_add_css_class (lb, "lb"); /* the class is used by CSS Selector */
38     gtk_window_set_child (win, lb);
39
40     GSimpleAction *act_maximize
41         = g_simple_action_new_stateful ("maximize", NULL, g_variant_new_boolean (FALSE));
42     g_signal_connect (act_maximize, "change-state", G_CALLBACK (maximize_state_changed), win);
43     g_action_map_add_action (G_ACTION_MAP (win), G_ACTION (act_maximize));
44
45     gtk_application_window_set_show_menubar (GTK_APPLICATION_WINDOW (win), TRUE);
46
47     gtk_window_present (win);
48 }
49
50 static void
51 app_startup (GApplication *app) {
52     GVariantType *vtype = g_variant_type_new("s");
53     GSimpleAction *act_color
54         = g_simple_action_new_stateful ("color", vtype, g_variant_new_string ("red"));
55     g_variant_type_free (vtype);
56     GSimpleAction *act_quit
57         = g_simple_action_new ("quit", NULL);

```

```

58 g_signal_connect (act_color, "activate", G_CALLBACK (color_activated), NULL);
59 g_signal_connect_swapped (act_quit, "activate", G_CALLBACK (g_application_quit), app);
60 g_action_map_add_action (G_ACTION_MAP (app), G_ACTION (act_color));
61 g_action_map_add_action (G_ACTION_MAP (app), G_ACTION (act_quit));
62
63 GMenu *menubar = g_menu_new ();
64 GMenu *menu = g_menu_new ();
65 GMenu *section1 = g_menu_new ();
66 GMenu *section2 = g_menu_new ();
67 GMenu *section3 = g_menu_new ();
68 GMenuItem *menu_item_maximize = g_menu_item_new ("Maximized", "win.maximize");
69 GMenuItem *menu_item_red = g_menu_item_new ("Red", "app.color:red");
70 GMenuItem *menu_item_green = g_menu_item_new ("Green", "app.color:green");
71 GMenuItem *menu_item_blue = g_menu_item_new ("Blue", "app.color:blue");
72 GMenuItem *menu_item_quit = g_menu_item_new ("Quit", "app.quit");
73
74 g_menu_append_item (section1, menu_item_maximize);
75 g_menu_append_item (section2, menu_item_red);
76 g_menu_append_item (section2, menu_item_green);
77 g_menu_append_item (section2, menu_item_blue);
78 g_menu_append_item (section3, menu_item_quit);
79 g_object_unref (menu_item_red);
80 g_object_unref (menu_item_green);
81 g_object_unref (menu_item_blue);
82 g_object_unref (menu_item_maximize);
83 g_object_unref (menu_item_quit);
84
85 g_menu_append_section (menu, NULL, G_MENU_MODEL (section1));
86 g_menu_append_section (menu, "Color", G_MENU_MODEL (section2));
87 g_menu_append_section (menu, NULL, G_MENU_MODEL (section3));
88 g_menu_append_submenu (menubar, "Menu", G_MENU_MODEL (menu));
89 g_object_unref (section1);
90 g_object_unref (section2);
91 g_object_unref (section3);
92 g_object_unref (menu);
93
94 gtk_application_set_menubar (GTK_APPLICATION (app), G_MENU_MODEL (menubar));
95
96 provider = gtk_css_provider_new ();
97 /* Initialize the css data */
98 gtk_css_provider_load_from_string (provider, "label.lb {background-color: red;}");
99 /* Add CSS to the default GdkDisplay. */
100 gtk_style_context_add_provider_for_display (gdk_display_get_default (),
101      GTK_STYLE_PROVIDER (provider), GTK_STYLE_PROVIDER_PRIORITY_APPLICATION);
102 g_signal_connect (app, "shutdown", G_CALLBACK (app_shutdown), provider);
103 g_object_unref (provider); /* release provider, but it's still alive because the display owns it */
104 }
105
106 #define APPLICATION_ID "com.github.ToshioCP.menu2"
107
108 int
109 main (int argc, char **argv) {
110     GtkApplication *app;

```

```

111  int stat;
112
113  app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
114  g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
115  g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
116
117  stat = g_application_run (G_APPLICATION (app), argc, argv);
118  g_object_unref (app);
119  return stat;
120 }

```

- 6-23: Action signal handlers.
- 25-28: The handler `app_shutdown` is called when the application quits. It removes the provider from the display.
- 30-48: An activate signal handler.
- 32-34: A new window is created and assigned to `win`. Its title and default size are set to “menu2” and 400x300 respectively.
- 36-38: A new label is created and assigned to `lb`. The label is given a CSS class “lb”. It is added to `win` as a child.
- 40-43: A toggle action is created and assigned to `act_maximize`. It’s connected to the signal handler `maximize_state_changed`. It’s added to the window, so the action scope is “win”.
- 45: The function `gtk_application_window_set_show_menubar` adds a menubar to the window.
- 47: Shows the window.
- 50-104: A startup signal handler.
- 52-61: Two actions `act_color` and `act_quit` are created. There is only one instance of each of these actions, because the startup handler is called once. They are connected to their handlers and added to the application. Their scopes are “app”.
- 63-92: Menus are built.
- 94: The menubar is added to the application.
- 96-103: A CSS provider is created with the CSS data and added to the default display. The “shutdown” signal on the application is connected to a handler “`app_shutdown`”. The provider is removed from the display and freed when the application quits.

2.16.4 Compile

Change your current directory to `src/menu`.

```

$ comp menu2
$ ./a.out

```

Then, you will see a window and the background color of the content is red. You can change the size to maximum and change back to the original size. You can change the background color to green or blue.

If you run the second application while the first application is running, another window will appear in the same screen. Both of the windows have the same background color. Because the `act_color` action has “app” scope and the CSS is applied to the default display shared by the windows.

```

$ ./a.out & # Run the first application
[1] 82113
$ ./a.out # Run the second application
$

```

2.17 Ui File for Menu and Action Entries

2.17.1 Ui File for Menu

You may have thought that building menus was really bothersome. Yes, the program was complicated and it needs lots of time to code them. The situation is similar to building widgets. When we built widgets, using UI files was a good way to avoid such complication. The same goes for menus.

The UI file for menus has interface and `<menu>` tags. The file starts and ends with interface tags.

```
<interface>
  <menu id="menubar">
    </menu>
  </interface>
```

The `<menu>` tag corresponds to a `GMenu` object. The `id` attribute defines the name of the object. It will be referred by `GtkBuilder`.

```
<submenu>
  <attribute name="label">File</attribute>
  <item>
    <attribute name="label">New</attribute>
    <attribute name="action">win.new</attribute>
  </item>
</submenu>
```

The `<item>` tag corresponds to an item in the `GMenu` which has the same structure as `GMenuItem`. The item above has a `label` attribute. Its value is “New”. The item also has an `action` attribute and its value is “win.new”. “win” is a prefix and “new” is an action name. The `<submenu>` tag corresponds to both `GMenuItem` and `GMenu`. The `GMenuItem` has a link to `GMenu`.

The UI file above can be described as follows.

```
<item>
  <attribute name="label">File</attribute>
  <link name="submenu">
    <item>
      <attribute name="label">New</attribute>
      <attribute name="action">win.new</attribute>
    </item>
  </link>
</item>
```

The `<link>` tag expresses the link to submenu. And at the same time it also expresses the submenu itself. This file illustrates the relationship between the menus and items better than the prior UI file. But `<submenu>` tags are simple and easy to understand. So, we usually prefer the former UI style.

For further information, see [GTK 4 API reference – PopoverMenu](#).

The following is a screenshot of the sample program `menu3`. It is located in the directory `/src/menu3`.

The following is the UI file for `menu3`.

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <interface>
3   <menu id="menubar">
4     <submenu>
5       <attribute name="label">File</attribute>
6       <section>
7         <item>
```

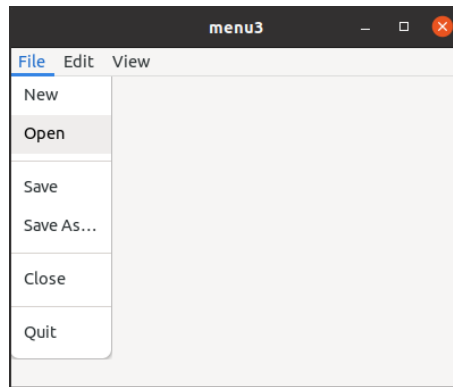


Figure 2.25: menu3

```

8         <attribute name="label">New</attribute>
9         <attribute name="action">app.new</attribute>
10    </item>
11    <item>
12        <attribute name="label">Open</attribute>
13        <attribute name="action">app.open</attribute>
14    </item>
15 </section>
16 <section>
17     <item>
18         <attribute name="label">Save</attribute>
19         <attribute name="action">win.save</attribute>
20     </item>
21     <item>
22         <attribute name="label">Save As...</attribute>
23         <attribute name="action">win.saveas</attribute>
24     </item>
25 </section>
26 <section>
27     <item>
28         <attribute name="label">Close</attribute>
29         <attribute name="action">win.close</attribute>
30     </item>
31 </section>
32 <section>
33     <item>
34         <attribute name="label">Quit</attribute>
35         <attribute name="action">app.quit</attribute>
36     </item>
37 </section>
38 </submenu>
39 <submenu>
40     <attribute name="label">Edit</attribute>
41     <section>
42         <item>
43             <attribute name="label">Cut</attribute>
44             <attribute name="action">app.cut</attribute>
45         </item>
46         <item>

```

```

47         <attribute name="label">Copy</attribute>
48         <attribute name="action">app.copy</attribute>
49     </item>
50     <item>
51         <attribute name="label">Paste</attribute>
52         <attribute name="action">app.paste</attribute>
53     </item>
54 </section>
55 <section>
56     <item>
57         <attribute name="label">Select All</attribute>
58         <attribute name="action">app.selectall</attribute>
59     </item>
60 </section>
61 </submenu>
62 <submenu>
63     <attribute name="label">View</attribute>
64     <section>
65         <item>
66             <attribute name="label">Maximized</attribute>
67             <attribute name="action">win.maximize</attribute>
68         </item>
69     </section>
70 </submenu>
71 </menu>
72 </interface>

```

The UI file is converted to the resource by the resource compiler `glib-compile-resources` with xml file.

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <gresources>
3     <gresource prefix="/com/github/ToshioCP/menu3">
4         <file>menu3.ui</file>
5     </gresource>
6 </gresources>

```

GtkBuilder builds menus from the resource.

```

GtkBuilder *builder = gtk_builder_new_from_resource ("/com/github/ToshioCP/menu3/menu3.ui");
GMenuModel *menubar = G_MENU_MODEL (gtk_builder_get_object (builder, "menubar"));

```

```

gtk_application_set_menubar (GTK_APPLICATION (app), menubar);
g_object_unref (builder);

```

The builder instance is freed after the `GMenuModel menubar` is inserted to the application. If you do it before the insertion, bad thing will happen – your computer might freeze. It is because you don't own the `menubar` instance. The function `gtk_builder_get_object` just returns the pointer to `menubar` and doesn't increase the reference count of `menubar`. So, if you released builder before `gtk_application_set_menubar`, builder would be destroyed and `menubar` as well.

2.17.2 Action Entry

The coding for building actions and signal handlers is complex work as well. Therefore, it should be automated. You can implement them easily with `GActionEntry` structure and `g_action_map_add_action_entries` function.

`GActionEntry` contains action name, signal handlers, parameter and state.

```
typedef struct _GActionEntry GActionEntry;

struct _GActionEntry
{
    /* action name */
    const char *name;
    /* activate handler */
    void (* activate) (GSimpleAction *action, GVariant *parameter, gpointer user_data);
    /* the type of the parameter given as a single GVariant type string */
    const char *parameter_type;
    /* initial state given in GVariant text format */
    const char *state;
    /* change-state handler */
    void (* change_state) (GSimpleAction *action, GVariant *value, gpointer user_data);
    /*< private >*/
    gsize padding[3];
};
```

For example, the actions in the previous section are:

```
{ "maximize", NULL, NULL, "false", maximize_state_changed }
{ "color", color_activated, "s", "'red'", NULL }
{ "quit", quit_activated, NULL, NULL, NULL },
```

- The “maximize” action is stateful, but doesn’t have parameters. So, the third element (parameter type) is NULL. [GVariant text format](#) provides “true” and “false” as boolean GVariant values. The initial state of the action is false (the fourth element). It doesn’t have activate handler, so the second element is NULL. Instead, it has change-state handler. The fifth element `maximize_state_changed` is the handler.
- The “color” action is stateful and has a parameter. The parameter type is string. [GVariant format strings](#) provides string formats to represent GVariant types. The third element “s” means GVariant string type. GVariant text format defines that strings are surrounded by single or double quotes. So, the string red is ‘red’ or “red”. The fourth element is `"'red'"`, which is a C string format and the string is ‘red’. You can write `"\"red\""` instead. The second element `color_activated` is the activate handler. The action doesn’t have change-state handler, so the fifth element is NULL.
- The “quit” action is non-stateful and has no parameter. So, the third and fourth elements are NULL. The second element `quit_activated` is the activate handler. The action doesn’t have change-state handler, so the fifth element is NULL.

The function `g_action_map_add_action_entries` does everything to create `GSimpleAction` instances and add them to a `GActionMap` (an application or window).

```
const GActionEntry app_entries[] = {
    { "color", color_activated, "s", "'red'", NULL },
    { "quit", quit_activated, NULL, NULL, NULL }
};
g_action_map_add_action_entries (G_ACTION_MAP (app), app_entries,
                                G_N_ELEMENTS (app_entries), app);
```

The code above does:

- Builds the “color” and the “quit” actions
- Connects the action and the “activate” signal handlers (`color_activated` and `quit_activated`).
- Adds the actions to the action map `app`.

The same goes for the other action.

```
const GActionEntry win_entries[] = {
```



```

    { "maximize", NULL, NULL, "false", maximize_state_changed }
};
g_action_map_add_action_entries (G_ACTION_MAP (win), win_entries,
                                G_N_ELEMENTS (win_entries), win);

```

The code above performs the following:

- Builds the “maximize” action.
- Connects the action and the signal handler `maximize_state_changed`
- Its initial state is set to false.
- Adds the action to the action map `win`.

2.17.3 Example

Source files are `menu3.c`, `menu3.ui`, `menu3.gresource.xml` and `meson.build`. They are in the directory `/src/menu3`. The following are `menu3.c` and `meson.build`.

```

1  #include <gtk/gtk.h>
2
3  static void
4  new_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
5  }
6
7  static void
8  open_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
9  }
10
11 static void
12 save_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
13 }
14
15 static void
16 saveas_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
17 }
18
19 static void
20 close_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
21     GtkWindow *win = GTK_WINDOW (user_data);
22
23     gtk_window_destroy (win);
24 }
25
26 static void
27 cut_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
28 }
29
30 static void
31 copy_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
32 }
33
34 static void
35 paste_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
36 }
37
38 static void

```

```

39 selectall_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
40 }
41
42 static void
43 maximize_state_changed (GSimpleAction *action, GVariant *state, gpointer user_data) {
44     GtkWidget *win = GTK_WINDOW (user_data);
45
46     if (g_variant_get_boolean (state))
47         gtk_window_maximize (win);
48     else
49         gtk_window_unmaximize (win);
50     g_simple_action_set_state (action, state);
51 }
52
53 static void
54 quit_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
55     GApplication *app = G_APPLICATION (user_data);
56
57     g_application_quit (app);
58 }
59
60 static void
61 app_activate (GApplication *app) {
62     GtkWidget *win = gtk_application_window_new (GTK_APPLICATION (app));
63
64     const GActionEntry win_entries[] = {
65         { "save", save_activated, NULL, NULL, NULL },
66         { "saveas", saveas_activated, NULL, NULL, NULL },
67         { "close", close_activated, NULL, NULL, NULL },
68         { "maximize", NULL, NULL, "false", maximize_state_changed }
69     };
70     g_action_map_add_action_entries (G_ACTION_MAP (win), win_entries, G_N_ELEMENTS (win_entries), win);
71
72     gtk_application_window_set_show_menubar (GTK_APPLICATION_WINDOW (win), TRUE);
73
74     gtk_window_set_title (GTK_WINDOW (win), "menu3");
75     gtk_window_set_default_size (GTK_WINDOW (win), 400, 300);
76     gtk_window_present (GTK_WINDOW (win));
77 }
78
79 static void
80 app_startup (GApplication *app) {
81     GtkBuilder *builder = gtk_builder_new_from_resource ("/com/github/ToshioCP/menu3/menu3.ui");
82     GMenuModel *menubar = G_MENU_MODEL (gtk_builder_get_object (builder, "menubar"));
83
84     gtk_application_set_menubar (GTK_APPLICATION (app), menubar);
85     g_object_unref (builder);
86
87     const GActionEntry app_entries[] = {
88         { "new", new_activated, NULL, NULL, NULL },
89         { "open", open_activated, NULL, NULL, NULL },
90         { "cut", cut_activated, NULL, NULL, NULL },
91         { "copy", copy_activated, NULL, NULL, NULL },

```

```

92     { "paste", paste_activated, NULL, NULL, NULL },
93     { "selectall", selectall_activated, NULL, NULL, NULL },
94     { "quit", quit_activated, NULL, NULL, NULL }
95 };
96 g_action_map_add_action_entries (G_ACTION_MAP (app), app_entries, G_N_ELEMENTS (app_entries), app)
97 }
98
99 #define APPLICATION_ID "com.github.ToshioCP.menu3"
100
101 int
102 main (int argc, char **argv) {
103     GtkApplication *app;
104     int stat;
105
106     app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
107     g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
108     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
109
110     stat = g_application_run (G_APPLICATION (app), argc, argv);
111     g_object_unref (app);
112     return stat;
113 }

```

meson.build

```

1 project('menu3', 'c')
2
3 gtkdep = dependency('gtk4')
4
5 gnome=import('gnome')
6 resources = gnome.compile_resources('resources','menu3.gresource.xml')
7
8 sourcefiles=files('menu3.c')
9
10 executable('menu3', sourcefiles, resources, dependencies: gtkdep)

```

Action handlers need to follow the following format.

```

static void
handler (GSimpleAction *action_name, GVariant *parameter, gpointer user_data) { ... .. }

```

You can't write, for example, "GApplication *app" instead of "gpointer user_data". Because `g_action_map_add_action_entries` expects that handlers follow the format above.

There are `menu2_ui.c` and `menu2.ui` under the `menu` directory. They are other examples to show a menu UI file and `g_action_map_add_action_entries`. It includes a stateful action with parameters.

```

<item>
  <attribute name="label">Red</attribute>
  <attribute name="action">app.color</attribute>
  <attribute name="target">red</attribute>
</item>

```

Action name and target are separated like this. Action attribute includes prefix and name only. You can't write like `<attribute name="action">app.color::red</attribute>`.

2.18 Composite Widgets and Alert Dialog

The source files are in the [Gtk4 tutorial GitHub repository](#). Download it and see `src/tfe6` directory.

2.18.1 An Outline of New Tfe Text Editor

Tfe text editor will be restructured. The program is divided into six parts.

- Main program: the C main function.
- TfeApplication object: It is like GtkApplication but keeps GSettings and CSS Provider.
- TfeWindow object: It is a window with buttons and a notebook.
- TfePref object: A preference dialog.
- TfeAlert object: An alert dialog.
- pdf2css.h and pdf2css.c: Font and CSS utility functions.

This section describes TfeAlert. Others will be explained in the following sections.

2.18.2 Composite Widgets

The alert dialog is like this:

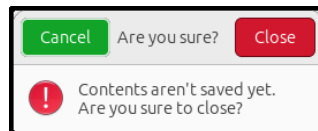


Figure 2.26: Alert dialog

Tfe uses it when a user quits the application or closes a notebook without saving data to files.

The dialog has a title, buttons, an icon and a message. Therefore, it consists of several widgets. Such dialog is called a composite widget.

Composite widgets are defined with template XMLs. The class is built in the class initialization function and the instances are built and disposed by the following functions.

- gtk_widget_init_template
- gtk_widget_dispose_template

TfeAlert is a good example to know composite widgets. It is defined with the three files.

- tfealert.ui: XML file
- tfealert.h: Header file
- tfealert.c: C program file

2.18.3 The XML File

A template tag is used in a composite widget XML.

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <interface>
3   <template class="TfeAlert" parent="GtkWindow">
4     <property name="resizable">FALSE</property>
5     <property name="modal">TRUE</property>
6     <property name="titlebar">
7       <object class="GtkHeaderBar">
8         <property name="show-title-buttons">FALSE</property>
9         <property name="title-widget">
10          <object class="GtkLabel" id="lb_title">
```

```

11         <property name="label">Are you sure?</property>
12         <property name="single-line-mode">True</property>
13     </object>
14 </property>
15 <child type="start">
16     <object class="GtkButton" id="btn_cancel">
17         <property name="label">Cancel</property>
18         <style>
19             <class name="suggested-action"/>
20         </style>
21         <signal name="clicked" handler="cancel_cb" swapped="TRUE" object="TfeAlert"></signal>
22     </object>
23 </child>
24 <child type="end">
25     <object class="GtkButton" id="btn_accept">
26         <property name="label">Close</property>
27         <style>
28             <class name="destructive-action"/>
29         </style>
30         <signal name="clicked" handler="accept_cb" swapped="TRUE" object="TfeAlert"></signal>
31     </object>
32 </child>
33 </object>
34 </property>
35 <child>
36     <object class="GtkBox">
37         <property name="orientation">GTK_ORIENTATION_HORIZONTAL</property>
38         <property name="spacing">12</property>
39         <property name="margin-top">12</property>
40         <property name="margin-bottom">12</property>
41         <property name="margin-start">12</property>
42         <property name="margin-end">12</property>
43         <child>
44             <object class="GtkImage">
45                 <property name="icon-name">dialog-warning</property>
46                 <property name="icon-size">GTK_ICON_SIZE_LARGE</property>
47             </object>
48         </child>
49         <child>
50             <object class="GtkLabel" id="lb_message">
51             </object>
52         </child>
53     </object>
54 </child>
55 </template>
56 </interface>

```

- 3: A template tag defines a composite widget. The class attribute tells the class name of the composite widget. The parent attribute tells the parent class of the composite widget. So, TfeAlert is a child class of GtkWindow. The parent attribute specifies the parent class of the composite widget. While it is technically optional in the XML, it is highly recommended to include it. Providing the parent class allows GtkBuilder to verify the class hierarchy at runtime and improves the readability of the UI file.
- 4-6: Its three properties are defined. These properties are inherited from GtkWindow. The titlebar

property has a widget for a custom title bar. The typical widget is `GtkHeaderBar`.

- 8: If the property “show-title-buttons” is `TRUE`, the title buttons like close, minimize and maximize are shown. Otherwise it is not shown. The `TfeAlert` object is not resizable. It is closed when either of the two buttons, cancel or accept, is clicked. Therefore the title buttons are not necessary and this property is set to `FALSE`.
- 9-14: The bar contains a title, implemented as a `GtkLabel` widget. The default title is “Are you sure?” but it can be replaced by an instance method.
- 15-32: The bar has two buttons, cancel and accept. The cancel button is on the left so the child tag has `type="start"` attribute. The accept button is on the right so the child tag has `type="end"` attribute. The dialog is shown when the user clicked the close button or the quit menu without saving the data. Therefore, it is safer for the user to click on the cancel button of the alert dialog. So, the cancel button has a “suggested-action” CSS class. Ubuntu colors the button green but the color can be blue or other appropriate one defined by the system. In the same way the accept button has a “destructive-action” CSS class and is colored red. Two buttons have signals which are defined by the signal tags. By setting `swapped="TRUE"`, the instance emitting the signal and the user data are swapped. Consequently, the `cancel_cb` handler receives the `TfeAlert` object as its first parameter.”
- 35-54: A horizontal box has an image icon and a label.
- 44-47: The `GtkImage` widget displays an image. The “icon-name” property is an icon name in the icon theme. The theme depends on your system. You can check it with an icon browser.

\$ `gtk4-icon-browser`

The “dialog-warning” icon is something like this.



Figure 2.27: dialog-warning icon is like ...

These were created manually. The real image on the alert dialog is nicer.

It is possible to define the alert widget as a child of `GtkDialog`. But `GtkDialog` is deprecated since GTK version 4.10. And users should use `GtkWindow` instead of `GtkDialog`.

2.18.4 The Header File

The header file is similar to the one of `TfeTextView`.

```

1  #pragma once
2
3  #include <gtk/gtk.h>
4
5  #define TFE_TYPE_ALERT tfe_alert_get_type ()
6  G_DECLARE_FINAL_TYPE (TfeAlert, tfe_alert, TFE, ALERT, GtkWindow)
7
8  /* "response" signal id */
9  enum TfeAlertResponseType
10 {
11     TFE_ALERT_RESPONSE_ACCEPT,
12     TFE_ALERT_RESPONSE_CANCEL
13 };
14
15 const char *
16 tfe_alert_get_title (TfeAlert *alert);
17
```

```

18  const char *
19  tfe_alert_get_message (TfeAlert *alert);
20
21  const char *
22  tfe_alert_get_button_label (TfeAlert *alert);
23
24  void
25  tfe_alert_set_title (TfeAlert *alert, const char *title);
26
27  void
28  tfe_alert_set_message (TfeAlert *alert, const char *message);
29
30  void
31  tfe_alert_set_button_label (TfeAlert *alert, const char *btn_label);
32
33  GtkWidget *
34  tfe_alert_new (void);
35
36  GtkWidget *
37  tfe_alert_new_with_data (const char *title, const char *message, const char *btn_label);

```

- 5-6: These two lines are always needed to define a new object. TFE_TYPE_ALERT is the type of TfeAlert object and it is a macro expanded into the call to tfe_alert_get_type (). G_DECLARE_FINAL_TYPE macro is expanded into:
 - The declaration of the function tfe_alert_get_type
 - TfeAlert is defined as a typedef of struct _TfeAlert, which is defined in the C file.
 - TFE_ALERT and TFE_IS_ALERT macro is defined as a cast and type check function.
 - TfeAlertClass structure is defined as a final class.
- 8-13: The TfeAlert class has a “response” signal. It has a parameter and the parameter type is defined as a TfeAlertResponseType enumerative constant.
- 15-31: Getter and setter methods.
- 33-37: Functions to create an instance. The function tfe_alert_new_with_data is a convenience function, which creates an instance and sets data at once.

2.18.5 The C File

2.18.5.1 Functions for Composite Widgets

The following codes are extracted from tfealert.c.

```

#include <gtk/gtk.h>
#include "tfealert.h"

struct _TfeAlert {
    GtkWidget parent;
    GtkLabel *lb_title;
    GtkLabel *lb_message;
    GtkButton *btn_accept;
    GtkButton *btn_cancel;
};

G_DECLARE_FINAL_TYPE (TfeAlert, tfe_alert, GTK_TYPE_WINDOW);

static void
cancel_cb (TfeAlert *alert) {

```

```

    ... ..
}

static void
accept_cb (TfeAlert *alert) {
    ... ..
}

static void
tfe_alert_dispose (GObject *gobject) { // gobject is actually a TfeAlert instance.
    gtk_widget_dispose_template (GTK_WIDGET (gobject), TFE_TYPE_ALERT);
    G_OBJECT_CLASS (tfe_alert_parent_class)->dispose (gobject);
}

static void
tfe_alert_init (TfeAlert *alert) {
    gtk_widget_init_template (GTK_WIDGET (alert));
}

static void
tfe_alert_class_init (TfeAlertClass *class) {
    G_OBJECT_CLASS (class)->dispose = tfe_alert_dispose;
    gtk_widget_class_set_template_from_resource (GTK_WIDGET_CLASS (class), "/com/github/ToshioCP/tfe/tfe-alert.ui");
    gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfeAlert, lb_title);
    gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfeAlert, lb_message);
    gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfeAlert, btn_accept);
    gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfeAlert, btn_cancel);
    gtk_widget_class_bind_template_callback (GTK_WIDGET_CLASS (class), cancel_cb);
    gtk_widget_class_bind_template_callback (GTK_WIDGET_CLASS (class), accept_cb);
    ... ..
}

GtkWidget *
tfe_alert_new (void) {
    return GTK_WIDGET (g_object_new (TFE_TYPE_ALERT, NULL));
}

```

- `G_DEFINE_FINAL_TYPE` was introduced in GLib 2.70 for final type classes. While `G_DEFINE_TYPE` still works, the new macro is preferred. It is expanded into:
 - The declaration of the functions `tfe_alert_init` and `tfe_alert_class_init`. They are defined in the following part of the C program.
 - The definition of the variable `tfe_alert_parent_class`.
 - The definition of the function `tfe_alert_get_type`.
- The names of the members of `_TfeAlert`, which are `lb_title`, `lb_message`, `btn_accept` and `btn_cancel`, must be the same as the id attribute in the XML file `tfealert.ui`.
- The function `tfe_alert_class_init` initializes the composite widget class.
 - The function `gtk_widget_class_set_template_from_resource` sets the template of the class. The template is built from the XML resource “tfealert.ui”. At this moment no instance is created. It just makes the class recognize the structure of the object. That’s why the top level tag is not object but template in the XML file.
 - The function macro `gtk_widget_class_bind_template_child` connects the member of `TfeAlert` and the object class in the template. So, for example, you can access to `lb_title` `GtkLabel` instance via `alert->lb_title` where `alert` is an instance of `TfeAlert` class.
 - The function `gtk_widget_class_bind_template_callback` connects the callback function

and the `handler` attribute of the signal tag in the XML. For example, the “clicked” signal on the cancel button has a handler named “cancel_cb” in the signal tag. And the function `cancel_cb` exists in the C file above. These two are connected so when the signal is emitted the function `cancel_cb` is called. You can add `static` storage class to the callback function thanks to this connection.

- The function `tfe_alert_init` initializes the newly created instance. You need to call `gtk_widget_init_template` to create and initialize the child widgets in the template.
- The function `tfe_alert_dispose` releases objects. The function `gtk_widget_dispose_template` clears the template children.
- The function `tfe_alert_new` creates the composite widget `TfeAlert` instance. It creates not only `TfeAlert` itself but also all the child widgets that the composite widget has.

2.18.5.2 Other Functions

The following is the full codes of `tfealert.c`.

```

1  #include <gtk/gtk.h>
2  #include "tfealert.h"
3
4  struct _TfeAlert {
5      GtkWidget parent;
6      GtkWidget *lb_title;
7      GtkWidget *lb_message;
8      GtkWidget *btn_accept;
9      GtkWidget *btn_cancel;
10 };
11
12 G_DEFINE_FINAL_TYPE (TfeAlert, tfe_alert, GTK_TYPE_WINDOW);
13
14 enum {
15     RESPONSE,
16     NUMBER_OF_SIGNALS
17 };
18
19 static guint tfe_alert_signals[NUMBER_OF_SIGNALS];
20
21 static void
22 cancel_cb (TfeAlert *alert) {
23     g_signal_emit (alert, tfe_alert_signals[RESPONSE], 0, TFE_ALERT_RESPONSE_CANCEL);
24     gtk_window_destroy (GTK_WINDOW (alert));
25 }
26
27 static void
28 accept_cb (TfeAlert *alert) {
29     g_signal_emit (alert, tfe_alert_signals[RESPONSE], 0, TFE_ALERT_RESPONSE_ACCEPT);
30     gtk_window_destroy (GTK_WINDOW (alert));
31 }
32
33 const char *
34 tfe_alert_get_title (TfeAlert *alert) {
35     return gtk_label_get_text (alert->lb_title);
36 }
37
38 const char *
```

```

39 tfe_alert_get_message (TfeAlert *alert) {
40     return gtk_label_get_text (alert->lb_message);
41 }
42
43 const char *
44 tfe_alert_get_button_label (TfeAlert *alert) {
45     return gtk_button_get_label (alert->btn_accept);
46 }
47
48 void
49 tfe_alert_set_title (TfeAlert *alert, const char *title) {
50     gtk_label_set_text (alert->lb_title, title);
51 }
52
53 void
54 tfe_alert_set_message (TfeAlert *alert, const char *message) {
55     gtk_label_set_text (alert->lb_message, message);
56 }
57
58 void
59 tfe_alert_set_button_label (TfeAlert *alert, const char *btn_label) {
60     gtk_button_set_label (alert->btn_accept, btn_label);
61 }
62
63 static void
64 tfe_alert_dispose (GObject *gobject) { // gobject is actually a TfeAlert instance.
65     gtk_widget_dispose_template (GTK_WIDGET (gobject), TFE_TYPE_ALERT);
66     G_OBJECT_CLASS (tfe_alert_parent_class)->dispose (gobject);
67 }
68
69 static void
70 tfe_alert_init (TfeAlert *alert) {
71     gtk_widget_init_template (GTK_WIDGET (alert));
72 }
73
74 static void
75 tfe_alert_class_init (TfeAlertClass *class) {
76     G_OBJECT_CLASS (class)->dispose = tfe_alert_dispose;
77     gtk_widget_class_set_template_from_resource (GTK_WIDGET_CLASS (class), "/com/github/ToshioCP/tfe/t.
78     gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfeAlert, lb_title);
79     gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfeAlert, lb_message);
80     gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfeAlert, btn_accept);
81     gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfeAlert, btn_cancel);
82     gtk_widget_class_bind_template_callback (GTK_WIDGET_CLASS (class), cancel_cb);
83     gtk_widget_class_bind_template_callback (GTK_WIDGET_CLASS (class), accept_cb);
84
85     tfe_alert_signals[RESPONSE] = g_signal_new ("response",
86     G_TYPE_FROM_CLASS (class),
87     G_SIGNAL_RUN_LAST | G_SIGNAL_NO_RECURSE | G_SIGNAL_NO_HOOKS,
88     0 /* class offset */,
89     NULL /* accumulator */,
90     NULL /* accumulator data */,
91     NULL /* C marshaller */,

```

```

92         G_TYPE_NONE /* return_type */,
93         1 /* n_params */,
94         G_TYPE_INT
95     );
96 }
97
98 GtkWidget *
99 tfe_alert_new (void) {
100     return GTK_WIDGET (g_object_new (TFE_TYPE_ALERT, NULL));
101 }
102
103 GtkWidget *
104 tfe_alert_new_with_data (const char *title, const char *message, const char *btn_label) {
105     GtkWidget *alert = tfe_alert_new ();
106     tfe_alert_set_title (TFE_ALERT (alert), title);
107     tfe_alert_set_message (TFE_ALERT (alert), message);
108     tfe_alert_set_button_label (TFE_ALERT (alert), btn_label);
109     return alert;
110 }

```

The function `tfe_alert_new_with_data` is used more often than `tfe_alert_new` to create a new instance. It creates the instance and sets three data at the same time. The following is the common process when you use the `TfeAlert` class.

- Call `tfe_alert_new_with_data` and create an instance.
- Call `gtk_window_set_transient_for` to set the transient parent window.
- Call `gtk_window_present` to show the `TfeAlert` dialog.
- Connect “response” signal and a handler.
- The user clicks on the cancel or accept button. Then the dialog emits the “response” signal and destroy itself.
- The user catches the signal and do something.

The rest of the program is:

- 14-19: An array for a signal id. You can use a variable instead of an array because the class has only one signal. But using an array is a common way.
- 21-31: Signal handlers. They emits the “response” signal and destroy the instance itself.
- 33-61: Getters and setters.
- 85-95: Creates the “response” signal.
- 103-110: A convenience function `tfe_alert_new_with_data` creates an instance and sets labels.

2.18.6 An Example

There’s an example in the `src/tfe6/example` directory. It shows how to use `TfeAlert`. The program is `src/example/ex_alert.c`.

```

1  #include <gtk/gtk.h>
2  #include "../tfealert.h"
3
4  static void
5  alert_response_cb (TfeAlert *alert, int response, gpointer user_data) {
6      if (response == TFE_ALERT_RESPONSE_ACCEPT)
7          g_print ("%s\n", tfe_alert_get_button_label (alert));
8      else if (response == TFE_ALERT_RESPONSE_CANCEL)
9          g_print ("Cancel\n");
10     else

```

```

11     g_print ("Unexpected error\n");
12 }
13
14 static void
15 app_activate (GApplication *application) {
16     GtkWidget *alert;
17     char *title, *message, *btn_label;
18
19     title = "Example for TfeAlert"; message = "Click on Cancel or Accept button"; btn_label = "Accept"
20     alert = tfe_alert_new_with_data (title, message, btn_label);
21     g_signal_connect (TFE_ALERT (alert), "response", G_CALLBACK (alert_response_cb), NULL);
22     gtk_window_set_application (GTK_WINDOW (alert), GTK_APPLICATION (application));
23     gtk_window_present (GTK_WINDOW (alert));
24 }
25
26 static void
27 app_startup (GApplication *application) {
28 }
29
30 #define APPLICATION_ID "com.github.ToshioCP.example_tfe_alert"
31
32 int
33 main (int argc, char **argv) {
34     GtkApplication *app;
35     int stat;
36
37     app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
38     g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
39     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
40     stat = g_application_run (G_APPLICATION (app), argc, argv);
41     g_object_unref (app);
42     return stat;
43 }

```

The “activate” signal handler `app_activate` initializes the alert dialog.

- A `TfeAlert` instance is created.
- Its “response” signal is connected to the handler `alert_response_cb`.
- `TfeAlert` class is a sub class of `GtkWindow` so it can be a top level window that is connected to an application instance. The function `gtk_window_set_application` does that.
- The dialog is shown.

A user clicks on either the cancel button or the accept button. Then, the “response” signal is emitted and the dialog is destroyed. The signal handler `alert_response_cb` checks the response and prints “Accept” or “Cancel”. If an error happens, it prints “Unexpected error”.

You can compile it with `meson` and `ninja`.

```

$ cd src/tfe6/example
$ meson setup _build
$ ninja -C _build
$ _build/ex_alert
Accept #<= if you clicked on the accept button

```

2.19 GtkFontDialogButton and Gsettings

2.19.1 The Preference Dialog

If the user clicks on the preference menu, a preference dialog appears.

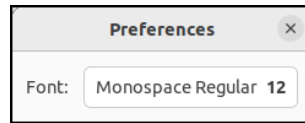


Figure 2.28: Preference dialog

It has only one button, which is a GtkFontDialogButton widget. You can add more widgets on the dialog but this simple dialog isn't so bad for the first example program.

If the button is clicked, a GtkFontDialog appears like this.

If the user chooses a font and clicks on the select button, the font is changed.

GtkFontDialogButton and GtkFontDialog are available since GTK version 4.10. They replace GtkFontButton and GtkFontChooserDialog, which are deprecated since 4.10.

2.19.2 Composite Widget

The preference dialog has GtkBox, GtkLabel and GtkFontButton in it and is defined as a composite widget. The following is the template UI file for TfePref.

```

1  <?xml version="1.0" encoding="UTF-8"?>
2  <interface>
3    <template class="TfePref" parent="GtkWindow">
4      <property name="title">Preferences</property>
5      <property name="resizable">FALSE</property>
6      <property name="modal">TRUE</property>
7      <child>
8        <object class="GtkBox">
9          <property name="orientation">GTK_ORIENTATION_HORIZONTAL</property>
10         <property name="spacing">12</property>
11         <property name="halign">GTK_ALIGN_CENTER</property>
12         <property name="margin-start">12</property>
13         <property name="margin-end">12</property>
14         <property name="margin-top">12</property>
15         <property name="margin-bottom">12</property>
16         <child>
17           <object class="GtkLabel">
18             <property name="label">Font:</property>
19             <property name="xalign">1</property>
20           </object>
21         </child>
22         <child>
23           <object class="GtkFontDialogButton" id="font_dialog_btn">
24             <property name="dialog">
25               <object class="GtkFontDialog"/>
26             </property>
27           </object>
28         </child>
29       </object>

```

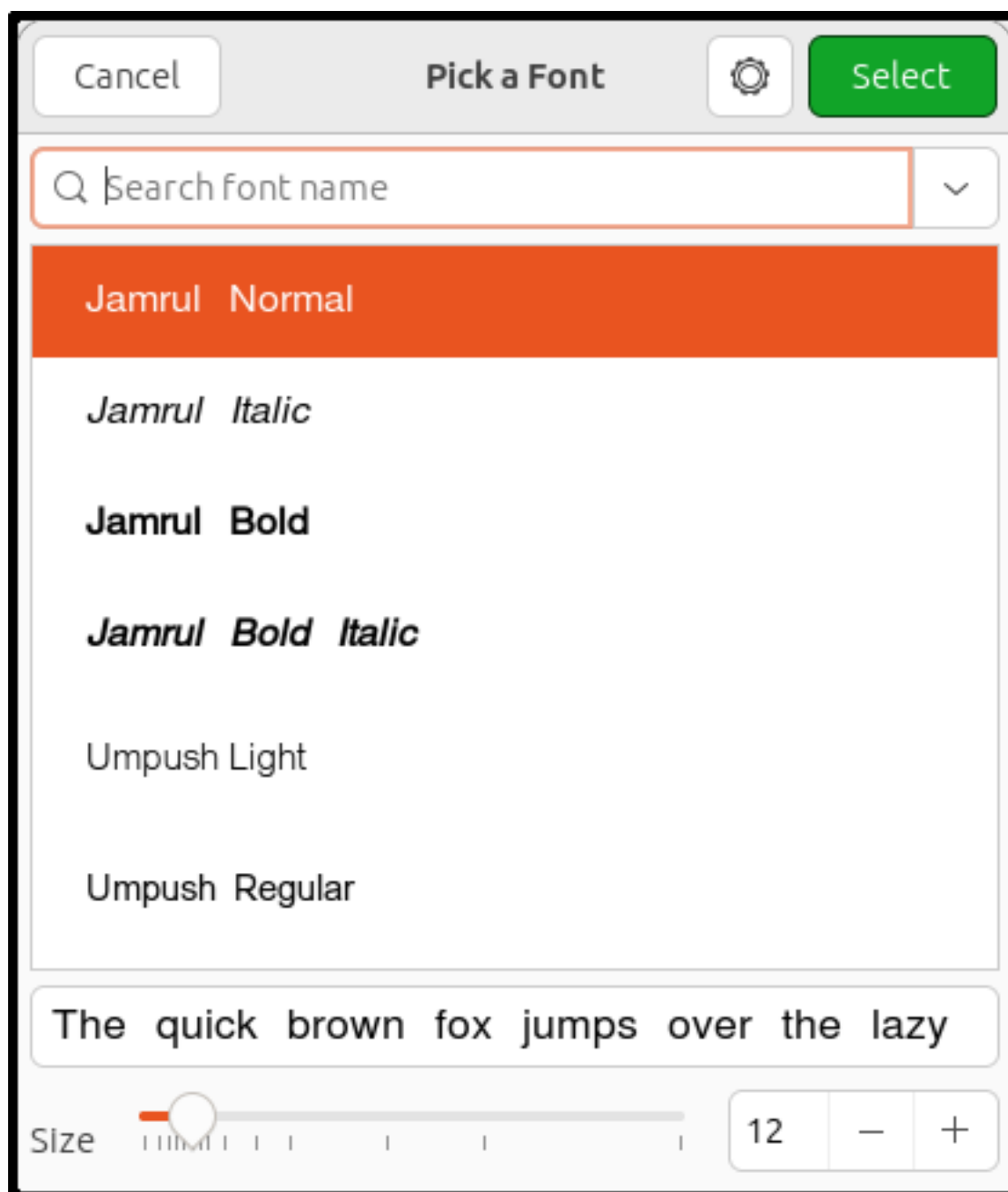


Figure 2.29: Font dialog

```

30     </child>
31 </template>
32 </interface>

```

- Template tag specifies a composite widget. The class attribute specifies the class name, which is “TfePref”. The parent attribute is `GtkWindow`. Therefore, `TfePref` is a child class of `GtkWindow`. A parent attribute is optional but it is recommended to write it explicitly. You can make `TfePref` as a child of `GtkDialog`, but `GtkDialog` is deprecated since version 4.10.
- There are three properties, title, resizable and modal.
- `TfePref` has a child widget `GtkBox` which is horizontal. The box has two children `GtkLabel` and `GtkFontDialogButton`.

2.19.3 The Header File

The file `tfepref.h` defines types and declares a public function.

```

1  #pragma once
2
3  #include <gtk/gtk.h>
4
5  #define TFE_TYPE_PREF tfe_pref_get_type ()
6  G_DECLARE_FINAL_TYPE (TfePref, tfe_pref, TFE, PREF, GtkWindow)
7
8  GtkWidget *
9  tfe_pref_new (void);

```

- 5: Defines the type `TFE_TYPE_PREF`, which is a macro replaced by `tfe_pref_get_type ()`.
- 6: The macro `G_DECLARE_FINAL_TYPE` expands to:
 - The function `tfe_pref_get_type ()` is declared.
 - `TfePref` type is defined as a typedef of `struct _TfePref`.
 - `TfePrefClass` type is defined as a typedef of `struct {GtkWindowClass *parent;}`.
 - Two functions `TFE_PREF ()` and `TFE_IS_PREF ()` are defined.
- 8-9: The function `tfe_pref_new` is declared. It creates a new `TfePref` instance.

2.19.4 The C File for the Composite Widget

The following codes are extracted from the file `tfepref.c`.

```

#include <gtk/gtk.h>
#include "tfepref.h"

struct _TfePref
{
    GtkWindow parent;
    GtkFontDialogButton *font_dialog_btn;
};

G_DEFINE_FINAL_TYPE (TfePref, tfe_pref, GTK_TYPE_WINDOW);

static void
tfe_pref_dispose (GObject *gobject) {
    TfePref *pref = TFE_PREF (gobject);
    gtk_widget_dispose_template (GTK_WIDGET (pref), TFE_TYPE_PREF);
    G_OBJECT_CLASS (tfe_pref_parent_class)->dispose (gobject);
}

```

```

static void
tfe_pref_init (TfePref *pref) {
    gtk_widget_init_template (GTK_WIDGET (pref));
}

static void
tfe_pref_class_init (TfePrefClass *class) {
    G_OBJECT_CLASS (class)->dispose = tfe_pref_dispose;
    gtk_widget_class_set_template_from_resource (GTK_WIDGET_CLASS (class), "/com/github/ToshioCP/tfe/tfe.ui");
    gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfePref, font_dialog_btn);
}

GtkWidget *
tfe_pref_new (void) {
    return GTK_WIDGET (g_object_new (TFE_TYPE_PREF, NULL));
}

```

- The structure `_TfePref` has `font_dialog_btn` member. It points to the `GtkFontDialogButton` object specified in the XML file “`tfe.pref.ui`”. The member name `font_dialog_btn` must be the same as the `GtkFontDialogButton` id attribute in the XML file.
- `G_DEFINE_FINAL_TYPE` macro expands to:
 - The declaration of the functions `tfe_pref_init` and `tfe_pref_class_init`. They are defined in the following part of the program.
 - The definition of the variable `tfe_pref_parent_class`.
 - The definition of the function `tfe_pref_get_type`.
- The function `tfe_pref_class_init` initializes the `TfePref` class. The function `gtk_widget_class_set_template_from_resource` initializes the composite widget template from the XML resource. The function `gtk_widget_class_bind_template_child` connects the `TfePref` structure member `font_dialog_btn` and the `GtkFontDialogButton` in the XML. The member name and the id attribute value must be the same.
- The function `tfe_pref_init` initializes a newly created instance. The function `gtk_widget_init_template` creates and initializes child widgets.
- The function `tfe_pref_dispose` releases objects. The function `gtk_widget_dispose_template` releases child widgets.

2.19.5 GtkFontDialogButton and Pango

If the `GtkFontDialogButton` button is clicked, the `GtkFontDialog` dialog appears. A user can choose a font on the dialog. If the user clicks on the “select” button, the dialog disappears. And the font information is given to the `GtkFontDialogButton` instance. The font data is retrieved with the method `gtk_font_dialog_button_get_font_desc`. It returns a pointer to the `PangoFontDescription` structure.

Pango is a text layout engine. The [documentation](#) is on the internet.

`PangoFontDescription` is a C structure and it isn’t allowed to be accessed directly. The document is [here](#). If you want to retrieve the font information, there are several functions.

- `pango_font_description_to_string` returns a string like “Jamrul Bold Italic Semi-Expanded 12”.
- `pango_font_description_get_family` returns a font family like “Jamrul”.
- `pango_font_description_get_weight` returns a `PangoWeight` constant like `PANGO_WEIGHT_BOLD`.
- `pango_font_description_get_style` returns a `PangoStyle` constant like `PANGO_STYLE_ITALIC`.
- `pango_font_description_get_stretch` returns a `PangoStretch` constant like `PANGO_STRETCH_SEMI_EXPANDED`.
- `pango_font_description_get_size` returns an integer like 12. Its unit is points or pixels (device units). The function `pango_font_description_get_size_is_absolute` returns `TRUE` if the unit is an absolute device unit. Otherwise the unit is point.

2.19.6 GSettings

We want to maintain the font data after the application quits. There are some ways to implement.

- Make a configuration file. For example, a text file “`~/.config/tfe/font_desc.cfg`” keeps font information.
- Use GSettings object. The basic idea of GSettings are similar to configuration file. Configuration information data is put into a database file.

GSettings is simple and easy to use but a bit hard to understand the concept. This subsection describes the concept first and then how to program it.

2.19.6.1 GSettings Schema

GSettings schema describes a set of keys, value types and some other information. GSettings object uses this schema and it writes/reads the value of a key to/from the right place in the database.

- A schema has an id. The id must be unique. We often use the same string as application id, but schema id and application id are different. You can use different name from application id. Schema id is a string delimited by periods. For example, “`com.github.ToshioCP.tfe`” is a correct schema id.
- A schema usually has a path. The path is a location in the database. Each key is stored under the path. For example, if a key `font-desc` is defined with a path `/com/github/ToshioCP/tfe/`, the key’s location in the database is `/com/github/ToshioCP/tfe/font-desc`. A path is a string begins with and ends with a slash (/). And it is delimited by slashes.
- GSettings save information as key-value style. A key is a string that begins with a lower case character followed by lower case letters, digits or dashes (-) and ends with a lower case letter or digit. No consecutive dashes are allowed. Values can be any type. GSettings stores values as GVariant type, which can be, for example, integer, double, boolean, string or complex types like an array. The type of values needs to be defined in the schema.
- A default value needs to be set for each key.
- A summary and description can be set for each key optionally.

Schemas are described in an XML format. For example,

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <schemalist>
3   <schema path="/com/github/ToshioCP/tfe/" id="com.github.ToshioCP.tfe">
4     <key name="font-desc" type="s">
5       <default>'Monospace 12'</default>
6       <summary>Font</summary>
7       <description>A font to be used for textview.</description>
8     </key>
9   </schema>
10 </schemalist>
```

- 4: The type attribute is “s”. It is GVariant type string. For GVariant type string, see [GLib API Reference – GVariant Type Strings](#). Other common types are:
 - “b”: gboolean
 - “i”: gint32.
 - “d”: double.

Further information is in:

- [GLib API Reference – GVariant Format Strings](#)
- [GLib API Reference – GVariant Text Format](#)
- [GLib API Reference – GVariant](#)
- [GLib API Reference – VariantType](#)

2.19.6.2 Gsettings Command

First, let's try `gsettings` application. It is a configuration tool for GSettings.

```
$ gsettings help
```

Usage:

```
gsettings --version
gsettings [--schemadir SCHEMADIR] COMMAND [ARGS?]
```

Commands:

<code>help</code>	Show this information
<code>list-schemas</code>	List installed schemas
<code>list-relocatable-schemas</code>	List relocatable schemas
<code>list-keys</code>	List keys in a schema
<code>list-children</code>	List children of a schema
<code>list-recursively</code>	List keys and values, recursively
<code>range</code>	Queries the range of a key
<code>describe</code>	Queries the description of a key
<code>get</code>	Get the value of a key
<code>set</code>	Set the value of a key
<code>reset</code>	Reset the value of a key
<code>reset-recursively</code>	Reset all values in a given schema
<code>writable</code>	Check if a key is writable
<code>monitor</code>	Watch for changes

Use "`gsettings help COMMAND`" to get detailed help.

List schemas.

```
$ gsettings list-schemas
org.gnome.rhythmbox.podcast
ca.desrt.dconf-editor.Demo.Empty
org.gnome.gedit.preferences.ui
org.gnome.evolution-data-server.calendar
org.gnome.rhythmbox.plugins.generic-player
... ..
```

Each line is an id of a schema. Each schema has a key-value configuration data. You can see them with `list-recursively` command. Let's look at the keys and values of `org.gnome.calculator` schema.

```
$ gsettings list-recursively org.gnome.calculator
org.gnome.calculator accuracy 9
org.gnome.calculator angle-units 'degrees'
org.gnome.calculator base 10
org.gnome.calculator button-mode 'basic'
org.gnome.calculator number-format 'automatic'
org.gnome.calculator precision 2000
org.gnome.calculator refresh-interval 604800
org.gnome.calculator show-thousands false
org.gnome.calculator show-zeroes false
org.gnome.calculator source-currency ''
org.gnome.calculator source-units 'degree'
org.gnome.calculator target-currency ''
org.gnome.calculator target-units 'radian'
org.gnome.calculator window-position (-1, -1)
```

```
org.gnome.calculator word-size 64
```

This schema is used by GNOME Calculator. Run the calculator and change the mode, then check the schema again.

```
$ gnome-calculator
```

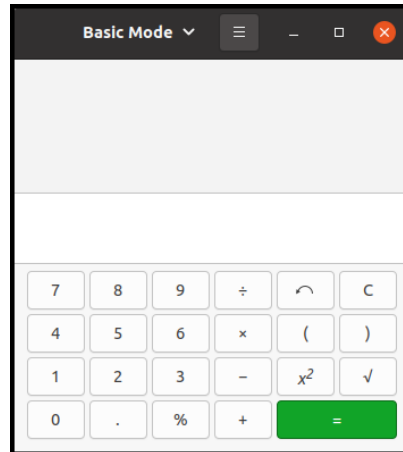


Figure 2.30: gnome-calculator basic mode

Change the mode to advanced and quit.

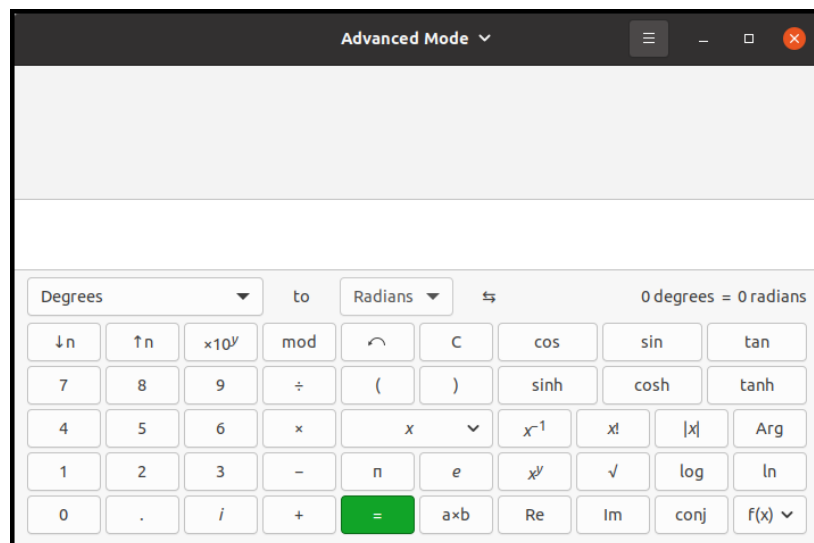


Figure 2.31: gnome-calculator advanced mode

Run gsettings and check the value of `button-mode`.

```
$ gsettings list-recursively org.gnome.calculator
```

```
... ..
```

```
org.gnome.calculator button-mode 'advanced'
```

```
... ..
```

Now we know that GNOME Calculator used gsettings and it has set `button-mode` key to “advanced”. The value remains even the calculator quits. So when the calculator runs again, it will appear as an advanced mode.

2.19.6.3 Glib-compile-schemas Utility

GSettings schemas are specified with an XML format. The XML schema files must have the filename extension `.gschema.xml`. The following is the XML schema file for the application `tfe`.

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <schemalist>
3   <schema path="/com/github/ToshioCP/tfe/" id="com.github.ToshioCP.tfe">
4     <key name="font-desc" type="s">
5       <default>'Monospace 12'</default>
6       <summary>Font</summary>
7       <description>A font to be used for textview.</description>
8     </key>
9   </schema>
10 </schemalist>

```

The filename is “com.github.ToshioCP.tfe.gschema.xml”. Schema XML filenames are usually the schema id followed by “gschema.xml” suffix. You can use different name from schema id, but it is not recommended.

- 2: The top level element is `<schemalist>`.
- 3: schema tag has `path` and `id` attributes. A path determines where the settings are stored in the conceptual global tree of settings. An id identifies the schema.
- 4: Key tag has two attributes. Name is the name of the key. Type is the type of the value of the key and it is a GVariant Format String.
- 5: default value of the key `font-desc` is `Monospace 12`.
- 6: Summary and description elements describe the key. They are optional, but it is recommended to add them in the XML file.

The XML file is compiled by `glib-compile-schemas`. When compiling, `glib-compile-schemas` compiles all the XML files which have “gschema.xml” file extension in the directory given as an argument. It converts the XML file into a binary file `gschemas.compiled`. Suppose the XML file above is under `tfe6` directory.

```
$ glib-compile-schemas tfe6
```

Then, `gschemas.compiled` is generated under `tfe6`. When you test your application, set `GSETTINGS_SCHEMA_DIR` environment variable so that GSettings objet can find `gschemas.compiled`.

```
$ GSETTINGS_SCHEMA_DIR=(the directory gschemas.compiled is located):$GSETTINGS_SCHEMA_DIR (your appl.
```

GSettings object looks for this file by the following process.

- It searches `glib-2.0/schemas` subdirectories of all the directories specified in the environment variable `XDG_DATA_DIRS`. Common diretores are `/usr/share/glib-2.0/schemas` and `/usr/local/share/glib-2.0/schemas`.
- If `$HOME/.local/share/glib-2.0/schemas` exists, it is also searched.
- If `GSETTINGS_SCHEMA_DIR` environment variable is defined, it searches all the directories specified in the variable. `GSETTINGS_SCHEMA_DIR` can specify multiple directories delimited by colon (:).

The directories above includes more than one `.gschema.xml` file. Therefore, when you install your application, follow the instruction below to install your schemas.

1. Make `.gschema.xml` file.
2. Copy it to one of the directories above. For example, `$HOME/.local/share/glib-2.0/schemas`.
3. Run `glib-compile-schemas` on the directory. It compiles all the schema files in the directory and creates or updates the database file `gschemas.compiled`.

2.19.6.4 GSettings Object and Binding

Now, we go on to the next topic, how to program GSettings.

You need to compile your schema file in advance.

Suppose id, key, class name and a property name are:

- GSettings id: com.github.ToshioCP.sample
- GSettings key: sample_key
- The class name: Sample
- The property to bind: sample_property

The example below uses `g_settings_bind`. If you use it, GSettings key and instance property must have the same the type. In the example, it is assumed that the type of “sample_key” and “sample_property” are the same.

```
GSettings *settings;
Sample *sample_object;

settings = g_settings_new ("com.github.ToshioCP.sample");
sample_object = sample_new ();
g_settings_bind (settings, "sample_key", sample_object, "sample_property", G_SETTINGS_BIND_DEFAULT);
```

The function `g_settings_bind` binds the GSettings value and the property of the instance. If the property value is changed, the GSettings value is also changed, and vice versa. The two values are always the same.

The function `g_settings_bind` is simple and easy but it isn’t always possible. The type of GSettings are restricted to the type GVariant has. Some property types are out of GVariant. For example, `GtkFontDialogButton` has “font-desc” property and its type is `PangoFontDescription`. `PangoFontDescription` is a C structure and it is wrapped in a boxed type `GValue` to store in the property. GVariant doesn’t support boxed type.

In that case, another function `g_settings_bind_with_mapping` is used. It binds GSettings GVariant value and object property via `GValue` with mapping functions. See [GIO documentation](#) for further details.

```
void
g_settings_bind_with_mapping (
    GSettings* settings,
    const gchar* key,
    GObject* object,
    const gchar* property,
    GSettingsBindFlags flags, // G_SETTINGS_BIND_DEFAULT is commonly used
    GSettingsBindGetMapping get_mapping, // GSettings => property, See the example below
    GSettingsBindSetMapping set_mapping, // property => GSettings, See the example below
    gpointer user_data, // NULL if unnecessary
    GDestroyNotify destroy //NULL if unnecessary
)
```

The mapping functions are defined like these:

```
gboolean
(* GSettingsBindGetMapping) (
    GValue* value,
    GVariant* variant,
    gpointer user_data
)

GVariant*
(* GSettingsBindSetMapping) (
    const GValue* value,
    const GVariantType* expected_type,
    gpointer user_data
)
```

The following codes are extracted from `tfepref.c`.

```

1  static gboolean // GSettings => property
2  get_mapping (GValue *value, GVariant *variant, gpointer user_data) {
3      const char *s = g_variant_get_string (variant, NULL);
4      PangoFontDescription *font_desc = pango_font_description_from_string (s);
5      g_value_take_boxed (value, font_desc);
6      return TRUE;
7  }
8
9  static GVariant * // Property => GSettings
10 set_mapping (const GValue *value, const GVariantType *expected_type, gpointer user_data) {
11     char *font_desc_string = pango_font_description_to_string (g_value_get_boxed (value));
12     return g_variant_new_take_string (font_desc_string);
13 }
14
15 static void
16 tfe_pref_init (TfePref *pref) {
17     gtk_widget_init_template (GTK_WIDGET (pref));
18     pref->settings = g_settings_new ("com.github.ToshioCP.tfe");
19     g_settings_bind_with_mapping (pref->settings, "font-desc", pref->font_dialog_btn, "font-desc", G_SETTINGS_BIND_DEFAULT,
20         get_mapping, set_mapping, NULL, NULL);
21 }

```

- 15-21: This function `tfe_pref_init` initializes the new `TfePref` instance.
- 18: Creates a new `GSettings` instance. The id is “com.github.ToshioCP.tfe”.
- 19-20: Binds the `GSettings` “font-desc” and the `GtkFontDialogButton` property “font-desc”. The mapping functions are `get_mapping` and `set_mapping`.
- 1-7: The mapping function from `GSettings` to the property. The first argument `value` is a `GValue` to be stored in the property. The second argument `variant` is a `GVariant` structure that comes from the `GSettings` value.
- 3: Retrieves a string from the `GVariant` structure.
- 4: Build a `PangoFontDescription` structure from the string and assigns its address to `font_desc`.
- 5: Puts `font_desc` into the `GValue value`. The ownership of `font_desc` moves to `value`.
- 6: Returns `TRUE` that means the mapping succeeds.
- 9-13: The mapping function from the property to `GSettings`. The first argument `value` holds the property data. The second argument `expected_type` is the type of `GVariant` that the `GSettings` value has. It isn’t used in this function.
- 11: Gets the `PangoFontDescription` structure from `value` and converts it to string.
- 12: The string is inserted to a `GVariant` structure. The ownership of the string `font_desc_string` moves to the returned value.

2.19.7 C File

The following is the full codes of `tfepref.c`

```

1  #include <gtk/gtk.h>
2  #include "tfepref.h"
3
4  struct _TfePref
5  {
6      GtkWindow parent;
7      GSettings *settings;
8      GtkFontDialogButton *font_dialog_btn;
9  };

```

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```
G_DEFINE_FINAL_TYPE (TfePref, tfe_pref, GTK_TYPE_WINDOW);

static void
tfe_pref_dispose (GObject *gobject) {
    TfePref *pref = TFE_PREF (gobject);

    /* GSetting bindings are automatically removed when the object is finalized, so it isn't necessary
    g_clear_object (&pref->settings);
    gtk_widget_dispose_template (GTK_WIDGET (pref), TFE_TYPE_PREF);
    G_OBJECT_CLASS (tfe_pref_parent_class)->dispose (gobject);
}

/* ----- get_mapping/set_mapping ----- */
static gboolean // GSettings => property
get_mapping (GValue *value, GVariant *variant, gpointer user_data) {
    const char *s = g_variant_get_string (variant, NULL);
    PangoFontDescription *font_desc = pango_font_description_from_string (s);
    g_value_take_boxed (value, font_desc);
    return TRUE;
}

static GVariant * // Property => GSettings
set_mapping (const GValue *value, const GVariantType *expected_type, gpointer user_data) {
    char *font_desc_string = pango_font_description_to_string (g_value_get_boxed (value));
    return g_variant_new_take_string (font_desc_string);
}

static void
tfe_pref_init (TfePref *pref) {
    gtk_widget_init_template (GTK_WIDGET (pref));
    pref->settings = g_settings_new ("com.github.ToshioCP.tfe");
    g_settings_bind_with_mapping (pref->settings, "font-desc", pref->font_dialog_btn, "font-desc", G_SETTINGS_BIND_DEFAULT,
    get_mapping, set_mapping, NULL, NULL);
}

static void
tfe_pref_class_init (TfePrefClass *class) {
    G_OBJECT_CLASS (class)->dispose = tfe_pref_dispose;
    gtk_widget_class_set_template_from_resource (GTK_WIDGET_CLASS (class), "/com/github/ToshioCP/tfe/tfe/tfe.ui");
    gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfePref, font_dialog_btn);
}

GtkWidget *
tfe_pref_new (void) {
    return GTK_WIDGET (g_object_new (TFE_TYPE_PREF, NULL));
}
```

2.19.8 Test Program

There's a test program located at `src/tfe6/test` directory.

```
1 #include <gtk/gtk.h>
2 #include "../tfepref.h"
```

```

3
4 GSettings *settings;
5
6 // "changed::font-desc" signal handler
7 static void
8 changed_font_desc_cb (GSettings *settings, char *key, gpointer user_data) {
9     char *s;
10    s = g_settings_get_string (settings, key);
11    g_print ("%s\n", s);
12    g_free (s);
13 }
14
15 static void
16 app_shutdown (GApplication *application) {
17     g_object_unref (settings);
18 }
19
20 static void
21 app_activate (GApplication *application) {
22     GtkWidget *pref = tfe_pref_new ();
23
24     gtk_window_set_application (GTK_WINDOW (pref), GTK_APPLICATION (application));
25     gtk_window_present (GTK_WINDOW (pref));
26 }
27
28 static void
29 app_startup (GApplication *application) {
30     settings = g_settings_new ("com.github.ToshioCP.tfe");
31     g_signal_connect (settings, "changed::font-desc", G_CALLBACK (changed_font_desc_cb), NULL);
32     g_print ("%s\n", "Change the font with the font button. Then the new font will be printed out.\n");
33 }
34
35 #define APPLICATION_ID "com.github.ToshioCP.test_tfe_pref"
36
37 int
38 main (int argc, char **argv) {
39     GtkApplication *app;
40     int stat;
41
42     app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
43     g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
44     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
45     g_signal_connect (app, "shutdown", G_CALLBACK (app_shutdown), NULL);
46     stat = g_application_run (GTK_APPLICATION (app), argc, argv);
47     g_object_unref (app);
48     return stat;
49 }

```

This program sets its active window to TfePref instance, which is a child object of GtkWidget.

It sets the “changed::font-desc” signal handler in the startup function. The process from the user’s font selection to the handler is:

- The user clicked on the GtkFontDialogButton and GtkFontDialog appears.
- He/she selects a new font.

- The “font-desc” property of the GtkFontDialogButton instance is changed.
- The value of “font-desc” key on the GSettings database is changed since it is bound to the property.
- The “changed::font-desc” signal on the GSettings instance is emitted.
- The handler is called.

The program building is divided into four steps.

- Compile the schema file
- Compile the XML file to a resource (C source file)
- Compile the C files
- Run the executable file

Commands are shown in the next four sub-subsections. You don’t need to try them. The final sub-subsection shows the meson-ninja way, which is the easiest.

2.19.8.1 Compile the Schema File

```
$ cd src/tef6/test
$ cp ../com.github.ToshioCP.tfe.gschema.xml com.github.ToshioCP.tfe.gschema.xml
$ glib-compile-schemas .
```

Be careful. The commands `glib-compile-schemas` has an argument “.”, which means the current directory. This results in creating `gschemas.compiled` file.

2.19.8.2 Compile the XML File

```
$ glib-compile-resources --sourcedir=.. --generate-source --target=resource.c ../tfe.gresource.xml
```

2.19.8.3 Compile the C File

```
$ gcc `pkg-config --cflags gtk4` test_pref.c ../tfepref.c resource.c `pkg-config --
libs gtk4`
```

2.19.8.4 Run the Executable File

```
$ GSETTINGS_SCHEMA_DIR=. ./a.out
```

```
Jamrul Italic Semi-Expanded 12 # <= select Jamrul Italic 12
Monospace 12 #<= select Monospace Regular 12
```

2.19.8.5 Meson-Ninja Way

Meson wraps up the commands above. Create the following text and save it to `meson.build`.

Note: Gtk4-tutorial repository has `meson.build` file that defines several tests. So you can try it instead of the following text.

```
project('tfe_pref_test', 'c')
```

```
gtkdep = dependency('gtk4')
```

```
gnome=import('gnome')
```

```
resources = gnome.compile_resources('resources','../tfe.gresource.xml', source_dir: '..')
```

```
gnome.compile_schemas(build_by_default: true, depend_files: 'com.github.ToshioCP.tfe.gschema.xml')
```

```
executable('test_pref', ['test_pref.c', '../tfepref.c'], resources, dependencies: gtkdep, export_dyn
```

- Project name is ‘tfe_pref_test’ and it is written in C language.
- It depends on GTK4 library.

- It uses GNOME module. Modules are prepared by Meson.
- GNOME module has `compile_resources` method. When you call this method, you need the prefix “gnome.”.
 - The target filename is resources.
 - The definition XML file is ‘./tfe.gresource.xml’.
 - The source dir is ‘.’. All the UI files are located there.
- GNOME module has `compile_schemas` method. It compiles the schema file ‘com.github.ToshioCP.tfe.gschema.xml’. You need to copy ‘./com.github.ToshioCP.tfe.gschema.xml’ to the current directory in advance.
- It creates an executable file ‘test_pref’. The source files are ‘test_pref.c’, ‘./tfepref.c’ and `resources`, which is made by `gnome.compile_resources`. It depends on `gtkdep`, which is GTK4 library. The symbols are exported and no installation support.

Type like this to build and test the program.

```
$ cd src/tef6/test
$ cp ../com.github.ToshioCP.tfe.gschema.xml com.github.ToshioCP.tfe.gschema.xml
$ meson setup _build
$ ninja -C _build
$ GSETTINGS_SCHEMA_DIR=_build _build/test_pref
```

A window appears and you can choose a font via `GtkFontDialog`. If you select a new font, the font string is output through the standard output.

2.20 TfeWindow

2.20.1 The Tfe Window and XML Files

The following is the window of Tfe.

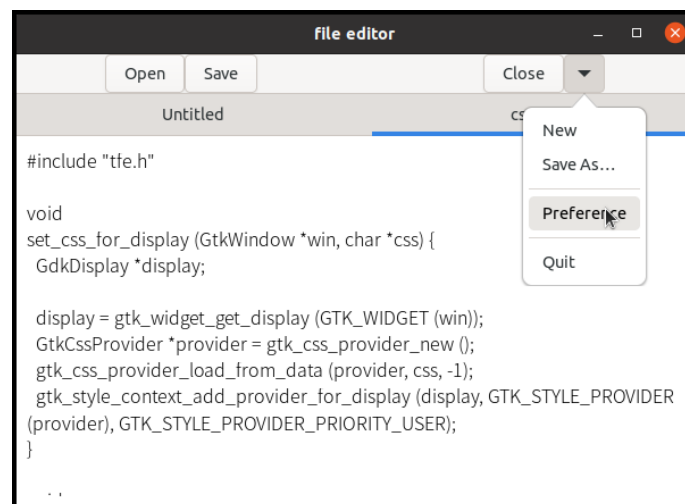


Figure 2.32: tfe6

- Open, save and close buttons are placed on the toolbar. In addition, `GtkMenuButton` is added to the toolbar. This button shows a popup menu when clicked on. Here, popup means widely, including pull-down menu.
- New, save-as, preference and quit items are put into the menu.

This makes the most frequently used operation bound to the tool bar buttons. And the others are stored behind the menus. So, it is more practical.

The window is a composite widget. The definition is described in the XML file `tfewindow.ui`.

```

1  <?xml version="1.0" encoding="UTF-8"?>
2  <interface>
3      <template class="TfeWindow" parent="GtkApplicationWindow">
4          <property name="title">Text File Editor</property>
5          <property name="default-width">600</property>
6          <property name="default-height">400</property>
7          <child>
8              <object class="GtkBox" id="boxv">
9                  <property name="orientation">GTK_ORIENTATION_VERTICAL</property>
10                 <child>
11                     <object class="GtkBox" id="boxh">
12                         <property name="orientation">GTK_ORIENTATION_HORIZONTAL</property>
13                         <child>
14                             <object class="GtkLabel">
15                                 <property name="width-chars">10</property>
16                             </object>
17                         </child>
18                         <child>
19                             <object class="GtkButton">
20                                 <property name="label">Open</property>
21                                 <property name="action-name">win.open</property>
22                             </object>
23                         </child>
24                         <child>
25                             <object class="GtkButton">
26                                 <property name="label">Save</property>
27                                 <property name="action-name">win.save</property>
28                             </object>
29                         </child>
30                         <child>
31                             <object class="GtkLabel">
32                                 <property name="hexpand">TRUE</property>
33                             </object>
34                         </child>
35                         <child>
36                             <object class="GtkButton">
37                                 <property name="label">Close</property>
38                                 <property name="action-name">win.close</property>
39                             </object>
40                         </child>
41                         <child>
42                             <object class="GtkMenuButton" id="btnm">
43                                 <property name="direction">down</property>
44                                 <property name="icon-name">open-menu-symbolic</property>
45                             </object>
46                         </child>
47                         <child>
48                             <object class="GtkLabel">
49                                 <property name="width-chars">10</property>
50                             </object>
51                         </child>
52                     </object>
53                 </child>

```

```

54     <child>
55         <object class="GtkNotebook" id="nb">
56             <property name="scrollable">TRUE</property>
57             <property name="hexexpand">TRUE</property>
58             <property name="vexpand">TRUE</property>
59         </object>
60     </child>
61 </object>
62 </child>
63 </template>
64 </interface>

```

- Three buttons “Open”, “Save” and “Close” are defined. You can use two ways to catch the button click event. One is “clicked” signal and the other is to register an action to the button. The first way is simple. You can connect the signal and your handler directly. The second way is like menu items. When the button is clicked, the corresponding action is activated. It is a bit complicated because you need to create an action and its “activate” handler in advance. But one advantage is you can connect two or more things to the action. For example, an accelerator can be connected to the action. Accelerators are keys that connect to actions. For example, Ctrl+O is often connected to a file open action. So, both open button and Ctrl+O activate an open action. The latter way is used in the XML file above.
- You can specify a theme icon to GtkMenuButton with “icon-name” property. The “open-menu-symbolic” is an image that is called hamburger menu.

The `menu.ui` XML file defines the menu for GtkMenuButton.

```

1  <?xml version="1.0" encoding="UTF-8"?>
2  <interface>
3      <menu id="menu">
4          <section>
5              <item>
6                  <attribute name="label">New</attribute>
7                  <attribute name="action">win.new</attribute>
8              </item>
9              <item>
10                 <attribute name="label">Save As...</attribute>
11                 <attribute name="action">win.saveas</attribute>
12             </item>
13         </section>
14         <section>
15             <item>
16                 <attribute name="label">Preference</attribute>
17                 <attribute name="action">win.pref</attribute>
18             </item>
19         </section>
20         <section>
21             <item>
22                 <attribute name="label">Quit</attribute>
23                 <attribute name="action">win.close-all</attribute>
24             </item>
25         </section>
26     </menu>
27 </interface>

```

There are four menu items and they are connected to actions.

2.20.2 The Header File

The following is the codes of `tfewindow.h`.

```

1  #pragma once
2
3  #include <gtk/gtk.h>
4
5  #define TFE_TYPE_WINDOW tfe_window_get_type ()
6  G_DECLARE_FINAL_TYPE (TfeWindow, tfe_window, TFE, WINDOW, GtkApplicationWindow)
7
8  void
9  tfe_window_notebook_page_new (TfeWindow *win);
10
11 void
12 tfe_window_notebook_page_new_with_files (TfeWindow *win, GFile **files, int n_files);
13
14 GtkWidget *
15 tfe_window_new (GtkApplication *app);

```

- 5-6: TFE_TYPE_WINDOW definition and the G_DECLARE_FINAL_TYPE macro.
- 8-15: Public functions. The first two functions create a notebook page and the last function creates a window.

2.20.3 C File

2.20.3.1 Composite Widget

The following codes are extracted from `tfewindow.c`.

```

#include <gtk/gtk.h>
#include "tfewindow.h"

struct _TfeWindow {
    GtkApplicationWindow parent;
    GtkMenuButton *btnm;
    GtkNotebook *nb;
    gboolean is_quit;
};

G_DEFINE_FINAL_TYPE (TfeWindow, tfe_window, GTK_TYPE_APPLICATION_WINDOW);

static void
tfe_window_dispose (GObject *gobject) {
    gtk_widget_dispose_template (GTK_WIDGET (gobject), TFE_TYPE_WINDOW);
    G_OBJECT_CLASS (tfe_window_parent_class)->dispose (gobject);
}

static void
tfe_window_init (TfeWindow *win) {
    GtkBuilder *build;
    GMenuModel *menu;

    gtk_widget_init_template (GTK_WIDGET (win));

    build = gtk_builder_new_from_resource ("/com/github/ToshioCP/tfe/menu.ui");

```

```

    menu = G_MENU_MODEL (gtk_builder_get_object (build, "menu"));
    gtk_menu_button_set_menu_model (win->btnm, menu);
    g_object_unref(build);
    ... ..
}

static void
tfe_window_class_init (TfeWindowClass *class) {
    GObjectClass *object_class = G_OBJECT_CLASS (class);

    object_class->dispose = tfe_window_dispose;
    gtk_widget_class_set_template_from_resource (GTK_WIDGET_CLASS (class), "/com/github/ToshioCP/tfe/tfe.ui");
    gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfeWindow, btnm);
    gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfeWindow, nb);
}

GtkWidget *
tfe_window_new (GtkApplication *app) {
    return GTK_WIDGET (g_object_new (TFE_TYPE_WINDOW, "application", app, NULL));
}

```

The program above is similar to `tfealert.c` and `tfepref.c`. It uses the same way to build a composite widget. But there's one thing new. It is menu. The menu is built from the XML resource `menu.ui` and inserted into the menu button. It is done in the instance initialization function `tfe_window_init`.

2.20.3.2 Actions

Actions can belong to an application or window. Tfe only has one top window and all the actions are registered in the window. For example, “close-all” action destroys the top level window and that brings the application to quit. You can make “app.quit” action instead of “win.close-all”. It's your choice. If your application has two or more windows, both “app.quit” and “win.close-all”, which closes all the notebook pages on the window, may be necessary. Anyway, you need to consider that each action should belong to the application or a window.

Actions are defined in the instance initialization function.

```

static void
tfe_window_init (TfeWindow *win) {
    ... ..
    /* ----- action ----- */
    const GActionEntry win_entries[] = {
        { "open", open_activated, NULL, NULL, NULL },
        { "save", save_activated, NULL, NULL, NULL },
        { "close", close_activated, NULL, NULL, NULL },
        { "new", new_activated, NULL, NULL, NULL },
        { "saveas", saveas_activated, NULL, NULL, NULL },
        { "pref", pref_activated, NULL, NULL, NULL },
        { "close-all", close_all_activated, NULL, NULL, NULL }
    };
    g_action_map_add_action_entries (G_ACTION_MAP (win), win_entries, G_N_ELEMENTS (win_entries), win);
    ... ..
}

```

Two things are necessary, an array and the `g_action_map_add_action_entries` function.

- The element of the array is the `GActionEntry` structure. The structure has the following members:

- an action name
- a handler for the activate signal
- the type of the parameter or NULL for no parameter
- the initial state for the action
- a handler for the change-state signal
- The actions above are stateless and have no parameters. So, the third parameter and after are all NULL.
- The function `g_action_map_add_action_entries` adds the actions in the `win_entries` array to the action map `win`. The last argument `win` is the `user_data`, which is the last argument of handlers.
- All the handlers are in `tfewindow.c` program and shown in the following subsections.

2.20.3.3 The Handlers of the Actions

2.20.3.3.1 `open_activated` The callback function `open_activated` is an activate signal handler on “open” action.

```

1 static void
2 open_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
3     TfeWindow *win = TFE_WINDOW (user_data);
4     GtkWidget *tv = tfe_text_view_new ();
5
6     g_signal_connect (tv, "open-response", G_CALLBACK (open_response_cb), win);
7     tfe_text_view_open (TFE_TEXT_VIEW (tv), GTK_WINDOW (win));
8 }

```

It connects the “open-response” signal on the newly created `TfeTextView` instance and just calls `tfe_text_view_open`. It leaves the rest of the task to the signal handler `open_response_cb`.

```

1 static void
2 open_response_cb (TfeTextView *tv, int response, gpointer user_data) {
3     TfeWindow *win = TFE_WINDOW (user_data);
4     GFile *file;
5     char *filename;
6
7     if (response != TFE_OPEN_RESPONSE_SUCCESS) {
8         g_object_ref_sink (tv);
9         g_object_unref (tv);
10    }else if (! G_IS_FILE (file = tfe_text_view_get_file (tv))) {
11        g_object_ref_sink (tv);
12        g_object_unref (tv);
13    }else {
14        filename = g_file_get_basename (file);
15        g_object_unref (file);
16        notebook_page_build (win, GTK_WIDGET (tv), filename);
17        g_free (filename);
18    }
19 }

```

If the `TfeTextView` instance failed to read a file, it destroys the instance with `g_object_ref_sink` and `g_object_unref`. Since newly created widgets have a floating reference, you need to convert the floating reference to the normal reference before you release it. The conversion is done with `g_object_ref_sink`.

If the instance successfully read the file, it calls `notebook_page_build` to build a notebook page and add it to the `GtkNotebook` object.

```

1 static void
2 notebook_page_build (TfeWindow *win, GtkWidget *tv, char *filename) {

```

```

3  // The arguments win, tv and filename are owned by the caller.
4  // If tv has a floating reference, it is consumed by the function.
5  GtkWidget *scr = gtk_scrolled_window_new ();
6  GtkTextBuffer *tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
7  GtkNotebookPage *nbp;
8  GtkWidget *lab;
9  int i;
10
11  gtk_text_view_set_wrap_mode (GTK_TEXT_VIEW (tv), GTK_WRAP_WORD_CHAR);
12  gtk_scrolled_window_set_child (GTK_SCROLLLED_WINDOW (scr), tv);
13  lab = gtk_label_new (filename);
14  i = gtk_notebook_append_page (win->nb, scr, lab);
15  nbp = gtk_notebook_get_page (win->nb, scr);
16  g_object_set (nbp, "tab-expand", TRUE, NULL);
17  gtk_notebook_set_current_page (win->nb, i);
18  g_signal_connect (tv, "change-file", G_CALLBACK (file_changed_cb), win->nb);
19  g_signal_connect (tb, "modified-changed", G_CALLBACK (modified_changed_cb), tv);
20 }

```

This function is a kind of library function and it is called from the different three places.

This function creates a new `GtkScrolledWindow` instance and sets its child to `tv`. Then it appends it to the `GtkNotebook` instance `win->nb`. And it sets the tab label to the filename.

After the building, it connects two signals and handlers.

- “change-file” signal and `file_changed_cb` handler. If the `TfeTextView` instance changes the file, the handler is called and the notebook page tab is updated.
- “modified-changed” signal and `modified_changed_cb` handler. If the text in the buffer of `TfeTextView` instance is modified, an asterisk is added at the beginning of the filename of the notebook page tab. If the text is saved to the file, the asterisk is removed. The asterisk tells the user that the text has been modified or not.

```

1  static void
2  file_changed_cb (TfeTextView *tv, gpointer user_data) {
3      GtkNotebook *nb = GTK_NOTEBOOK (user_data);
4      GtkWidget *scr;
5      GtkWidget *label;
6      GFile *file;
7      char *filename;
8
9      file = tfe_text_view_get_file (tv);
10     scr = gtk_widget_get_parent (GTK_WIDGET (tv));
11     if (G_IS_FILE (file)) {
12         filename = g_file_get_basename (file);
13         g_object_unref (file);
14     } else
15         filename = get_untitled ();
16     label = gtk_label_new (filename);
17     g_free (filename);
18     gtk_notebook_set_tab_label (GTK_NOTEBOOK (nb), scr, label);
19 }
20
21 static void
22 modified_changed_cb (GtkTextBuffer *tb, gpointer user_data) {
23     TfeTextView *tv = TFE_TEXT_VIEW (user_data);

```



```

24 GtkWidget *scr = gtk_widget_get_parent (GTK_WIDGET (tv));
25 GtkWidget *nb = gtk_widget_get_ancestor (GTK_WIDGET (tv), GTK_TYPE_NOTEBOOK);
26 GtkWidget *label;
27 GFile *file;
28 char *filename;
29 char *text;
30
31 file = tfe_text_view_get_file (tv);
32 filename = g_file_get_basename (file);
33 if (gtk_text_buffer_get_modified (tb))
34     text = g_strdup_printf ("%s", filename);
35 else
36     text = g_strdup (filename);
37 g_object_unref (file);
38 g_free (filename);
39 label = gtk_label_new (text);
40 g_free (text);
41 gtk_notebook_set_tab_label (GTK_NOTEBOOK (nb), scr, label);
42 }

```

2.20.3.3.2 save_activated The callback function `save_activated` is an activate signal handler on “save” action.

```

1 static void
2 save_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
3     TfeWindow *win = TFE_WINDOW (user_data);
4     TfeTextView *tv = get_current_textview (win->nb);
5
6     tfe_text_view_save (TFE_TEXT_VIEW (tv));
7 }

```

This function gets the current `TfeTextView` instance with the function `get_current_textview`. And it just calls `tfe_text_view_save`.

```

1 static TfeTextView *
2 get_current_textview (GtkNotebook *nb) {
3     int i;
4     GtkWidget *scr;
5     GtkWidget *tv;
6
7     i = gtk_notebook_get_current_page (nb);
8     scr = gtk_notebook_get_nth_page (nb, i);
9     tv = gtk_scrolled_window_get_child (GTK_SCROLLED_WINDOW (scr));
10    return TFE_TEXT_VIEW (tv);
11 }

```

2.20.3.3.3 close_activated The callback function `close_activated` is an activate signal handler on “close” action. It closes the current notebook page.

```

1 static void
2 close_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
3     TfeWindow *win = TFE_WINDOW (user_data);
4     TfeTextView *tv;
5     GtkTextBuffer *tb;
6     GtkWidget *alert;

```

```

7
8   tv = get_current_textview (win->nb);
9   tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
10  if (! gtk_text_buffer_get_modified (tb)) /* is saved? */
11      notebook_page_close (win);
12  else {
13      win->is_quit = FALSE;
14      alert = tfe_alert_new_with_data ("Are you sure?", "Contents aren't saved yet.\nAre you sure to c
15      gtk_window_set_transient_for (GTK_WINDOW (alert), GTK_WINDOW (win));
16      g_signal_connect (alert, "response", G_CALLBACK (alert_response_cb), win);
17      gtk_window_present (GTK_WINDOW (alert));
18  }
19 }

```

If the text in the current page has been saved, it calls `notebook_page_close` to close the page. Otherwise, it sets `win->is_quit` to `FALSE` and show the alert dialog. The “response” signal on the dialog is connected to the handler `alert_response_cb`.

```

1  static void
2  notebook_page_close (TfeWindow *win){
3      int i;
4
5      if (gtk_notebook_get_n_pages (win->nb) == 1)
6          gtk_window_destroy (GTK_WINDOW (win));
7      else {
8          i = gtk_notebook_get_current_page (win->nb);
9          gtk_notebook_remove_page (win->nb, i);
10     }
11 }

```

If the notebook has only one page, it destroys the window and the application quits. Otherwise, it removes the current page.

```

1  static void
2  alert_response_cb (TfeAlert *alert, int response_id, gpointer user_data) {
3      TfeWindow *win = TFE_WINDOW (user_data);
4
5      if (response_id == TFE_ALERT_RESPONSE_ACCEPT) {
6          if (win->is_quit)
7              gtk_window_destroy(GTK_WINDOW (win));
8          else
9              notebook_page_close (win);
10     }
11 }

```

If the user clicked on the cancel button, it does nothing. If the user clicked on the accept button, which is the same as close button, it calls `notebook_page_close`. Note that `win->is_quit` has been set to `FALSE` in the `close_activated` function.

2.20.3.3.4 new_activated The callback function `new_activated` is an activate signal handler on “new” action.

```

1  static void
2  new_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
3      TfeWindow *win = TFE_WINDOW (user_data);
4

```

```

5   tfe_window_notebook_page_new (win);
6 }

```

It just calls `tfe_window_notebook_page_new`, which is a public method of `TfeWindow`.

```

1  void
2  tfe_window_notebook_page_new (TfeWindow *win) {
3      GtkWidget *tv;
4      char *filename;
5
6      tv = tfe_text_view_new ();
7      filename = get_untitled ();
8      notebook_page_build (win, tv, filename);
9      g_free (filename);
10 }

```

This function creates a new `TfeTextView` instance, generates a string like “Untitled”, and calls `notebook_page_build`.

2.20.3.3.5 saveas__activated The callback function `saveas__activated` is an activate signal handler on “saveas” action.

```

1  static void
2  saveas_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
3      TfeWindow *win = TFE_WINDOW (user_data);
4      TfeTextView *tv = get_current_textview (win->nb);
5
6      tfe_text_view_saveas (TFE_TEXT_VIEW (tv));
7  }

```

This function gets the current page `TfeTextView` instance and calls `tfe_text_view_saveas`.

2.20.3.3.6 pref__activated The callback function `pref__activated` is an activate signal handler on “pref” action.

```

1  static void
2  pref_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
3      TfeWindow *win = TFE_WINDOW (user_data);
4      GtkWidget *pref;
5
6      pref = tfe_pref_new ();
7      gtk_window_set_transient_for (GTK_WINDOW (pref), GTK_WINDOW (win));
8      gtk_window_present (GTK_WINDOW (pref));
9  }

```

This function creates a `TfePref` instance, which is a dialog, and sets the transient parent window to `win`. And it shows the dialog.

2.20.3.3.7 close_all__activated The callback function `close_all__activated` is an activate signal handler on “close_all” action.

```

1  static void
2  close_all_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
3      TfeWindow *win = TFE_WINDOW (user_data);
4
5      if (close_request_cb (win) == FALSE)

```

```

6      gtk_window_destroy (GTK_WINDOW (win));
7  }

```

It first calls the function `close_request_cb`. It is a callback function for the “close-request” signal on the top window. It returns `FALSE` if all the texts have been saved. Otherwise it returns `TRUE`.

Therefore, function `close_all_activated` destroys the top window if all the texts have been saved. Otherwise it does nothing. But, the function `close_request_cb` shows an alert dialog and if the user clicks on the accept button, the window will be destroyed.

2.20.3.4 Window “close-request” Signal

`GtkWindow` has a “close-request” signal and it is emitted when the close button, which is the x-shaped button at the top right corner, is clicked on. And the user handler is called before the default handler. If the user handler returns `TRUE`, the rest of the close process is skipped. If it returns `FALSE`, the rest will go on and the window will be destroyed.

```

1  static gboolean
2  close_request_cb (TfeWindow *win) {
3      TfeAlert *alert;
4
5      if (is_saved_all (win->nb))
6          return FALSE;
7      else {
8          win->is_quit = TRUE;
9          alert = TFE_ALERT (tfe_alert_new_with_data ("Are you sure?", "Contents aren't saved yet.\nAre you
10             gtk_window_set_transient_for (GTK_WINDOW (alert), GTK_WINDOW (win));
11             g_signal_connect (alert, "response", G_CALLBACK (alert_response_cb), win);
12             gtk_window_present (GTK_WINDOW (alert));
13             return TRUE;
14         }
15     }

```

First, it calls `is_saved_all` and checks if the texts have been saved. If so, it returns `FALSE` and the close process continues. Otherwise, it sets `win->is_quit` to `TRUE` and shows an alert dialog. When the user clicks on the accept or cancel button, the dialog disappears and “response” signal is emitted. Then, the handler `alert_response_cb` is called. It destroys the top window if the user clicked on the accept button since `win->is_quit` is `TRUE`. Otherwise it does nothing.

```

1  static gboolean
2  is_saved_all (GtkNotebook *nb) {
3      int i, n;
4      GtkWidget *scr;
5      GtkWidget *tv;
6      GtkTextBuffer *tb;
7
8      n = gtk_notebook_get_n_pages (nb);
9      for (i = 0; i < n; ++i) {
10         scr = gtk_notebook_get_nth_page (nb, i);
11         tv = gtk_scrolled_window_get_child (GTK_SCROLLED_WINDOW (scr));
12         tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
13         if (gtk_text_buffer_get_modified (tb))
14             return FALSE;
15     }
16     return TRUE;
17 }

```

2.20.3.5 Public Functions

There are three public functions.

- void tfe_window_notebook_page_new (TfeWindow *win)
- void tfe_window_notebook_page_new_with_files (TfeWindow *win, GFile **files, int n_files)
- GtkWidget *tfe_window_new (GtkApplication *app)

The first function is called when the application emits the “activate” signal. The second is for “open” signal. It is given three arguments and they are owned by the caller.

```

1 void
2 tfe_window_notebook_page_new_with_files (TfeWindow *win, GFile **files, int n_files) {
3     int i;
4     GtkWidget *tv;
5     char *filename;
6
7     for (i = 0; i < n_files; i++)
8         if ((tv = tfe_text_view_new_with_file (*(files+i))) != NULL) {
9             filename = g_file_get_basename (*(files+i));
10            notebook_page_build (win, tv, filename);
11            g_free (filename);
12        }
13    if (gtk_notebook_get_n_pages (win->nb) == 0)
14        tfe_window_notebook_page_new (win);
15 }
```

This function has a loop for the array `files`. It creates `TfeTextView` instance with the text from each file. And build a page with it.

If an error happens and no page is created, it creates a new empty page.

2.20.3.6 Full Codes of tfewindow.c

The following is the full source codes of `tfewindow.c`.

```

1 #include <gtk/gtk.h>
2 #include "tfewindow.h"
3 #include "tfepref.h"
4 #include "tfealert.h"
5 #include "../tfetextview/tfetextview.h"
6
7 struct _TfeWindow {
8     GtkApplicationWindow parent;
9     GtkMenuButton *btm;
10    GtkNotebook *nb;
11    gboolean is_quit;
12 };
13
14 G_DEFINE_FINAL_TYPE (TfeWindow, tfe_window, GTK_TYPE_APPLICATION_WINDOW);
15
16 /* Low level functions */
17
18 /* Create a new untitled string */
19 /* The returned string should be freed with g_free() when no longer needed. */
20 static char *
```

```

21 get_untitled () {
22     static int c = -1;
23     if (++c == 0)
24         return g_strdup_printf("Untitled");
25     else
26         return g_strdup_printf ("Untitled%u", c);
27 }
28
29 /* The returned object is owned by the scrolled window. */
30 /* The caller won't get the ownership and mustn't release it. */
31 static TfeTextView *
32 get_current_textview (GtkNotebook *nb) {
33     int i;
34     GtkWidget *scr;
35     GtkWidget *tv;
36
37     i = gtk_notebook_get_current_page (nb);
38     scr = gtk_notebook_get_nth_page (nb, i);
39     tv = gtk_scrolled_window_get_child (GTK_SCROLLED_WINDOW (scr));
40     return TFE_TEXT_VIEW (tv);
41 }
42
43 /* This call back is called when a TfeTextView instance emits a "change-file" signal. */
44 static void
45 file_changed_cb (TfeTextView *tv, gpointer user_data) {
46     GtkNotebook *nb = GTK_NOTEBOOK (user_data);
47     GtkWidget *scr;
48     GtkWidget *label;
49     GFile *file;
50     char *filename;
51
52     file = tfe_text_view_get_file (tv);
53     scr = gtk_widget_get_parent (GTK_WIDGET (tv));
54     if (G_IS_FILE (file)) {
55         filename = g_file_get_basename (file);
56         g_object_unref (file);
57     } else
58         filename = get_untitled ();
59     label = gtk_label_new (filename);
60     g_free (filename);
61     gtk_notebook_set_tab_label (GTK_NOTEBOOK (nb), scr, label);
62 }
63
64 static void
65 modified_changed_cb (GtkTextBuffer *tb, gpointer user_data) {
66     TfeTextView *tv = TFE_TEXT_VIEW (user_data);
67     GtkWidget *scr = gtk_widget_get_parent (GTK_WIDGET (tv));
68     GtkWidget *nb = gtk_widget_get_ancestor (GTK_WIDGET (tv), GTK_TYPE_NOTEBOOK);
69     GtkWidget *label;
70     GFile *file;
71     char *filename;
72     char *text;
73

```

```

74     file = tfe_text_view_get_file (tv);
75     filename = g_file_get_basename (file);
76     if (gtk_text_buffer_get_modified (tb))
77         text = g_strdup_printf ("%s", filename);
78     else
79         text = g_strdup (filename);
80     g_object_unref (file);
81     g_free (filename);
82     label = gtk_label_new (text);
83     g_free (text);
84     gtk_notebook_set_tab_label (GTK_NOTEBOOK (nb), scr, label);
85 }
86
87 static gboolean
88 is_saved_all (GtkNotebook *nb) {
89     int i, n;
90     GtkWidget *scr;
91     GtkWidget *tv;
92     GtkTextBuffer *tb;
93
94     n = gtk_notebook_get_n_pages (nb);
95     for (i = 0; i < n; ++i) {
96         scr = gtk_notebook_get_nth_page (nb, i);
97         tv = gtk_scrolled_window_get_child (GTK_SCROLLED_WINDOW (scr));
98         tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
99         if (gtk_text_buffer_get_modified (tb))
100             return FALSE;
101     }
102     return TRUE;
103 }
104
105 static void
106 notebook_page_close (TfeWindow *win){
107     int i;
108
109     if (gtk_notebook_get_n_pages (win->nb) == 1)
110         gtk_window_destroy (GTK_WINDOW (win));
111     else {
112         i = gtk_notebook_get_current_page (win->nb);
113         gtk_notebook_remove_page (win->nb, i);
114     }
115 }
116
117 static void
118 notebook_page_build (TfeWindow *win, GtkWidget *tv, char *filename) {
119     // The arguments win, tv and filename are owned by the caller.
120     // If tv has a floating reference, it is consumed by the function.
121     GtkWidget *scr = gtk_scrolled_window_new ();
122     GtkTextBuffer *tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
123     GtkNotebookPage *nbp;
124     GtkWidget *lab;
125     int i;
126

```

```

127  gtk_text_view_set_wrap_mode (GTK_TEXT_VIEW (tv), GTK_WRAP_WORD_CHAR);
128  gtk_scrolled_window_set_child (GTK_SCROLLED_WINDOW (scr), tv);
129  lab = gtk_label_new (filename);
130  i = gtk_notebook_append_page (win->nb, scr, lab);
131  nbp = gtk_notebook_get_page (win->nb, scr);
132  g_object_set (nbp, "tab-expand", TRUE, NULL);
133  gtk_notebook_set_current_page (win->nb, i);
134  g_signal_connect (tv, "change-file", G_CALLBACK (file_changed_cb), win->nb);
135  g_signal_connect (tb, "modified-changed", G_CALLBACK (modified_changed_cb), tv);
136 }
137
138 static void
139 open_response_cb (TfeTextView *tv, int response, gpointer user_data) {
140     TfeWindow *win = TFE_WINDOW (user_data);
141     GFile *file;
142     char *filename;
143
144     if (response != TFE_OPEN_RESPONSE_SUCCESS) {
145         g_object_ref_sink (tv);
146         g_object_unref (tv);
147     } else if (! G_IS_FILE (file = tfe_text_view_get_file (tv))) {
148         g_object_ref_sink (tv);
149         g_object_unref (tv);
150     } else {
151         filename = g_file_get_basename (file);
152         g_object_unref (file);
153         notebook_page_build (win, GTK_WIDGET (tv), filename);
154         g_free (filename);
155     }
156 }
157
158 /* alert response signal handler */
159 static void
160 alert_response_cb (TfeAlert *alert, int response_id, gpointer user_data) {
161     TfeWindow *win = TFE_WINDOW (user_data);
162
163     if (response_id == TFE_ALERT_RESPONSE_ACCEPT) {
164         if (win->is_quit)
165             gtk_window_destroy(GTK_WINDOW (win));
166         else
167             notebook_page_close (win);
168     }
169 }
170
171 /* ----- Close request on the top window ----- */
172 /* ----- The signal is emitted when the close button is clicked. ----- */
173 static gboolean
174 close_request_cb (TfeWindow *win) {
175     TfeAlert *alert;
176
177     if (is_saved_all (win->nb))
178         return FALSE;
179     else {

```



```

180     win->is_quit = TRUE;
181     alert = TFE_ALERT (tfe_alert_new_with_data ("Are you sure?", "Contents aren't saved yet.\nAre you
182     gtk_window_set_transient_for (GTK_WINDOW (alert), GTK_WINDOW (win));
183     g_signal_connect (alert, "response", G_CALLBACK (alert_response_cb), win);
184     gtk_window_present (GTK_WINDOW (alert));
185     return TRUE;
186 }
187 }
188
189 /* ----- action activated handlers ----- */
190 static void
191 open_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
192     TfeWindow *win = TFE_WINDOW (user_data);
193     GtkWidget *tv = tfe_text_view_new ();
194
195     g_signal_connect (tv, "open-response", G_CALLBACK (open_response_cb), win);
196     tfe_text_view_open (TFE_TEXT_VIEW (tv), GTK_WINDOW (win));
197 }
198
199 static void
200 save_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
201     TfeWindow *win = TFE_WINDOW (user_data);
202     TfeTextView *tv = get_current_textview (win->nb);
203
204     tfe_text_view_save (TFE_TEXT_VIEW (tv));
205 }
206
207 static void
208 close_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
209     TfeWindow *win = TFE_WINDOW (user_data);
210     TfeTextView *tv;
211     GtkTextBuffer *tb;
212     GtkWidget *alert;
213
214     tv = get_current_textview (win->nb);
215     tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
216     if (! gtk_text_buffer_get_modified (tb)) /* is saved? */
217         notebook_page_close (win);
218     else {
219         win->is_quit = FALSE;
220         alert = tfe_alert_new_with_data ("Are you sure?", "Contents aren't saved yet.\nAre you sure to c
221         gtk_window_set_transient_for (GTK_WINDOW (alert), GTK_WINDOW (win));
222         g_signal_connect (alert, "response", G_CALLBACK (alert_response_cb), win);
223         gtk_window_present (GTK_WINDOW (alert));
224     }
225 }
226
227 static void
228 new_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
229     TfeWindow *win = TFE_WINDOW (user_data);
230
231     tfe_window_notebook_page_new (win);
232 }

```

```

233
234 static void
235 saveas_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
236     TfeWindow *win = TFE_WINDOW (user_data);
237     TfeTextView *tv = get_current_textview (win->nb);
238
239     tfe_text_view_saveas (TFE_TEXT_VIEW (tv));
240 }
241
242 static void
243 pref_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
244     TfeWindow *win = TFE_WINDOW (user_data);
245     GtkWidget *pref;
246
247     pref = tfe_pref_new ();
248     gtk_window_set_transient_for (GTK_WINDOW (pref), GTK_WINDOW (win));
249     gtk_window_present (GTK_WINDOW (pref));
250 }
251
252 static void
253 close_all_activated (GSimpleAction *action, GVariant *parameter, gpointer user_data) {
254     TfeWindow *win = TFE_WINDOW (user_data);
255
256     if (close_request_cb (win) == FALSE)
257         gtk_window_destroy (GTK_WINDOW (win));
258 }
259
260 /* --- public functions --- */
261
262 void
263 tfe_window_notebook_page_new (TfeWindow *win) {
264     GtkWidget *tv;
265     char *filename;
266
267     tv = tfe_text_view_new ();
268     filename = get_untitled ();
269     notebook_page_build (win, tv, filename);
270     g_free (filename);
271 }
272
273 void
274 tfe_window_notebook_page_new_with_files (TfeWindow *win, GFile **files, int n_files) {
275     int i;
276     GtkWidget *tv;
277     char *filename;
278
279     for (i = 0; i < n_files; i++)
280         if ((tv = tfe_text_view_new_with_file (*(files+i))) != NULL) {
281             filename = g_file_get_basename (*(files+i));
282             notebook_page_build (win, tv, filename);
283             g_free (filename);
284         }
285     if (gtk_notebook_get_n_pages (win->nb) == 0)

```

```

286     tfe_window_notebook_page_new (win);
287 }
288
289 static void
290 tfe_window_dispose (GObject *gobject) {
291     gtk_widget_dispose_template (GTK_WIDGET (gobject), TFE_TYPE_WINDOW);
292     G_OBJECT_CLASS (tfe_window_parent_class)->dispose (gobject);
293 }
294
295 static void
296 tfe_window_init (TfeWindow *win) {
297     GtkBuilder *build;
298     GMenuModel *menu;
299
300     gtk_widget_init_template (GTK_WIDGET (win));
301
302     build = gtk_builder_new_from_resource ("/com/github/ToshioCP/tfe/menu.ui");
303     menu = G_MENU_MODEL (gtk_builder_get_object (build, "menu"));
304     gtk_menu_button_set_menu_model (win->btnm, menu);
305     g_object_unref(build);
306
307     /* ----- action ----- */
308     const GActionEntry win_entries[] = {
309         { "open", open_activated, NULL, NULL, NULL },
310         { "save", save_activated, NULL, NULL, NULL },
311         { "close", close_activated, NULL, NULL, NULL },
312         { "new", new_activated, NULL, NULL, NULL },
313         { "saveas", saveas_activated, NULL, NULL, NULL },
314         { "pref", pref_activated, NULL, NULL, NULL },
315         { "close-all", close_all_activated, NULL, NULL, NULL }
316     };
317     g_action_map_add_action_entries (G_ACTION_MAP (win), win_entries, G_N_ELEMENTS (win_entries), win);
318
319     g_signal_connect (win, "close-request", G_CALLBACK (close_request_cb), NULL);
320 }
321
322 static void
323 tfe_window_class_init (TfeWindowClass *class) {
324     GObjectClass *object_class = G_OBJECT_CLASS (class);
325
326     object_class->dispose = tfe_window_dispose;
327     gtk_widget_class_set_template_from_resource (GTK_WIDGET_CLASS (class), "/com/github/ToshioCP/tfe/tfe.ui");
328     gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfeWindow, btnm);
329     gtk_widget_class_bind_template_child (GTK_WIDGET_CLASS (class), TfeWindow, nb);
330 }
331
332 GtkWidget *
333 tfe_window_new (GtkApplication *app) {
334     return GTK_WIDGET (g_object_new (TFE_TYPE_WINDOW, "application", app, NULL));
335 }

```

2.21 Pango, CSS and Application

2.21.1 PangoFontDescription

PangoFontDescription is a C structure for a font. You can get font family, style, weight and size. You can also get a string that includes font attributes. For example, suppose that the PangoFontDescription has a font of “Noto Sans Mono”, “Bold”, “Italic” and 12 points of size. Then the string converted from the PangoFontDescription is “Noto Sans Mono Bold Italic 12”.

- Font family is “Noto Sans Mono”.
- Font style is “Italic”.
- Font weight is “Bold”, or 700.
- Font size is 12 pt.

The font in CSS is different from the string from PangoFontDescription.

- font: bold italic 12pt "Noto Sans Mono"
- Noto Sans Mono Bold Italic 12

So, it may be easier to use each property, i.e. font-family, font-style, font-weight and font-size, to convert a PangoFontDescription data to CSS.

Refer to [Pango document](#) and [W3C CSS Fonts Module Level 3](#) for further information.

2.21.2 Converter from PangoFontDescription to CSS

Two files pfd2css.h and pfd2css.c include the converter from PangoFontDescription to CSS.

```

1  #pragma once
2
3  #include <pango/pango.h>
4
5  // Pango font description to CSS style string
6  // The returned string is owned by the caller. The caller should free it when it is no longer needed
7
8  char*
9  pfd2css (PangoFontDescription *pango_font_desc);
10
11 // Each element (family, style, weight and size)
12
13 const char*
14 pfd2css_family (PangoFontDescription *pango_font_desc);
15
16 const char*
17 pfd2css_style (PangoFontDescription *pango_font_desc);
18
19 int
20 pfd2css_weight (PangoFontDescription *pango_font_desc);
21
22 // Returned string is owned by the caller. The caller should free it when it is no longer needed.
23 char *
24 pfd2css_size (PangoFontDescription *pango_font_desc);

```

The five functions are public. The first function is a convenient function to set other four CSS properties at once.

```

1  #include <pango/pango.h>
2  #include "pfd2css.h"

```

```

3
4 // Pango font description to CSS style string
5 // The returned string is owned by caller. The caller should free it when it is no longer needed.
6
7 char *
8 pfd2css (PangoFontDescription *pango_font_desc) {
9     char *fontsize;
10    char *result;
11
12    fontsize = pfd2css_size (pango_font_desc);
13    result = g_strdup_printf ("font-family: \"%s\"; font-style: %s; font-weight: %d; font-size: %s;",
14                             pfd2css_family (pango_font_desc), pfd2css_style (pango_font_desc),
15                             pfd2css_weight (pango_font_desc), fontsize);
16    g_free (fontsize);
17    return result;
18 }
19
20 // Each element (family, style, weight and size)
21
22 const char *
23 pfd2css_family (PangoFontDescription *pango_font_desc) {
24     return pango_font_description_get_family (pango_font_desc);
25 }
26
27 const char *
28 pfd2css_style (PangoFontDescription *pango_font_desc) {
29     PangoStyle pango_style = pango_font_description_get_style (pango_font_desc);
30     switch (pango_style) {
31     case PANGO_STYLE_NORMAL:
32         return "normal";
33     case PANGO_STYLE_ITALIC:
34         return "italic";
35     case PANGO_STYLE_OBLIQUE:
36         return "oblique";
37     default:
38         return "normal";
39     }
40 }
41
42 int
43 pfd2css_weight (PangoFontDescription *pango_font_desc) {
44     PangoWeight pango_weight = pango_font_description_get_weight (pango_font_desc);
45     switch (pango_weight) {
46     case PANGO_WEIGHT_THIN:
47         return 100;
48     case PANGO_WEIGHT_ULTRALIGHT:
49         return 200;
50     case PANGO_WEIGHT_LIGHT:
51         return 300;
52     case PANGO_WEIGHT_SEMILIGHT:
53         return 350;
54     case PANGO_WEIGHT_BOOK:
55         return 380;

```

```

56  case PANGO_WEIGHT_NORMAL:
57      return 400; /* or "normal" */
58  case PANGO_WEIGHT_MEDIUM:
59      return 500;
60  case PANGO_WEIGHT_SEMIBOLD:
61      return 600;
62  case PANGO_WEIGHT_BOLD:
63      return 700; /* or "bold" */
64  case PANGO_WEIGHT_ULTRABOLD:
65      return 800;
66  case PANGO_WEIGHT_HEAVY:
67      return 900;
68  case PANGO_WEIGHT_ULTRAHEAVY:
69      return 900; /* 1000 is available since CSS Fonts level 4 but GTK currently supports level 3. */
70  default:
71      return 400; /* "normal" */
72  }
73 }
74
75 char *
76 pfd2css_size (PangoFontDescription *pango_font_desc) {
77     if (pango_font_description_get_size_is_absolute (pango_font_desc))
78         return g_strdup_printf ("%dpx", pango_font_description_get_size (pango_font_desc) / PANGO_SCALE)
79     else
80         return g_strdup_printf ("%dpt", pango_font_description_get_size (pango_font_desc) / PANGO_SCALE)
81 }

```

- The function `pfd2css_family` returns font family.
- The function `pfd2css_style` returns font style which is one of “normal”, “italic” or “oblique”.
- The function `pfd2css_weight` returns font weight in integer. See the list below.
- The function `pfd2css_size` returns font size.
 - If the font description size is absolute, it returns the size in device units, which is pixel. Otherwise the unit is point.
 - The function `pango_font_description_get_size` returns the integer of the size but it is multiplied by `PANGO_SCALE`. So, you need to divide it by `PANGO_SCALE`. The `PANGO_SCALE` is currently 1024, but this might be changed in the future. If the font size is 12pt, the size in pango is $12 * PANGO_SCALE = 12 * 1024 = 12288$.
- The function `pfd2css` returns a string of the font. For example, if a font “Noto Sans Mono Bold Italic 12” is given, it returns “font-family: Noto Sans Mono; font-style: italic; font-weight: 700; font-size: 12pt;”.

The font weight number is one of:

- 100 - Thin
- 200 - Extra Light (Ultra Light)
- 300 - Light
- 400 - Normal
- 500 - Medium
- 600 - Semi Bold (Demi Bold)
- 700 - Bold
- 800 - Extra Bold (Ultra Bold)
- 900 - Black (Heavy)

2.21.3 Application Object

2.21.3.1 TfeApplication Class

TfeApplication class is a child of GtkApplication. It has some instance variables. The header file defines the type macro and a public function.

```

1  #pragma once
2
3  #include <gtk/gtk.h>
4
5  #define TFE_TYPE_APPLICATION tfe_application_get_type ()
6  G_DECLARE_FINAL_TYPE (TfeApplication, tfe_application, TFE, APPLICATION, GtkApplication)
7
8  TfeApplication *
9  tfe_application_new (const char *application_id, GApplicationFlags flag);

```

The following code is extracted from tfeapplication.c. It builds TfeApplication class and instance.

```

#include <gtk/gtk.h>
#include "tfeapplication.h"

struct _TfeApplication {
    GtkApplication parent;
    TfeWindow *win;
    GSettings *settings;
    GtkCssProvider *provider;
};

G_DEFINE_FINAL_TYPE (TfeApplication, tfe_application, GTK_TYPE_APPLICATION)

static void
tfe_application_dispose (GObject *gobject) {
    TfeApplication *app = TFE_APPLICATION (gobject);

    g_clear_object (&app->settings);
    g_clear_object (&app->provider);
    G_OBJECT_CLASS (tfe_application_parent_class)->dispose (gobject);
}

static void
tfe_application_init (TfeApplication *app) {
    app->settings = g_settings_new ("com.github.ToshioCP.tfe");
    g_signal_connect (app->settings, "changed::font-desc", G_CALLBACK (changed_font_cb), app);
    app->provider = gtk_css_provider_new ();
}

static void
tfe_application_class_init (TfeApplicationClass *class) {
    G_OBJECT_CLASS (class)->dispose = tfe_application_dispose;
    G_APPLICATION_CLASS (class)->startup = app_startup;
    G_APPLICATION_CLASS (class)->activate = app_activate;
    G_APPLICATION_CLASS (class)->open = app_open;
}

TfeApplication *

```

```
tfe_application_new (const char *application_id, GApplicationFlags flag) {
    return TFE_APPLICATION (g_object_new (TFE_TYPE_APPLICATION, "application-id", application_id, "flags", flag, NULL));
}
```

- The structure `_TfeApplication` is defined. It has four members. One is its parents and the others are kinds of instance variables. The members are usually initialized in the instance initialization function. And they are released in the disposal function or freed in the finalization function. The members are:
 - `win`: main window instance
 - `settings`: `GSettings` instance. it is bound to “font-desc” item in the `GSettings`
 - `provider`: a provider for the font of the textview.
- The macro `G_DEFINE_FINAL_TYPE` defines `tfe_application_get_type` function and some other useful things.
- The function `tfe_application_class_init` initializes the `TfeApplication` class. It overrides four class methods. Three class methods `startup`, `activate` and `open` point to the default signal handlers. The overriding replaces the default handlers. You can connect the handlers with `g_signal_connect` macro but the result is different. The macro connects a user handler to the signal. The default handler still exists and no change is made to them.
- The function `tfe_application_init` initializes an instance.
 - Creates a new `GSettings` instance and make `app->settings` point to it. Then connects the handler `changed_font_cb` to the “changed::font-desc” signal.
 - Creates a new `GtkCssProvider` instance and make `app->provider` point to it.
- The function `tfe_application_dispose` releases the `GSettings` and `GtkCssProvider` instances. Then, it calls the parent’s dispose handler. It is called “chaining up”. See [GObject document](#).

2.21.3.2 Startup Signal Handlers

```
1 static void
2 app_startup (GApplication *application) {
3     TfeApplication *app = TFE_APPLICATION (application);
4     int i;
5     GtkCssProvider *provider = gtk_css_provider_new ();
6     GdkDisplay *display;
7
8     G_APPLICATION_CLASS (tfe_application_parent_class)->startup (application);
9
10    app->win = TFE_WINDOW (tfe_window_new (GTK_APPLICATION (app)));
11
12    gtk_css_provider_load_from_string (provider, "textview {padding: 10px;}");
13    display = gdk_display_get_default ();
14    gtk_style_context_add_provider_for_display (display, GTK_STYLE_PROVIDER (provider),
15                                              GTK_STYLE_PROVIDER_PRIORITY_APPLICATION);
16    g_object_unref (provider);
17    gtk_style_context_add_provider_for_display (display, GTK_STYLE_PROVIDER (app->provider),
18                                              GTK_STYLE_PROVIDER_PRIORITY_APPLICATION);
19
20    changed_font_cb (app->settings, "font-desc", app); // Sets the text view font to the font from the
21
22    /* ----- accelerator ----- */
23    struct {
24        const char *action;
25        const char *accels[2];
26    } action_accels[] = {
27        { "win.open", { "<Control>o", NULL } },
```



```

28     { "win.save", { "<Control>s", NULL } },
29     { "win.close", { "<Control>w", NULL } },
30     { "win.new", { "<Control>n", NULL } },
31     { "win.saveas", { "<Shift><Control>s", NULL } },
32     { "win.close-all", { "<Control>q", NULL } },
33 };
34
35 for (i = 0; i < G_N_ELEMENTS(action_accels); i++)
36     gtk_application_set_accels_for_action(GTK_APPLICATION(app), action_accels[i].action, action_accels[i].accels);
37 }

```

The function `app_startup` replace the default signal handlers. It does five things.

- Calls the parent's startup handler. It is called "chaining up". The "startup" default handler runs before user handlers. So the call for the parent's handler must be done at the beginning.
- Creates the main window. This application has only one top level window. In that case, it is a good way to create the window in the startup handler, which is called only once. Activate or open handlers can be called twice or more. Therefore, if you create a window in the activate or open handler, two or more windows can be created.
- Sets the default display CSS to "textview {padding: 10px;}". It sets the `GtkTextView`, or `TfeTextView`, padding to 10px and makes the text easier to read. This CSS is fixed and never changed through the application life.
- Adds another CSS provider, which is pointed to by `app->provider`, to the default display. This CSS depends on the `GSettings` "font-desc" value and it can be changed during the application life time. And calls `changed_font_cb` to update the font CSS setting.
- Sets application accelerator with the function `gtk_application_set_accels_for_action`. Accelerators are kinds of short cut key functions. For example, `Ctrl+O` is an accelerator to activate "open" action. Accelerators are written in the array `action_accels[]`. Its element is a structure `struct {const char *action; const char *accels[2];}`. The member `action` is an action name. The member `accels` is an array of two pointers. For example, `{"win.open", { "<Control>o", NULL }}` tells that the accelerator `Ctrl+O` is connected to the "win.open" action. The second element of `accels` is `NULL` which is the end mark. You can define more than one accelerator keys and the list must end with `NULL` (zero). If you want to do so, the array length needs to be three or more. For example, `{"win.open", { "<Control>o", "<Alt>o", NULL }}` means two accelerators `Ctrl+O` and `Alt+O` is connected to the "win.open" action. The parser recognizes "<control>o", "<Shift><Alt>F2", "<Ctrl>minus" and so on. If you want to use symbol key like "<Ctrl>-", use "<Ctrl>minus" instead. Such relation between lower case and symbol (character code) is specified in [gdkkeysyms.h](#) in the GTK 4 source code.

2.21.3.3 Activate and Open Signal Handlers

Two functions `app_activate` and `app_open` replace the default signal handlers.

```

1  static void
2  app_activate (GApplication *application) {
3      TfeApplication *app = TFE_APPLICATION (application);
4
5      tfe_window_notebook_page_new (app->win);
6      gtk_window_present (GTK_WINDOW (app->win));
7  }
8
9  static void
10 app_open (GApplication *application, GFile ** files, gint n_files, const gchar *hint) {
11     TfeApplication *app = TFE_APPLICATION (application);
12

```

```

13     tfe_window_notebook_page_new_with_files (app->win, files, n_files);
14     gtk_window_present (GTK_WINDOW (app->win));
15 }

```

The original default handlers don't do useful works and you don't need to chain up to the parent's default handlers. They just create notebook pages and show the top level window.

2.21.3.4 CSS Font Setting

```

1  static void
2  changed_font_cb (GSettings *settings, char *key, gpointer user_data) {
3      TfeApplication *app = TFE_APPLICATION (user_data);
4      char *font, *s, *css;
5      PangoFontDescription *pango_font_desc;
6
7      if (g_strcmp0(key, "font-desc") != 0)
8          return;
9      font = g_settings_get_string (app->settings, "font-desc");
10     pango_font_desc = pango_font_description_from_string (font);
11     g_free (font);
12     s = pfd2css (pango_font_desc); // converts Pango Font Description into CSS style string
13     pango_font_description_free (pango_font_desc);
14     css = g_strdup_printf ("textview {%s}", s);
15     gtk_css_provider_load_from_string (app->provider, css);
16     g_free (s);
17     g_free (css);
18 }

```

The function `changed_font_cb` is a handler for “changed::font-desc” signal on the `GSettings` instance. The signal name is “changed” and “font-desc” is a key name. This signal is emitted when the value of the “font-desc” key is changed. The value is bound to the “font-desc” property of the `GtkFontDialogButton` instance. Therefore, the handler `changed_font_cb` is called when the user selects and updates a font through the font dialog.

A string is retrieved from the `GSetting` database and converts it into a pango font description. And a CSS string is made by the function `pfd2css` and `g_strdup_printf`. Then the css provider is set to the string. The provider has been inserted to the current display in advance. So, the font is applied to the display.

2.21.4 Other Files

main.c

```

1  #include <gtk/gtk.h>
2  #include "tfeapplication.h"
3
4  #define APPLICATION_ID "com.github.ToshioCP.tfe"
5
6  int
7  main (int argc, char **argv) {
8      TfeApplication *app;
9      int stat;
10
11     app = tfe_application_new (APPLICATION_ID, G_APPLICATION_HANDLES_OPEN);
12     stat = g_application_run (G_APPLICATION (app), argc, argv);
13     g_object_unref (app);

```

```

14     return stat;
15 }

```

Resource XML file.

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <gresources>
3   <gresource prefix="/com/github/ToshioCP/tfe">
4     <file>tfewindow.ui</file>
5     <file>tfepref.ui</file>
6     <file>tfealert.ui</file>
7     <file>menu.ui</file>
8   </gresource>
9 </gresources>

```

GSchema XML file

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <schemalist>
3   <schema path="/com/github/ToshioCP/tfe/" id="com.github.ToshioCP.tfe">
4     <key name="font-desc" type="s">
5       <default>'Monospace 12'</default>
6       <summary>Font</summary>
7       <description>A font to be used for textview.</description>
8     </key>
9   </schema>
10 </schemalist>

```

Meson.build

```

1 project('tfe', 'c', license : 'GPL-3.0-or-later', meson_version: '>=1.0.1', version: '0.5')
2
3 gtkdep = dependency('gtk4')
4
5 gnome = import('gnome')
6 resources = gnome.compile_resources('resources', 'tfe.gresource.xml')
7 gnome.compile_schemas(depend_files: 'com.github.ToshioCP.tfe.gschema.xml')
8
9 sourcefiles = files('main.c', 'tfeapplication.c', 'tfewindow.c', 'tfepref.c', 'tfealert.c', 'pfd2css')
10
11 executable(meson.project_name(), sourcefiles, resources, dependencies: gtkdep, export_dynamic: true,
12
13 schema_dir = get_option('prefix') / get_option('datadir') / 'glib-2.0/schemas/'
14 install_data('com.github.ToshioCP.tfe.gschema.xml', install_dir: schema_dir)
15 gnome.post_install (glib_compile_schemas: true)

```

- The function `project` defines project and initialize meson. The first argument is the project name and the second is the language name. The other arguments are keyword arguments.
- The function `dependency` defines the dependent library. Tfe depends on GTK4. This is used to create `pkg-config` option in the command line of C compiler to include header files and link libraries. The returned object `gtkdep` is used as an argument to the `executable` function later.
- The function `import` imports an extension module. The GNOME module has some convenient methods like `gnome.compile_resources` and `gnome.compile_schemas`.
- The method `gnome.compile_resources` compiles and creates resource files. The first argument is the resource name without extension and the second is the name of XML file. The returned value is an array `['resources.c', 'resources.h']`.

- The function `gnome.compile_schemas` compiles the schema files in the current directory. This just creates `gschemas.compiled` in the build directory. It is used to test the executable binary in the build directory. The function doesn't install the schema file.
- The function `files` creates a File Object.
- The function `executable` defines the compilation elements such as target name, source files, dependencies and installation. The target name is "tfe". The source files are elements of 'sourcefiles' and 'resources'. It uses GTK4 libraries. It can be installed.
- The last three lines are post install work. The variable `schema_dir` is the directory stored the schema file. If meson runs with `--prefix=$HOME/.local` argument, it is `$HOME/.local/share/glib-2.9/schemas`. The function `install_data` copies the file specified in the first argument into the directory specified in the second argument. The method `gnome.post_install` runs `glib-compile-schemas` and updates `gschemas.compiled` file.

2.21.5 Compilation and Installation.

If you want to install it to your local area, use `--prefix=$HOME/.local` or `--prefix=$HOME` option. If you want to install it to the system area, no option is needed. It will be installed under `/usr/local` directory.

```
$ meson setup --prefix=$HOME/.local _build
$ ninja -C _build
$ ninja -C _build install
```

You need root privilege to install it to the system area..

```
$ meson setup _build
$ ninja -C _build
$ sudo ninja -C _build install
```

Source files are in `/src/tfe6` directory.

We made a very small text editor. You can add features to this editor. When you add a new feature, be careful about the structure of the program. Maybe you need to divide a file into several files. It isn't good to put many things into one file. And it is important to think about the relationship between source files and widget structures.

The source files are in the [Gtk4 tutorial GitHub repository](#). Download it and see `/src/tfe6` directory.

Chapter 3

GtkDrawingArea

3.1 GtkDrawingArea and Cairo

If you want to draw shapes or paint images dynamically on the screen, use the GtkDrawingArea widget.

GtkDrawingArea provides a Cairo drawing context. You can draw images with Cairo library functions. This section describes:

1. A brief introduction to Cairo
2. GtkDrawingArea, with a very simple example.

3.1.1 Cairo

Cairo is a drawing library for two dimensional graphics. There are a lot of documents on [Cairo's website](#). If you aren't familiar with Cairo, it is worth reading the [tutorial](#).

The following is an introduction to the Cairo library. First, you need to know surfaces, sources, masks, destinations, cairo context and transformations.

- A surface represents an image. It is like a canvas. We can draw shapes and images with different colors on surfaces.
- The source pattern, or simply source, is like paint, which will be transferred to destination surface by cairo functions.
- The mask describes the area to be used in the copy;
- The destination is a target surface;
- The cairo context manages the transfer from source to destination, through mask with its functions; For example, `cairo_stroke` is a function to draw a path to the destination by the transfer.
- A transformation can be applied before the transfer completes. The type of the transformation is called affine, which is a mathematical term meaning transformations that preserve straight lines. Scaling, rotating, reflecting, shearing and translating are examples of affine transformations. They are mathematically represented by matrix multiplication and vector addition. In this section we don't use it, instead we will only use the identity transformation. This means that the coordinates in the source and mask map directly to the same coordinates in the destination.

The instruction is as follows:

1. Create a surface. This will be the destination.
2. Create a cairo context with the surface. This surface will be the destination of the context.
3. Create a source pattern within the context.
4. Create paths, which are lines, rectangles, arcs, texts or more complicated shapes in the mask.
5. Use a drawing operator such as `cairo_stroke` to transfer the paint in the source to the destination.
6. Save the destination surface to a file if necessary.

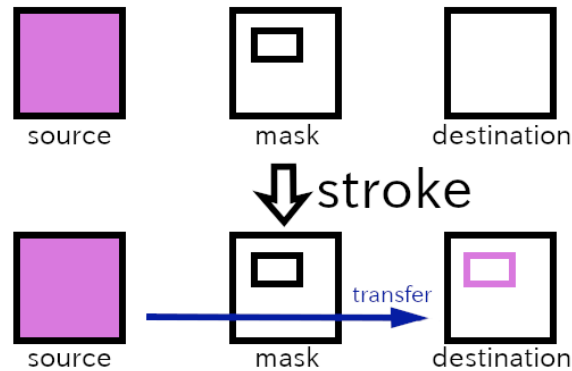


Figure 3.1: Stroke a rectangle

Here's a simple example program that draws a small square and saves it as a png file. The path of the file is `/src/misc/cairo.c`.

```

1  #include <cairo.h>
2
3  int
4  main (int argc, char **argv)
5  {
6      cairo_surface_t *surface;
7      cairo_t *cr;
8      int width = 100;
9      int height = 100;
10     int square_size = 40.0;
11
12     /* Create surface and cairo */
13     surface = cairo_image_surface_create (CAIRO_FORMAT_RGB24, width, height);
14     cr = cairo_create (surface);
15
16     /* Drawing starts here. */
17     /* Paint the background white */
18     cairo_set_source_rgb (cr, 1.0, 1.0, 1.0);
19     cairo_paint (cr);
20     /* Draw a black rectangle */
21     cairo_set_source_rgb (cr, 0.0, 0.0, 0.0);
22     cairo_set_line_width (cr, 2.0);
23     cairo_rectangle (cr,
24                     width/2.0 - square_size/2,
25                     height/2.0 - square_size/2,
26                     square_size,
27                     square_size);
28     cairo_stroke (cr);
29
30     /* Write the surface to a png file and clean up cairo and surface. */
31     cairo_surface_write_to_png (surface, "rectangle.png");
32     cairo_destroy (cr);
33     cairo_surface_destroy (surface);
34 
```

```

35     return 0;
36 }

```

- 1: Includes the header file of Cairo.
- 6: `cairo_surface_t` is the type of a surface.
- 7: `cairo_t` is the type of a cairo context.
- 8-10: `width` and `height` are the size of `surface`. `square_size` is the size of a square to be drawn on the surface.
- 13: `cairo_image_surface_create` creates an image surface. `CAIRO_FORMAT_RGB24` is a constant that means each pixel has red, green, and blue components, and each component is an 8-bit number (24 bits in total). Modern displays have this type of color depth. Width and height are in pixels and given as integers.
- 14: Creates cairo context. The surface given as an argument will be the destination of the context.
- 18: `cairo_set_source_rgb` creates a source pattern, which is a solid white paint. The second to fourth arguments are red, green and blue color values respectively, and they are of type float. The values are between zero (0.0) and one (1.0). Black is (0.0,0.0,0.0) and white is (1.0,1.0,1.0).
- 19: `cairo_paint` copies everywhere in the source to destination. The destination is filled with white pixels with this command.
- 21: Sets the source color to black.
- 22: `cairo_set_line_width` sets the width of lines. In this case, the line width is set to be two pixels and will result in that size. (It is because the transformation is identity. If the transformation isn't identity, for example scaling with the factor three, the actual width in destination will be six (2x3=6) pixels.)
- 23: Draws a rectangle (square) on the mask. The square is located at the center.
- 24: `cairo_stroke` transfers the source to destination through the rectangle in the mask.
- 31: Outputs the image to a png file `rectangle.png`.
- 32: Destroys the context. At the same time the source is destroyed.
- 33: Destroys the surface.

To compile this, change your current directory to `/src/misc` and type the following.

```
$ gcc `pkg-config --cflags cairo` cairo.c `pkg-config --libs cairo`
```

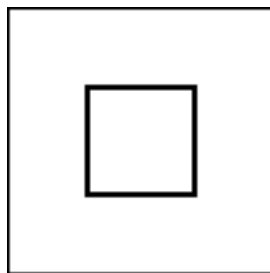


Figure 3.2: rectangle.png

See the [Cairo's website](#) for further information.

3.1.2 GtkDrawingArea

The following is a very simple example.

```

1  #include <gtk/gtk.h>
2
3  static void
4  draw_function (GtkDrawingArea *area, cairo_t *cr, int width, int height, gpointer user_data) {
5      int square_size = 40.0;
6

```

```

7   cairo_set_source_rgb (cr, 1.0, 1.0, 1.0); /* white */
8   cairo_paint (cr);
9   cairo_set_line_width (cr, 2.0);
10  cairo_set_source_rgb (cr, 0.0, 0.0, 0.0); /* black */
11  cairo_rectangle (cr,
12                  width/2.0 - square_size/2,
13                  height/2.0 - square_size/2,
14                  square_size,
15                  square_size);
16  cairo_stroke (cr);
17  }
18
19  static void
20  app_activate (GApplication *app, gpointer user_data) {
21      GtkWidget *win = gtk_application_window_new (GTK_APPLICATION (app));
22      GtkWidget *area = gtk_drawing_area_new ();
23
24      gtk_window_set_title (GTK_WINDOW (win), "da1");
25      gtk_drawing_area_set_draw_func (GTK_DRAWING_AREA (area), draw_function, NULL, NULL);
26      gtk_window_set_child (GTK_WINDOW (win), area);
27
28      gtk_window_present (GTK_WINDOW (win));
29  }
30
31  #define APPLICATION_ID "com.github.ToshioCP.da1"
32
33  int
34  main (int argc, char **argv) {
35      GtkApplication *app;
36      int stat;
37
38      app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
39      g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
40      stat = g_application_run (G_APPLICATION (app), argc, argv);
41      g_object_unref (app);
42      return stat;
43  }

```

The `main` function is almost the same as those in the previous sections. The two functions `app_activate` and `draw_function` are important in this example.

- 22: Creates a `GtkDrawingArea` instance.
- 25: Sets a drawing function of the widget. `GtkDrawingArea` widget uses the function `draw_function` to draw the contents of itself whenever it is necessary. For example, when a user drag a mouse pointer and resize a top-level window, `GtkDrawingArea` also changes the size. Then, the whole window needs to be redrawn. For the information of `gtk_drawing_area_set_draw_func`, see [Gtk API Reference – gtk_drawing_area_set_draw_func](#).

The drawing function has five parameters.

```
void drawing_function (GtkDrawingArea *drawing_area, cairo_t *cr, int width, int height,
                      gpointer user_data);
```

The first parameter is the `GtkDrawingArea` widget. You can't change any properties, for example `content-width` or `content-height`, in this function. The second parameter is a cairo context given by the widget. The destination surface of the context is connected to the contents of the widget. What you

draw to this surface will appear in the widget on the screen. The third and fourth parameters are the size of the destination surface. Now, look at the program again.

- 3-17: The drawing function.
- 7-8: Sets the source to be white and paint the destination white.
- 9: Sets the line width to be 2.
- 10: Sets the source to be black.
- 11-15: Adds a rectangle to the mask.
- 16: Draws the rectangle with black color to the destination.

The program is `/src/misc/da1.c`. Compile and run it, then a window with a black rectangle (square) appears. When you resize the window, the square always appears at the center of the window because the drawing function is invoked each time the window is resized.

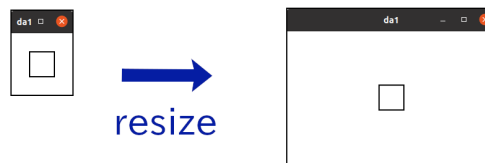


Figure 3.3: Square in the window

3.2 Periodic Events

This chapter was written by Paul Schulz paul@mawsonlakes.org.

3.2.1 How do we create an animation?

In this section we will continue to build on our previous work. We will create an analog clock application. By adding a function which periodically redraws `GtkDrawingArea`, the clock will be able to continuously display the time.

The application uses a compiled in ‘resource’ file, so if the GTK4 libraries and their dependencies are installed and available, the application will run from anywhere.

The program also makes use of some standard mathematical and time handling functions.

The clocks mechanics were taken from a Cairo drawing example, using `gtkmm4`, which can be found [here](#).

The complete code is at the end.

3.2.2 Drawing the clock face, hour, minute and second hands

The `draw_clock()` function does all the work. See the in-file comments for an explanation of how the Cairo drawing works.

For a detailed reference of what each of the Cairo functions does see the [cairo_t reference](#).

```

1 static void
2 draw_clock (GtkDrawingArea *area, cairo_t *cr, int width, int height, gpointer user_data) {
3
4     // Scale to unit square and translate (0, 0) to be (0.5, 0.5), i.e.
5     // the center of the window
6     cairo_scale(cr, width, height);
7     cairo_translate(cr, 0.5, 0.5);

```

```

8
9 // Set the line width and save the cairo drawing state.
10 cairo_set_line_width(cr, m_line_width);
11 cairo_save(cr);
12
13 // Set the background to a slightly transparent green.
14 cairo_set_source_rgba(cr, 0.337, 0.612, 0.117, 0.9); // green
15 cairo_paint(cr);
16
17 // Restore back to previous drawing state and draw the circular path
18 // representing the clockface. Save this state (including the path) so we
19 // can reuse it.
20 cairo_restore(cr);
21 cairo_arc(cr, 0.0, 0.0, m_radius, 0.0, 2.0 * M_PI);
22 cairo_save(cr);
23
24 // Fill the clockface with white
25 cairo_set_source_rgba(cr, 1.0, 1.0, 1.0, 0.8);
26 cairo_fill_preserve(cr);
27 // Restore the path, paint the outside of the clock face.
28 cairo_restore(cr);
29 cairo_stroke_preserve(cr);
30 // Set the 'clip region' to the inside of the path (fill region).
31 cairo_clip(cr);
32
33 // Clock ticks
34 for (int i = 0; i < 12; i++)
35 {
36     // Major tick size
37     double inset = 0.05;
38
39     // Save the graphics state, restore after drawing tick to maintain pen
40     // size
41     cairo_save(cr);
42     cairo_set_line_cap(cr, CAIRO_LINE_CAP_ROUND);
43
44     // Minor ticks are shorter, and narrower.
45     if(i % 3 != 0)
46     {
47         inset *= 0.8;
48         cairo_set_line_width(cr, 0.03);
49     }
50
51     // Draw tick mark
52     cairo_move_to(
53         cr,
54         (m_radius - inset) * cos (i * M_PI / 6.0),
55         (m_radius - inset) * sin (i * M_PI / 6.0));
56     cairo_line_to(
57         cr,
58         m_radius * cos (i * M_PI / 6.0),
59         m_radius * sin (i * M_PI / 6.0));
60     cairo_stroke(cr);

```

```

61     cairo_restore(cr); /* stack-pen-size */
62 }
63
64 // Draw the analog hands
65
66 // Get the current Unix time, convert to the local time and break into time
67 // structure to read various time parts.
68 time_t rawtime;
69 time(&rawtime);
70 struct tm * timeinfo = localtime (&rawtime);
71
72 // Calculate the angles of the hands of our clock
73 double hours    = timeinfo->tm_hour * M_PI / 6.0;
74 double minutes  = timeinfo->tm_min * M_PI / 30.0;
75 double seconds  = timeinfo->tm_sec * M_PI / 30.0;
76
77 // Save the graphics state
78 cairo_save(cr);
79 cairo_set_line_cap(cr, CAIRO_LINE_CAP_ROUND);
80
81 cairo_save(cr);
82
83 // Draw the seconds hand
84 cairo_set_line_width(cr, m_line_width / 3.0);
85 cairo_set_source_rgba(cr, 0.7, 0.7, 0.7, 0.8); // gray
86 cairo_move_to(cr, 0.0, 0.0);
87 cairo_line_to(cr,
88               sin(seconds) * (m_radius * 0.9),
89               -cos(seconds) * (m_radius * 0.9));
90 cairo_stroke(cr);
91 cairo_restore(cr);
92
93 // Draw the minutes hand
94 cairo_set_source_rgba(cr, 0.117, 0.337, 0.612, 0.9); // blue
95 cairo_move_to(cr, 0, 0);
96 cairo_line_to(cr,
97               sin(minutes + seconds / 60) * (m_radius * 0.8),
98               -cos(minutes + seconds / 60) * (m_radius * 0.8));
99 cairo_stroke(cr);
100
101 // draw the hours hand
102 cairo_set_source_rgba(cr, 0.337, 0.612, 0.117, 0.9); // green
103 cairo_move_to(cr, 0.0, 0.0);
104 cairo_line_to(cr,
105               sin(hours + minutes / 12.0) * (m_radius * 0.5),
106               -cos(hours + minutes / 12.0) * (m_radius * 0.5));
107 cairo_stroke(cr);
108 cairo_restore(cr);
109
110 // Draw a little dot in the middle
111 cairo_arc(cr, 0.0, 0.0, m_line_width / 3.0, 0.0, 2.0 * M_PI);
112 cairo_fill(cr);
113 }

```

In order for the clock to be drawn, the drawing function `draw_clock()` needs to be registered with GTK4. This is done in the `app_activate()` function (on line 24).

Whenever the application needs to redraw the `GtkDrawingArea`, it will now call `draw_clock()`.

There is still a problem though. In order to animate the clock we need to also tell the application that the clock needs to be redrawn every second. This process starts by registering (on the next line, line 15) a timeout function with `g_timeout_add()` that will wakeup and run another function `time_handler`, every second (or 1000ms).

```

1  static void
2  app_activate (GApplication *app, gpointer user_data) {
3      GtkWidget *win;
4      GtkWidget *clock;
5      GtkBuilder *build;
6
7      build = gtk_builder_new_from_resource ("/com/github/ToshioCP/tfc/tfc.ui");
8      win = GTK_WIDGET (gtk_builder_get_object (build, "win"));
9      gtk_window_set_application (GTK_WINDOW (win), GTK_APPLICATION (app));
10
11     clock = GTK_WIDGET (gtk_builder_get_object (build, "clock"));
12     g_object_unref(build);
13
14     gtk_drawing_area_set_draw_func(GTK_DRAWING_AREA (clock), draw_clock, NULL, NULL);
15     g_timeout_add(1000, (GSourceFunc) time_handler, (gpointer) clock);
16     gtk_window_present(GTK_WINDOW (win));
17
18 }
```

Our `time_handler()` function is very simple, as it just calls `gtk_widget_queue_draw()` which schedules a redraw of the widget.

```

1  gboolean
2  time_handler(GtkWidget* widget) {
3      gtk_widget_queue_draw(widget);
4
5      return TRUE;
6  }
```

.. and that is all there is to it. If you compile and run the example you will get a ticking analog clock.

If you get this working, you can try modifying some of the code in `draw_clock()` to tweak the application (such as change the color or size and length of the hands) or even add text, or create a digital clock.

3.2.3 The Complete code

You can find the source files in the `tfc` directory. it can be compiled with `./comp tfc`.

`tfc.c`

```

1  #include <gtk/gtk.h>
2  #include <math.h>
3  #include <time.h>
4
5  float m_radius      = 0.42;
6  float m_line_width  = 0.05;
7
8  static void
```

```

9  draw_clock (GtkDrawingArea *area, cairo_t *cr, int width, int height, gpointer user_data) {
10
11      // Scale to unit square and translate (0, 0) to be (0.5, 0.5), i.e.
12      // the center of the window
13      cairo_scale(cr, width, height);
14      cairo_translate(cr, 0.5, 0.5);
15
16      // Set the line width and save the cairo drawing state.
17      cairo_set_line_width(cr, m_line_width);
18      cairo_save(cr);
19
20      // Set the background to a slightly transparent green.
21      cairo_set_source_rgba(cr, 0.337, 0.612, 0.117, 0.9); // green
22      cairo_paint(cr);
23
24      // Restore back to previous drawing state and draw the circular path
25      // representing the clockface. Save this state (including the path) so we
26      // can reuse it.
27      cairo_restore(cr);
28      cairo_arc(cr, 0.0, 0.0, m_radius, 0.0, 2.0 * M_PI);
29      cairo_save(cr);
30
31      // Fill the clockface with white
32      cairo_set_source_rgba(cr, 1.0, 1.0, 1.0, 0.8);
33      cairo_fill_preserve(cr);
34      // Restore the path, paint the outside of the clock face.
35      cairo_restore(cr);
36      cairo_stroke_preserve(cr);
37      // Set the 'clip region' to the inside of the path (fill region).
38      cairo_clip(cr);
39
40      // Clock ticks
41      for (int i = 0; i < 12; i++)
42      {
43          // Major tick size
44          double inset = 0.05;
45
46          // Save the graphics state, restore after drawing tick to maintain pen
47          // size
48          cairo_save(cr);
49          cairo_set_line_cap(cr, CAIRO_LINE_CAP_ROUND);
50
51          // Minor ticks are shorter, and narrower.
52          if(i % 3 != 0)
53          {
54              inset *= 0.8;
55              cairo_set_line_width(cr, 0.03);
56          }
57
58          // Draw tick mark
59          cairo_move_to(
60              cr,
61              (m_radius - inset) * cos (i * M_PI / 6.0),

```

```

62         (m_radius - inset) * sin (i * M_PI / 6.0));
63     cairo_line_to(
64         cr,
65         m_radius * cos (i * M_PI / 6.0),
66         m_radius * sin (i * M_PI / 6.0));
67     cairo_stroke(cr);
68     cairo_restore(cr); /* stack-pen-size */
69 }
70
71 // Draw the analog hands
72
73 // Get the current Unix time, convert to the local time and break into time
74 // structure to read various time parts.
75 time_t rawtime;
76 time(&rawtime);
77 struct tm * timeinfo = localtime (&rawtime);
78
79 // Calculate the angles of the hands of our clock
80 double hours    = timeinfo->tm_hour * M_PI / 6.0;
81 double minutes  = timeinfo->tm_min * M_PI / 30.0;
82 double seconds  = timeinfo->tm_sec * M_PI / 30.0;
83
84 // Save the graphics state
85 cairo_save(cr);
86 cairo_set_line_cap(cr, CAIRO_LINE_CAP_ROUND);
87
88 cairo_save(cr);
89
90 // Draw the seconds hand
91 cairo_set_line_width(cr, m_line_width / 3.0);
92 cairo_set_source_rgba(cr, 0.7, 0.7, 0.7, 0.8); // gray
93 cairo_move_to(cr, 0.0, 0.0);
94 cairo_line_to(cr,
95         sin(seconds) * (m_radius * 0.9),
96         -cos(seconds) * (m_radius * 0.9));
97 cairo_stroke(cr);
98 cairo_restore(cr);
99
100 // Draw the minutes hand
101 cairo_set_source_rgba(cr, 0.117, 0.337, 0.612, 0.9); // blue
102 cairo_move_to(cr, 0, 0);
103 cairo_line_to(cr,
104         sin(minutes + seconds / 60) * (m_radius * 0.8),
105         -cos(minutes + seconds / 60) * (m_radius * 0.8));
106 cairo_stroke(cr);
107
108 // draw the hours hand
109 cairo_set_source_rgba(cr, 0.337, 0.612, 0.117, 0.9); // green
110 cairo_move_to(cr, 0.0, 0.0);
111 cairo_line_to(cr,
112         sin(hours + minutes / 12.0) * (m_radius * 0.5),
113         -cos(hours + minutes / 12.0) * (m_radius * 0.5));
114 cairo_stroke(cr);

```

```

115     cairo_restore(cr);
116
117     // Draw a little dot in the middle
118     cairo_arc(cr, 0.0, 0.0, m_line_width / 3.0, 0.0, 2.0 * M_PI);
119     cairo_fill(cr);
120 }
121
122
123 gboolean
124 time_handler(GtkWidget* widget) {
125     gtk_widget_queue_draw(widget);
126
127     return TRUE;
128 }
129
130
131 static void
132 app_activate (GApplication *app, gpointer user_data) {
133     GtkWidget *win;
134     GtkWidget *clock;
135     GtkBuilder *build;
136
137     build = gtk_builder_new_from_resource ("/com/github/ToshioCP/tfc/tfc.ui");
138     win = GTK_WIDGET (gtk_builder_get_object (build, "win"));
139     gtk_window_set_application (GTK_WINDOW (win), GTK_APPLICATION (app));
140
141     clock = GTK_WIDGET (gtk_builder_get_object (build, "clock"));
142     g_object_unref(build);
143
144     gtk_drawing_area_set_draw_func(GTK_DRAWING_AREA (clock), draw_clock, NULL, NULL);
145     g_timeout_add(1000, (GSourceFunc) time_handler, (gpointer) clock);
146     gtk_window_present(GTK_WINDOW (win));
147
148 }
149
150 static void
151 app_open (GApplication *app, GFile **files, gint n_files, gchar *hint, gpointer user_data) {
152     app_activate(app,user_data);
153 }
154
155 int
156 main (int argc, char **argv) {
157     GtkApplication *app;
158     int stat;
159
160     app = gtk_application_new ("com.github.ToshioCP.tfc", G_APPLICATION_HANDLES_OPEN);
161     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
162     g_signal_connect (app, "open", G_CALLBACK (app_open), NULL);
163     stat = g_application_run (G_APPLICATION (app), argc, argv);
164     g_object_unref (app);
165     return stat;
166 }

```

tfc.ui

```

1  <?xml version="1.0" encoding="UTF-8"?>
2  <interface>
3      <object class="GtkApplicationWindow" id="win">
4          <property name="title">Clock</property>
5          <property name="default-width">200</property>
6          <property name="default-height">200</property>
7          <child>
8              <object class="GtkDrawingArea" id="clock">
9                  <property name="hexpand">TRUE</property>
10                 <property name="vexpand">TRUE</property>
11             </object>
12         </child>
13     </object>
14 </interface>

tfc.gresource.xml

1  <?xml version="1.0" encoding="UTF-8"?>
2  <gresources>
3      <gresource prefix="/com/github/ToshioCP/tfc">
4          <file>tfc.ui</file>
5      </gresource>
6  </gresources>

comp

1  glib-compile-resources $1.gresource.xml --target=$1.gresource.c --generate-source
2  gcc `pkg-config --cflags gtk4` $1.gresource.c $1.c `pkg-config --libs gtk4` -lm

```

3.3 Custom Drawing

Custom drawing is to draw shapes dynamically. This section shows an example of custom drawing. You can draw rectangles by dragging the mouse.

Press down the button.

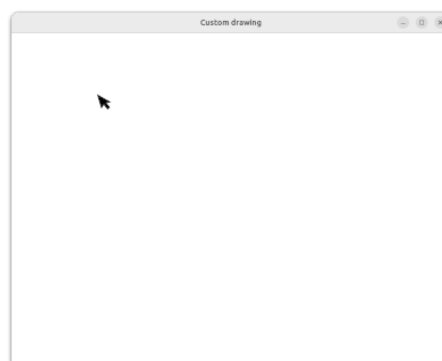


Figure 3.4: Press down the button

Move the mouse.

Release the button.

The programs are at `src/custom_drawing` directory. Download the [repository](#) and see the directory. There are four files.

- `meson.build`

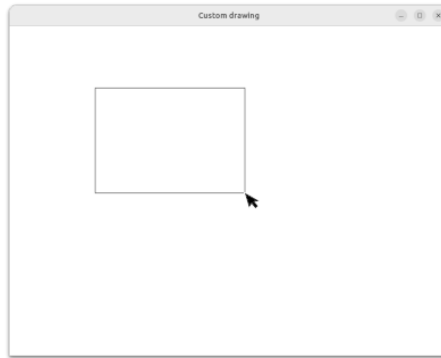


Figure 3.5: Move the mouse

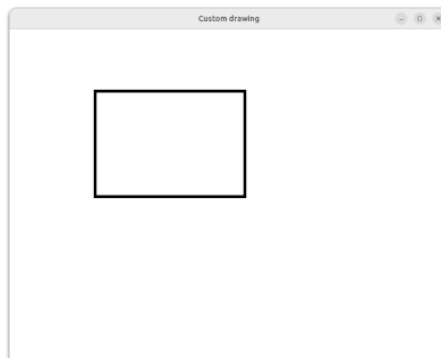


Figure 3.6: Release the button

- rect.c
- rect.gresource.xml
- rect.ui

3.3.1 rect.gresource.xml

This file describes a UI file to compile. The compiler glib-compile-resources uses it.

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <gresources>
3   <gresource prefix="/com/github/ToshioCP/rect">
4     <file>rect.ui</file>
5   </gresource>
6 </gresources>

```

The prefix is `/com/github/ToshioCP/rect` and the file is `rect.ui`. Therefore, GtkBuilder reads the resource from `/com/github/ToshioCP/rect/rect.ui`.

3.3.2 rect.ui

The following is the UI file that defines the widgets. There are two widgets which are `GtkApplicationWindow` and `GtkDrawingArea`. The ids are `win` and `da` respectively.

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <interface>
3   <object class="GtkApplicationWindow" id="win">
4     <property name="default-width">800</property>
5     <property name="default-height">600</property>

```

```

6     <property name="resizable">FALSE</property>
7     <property name="title">Custom drawing</property>
8     <child>
9         <object class="GtkDrawingArea" id="da">
10             <property name="hexpand">TRUE</property>
11             <property name="vexpand">TRUE</property>
12         </object>
13     </child>
14 </object>
15 </interface>

```

3.3.3 rect.c

3.3.3.1 GtkApplication

This program uses `GtkApplication`. The application ID is `com.github.ToshioCP.rect`.

```
#define APPLICATION_ID "com.github.ToshioCP.rect"
```

See [GNOME Developer Documentation](#) for further information.

The function `main` is called at the beginning of the application.

```

1  int
2  main (int argc, char **argv) {
3      GtkApplication *app;
4      int stat;
5
6      app = gtk_application_new (APPLICATION_ID, G_APPLICATION_HANDLES_OPEN);
7      g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
8      g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
9      g_signal_connect (app, "shutdown", G_CALLBACK (app_shutdown), NULL);
10     stat = g_application_run (G_APPLICATION (app), argc, argv);
11     g_object_unref (app);
12     return stat;
13 }

```

It connects three signals and handlers.

- startup: It is emitted after the application is registered to the system.
- activate: It is emitted when the application is activated.
- shutdown: It is emitted just before the application quits.

```

1  static void
2  app_startup (GApplication *application) {
3      GtkApplication *app = GTK_APPLICATION (application);
4      GtkBuilder *build;
5      GtkWindow *win;
6      GtkDrawingArea *da;
7      GtkGesture *drag;
8
9      build = gtk_builder_new_from_resource ("/com/github/ToshioCP/rect/rect.ui");
10     win = GTK_WINDOW (gtk_builder_get_object (build, "win"));
11     da = GTK_DRAWING_AREA (gtk_builder_get_object (build, "da"));
12     gtk_window_set_application (win, app);
13     g_object_unref (build);
14
15     gtk_drawing_area_set_draw_func (da, draw_cb, NULL, NULL);

```

```

16 g_signal_connect_after (da, "resize", G_CALLBACK (resize_cb), NULL);
17
18 drag = gtk_gesture_drag_new ();
19 gtk_gesture_single_set_button (GTK_GESTURE_SINGLE (drag), GDK_BUTTON_PRIMARY);
20 gtk_widget_add_controller (GTK_WIDGET (da), GTK_EVENT_CONTROLLER (drag));
21 g_signal_connect (drag, "drag-begin", G_CALLBACK (drag_begin), NULL);
22 g_signal_connect (drag, "drag-update", G_CALLBACK (drag_update), da);
23 g_signal_connect (drag, "drag-end", G_CALLBACK (drag_end), da);
24 }

```

The startup handler performs the following three tasks:

- Builds the widgets.
- Initializes the `GtkDrawingArea` instance.
 - Sets the drawing function
 - Connects the “resize” signal and the handler.
- Creates the `GtkGestureDrag` instance and initializes it. Gestures will be explained in this section later.

```

1 static void
2 app_activate (GApplication *application) {
3     GtkApplication *app = GTK_APPLICATION (application);
4     GtkWidget *win;
5
6     win = gtk_application_get_active_window (app);
7     gtk_window_present (win);
8 }

```

The activate handler just shows the window.

3.3.3.2 GtkDrawingArea

The program has two cairo surfaces and they are pointed to by the global variables.

```

static cairo_surface_t *surface = NULL;
static cairo_surface_t *surface_save = NULL;

```

The drawing process is as follows.

- Creates an image on `surface`.
- Copies `surface` to the cairo surface of the `GtkDrawingArea`.
- Calls `gtk_widget_queue_draw (da)` to draw it if necessary.

They are created in the “resize” signal handler.

```

1 static void
2 resize_cb (GtkWidget *widget, int width, int height, gpointer user_data) {
3     cairo_t *cr;
4
5     if (surface)
6         cairo_surface_destroy (surface);
7     surface = cairo_image_surface_create (CAIRO_FORMAT_RGB24, width, height);
8     if (surface_save)
9         cairo_surface_destroy (surface_save);
10    surface_save = cairo_image_surface_create (CAIRO_FORMAT_RGB24, width, height);
11    /* Paint the surface white. It is the background color. */
12    cr = cairo_create (surface);
13    cairo_set_source_rgb (cr, 1.0, 1.0, 1.0);

```

```

14     cairo_paint (cr);
15     cairo_destroy (cr);
16 }

```

This callback is called when the GtkDrawingArea is shown. This is called only once since the window is not resizable.

It creates image surfaces for `surface` and `surface_save`. The `surface` surface is painted white, which is the background color.

The drawing function copies `surface` to the GtkDrawingArea surface.

```

1  static void
2  draw_cb (GtkDrawingArea *da, cairo_t *cr, int width, int height, gpointer user_data) {
3      if (surface) {
4          cairo_set_source_surface (cr, surface, 0.0, 0.0);
5          cairo_paint (cr);
6      }
7  }

```

This function is called by the system when it needs to redraw the drawing area.

Two surfaces `surface` and `surface_save` are destroyed before the application quits.

```

1  static void
2  app_shutdown (GApplication *application) {
3      if (surface)
4          cairo_surface_destroy (surface);
5      if (surface_save)
6          cairo_surface_destroy (surface_save);
7  }

```

3.3.3.3 GtkGestureDrag

Gesture class is used to recognize human gestures such as click, drag, pan, swipe and so on. It is a subclass of GtkEventController. GtkGesture class is abstract and there are several implementations.

- GtkGestureClick
- GtkGestureDrag
- GtkGesturePan
- GtkGestureSwipe
- And others

The program `rect.c` uses GtkGestureDrag. It is the implementation for drag gestures. The parent-child relationship is as follows.

```
GObject -- GtkEventController -- GtkGesture -- GtkGestureSingle -- GtkGestureDrag
```

GtkGestureSingle is a subclass of GtkGesture and optimized for single-touch and mouse interactions.

A GtkGestureDrag instance is created and initialized in the startup signal handler in `rect.c`. See line 18 to 23 in the following.

```

1  static void
2  app_startup (GApplication *application) {
3      GtkApplication *app = GTK_APPLICATION (application);
4      GtkBuilder *build;
5      GtkWindow *win;
6      GtkDrawingArea *da;
7      GtkGesture *drag;
8

```

```

9   build = gtk_builder_new_from_resource ("/com/github/ToshioCP/rect/rect.ui");
10  win = GTK_WINDOW (gtk_builder_get_object (build, "win"));
11  da = GTK_DRAWING_AREA (gtk_builder_get_object (build, "da"));
12  gtk_window_set_application (win, app);
13  g_object_unref (build);
14
15  gtk_drawing_area_set_draw_func (da, draw_cb, NULL, NULL);
16  g_signal_connect_after (da, "resize", G_CALLBACK (resize_cb), NULL);
17
18  drag = gtk_gesture_drag_new ();
19  gtk_gesture_single_set_button (GTK_GESTURE_SINGLE (drag), GDK_BUTTON_PRIMARY);
20  gtk_widget_add_controller (GTK_WIDGET (da), GTK_EVENT_CONTROLLER (drag));
21  g_signal_connect (drag, "drag-begin", G_CALLBACK (drag_begin), NULL);
22  g_signal_connect (drag, "drag-update", G_CALLBACK (drag_update), da);
23  g_signal_connect (drag, "drag-end", G_CALLBACK (drag_end), da);
24 }

```

- The function `gtk_gesture_drag_new` creates a new `GtkGestureDrag` instance.
- The function `gtk_gesture_single_set_button` sets the button number to listen to. The constant `GDK_BUTTON_PRIMARY` is the left button of a mouse.
- The function `gtk_widget_add_controller` adds an event controller, which is a base class for gestures.
- Three signals and handlers are connected.
 - `drag-begin`: Emitted when dragging starts.
 - `drag-update`: Emitted when the dragging pointer moves.
 - `drag-end`: Emitted when the dragging ends.

The process during the drag is as follows.

- start: save the surface and start points
- update: restore the surface and draw a thin rectangle between the start point and the current point of the mouse
- end: restore the surface and draw a thick rectangle between the start and end points.

We need two global variables for the start point.

```

static double start_x;
static double start_y;

```

The following is the handler for the “drag-begin” signal.

```

1  static void
2  copy_surface (cairo_surface_t *src, cairo_surface_t *dst) {
3      if (!src || !dst)
4          return;
5      cairo_t *cr = cairo_create (dst);
6      cairo_set_source_surface (cr, src, 0.0, 0.0);
7      cairo_paint (cr);
8      cairo_destroy (cr);
9  }
10
11  static void
12  drag_begin (GtkGestureDrag *gesture, double x, double y, gpointer user_data) {
13      // save the surface and record (x, y)
14      copy_surface (surface, surface_save);
15      start_x = x;
16      start_y = y;

```

```
17 }
```

- Copies `surface` to `surface_save`, which is an image just before the dragging.
- Stores the points to `start_x` and `start_y`.

```
1 static void
2 drag_update (GtkGestureDrag *gesture, double offset_x, double offset_y, gpointer user_data) {
3     GtkWidget *da = GTK_WIDGET (user_data);
4     cairo_t *cr;
5
6     copy_surface (surface_save, surface);
7     cr = cairo_create (surface);
8     cairo_rectangle (cr, start_x, start_y, offset_x, offset_y);
9     cairo_set_line_width (cr, 1.0);
10    cairo_stroke (cr);
11    cairo_destroy (cr);
12    gtk_widget_queue_draw (da);
13 }
```

- Restores `surface` from `surface_save`.
- Draws a rectangle with thin lines.
- Calls `gtk_widget_queue_draw` to add the `GtkDrawingArea` to the queue to redraw.

```
1 static void
2 drag_end (GtkGestureDrag *gesture, double offset_x, double offset_y, gpointer user_data) {
3     GtkWidget *da = GTK_WIDGET (user_data);
4     cairo_t *cr;
5
6     copy_surface (surface_save, surface);
7     cr = cairo_create (surface);
8     cairo_rectangle (cr, start_x, start_y, offset_x, offset_y);
9     cairo_set_line_width (cr, 6.0);
10    cairo_stroke (cr);
11    cairo_destroy (cr);
12    gtk_widget_queue_draw (da);
13 }
```

- Restores `surface` from `surface_save`.
- Draws a rectangle with thick lines.
- Calls `gtk_widget_queue_draw` to add the `GtkDrawingArea` to the queue to redraw.

3.3.4 Build and Run

Download the [repository](#). Change your current directory to `src/custom_drawing`. Run `meson` and `ninja` to build the program. Type `_build/rect` to run the program. Try to draw rectangles.

```
$ cd src/custom_drawing
$ meson setup _build
$ ninja -C _build
$ _build/rect
```

3.4 Tiny Turtle Graphics Interpreter

The Turtle program is an example with the combination of `TfeTextView` and `GtkDrawingArea` objects. It is a very small interpreter but you can draw fractal curves with it. The following diagram is a Koch curve, which is one of the famous fractal curves.

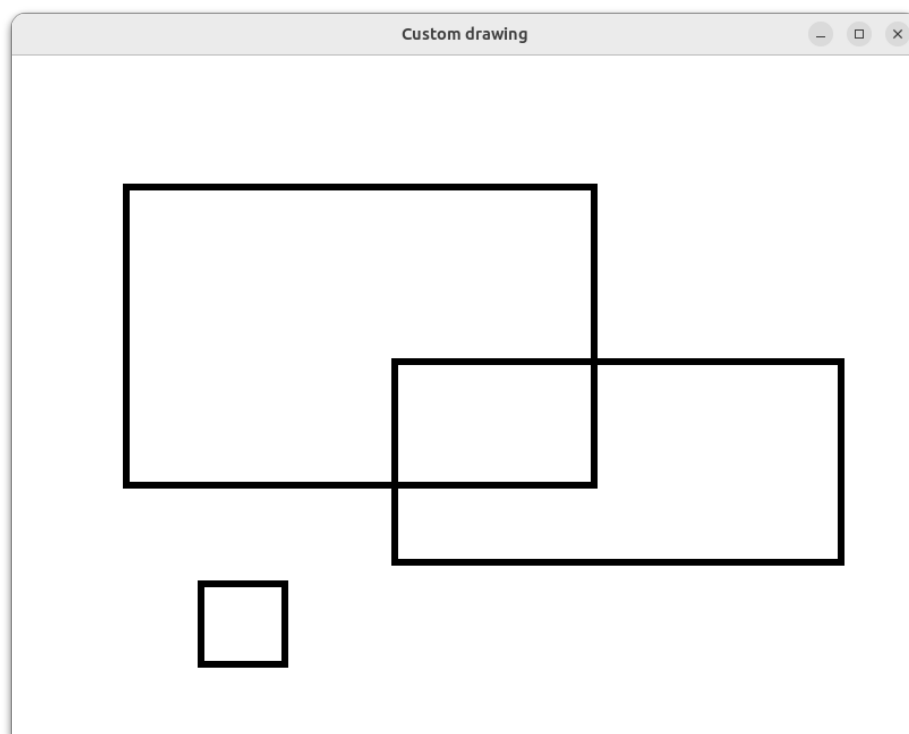


Figure 3.7: The screen of rect program

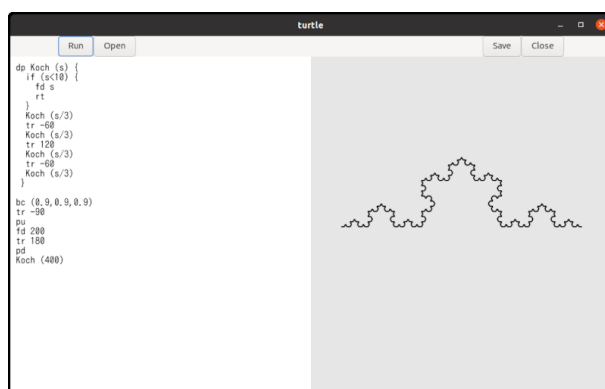


Figure 3.8: Koch curve

The following is a snow-crystal-shaped curve. It is composed of six Koch curves.

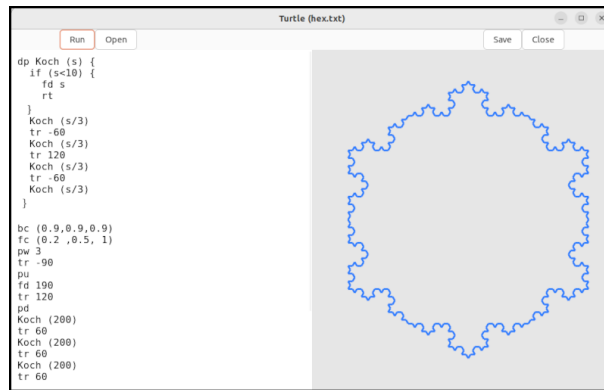


Figure 3.9: Snow

This program uses flex and bison. Flex is a lexical analyzer. Bison is a parser generator. These two programs are similar to lex and yacc which are proprietary software developed in Bell Laboratory. However, flex and bison are open source software. This section describes them and they are not the topics about GTK 4. So, readers can skip this section.

- [Flex GitHub repository](#)
- [Flex Documentation](#)
- [Bison Project Page](#)

3.4.1 How to Use Turtle

The turtle document is [here](#). I'll show you a simple example.

```
fc (1,0,0) # Foreground color is red, rgb = (1,0,0).
pd        # Pen down.
rp (4) {   # Repeat four times.
    fd 100  # Go forward by 100 pixels.
    tr 90   # Turn right by 90 degrees.
}
```

1. Compile and install `turtle` (See the documentation above). Then, run `turtle`.
2. Type the program above in the editor (left part of the window).
3. Click on the Run button, then a red square appears on the right part of the window. Each side of the square is 100 pixels long.

In the same way, you can draw other curves. The turtle document includes some fractal curves such as tree, snow and square-koch. The source codes are located at [/src/turtle/example](#) directory. You can read these files into `turtle` editor by clicking on the Open button.

3.4.2 Combination of TfeTextView and GtkDrawingArea Objects

Turtle uses TfeTextView and GtkDrawingArea.

1. A user inputs/reads a turtle program into the buffer in the TfeTextView instance.
2. The user clicks on the “Run” button.
3. The parser reads the program and generates tree-structured data.
4. The interpreter reads the data and executes it step by step. And it draws shapes on a surface. The surface isn't the one in the GtkDrawingArea widget.
5. The widget is added to the queue. It will be redrawn with the drawing function, which just copies the surface into the one in the GtkDrawingArea.

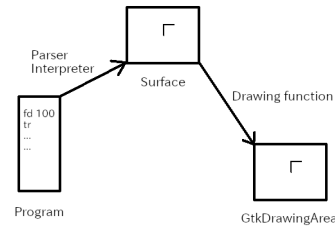


Figure 3.10: Parser, interpreter and drawing function

The body of the interpreter is written with flex and bison. The codes are not thread safe. So the callback function `run_cb`, which is the handler of “clicked” signal on the Run button, prevents reentering.

```

1  void
2  run_cb (GtkWidget *btnr) {
3      GtkTextBuffer *tb = gtk_text_view_get_buffer (GTK_TEXT_VIEW (tv));
4      GtkTextIter start_iter;
5      GtkTextIter end_iter;
6      char *contents;
7      int stat;
8      static gboolean busy = FALSE; /* initialized only once */
9      cairo_t *cr;
10
11     /* yyparse() and run() are NOT thread safe. */
12     /* The variable busy avoids reentrance. */
13     if (busy)
14         return;
15     busy = TRUE;
16     gtk_text_buffer_get_bounds (tb, &start_iter, &end_iter);
17     contents = gtk_text_buffer_get_text (tb, &start_iter, &end_iter, FALSE);
18     if (surface && contents[0] != '\0') {
19         init_flex (contents);
20         stat = yyparse ();
21         if (stat == 0) { /* No error */
22             run ();
23         }
24         finalize_flex ();
25     } else if (surface) {
26         cr = cairo_create (surface);
27         cairo_set_source_rgb (cr, 1.0, 1.0, 1.0);
28         cairo_paint (cr);
29         cairo_destroy (cr);
30     }
31     g_free (contents);
32     gtk_widget_queue_draw (GTK_WIDGET (da));
33     busy = FALSE;
34 }
35
36 static void
37 resize_cb (GtkDrawingArea *drawing_area, int width, int height, gpointer user_data) {
38
39     if (surface)
40         cairo_surface_destroy (surface);

```

```

41 surface = cairo_image_surface_create (CAIRO_FORMAT_ARGB32, width, height);
42 run_cb (NULL); // NULL is a fake (run button).
43 }

```

- 8, 13-15: The static value `busy` holds a status of the interpreter. If it is `TRUE`, the interpreter is running and it is not possible to call the interpreter because it's not a re-entrant program. If it is `FALSE`, it is safe to call the interpreter and set the variable `busy` to `TRUE`.
- 16-17: Gets the contents of `tb`.
- 18-30: The variable `surface` is a static variable. It points to a `cairo_surface_t` instance. It is created when the `GtkDrawingArea` instance is realized and whenever it is resized. Therefore, `surface` isn't `NULL` usually. But if it is `NULL`, the interpreter won't be called.
- 18-24: If `surface` points to a surface instance and the string `contents` isn't empty, it calls the interpreter.
 - Initializes the lexical analyzer.
 - Calls the parser. The parser analyzes the program codes syntactically and generates a tree structured data.
 - If the parser successfully parsed, it calls the runtime routine 'run'.
 - Finalizes the lexical analyzer.
- 25-29: If `surface` points to a surface instance and the string `contents` is empty, it clears the surface `surface`.
- 31: Frees `contents`.
- 32: Adds the drawing area widget to the queue to draw.
- 33: Sets the variable `busy` to `FALSE`.
- 36-43: The "resized" signal handler. If the `surface` isn't `NULL`, it is destroyed. A new surface is created. Its size is the same as the surface of the `GtkDrawingArea` instance. It calls the callback function `run_cb` to redraw the shape on the drawing area.

If the open button is clicked and a file is read, the filename will be shown on the header bar.

```

1 static void
2 show_filename (TfeTextView *tv) {
3     GFile *file;
4     char *filename;
5     char *title;
6
7     file = tfe_text_view_get_file (tv);
8     if (G_IS_FILE (file)) {
9         filename = g_file_get_basename (file);
10        title = g_strdup_printf ("Turtle (%s)", filename);
11        g_free (filename);
12        g_object_unref (file);
13    } else
14        title = g_strdup ("Turtle");
15    gtk_window_set_title (GTK_WINDOW (win), title);
16    g_free (title);
17 }

```

This function is the callback function of the "change-file" signal on the `TfeTextView` instance. It calls `tfe_text_view_get_file`.

- If the return value is a `GFile` instance, the title will be "Turtle (the filename)".
- Otherwise, the title will be "Turtle".

Other part of `turtleapplication.c` is very simple and similar to the codes in the former applications. The codes of `turtleapplication.c` is in the [turtle directory](#).

3.4.3 What Does the Interpreter Do?

Suppose that the turtle application runs with the following program.

```
distance = 100
fd distance*2
```

The application recognizes the program and works as follows.

- Generally, a program consists of tokens. Tokens are “distance”, “=”, “100”, “fd”, “*” and “2” in the above example..
- The parser calls a function `yylex` to read a token in the source file. `yylex` returns a code which is called “token kind” and sets a global variable `yylval` to a value, which is called a semantic value. The type of `yylval` is union. The type of `yylval.ID` and `yylval.NUM` are string and double respectively. There are seven tokens in the program so `yylex` is called seven times.

	token kind	yylval.ID	yylval.NUM
1	ID	distance	
2	=		
3	NUM		100
4	FD		
5	ID	distance	
6	*		
7	NUM		2

- The function `yylex` returns a token kind every time, but it doesn’t set `yylval.ID` or `yylval.NUM` every time. It is because keywords (FD) and symbols (= and *) don’t have any semantic values. The function `yylex` is called a lexical analyzer or a scanner.
- The application `turtle` makes a tree structured data. This part of `turtle` is called parser.

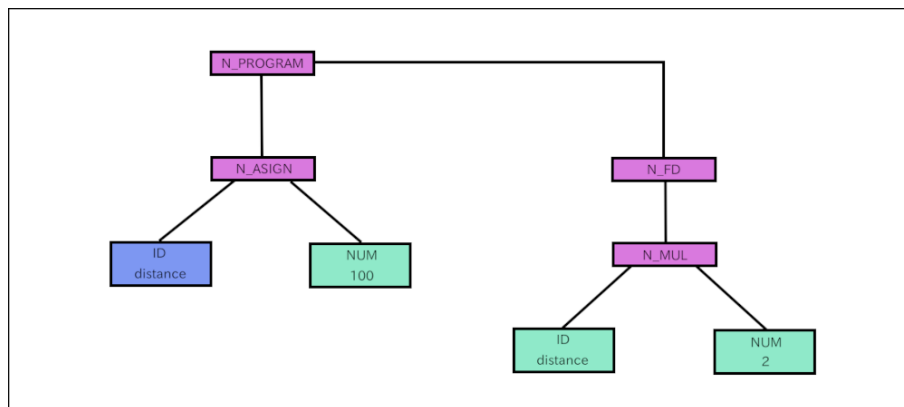


Figure 3.11: turtle parser tree

- **Turtle** analyzes the tree and executes it. This part of **turtle** is called runtime routine or interpreter. The tree consists of rectangles and line segments between the rectangles. The rectangles are called nodes. For example, `N_PROGRAM`, `N_ASSIGN`, `N_FD` and `N_MUL` are nodes.
 1. Goes down from `N_PROGRAM` to `N_ASSIGN`.
 2. `N_ASSIGN` node has two children, `ID` and `NUM`. This node comes from “distance = 100” which is “ID = NUM” syntactically. First, **turtle** checks if the first child is `ID`. If it’s `ID`, then **turtle** looks for the variable in the variable table. If it doesn’t exist, it registers the `ID` (`distance`) to the table. Then go back to the `N_ASSIGN` node.
 3. **Turtle** calculates the second child. In this case its a number 100. Saves 100 to the variable table at the `distance` record.

4. **Turtle** goes back to **N_PROGRAM** then go to the next node **N_FD**. It has only one child. Goes down to the child **N_MUL**.
 5. The first child is **ID** (distance). Searches the variable table for the variable **distance** and gets the value 100. The second child is a number 2. Multiplies 100 by 2 and gets 200. Then **turtle** goes back to **N_FD**.
 6. Now **turtle** knows the distance is 200. It moves the cursor forward by 200 pixels. The segment is drawn on the **surface**.
 7. There is no following node. Runtime routine returns to the function **run_cb**.
- The function **run_cb** calls **gtk_widget_queue_draw** and put the **GtkDrawingArea** widget to the queue.
 - The system redraws the widget. At that time drawing function **draw_func** is called. The function copies the **surface** to the surface in the **GtkDrawingArea**.

Actual turtle program is more complicated than the example above. However, what turtle does is basically the same. Interpretation consists of three parts.

- Lexical analysis
- Syntax Parsing and tree generation
- Interpretation and execution of the tree.

3.4.4 Compilation Flow

The source files are:

- flex source file => **turtle.lex**
- bison source file => **turtle.y**
- C header file => **turtle_lex.h**
- C source file => **turtleapplication.c**
- other files => **turtle.ui**, **turtle.gresources.xml** and **meson.build**

The compilation process is a bit complicated.

1. glib-compile-resources compiles **turtle.ui** to **resources.c** according to **turtle.gresource.xml**. It also generates **resources.h**.
2. bison compiles **turtle.y** to **turtle_parser.c** and generates **turtle_parser.h**
3. flex compiles **turtle.lex** to **turtle_lex.c**.
4. gcc compiles **application.c**, **resources.c**, **turtle_parser.c** and **turtle_lex.c** with **turtle_lex.h**, **resources.h** and **turtle_parser.h**. It generates an executable file **turtle**.

Meson controls the process. The instruction is described in **meson.build**.

```

1 project('turtle', 'c')
2
3 compiler = meson.get_compiler('c')
4 mathdep = compiler.find_library('m', required : true)
5
6 gtkdep = dependency('gtk4')
7
8 gnome=import('gnome')
9 resources = gnome.compile_resources('resources','turtle.gresource.xml')
10
11 flex = find_program('flex')
12 bison = find_program('bison')
13 turtleparser = custom_target('turtleparser', input: 'turtle.y', output: ['turtle_parser.c', 'turtle_
14 turtlelexer = custom_target('turtlelexer', input: 'turtle.lex', output: 'turtle_lex.c', command: [fl
15
16 sourcefiles=files('turtleapplication.c', '../tfetextview/tfetextview.c')
```

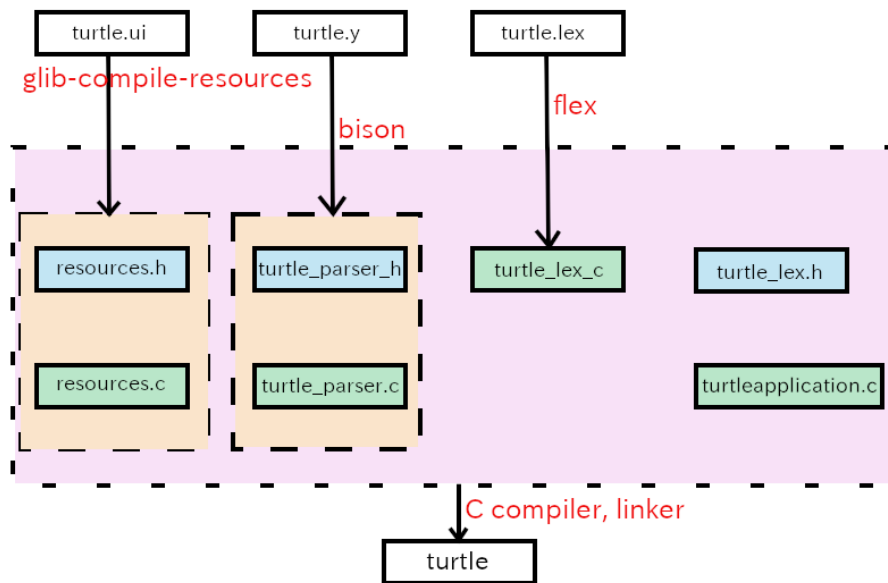


Figure 3.12: compile process

17

18

```
executable('turtle', sourcefiles, resources, turtleparser, turtlelexer, turtleparser[1], dependencies
```

- 1: The project name is “turtle” and the program language is C.
- 3: Gets C compiler. It is usually gcc in linux.
- 4: Gets math library. This program uses trigonometric functions. They are defined in the math library, but the library is optional. So, it is necessary to include it by `#include <math.h>` and also link the library with the linker.
- 6: Gets gtk4 library.
- 8: Gets gnome module. See [Meson build system website – GNOME module](#) for further information.
- 9: Compiles the UI file to C source file according to the XML file `turtle.gresource.xml`.
- 11: Gets flex.
- 12: Gets bison.
- 13: Compiles `turtle.y` to `turtle_parser.c` and `turtle_parser.h` by bison. The function `custom_target` creates a custom top level target. See [Meson build system website – custom target](#) for further information.
- 14: Compiles `turtle.lex` to `turtle_lex.c` by flex.
- 16: The variable `sourcefiles` is a file object created with the C source files.
- 18: Compiles C source files including generated files by glib-compile-resources, bison and flex. The argument `turtleparser[1]` refers to `turtle_parser.h` which is the second output in the line 13.

3.4.5 Turtle.lex

3.4.5.1 What Does Flex Do?

Flex creates lexical analyzer from flex source file. Flex source file is a text file. Its syntactic rule will be explained later. Generated lexical analyzer is a C source file. It is also called scanner. It reads a text file, which is a source file of a program language, and gets variable names, numbers and symbols. Suppose here is a turtle source file.

```
fc (1,0,0) # Foreground color is red, rgb = (1,0,0).
pd          # Pen down.
```

```

distance = 100
angle = 90
fd distance    # Go forward by distance (100) pixels.
tr angle      # Turn right by angle (90) degrees.

```

The content of the text file is separated into `fc`, `(`, `1` and so on. The words `fc`, `pd`, `distance`, `angle`, `tr`, `1`, `0`, `100` and `90` are called tokens. The characters `'(` (left parenthesis), `'.'` (comma), `')` (right parenthesis) and `'='` (equal sign) are called symbols. (Sometimes those symbols called tokens, too.)

Flex reads `turtle.lex` and generates the C source file of a scanner. The file `turtle.lex` specifies tokens, symbols and the behavior which corresponds to each token or symbol. Turtle.lex isn't a big program.

```

1  %top{
2  #include <string.h>
3  #include <stdlib.h>
4  #include <glib.h>
5  #include "turtle_parser.h"
6
7  static int nline = 1;
8  static int ncolumn = 1;
9  static void get_location (char *text);
10
11  /* Dynamically allocated memories are added to the single list. They will be freed in the finalize
12  extern GSList *list;
13  }
14
15  %option noyywrap
16
17  REAL_NUMBER (0|[1-9][0-9]*)(\.[0-9]+)?
18  IDENTIFIER [a-zA-Z][a-zA-Z0-9]*
19  %%
20  /* rules */
21  #.*          ; /* comment. Be careful. Dot symbol (.) matches any character but new line. */
22  [ ]          ncolumn++; /* white space. [ and ] is a "character class". */
23  \t          ncolumn += 8; /* assume that tab is 8 spaces. */
24  \n          nline++; ncolumn = 1;
25  /* reserved keywords */
26  pu          get_location (yytext); return PU; /* pen up */
27  pd          get_location (yytext); return PD; /* pen down */
28  pw          get_location (yytext); return PW; /* pen width = line width */
29  fd          get_location (yytext); return FD; /* forward */
30  tr          get_location (yytext); return TR; /* turn right */
31  tl          get_location (yytext); return TL; /* turn left, since ver 0.5 */
32  bc          get_location (yytext); return BC; /* background color */
33  fc          get_location (yytext); return FC; /* foreground color */
34  dp          get_location (yytext); return DP; /* define procedure */
35  if          get_location (yytext); return IF; /* if statement */
36  rt          get_location (yytext); return RT; /* return statement */
37  rs          get_location (yytext); return RS; /* reset the status */
38  rp          get_location (yytext); return RP; /* repeat, since ver 0.5 */
39  /* constant */
40  {REAL_NUMBER} get_location (yytext); yylval.NUM = atof (yytext); return NUM;
41  /* identifier */
42  {IDENTIFIER}  { get_location (yytext); yylval.ID = g_strdup(yytext);
43                list = g_slist_prepend (list, yylval.ID);

```

```

44         return ID;
45     }
46     "=" get_location (yytext); return '=';
47     ">" get_location (yytext); return '>';
48     "<" get_location (yytext); return '<';
49     "+" get_location (yytext); return '+';
50     "-" get_location (yytext); return '-';
51     "*" get_location (yytext); return '*';
52     "/" get_location (yytext); return '/';
53     "(" get_location (yytext); return '(';
54     ")" get_location (yytext); return ')';
55     "{" get_location (yytext); return '{';
56     "}" get_location (yytext); return '}';
57     "," get_location (yytext); return ',';
58     "." ncolumn++; return YYUNDEF;
59 %%
60
61 static void
62 get_location (char *text) {
63     yylloc.first_line = yylloc.last_line = nline;
64     yylloc.first_column = ncolumn;
65     yylloc.last_column = (ncolumn += strlen(text)) - 1;
66 }
67
68 static YY_BUFFER_STATE state;
69
70 void
71 init_flex (const char *text) {
72     state = yy_scan_string (text);
73 }
74
75 void
76 finalize_flex (void) {
77     yy_delete_buffer (state);
78 }

```

The file consists of three sections which are separated by “%%” (line 19 and 59). They are definitions, rules and user code sections.

3.4.5.2 Definitions Section

- 1-12: Lines between “%top{” and “}” are C source codes. They will be copied to the top of the generated C source file.
- 2-3: This program uses two functions `strlen` (l.65) and `atoi` (l.40). They are defined in `string.h` and `stdlib.h` respectively. These two header files are included here.
- 4: This program uses some GLib functions and structures like `g_strdup` and `GSList`. GLib header file is `glib.h` and it is included here.
- 5: The header file “turtle_parser.h” is generated from “turtle.y” by bison. It defines some constants and functions like `PU` and `yylloc`. The header file is included here.
- 7-9: The current input position is pointed to by `nline` and `ncolumn`. The function `get_location` is declared here so that it can be called before the function is defined (l.61-65).
- 12: `GSList` is a structure for a singly-linked list. The variable `list` is defined in `turtle.y` so its class is `extern`. It is the start point of the list. The list is used to keep allocated memories.
- 15: This option `%option noyywrap` must be specified when you have only single source file to the

scanner. Refer to “9 The Generated Scanner” in the flex documentation in your distribution.

- 17-18: `REAL_NUMBER` and `IDENTIFIER` are names. A name begins with a letter or an underscore followed by zero or more letters, digits, underscores (`_`) or dashes (`-`). They are followed by regular expressions which are their definitions. They will be used in rules section and will expand to the definition.

3.4.5.3 Rules Section

This section is the most important part. Rules consist of patterns and actions. The patterns are regular expressions or names surrounded by braces. The names must be defined in the definitions section. The definition of the regular expression is written in the flex documentation.

For example, line 40 is a rule.

- `{REAL_NUMBER}` is a pattern
- `get_location (yytext); yylval.NUM = atof (yytext); return NUM;` is an action.

`{REAL_NUMBER}` is defined in the line 17, so it expands to `(0|[1-9][0-9]*)(\.[0-9]+)?`. This regular expression matches numbers like 0, 12 and 1.5. If an input is a number, it matches the pattern in line 40. Then the matched text is assigned to `yytext` and corresponding action is executed. A function `get_location` changes the location variables to the position at the text. It assigns `atof (yytext)`, which is double sized number converted from `yytext`, to `yylval.NUM` and return `NUM`. `NUM` is a token kind and it represents (double type) numbers. It is defined in `turtle.y`.

The scanner generated by flex has `yylex` function. If `yylex` is called and the input is “123.4”, then it works as follows.

1. A string “123.4” matches `{REAL_NUMBER}`.
2. Updates the location variable `ncolumn`. The structure `yylloc` is set by `get_location`.
3. The function `atof` converts the string “123.4” to double type number 123.4.
4. It is assigned to `yylval.NUM`.
5. `yylex` returns `NUM` to the caller.

Then the caller knows the input is a number (`NUM`), and its value is 123.4.

- 20-58: Rules section.
- 21: The symbol `.` (dot) matches any character except newline. Therefore, a comment begins `#` followed by any characters except newline. No action happens. That means that comments are ignored.
- 22: White space just increases the variable `ncolumn` by one.
- 23: Tab is assumed to be equal to eight spaces.
- 24: New line increases a variable `nline` by one and resets `ncolumn`.
- 26-38: Keywords updates the location variables `ncolumn` and `yylloc`, and returns the token kinds of the keywords.
- 40: Real number constant. The action converts the text `yytext` to a double type number, puts it into `yylval.NUM` and returns `NUM`.
- 42: `IDENTIFIER` is defined in line 18. The identifier is a name of variable or procedure. It begins with a letter and followed by letters or digits. The location variables are updated and the name of the identifier is assigned to `yylval.ID`. The memory of the name is allocated by the function `g_strdup`. The memory is registered to the list (GList type list). The memory will be freed after the runtime routine finishes. A token kind `ID` is returned.
- 46-57: Symbols update the location variable and return the token kinds. The token kind is the same as the symbol itself.
- 58: If the input doesn’t match the patterns, then it is an error. A special token kind `YYUNDEF` is returned.

3.4.5.4 User Code Section

This section is just copied to C source file.

- 61-66: A function `get_location`. The location of the input is recorded to `nline` and `ncolumn`. A variable `yylloc` is referred by the parser. It is a C structure and has four members, `first_line`, `first_column`, `last_line` and `last_column`. They point to the start and end of the current input text.
- 68: `YY_BUFFER_STATE` is a pointer points to the input buffer. Flex makes the definition of `YY_BUFFER_STATE` in the C file (scanner source file `turtle_lex.c`). See your flex document, Section 11 Multiple Input Buffers, for further information.
- 70-73: A function `init_flex` is called by `run_cb` which is a “clicked” signal handler on the Run button. It has one string type parameter. The caller assigns it with the content of the `GtkTextBuffer` instance. A function `yy_scan_string` sets the input buffer for the scanner.
- 75-78: A function `finalize_flex` is called after runtime routine finishes. It deletes the input buffer.

3.4.6 Turtle.y

Turtle.y has more than 800 lines so it is difficult to explain all the source code. So I will explain the key points and leave out other less important parts.

3.4.6.1 What Does Bison Do?

Bison creates C source file of a parser from a bison source file. The bison source file is a text file. A parser analyzes a program source code according to its grammar. Suppose here is a turtle source file.

```
fc (1,0,0) # Foreground color is red, rgb = (1,0,0).
pd        # Pen down.
distance = 100
angle = 90
fd distance    # Go forward by distance (100) pixels.
tr angle      # Turn right by angle (90) degrees.
```

The parser calls `yylex` to get a token. The token consists of its type (token kind) and value (semantic value). So, the parser gets items in the following table whenever it calls `yylex`.

	token kind	yylval.ID	yylval.NUM
1	FC		
2	(		
3	NUM		1.0
4	,		
5	NUM		0.0
6	,		
7	NUM		0.0
8	)		
9	PD		
10	ID	distance	
11	=		
12	NUM		100.0
13	ID	angle	
14	=		
15	NUM		90.0
16	FD		
17	ID	distance	
18	TR		

	token kind	yylval.ID	yylval.NUM
19	ID	angle	

Bison source code specifies the grammar rules of the turtle language. For example, `fc (1,0,0)` is called primary procedure. A procedure is like a void type C function. It doesn't return any values. Programmers can define their own procedures. On the other hand, `fc` is a built-in procedure. Such procedures are called primary procedures. It is described in bison source code like:

```
primary_procedure: FC '(' expression ',' expression ',' expression ')';
expression: ID | NUM;
```

This means:

- Primary procedure is FC followed by '(', expression, ',', expression, ',', expression and ')'.
- expression is ID or NUM.

The description above is called BNF (Backus-Naur form). Precisely speaking, it is not exactly the same as BNF. But the difference is small.

The first line is:

```
FC '(' NUM ',' NUM ',' NUM ')';
```

The parser analyzes the turtle source code and if the input matches the definition above, the parser recognizes it as a primary procedure.

The grammar of turtle is described in the [Turtle manual](#). The following is an extract from the document.

```
program:
    statement
| program statement
;

statement:
    primary_procedure
| procedure_definition
;

primary_procedure:
    PU
| PD
| PW expression
| FD expression
| TR expression
| TL expression
| BC '(' expression ',' expression ',' expression ')'
| FC '(' expression ',' expression ',' expression ')'
| ID '=' expression
| IF '(' expression ')' '{' primary_procedure_list '}'
| RT
| RS
| RP '(' expression ')' '{' primary_procedure_list '}'
| ID '(' ')'
| ID '(' argument_list ')'
;

procedure_definition:
```

```

    DP ID '(' ')' '{' primary_procedure_list '}'
| DP ID '(' parameter_list ')' '{' primary_procedure_list '}'
;

parameter_list:
    ID
| parameter_list ',' ID
;

argument_list:
    expression
| argument_list ',' expression
;

primary_procedure_list:
    primary_procedure
| primary_procedure_list primary_procedure
;

expression:
    expression '=' expression
| expression '>' expression
| expression '<' expression
| expression '+' expression
| expression '-' expression
| expression '*' expression
| expression '/' expression
| '-' expression %prec UMINUS
| '(' expression ')'
| ID
| NUM
;

```

The grammar rule defines **program** first.

- **program** is a statement or a program followed by a statement.

The definition is recursive.

- **statement** is **program**.
- **statement statement** is **program statement**. Therefore, it is **program**.
- **statement statement statement** is **program statement** because the first two statements are **program**. Therefore, it is **program**.

You can find that a sequence of statements is **program** as well.

The symbols **program** and **statement** aren't tokens. They don't appear in the input. They are called non terminal symbols. On the other hand, tokens are called terminal symbols. The word "token" used here has wide meaning, it includes tokens and symbols which appear in the input. Non terminal symbols are often shortened to nterm.

Let's analyze the program above as bison does.

	token kind	yylval.ID	yylval.NUM	parse	S/R
1	FC			FC	S
2	(			FC(	S

	token kind	yylval.ID	yylval.NUM	parse	S/R
3	NUM		1.0	FC(NUM	S
				FC(expression	R
4	,			FC(expression,	S
5	NUM		0.0	FC(expression,NUM	S
				FC(expression,expression	R
6	,			FC(expression,expression,	S
7	NUM		0.0	FC(expression,expression,NUM	S
				FC(expression,expression,expression	R
8	)			FC(expression,expression,expression)	S
				primary_procedure	R
				statement	R
				program	R
9	PD			program PD	S
				program primary_procedure	R
				program statement	R
				program	R
10	ID	distance		program ID	S
11	=			program ID=	S
12	NUM		100.0	program ID=NUM	S
				program ID=expression	R
				program primary_procedure	R
				program statement	R
				program	R
13	ID	angle		program ID	S
14	=			program ID=	S
15	NUM		90.0	program ID=NUM	S
				program ID=expression	R
				program primary_procedure	R
				program statement	R
				program	R
16	FD			program FD	S
17	ID	distance		program FD ID	S
				program FD expression	R
				program primary_procedure	R
				program statement	R
				program	R
18	TR			program TR	S
19	ID	angle		program TR ID	S
				program TR expression	R
				program primary_procedure	R
				program statement	R
				program	R

The right most column shows shift/reduce. Shift is appending an input to the buffer. Reduce is substituting a higher nterm for the pattern in the buffer. For example, NUM is replaced by expression in the forth row. This substitution is “reduce”.

Bison repeats shift and reduction until the end of the input. If the result is reduced to **program**, the input is syntactically valid. Bison executes an action whenever a reduction occurs. Actions build a tree. The tree is analyzed and executed by runtime routine later.

Bison source files are called bison grammar files. A bison grammar file consists of four sections, prologue,

declarations, rules and epilogue. The format is as follows.

```
%{
prologue
%}
declarations
%%
rules
%%
epilogue
```

3.4.6.2 Prologue

Prologue section consists of C codes and the codes are copied to the parser implementation file. You can use `%code` directives to qualify the prologue and identifies the purpose explicitly. The following is an extract from `turtle.y`.

```
%code top{
    #include <stdarg.h>
    #include <setjmp.h>
    #include <math.h>
    #include <glib.h>
    #include <cairo.h>
    #include "turtle_parser.h"

    /* The following line defines 'debug' so that debug information is printed out during the run time
    /* However it makes the program slow. */
    /* If you want to debug on, uncomment the line. */

    /* #define debug 1 */

    extern cairo_surface_t *surface;

    /* error reporting */
    static void yyerror (char const *s) { /* for syntax error */
        g_printerr ("%s from line %d, column %d to line %d, column %d\n",s, yylloc.first_line, yylloc.fi
    }
    /* Node type */
    enum {
        N_PU,
        N_PD,
        N_PW,
        ... ..
    };
}
```

The directive `%code top` copies its contents to the top of the parser implementation file. It usually includes `#include` directives, declarations of functions and definitions of constants. A function `yyerror` reports a syntax error and is called by the parser. Node type identifies a node in the tree.

Another directive `%code requires` copies its contents to both the parser implementation file and header file. The header file is read by the scanner C source file and other files.

```
%code requires {
    int yylex (void);
    int yyparse (void);
```

```

void run (void);

/* semantic value type */
typedef struct _node_t node_t;
struct _node_t {
    int type;
    union {
        struct {
            node_t *child1, *child2, *child3;
        } child;
        char *name;
        double value;
    } content;
};
}

```

- `yylex` is shared by the parser implementation file and scanner file.
- `yyparse` and `run` is called by `run_cb` in `turtleapplication.c`.
- `node_t` is the type of the semantic value of nterms. The header file defines `YYSTYPE`, which is the semantic value type, with all the token and nterm value types. The following is extracted from the header file.

```

/* Value type. */
#if ! defined YYSTYPE && ! defined YYSTYPE_IS_DECLARED
union YYSTYPE
{
    char * ID; /* ID */
    double NUM; /* NUM */
    node_t * program; /* program */
    node_t * statement; /* statement */
    node_t * primary_procedure; /* primary_procedure */
    node_t * primary_procedure_list; /* primary_procedure_list */
    node_t * procedure_definition; /* procedure_definition */
    node_t * parameter_list; /* parameter_list */
    node_t * argument_list; /* argument_list */
    node_t * expression; /* expression */
};

```

Other useful macros and declarations are put into the `%code` directive.

```

%code {
/* The following macro is convenient to get the member of the node. */
#define child1(n) (n)->content.child.child1
#define child2(n) (n)->content.child.child2
#define child3(n) (n)->content.child.child3
#define name(n) (n)->content.name
#define value(n) (n)->content.value

/* start of nodes */
static node_t *node_top = NULL;
/* functions to generate trees */
static node_t *tree1 (int type, node_t *child1, node_t *child2, node_t *child3);
static node_t *tree2 (int type, double value);
static node_t *tree3 (int type, char *name);
}

```

3.4.6.3 Bison declarations

Bison declarations defines terminal and non-terminal symbols. It also specifies some directives.

```
%locations
#define api.value.type union /* YYSTYPE, the type of semantic values, is union of following types */
/* key words */
%token PU
%token PD
%token PW
%token FD
%token TR
%token TL /* ver 0.5 */
%token BC
%token FC
%token DP
%token IF
%token RT
%token RS
%token RP /* ver 0.5 */
/* constant */
%token <double> NUM
/* identifier */
%token <char *> ID
/* non terminal symbol */
%nterm <node_t *> program
%nterm <node_t *> statement
%nterm <node_t *> primary_procedure
%nterm <node_t *> primary_procedure_list
%nterm <node_t *> procedure_definition
%nterm <node_t *> parameter_list
%nterm <node_t *> argument_list
%nterm <node_t *> expression
/* logical relation symbol */
%left '=' '<' '>'
/* arithmetic symbol */
%left '+' '-'
%left '*' '/'
%precedence UMINUS /* unary minus */
```

%locations directive inserts the location structure into the header file. It is like this.

```
typedef struct YYLTYPE YYLTYPE;
struct YYLTYPE
{
    int first_line;
    int first_column;
    int last_line;
    int last_column;
};
```

This type is shared by the scanner file and the parser implementation file. The error report function `yyerror` uses it so that it can inform the location that error occurs.

%define api.value.type union generates semantic value type with tokens and nterms and inserts it to the header file. The inserted part is shown in the previous subsection as the extracts that show the value

type (YYSTYPE).

%token and %nterm directives define tokens and non terminal symbols respectively.

```
%token PU
... ..
%token <double> NUM
```

These directives define a token PU and NUM. The values of token kinds PU and NUM are defined as an enumeration constant in the header file.

```
enum yytokentype
{
    ... ..
    PU = 258,                /* PU */
    ... ..
    NUM = 269,              /* NUM */
    ... ..
};
typedef enum yytokentype yytoken_kind_t;
```

In addition, the type of the semantic value of NUM is defined as double in the header file because of <double> tag.

```
union YYSTYPE
{
    char * ID;                /* ID */
    double NUM;              /* NUM */
    ... ..
}
```

All the nterm symbols have the same type `*node_t` of the semantic value.

%left and %precedence directives define the precedence of operation symbols.

```
/* logical relation symbol */
%left '=' '<' '>'
/* arithmetic symbol */
%left '+' '-'
%left '*' '/'
%precedence UMINUS /* unary minus */
```

%left directive defines the following symbols as left-associated operators. If an operator + is left-associated, then

$$A + B + C = (A + B) + C$$

That is, the calculation is carried out the left operator first, then the right operator. If an operator * is right-associated, then:

$$A * B * C = A * (B * C)$$

The definition above decides the behavior of the parser. Addition and multiplication hold associative law so the result of $(A+B)+C$ and $A+(B+C)$ are equal in terms of mathematics. However, the parser will be confused if left (or right) associativity is not specified.

%left and %precedence directives show the precedence of operators. Later declared operators have higher precedence than former declared ones. The declaration above says, for example,

`v=w+z*5+7` is the same as `v=((w+(z*5))+7)`

Be careful. The operator = above is an assignment. Assignment is not expression in turtle language. It is `primary_procedure`. But if = appears in an expression, it is a logical operator, not an assignment. The logical equal '=' usually used in the conditional expression, for example, in `if` statement. (Turtle language uses '=' instead of '==' in C language).

3.4.6.4 Grammar rules

Grammar rules section defines the syntactic grammar of the language. It is similar to BNF form.

`result: components { action };`

- `result` is a nterm.
- `components` are list of tokens or nterms.
- `action` is C codes. It is executed whenever the components are reduced to the result. Action can be left out.

The following is a part of the grammar rule in `turtle.y`. But it is not exactly the same.

```
program:
    statement { node_top = $$ = $1; }
;
statement:
    primary_procedure
;
primary_procedure:
    FD expression    { $$ = tree1 (N_FD, $2, NULL, NULL); }
;
expression:
    NUM    { $$ = tree2 (N_NUM, $1); }
;
```

- The first two lines tell that `program` is `statement`.
- Whenever `statement` is reduced to `program`, an action `node_top=$$=$1;` is executed.
- `node_top` is a static variable. It points to the top node of the tree.
- The symbol `$$` is a semantic value of the result. For example, `$$` in line 2 is the semantic value of `program`. It is a pointer to a `node_t` type structure.
- The symbol `$1` is a semantic value of the first component. For example, `$1` in line 2 is the semantic value of `statement`. It is also a pointer to `node_t`.
- The next rule is that `statement` is `primary_procedure`. There's no action specified. Then, the default action `$$ = $1` is executed.
- The next rule is that `primary_procedure` is `FD` followed by `expression`. The action calls `tree1` and assigns its return value to `$$`. The function `tree1` makes a tree node. The tree node has type and union of three pointers to children nodes, string or double.

```
node ---- type
    +--- union contents
            +---struct {node_t *child1, *child2, *child3;};
            +---char *name
            +---double value
```

- `tree1` assigns the four arguments to type, child1, child2 and child3 members.
- The last rule is that `expression` is `NUM`.
- `tree2` makes a tree node. The parameters of `tree2` are a type and a semantic value.

Suppose the parser reads the following program.

```
fd 100
```

What does the parser do?

1. The parser recognizes the input is `FD`. Maybe it is the start of `primary_procedure`, but parser needs to read the next token.
2. `yylex` returns the token kind `NUM` and sets `yylval.NUM` to 100.0 (the type is double). The parser reduces `NUM` to `expression`. At the same time, it sets the semantic value of the `expression` to point to a new node. The node has the type `N_NUM` and a semantic value 100.0.
3. After the reduction, the buffer has `FD` and `expression`. The parser reduces it to `primary_procedure`. And it sets the semantic value of the `primary_procedure` to point to a new node. The node has the type `N_FD` and its member `child1` points to the node of `expression`, whose type is `N_NUM`.
4. The parser reduces `primary_procedure` to `statement`. The semantic value of `statement` is the same as the one of `primary_procedure`, which points to the node `N_FD`.
5. The parser reduces `statement` to `program`. The semantic value of `statement` is assigned to the one of `program` and the static variable `node_top`.
6. Finally `node_top` points to the node `N_FD` and the node `N_FD` points to the node `N_NUM`.

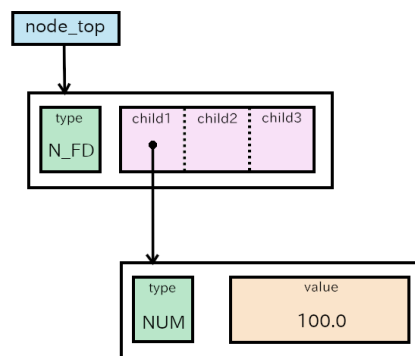


Figure 3.13: tree

The following is the grammar rule extracted from `turtle.y`. The rules there are based on the same idea above. I don't want to explain the whole rules below. Please look into each line carefully so that you will understand all the rules and actions.

```

program:
    statement { node_top = $$ = $1; }
| program statement {
    node_top = $$ = tree1 (N_program, $1, $2, NULL);
#ifdef debug
if (node_top == NULL) g_printerr("program: node_top is NULL.\n"); else g_printerr ("program: node_top is %s\n", node_top->type);
#endif
    }
;

statement:
    primary_procedure
| procedure_definition
;

primary_procedure:
    PU    { $$ = tree1 (N_PU, NULL, NULL, NULL); }
| PD    { $$ = tree1 (N_PD, NULL, NULL, NULL); }
| PW expression    { $$ = tree1 (N_PW, $2, NULL, NULL); }
| FD expression    { $$ = tree1 (N_FD, $2, NULL, NULL); }
| TR expression    { $$ = tree1 (N_TR, $2, NULL, NULL); }

```

```

| TL expression    { $$ = tree1 (N_TL, $2, NULL, NULL); } /* ver 0.5 */
| BC '(' expression ',' expression ',' expression ')' { $$ = tree1 (N_BC, $3, $5, $7); }
| FC '(' expression ',' expression ',' expression ')' { $$ = tree1 (N_FC, $3, $5, $7); }
/* assignment */
| ID '=' expression { $$ = tree1 (N_ASSIGN, tree3 (N_ID, $1), $3, NULL); }
/* control flow */
| IF '(' expression ')' '{' primary_procedure_list '}' { $$ = tree1 (N_IF, $3, $6, NULL); }
| RT { $$ = tree1 (N_RT, NULL, NULL, NULL); }
| RS { $$ = tree1 (N_RS, NULL, NULL, NULL); }
| RP '(' expression ')' '{' primary_procedure_list '}' { $$ = tree1 (N_RP, $3, $6, NULL); }
/* user defined procedure call */
| ID '(' ')' { $$ = tree1 (N_procedure_call, tree3 (N_ID, $1), NULL, NULL); }
| ID '(' argument_list ')' { $$ = tree1 (N_procedure_call, tree3 (N_ID, $1), $3, NULL); }
;

procedure_definition:
    DP ID '(' ')' '{' primary_procedure_list '}' {
        $$ = tree1 (N_procedure_definition, tree3 (N_ID, $2), NULL, $6);
    }
| DP ID '(' parameter_list ')' '{' primary_procedure_list '}' {
    $$ = tree1 (N_procedure_definition, tree3 (N_ID, $2), $4, $7);
}
;

parameter_list:
    ID { $$ = tree3 (N_ID, $1); }
| parameter_list ',' ID { $$ = tree1 (N_parameter_list, $1, tree3 (N_ID, $3), NULL); }
;

argument_list:
    expression
| argument_list ',' expression { $$ = tree1 (N_argument_list, $1, $3, NULL); }
;

primary_procedure_list:
    primary_procedure
| primary_procedure_list primary_procedure {
    $$ = tree1 (N_primary_procedure_list, $1, $2, NULL);
}
;

expression:
    expression '=' expression { $$ = tree1 (N_EQ, $1, $3, NULL); }
| expression '>' expression { $$ = tree1 (N_GT, $1, $3, NULL); }
| expression '<' expression { $$ = tree1 (N_LT, $1, $3, NULL); }
| expression '+' expression { $$ = tree1 (N_ADD, $1, $3, NULL); }
| expression '-' expression { $$ = tree1 (N_SUB, $1, $3, NULL); }
| expression '*' expression { $$ = tree1 (N_MUL, $1, $3, NULL); }
| expression '/' expression { $$ = tree1 (N_DIV, $1, $3, NULL); }
| '-' expression %prec UMINUS { $$ = tree1 (N_UMINUS, $2, NULL, NULL); }
| '(' expression ')' { $$ = $2; }
| ID { $$ = tree3 (N_ID, $1); }
| NUM { $$ = tree2 (N_NUM, $1); }

```

```
;
```

3.4.6.5 Epilogue

The epilogue is written in C language and copied to the parser implementation file. Generally, you can put anything into the epilogue. In the case of turtle interpreter, the runtime routine and some other functions are in the epilogue.

3.4.6.5.1 Functions to create tree nodes There are three functions, `tree1`, `tree2` and `tree3`.

- `tree1` creates a node and sets the node type and pointers to its three children (NULL is possible).
- `tree2` creates a node and sets the node type and a value (double).
- `tree3` creates a node and sets the node type and a pointer to a string.

Each function gets memories first and build a node on them. The memories are inserted to the list. They will be freed when runtime routine finishes.

The three functions are called in the actions in the rules section.

/ Dynamically allocated memories are added to the single list. They will be freed in the finalize f*
`GSList *list = NULL;`

```
node_t *
tree1 (int type, node_t *child1, node_t *child2, node_t *child3) {
    node_t *new_node;

    list = g_slist_prepend (list, g_malloc (sizeof (node_t)));
    new_node = (node_t *) list->data;
    new_node->type = type;
    child1(new_node) = child1;
    child2(new_node) = child2;
    child3(new_node) = child3;
    return new_node;
}
```

```
node_t *
tree2 (int type, double value) {
    node_t *new_node;

    list = g_slist_prepend (list, g_malloc (sizeof (node_t)));
    new_node = (node_t *) list->data;
    new_node->type = type;
    value(new_node) = value;
    return new_node;
}
```

```
node_t *
tree3 (int type, char *name) {
    node_t *new_node;

    list = g_slist_prepend (list, g_malloc (sizeof (node_t)));
    new_node = (node_t *) list->data;
    new_node->type = type;
    name(new_node) = name;
    return new_node;
}
```

3.4.6.5.2 Symbol Table Variables and user defined procedures are registered in the symbol table. This table is a C array. It should be replaced by better algorithm and data structure, for example hash, in the future version

- Variables are registered with its name and value.
- Procedures are registered with its name and a pointer to the node of the procedure.

Therefore the table has the following fields.

- type to identify variable or procedure
- name
- value or pointer to a node

```
#define MAX_TABLE_SIZE 100
enum {
    PROC,
    VAR
};

struct {
    int type;
    char *name;
    union {
        node_t *node;
        double value;
    } object;
} table[MAX_TABLE_SIZE];
int tp;

void
init_table (void) {
    tp = 0;
}
```

The function `init_table` initializes the table. This must be called before registrations.

There are five functions to access the table,

- `proc_install` installs a procedure.
- `var_install` installs a variable.
- `proc_lookup` looks up a procedure. If the procedure is found, it returns a pointer to the node. Otherwise it returns NULL.
- `var_lookup` looks up a variable. If the variable is found, it returns TRUE and sets the pointer (argument) to point to the value. Otherwise it returns FALSE.
- `var_replace` replaces the value of a variable. If the variable hasn't registered yet, it installs the variable.

```
int
tbl_lookup (int type, char *name) {
    int i;

    if (tp == 0)
        return -1;
    for (i=0; i<tp; ++i)
        if (type == table[i].type && strcmp(name, table[i].name) == 0)
            return i;
    return -1;
}
```

```

}

void
tbl_install (int type, char *name, node_t *node, double value) {
    if (tp >= MAX_TABLE_SIZE)
        runtime_error ("Symbol table overflow.\n");
    else if (tbl_lookup (type, name) >= 0)
        runtime_error ("Name %s is already registered.\n", name);
    else {
        table[tp].type = type;
        table[tp].name = name;
        if (type == PROC)
            table[tp++].object.node = node;
        else
            table[tp++].object.value = value;
    }
}

void
proc_install (char *name, node_t *node) {
    tbl_install (PROC, name, node, 0.0);
}

void
var_install (char *name, double value) {
    tbl_install (VAR, name, NULL, value);
}

void
var_replace (char *name, double value) {
    int i;
    if ((i = tbl_lookup (VAR, name)) >= 0)
        table[i].object.value = value;
    else
        var_install (name, value);
}

node_t *
proc_lookup (char *name) {
    int i;
    if ((i = tbl_lookup (PROC, name)) < 0)
        return NULL;
    else
        return table[i].object.node;
}

gboolean
var_lookup (char *name, double *value) {
    int i;
    if ((i = tbl_lookup (VAR, name)) < 0)
        return FALSE;
    else {
        *value = table[i].object.value;
    }
}

```

```

    return TRUE;
}
}

```

3.4.6.5.3 Stack for Parameters and Arguments Stack is a last-in first-out data structure. It is shortened to LIFO. Turtle uses a stack to keep parameters and arguments. They are like `auto` class variables in C language. They are pushed to the stack whenever the procedure is called. LIFO structure is useful for recursive calls.

Each element of the stack has name and value.

```

#define MAX_STACK_SIZE 500
struct {
    char *name;
    double value;
} stack[MAX_STACK_SIZE];
int sp, sp_biggest;

void
init_stack (void) {
    sp = sp_biggest = 0;
}

```

`sp` is a stack pointer. It is an index of the array `stack` and it always points to an element of the array to store the next data. `sp_biggest` is the biggest number assigned to `sp`. We can know the amount of elements used in the array during the runtime. The purpose of the variable is to find appropriate `MAX_STACK_SIZE`. It will be unnecessary in the future version if the stack is implemented with better data structure and memory allocation.

The runtime routine push data to the stack when it executes a node of a procedure call. (The type of the node is `N_procedure_call`.)

```

dp drawline (angle, distance) { ... .. }
drawline (90, 100)

```

- The first line defines a procedure `drawline`. The runtime routine stores the name `drawline` and the node of the procedure to the symbol table.
- The second line calls the procedure. First, it looks for the procedure in the symbol table and gets its node. Then it searches the node for the parameters and gets `angle` and `distance`.
- It pushes (“angle”, 90.0) to the stack.
- It pushes (“distance”, 100.0) to the stack.
- It pushes (NULL, 2.0) to the stack. The number 2.0 is the number of parameters (or arguments). It is used when the procedure returns.

The following diagram shows the structure of the stack. First, **procedure 1** is called. The procedure has two parameters. In the **procedure 1**, another procedure **procedure 2** is called. It has one parameter. In the **procedure 2**, another procedure **procedure 3** is called. It has three parameters. These three procedures are nested.

Programs push data to a stack from a low address memory to a high address memory. In the following diagram, the lowest address is at the top and the highest address is at the bottom. That is the order of the address. However, “the top of the stack” is the last pushed data and “the bottom of the stack” is the first pushed data. Therefore, “the top of the stack” is the bottom of the rectangle in the diagram and “the bottom of the stack” is the top of the rectangle.

There are four functions to access the stack.

- `stack_push` pushes data to the stack.

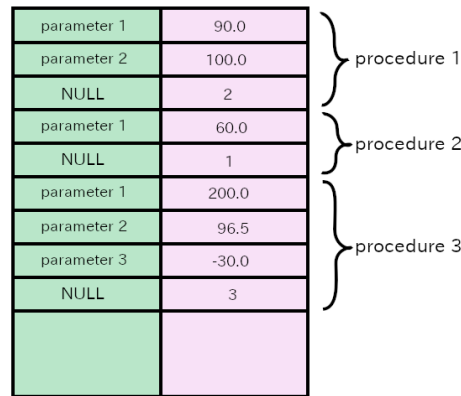


Figure 3.14: Stack

- `stack_lookup` searches the stack for the variable given its name as an argument. It searches only the parameters of the latest procedure. It returns TRUE and sets the argument `value` to point to the value, if the variable has been found. Otherwise it returns FALSE.
- `stack_replace` replaces the value of the variable in the stack. If it succeeds, it returns TRUE. Otherwise returns FALSE.
- `stack_return` throws away the latest parameters. The stack pointer goes back to the point before the latest procedure call so that it points to parameters of the previously called procedure.

```
void
stack_push (char *name, double value) {
    if (sp >= MAX_STACK_SIZE)
        runtime_error ("Stack overflow.\n");
    else {
        stack[sp].name = name;
        stack[sp++].value = value;
        sp_biggest = sp > sp_biggest ? sp : sp_biggest;
    }
}
```

```
int
stack_search (char *name) {
    int depth, i;

    if (sp == 0)
        return -1;
    depth = (int) stack[sp-1].value;
    if (depth + 1 > sp) /* something strange */
        runtime_error ("Stack error.\n");
    for (i=0; i<depth; ++i)
        if (strcmp(name, stack[sp-(i+2)].name) == 0) {
            return sp-(i+2);
        }
    return -1;
}
```

```
gboolean
stack_lookup (char *name, double *value) {
    int i;
```



```

    if ((i = stack_search (name)) < 0)
        return FALSE;
    else {
        *value = stack[i].value;
        return TRUE;
    }
}

gboolean
stack_replace (char *name, double value) {
    int i;

    if ((i = stack_search (name)) < 0)
        return FALSE;
    else {
        stack[i].value = value;
        return TRUE;
    }
}

void
stack_return(void) {
    int depth;

    if (sp <= 0)
        return;
    depth = (int) stack[sp-1].value;
    if (depth + 1 > sp) /* something strange */
        runtime_error ("Stack error.\n");
    sp -= depth + 1;
}

```

3.4.6.5.4 Surface and Cairo A global variable `surface` is shared by `turtleapplication.c` and `turtle.y`. It is initialized in `turtleapplication.c`.

The runtime routine has its own cairo context. This is different from the cairo in the `GtkDrawingArea` instance. The runtime routine draws a shape on the `surface` with the cairo context. After runtime routine returns to `run_cb`, `run_cb` adds the `GtkDrawingArea` widget to the queue to redraw. When the widget is redraw, the drawing function `draw_func` is called. It copies the `surface` to the surface in the `GtkDrawingArea` object.

`turtle.y` has two functions `init_cairo` and `destroy_cairo`.

- `init_cairo` initializes static variables and cairo context. The variables keep pen status (up or down), direction, initial location, line width and color. The size of the `surface` changes according to the size of the window. Whenever a user drags and resizes the window, the `surface` is also resized. `init_cairo` gets the size first and sets the initial location of the turtle (center of the surface) and the transformation matrix.
- `destroy_cairo` just destroys the cairo context.

Turtle has its own coordinate. The origin is at the center of the surface, and positive direction of x and y axes are right and up respectively. But surfaces have its own coordinate. Its origin is at the top-left corner of the surface and positive direction of x and y are right and down respectively. A plane with the turtle's coordinate is called user space, which is the same as cairo's user space. A plane with the surface's

coordinate is called device space.

Cairo provides a transformation which is an affine transformation. It transforms a user-space coordinate (x, y) into a device-space coordinate (z, w).

$$\begin{bmatrix} z \\ w \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} p \\ q \end{bmatrix}$$

Figure 3.15: transformation

The function `init_cairo` gets the width and height of the `surface` (See the program below).

- The center of the surface is (0,0) in the user-space coordinate, while it is (width/2, height/2) in the device-space coordinate.
- The positive direction of x axis in the two spaces are the same. So, (1,0) is transformed into (1+width/2,height/2).
- The positive direction of y axis in the two spaces are opposite. So, (0,1) is transformed into (width/2,-1+height/2).

You can determine a, b, c, d, p and q by substituting the numbers above for x, y, z and w in the equation above. The solution of the simultaneous equations is:

a = 1, b = 0, c = 0, d = -1, p = width/2, q = height/2

Cairo provides a structure `cairo_matrix_t`. The function `init_cairo` uses it and sets the cairo transformation (See the program below). Once the matrix is set, the transformation always performs whenever `cairo_stroke` function is invoked.

```
/* status of the surface */
static gboolean pen = TRUE;
static double angle = 90.0; /* angle starts from x axis and measured counterclockwise */
/* Initially facing to the north */
static double cur_x = 0.0;
static double cur_y = 0.0;
static double line_width = 2.0;

struct color {
    double red;
    double green;
    double blue;
};

static struct color bc = {0.95, 0.95, 0.95}; /* white */
static struct color fc = {0.0, 0.0, 0.0}; /* black */

/* cairo */
static cairo_t *cr;
gboolean
init_cairo (void) {
    int width, height;
    cairo_matrix_t matrix;

    pen = TRUE;
    angle = 90.0;
    cur_x = 0.0;
```

```

cur_y = 0.0;
line_width = 2.0;
bc.red = 0.95; bc.green = 0.95; bc.blue = 0.95;
fc.red = 0.0; fc.green = 0.0; fc.blue = 0.0;

if (surface) {
    width = cairo_image_surface_get_width (surface);
    height = cairo_image_surface_get_height (surface);
    matrix.xx = 1.0; matrix.xy = 0.0; matrix.x0 = (double) width / 2.0;
    matrix.yx = 0.0; matrix.yy = -1.0; matrix.y0 = (double) height / 2.0;

    cr = cairo_create (surface);
    cairo_transform (cr, &matrix);
    cairo_set_source_rgb (cr, bc.red, bc.green, bc.blue);
    cairo_paint (cr);
    cairo_set_source_rgb (cr, fc.red, fc.green, fc.blue);
    cairo_move_to (cr, cur_x, cur_y);
    return TRUE;
} else
    return FALSE;
}

void
destroy_cairo () {
    cairo_destroy (cr);
}

```

3.4.6.5.5 Eval Function A function `eval` evaluates an expression and returns the value of the expression. It calls itself recursively. For example, if the node is `N_ADD`, then:

1. Calls `eval(child1(node))` and gets the value1.
2. Calls `eval(child2(node))` and gets the value2.
3. Returns `value1+value2`.

This is performed by a macro `calc` defined in the sixth line in the following program.

```

double
eval (node_t *node) {
double value = 0.0;
    if (node == NULL)
        runtime_error ("No expression to evaluate.\n");
#define calc(op) eval (child1(node)) op eval (child2(node))
    switch (node->type) {
        case N_EQ:
            value = (double) calc(==);
            break;
        case N_GT:
            value = (double) calc(>);
            break;
        case N_LT:
            value = (double) calc(<);
            break;
        case N_ADD:
            value = calc(+);
            break;

```

```

    case N_SUB:
        value = calc(-);
        break;
    case N_MUL:
        value = calc(*);
        break;
    case N_DIV:
        if (eval (child2(node)) == 0.0)
            runtime_error ("Division by zero.\n");
        else
            value = calc(/);
        break;
    case N_UMINUS:
        value = -(eval (child1(node)));
        break;
    case N_ID:
        if (! (stack_lookup (name(node), &value)) && ! var_lookup (name(node), &value) )
            runtime_error ("Variable %s not defined.\n", name(node));
        break;
    case N_NUM:
        value = value(node);
        break;
    default:
        runtime_error ("Illegal expression.\n");
}
return value;
}

```

3.4.6.5.6 Execute Function Primary procedures and procedure definitions are analyzed and executed by the function `execute`. It doesn't return any values. It calls itself recursively. The process of `N_RT` and `N_procedure_call` is complicated. It will be explained after the following program. Other parts are not so difficult. Read the program below carefully so that you will understand the process.

```

/* procedure - return status */
static int proc_level = 0;
static int ret_level = 0;

void
execute (node_t *node) {
    double d, x, y;
    char *name;
    int n, i;

    if (node == NULL)
        runtime_error ("Node is NULL.\n");
    if (proc_level > ret_level)
        return;
    switch (node->type) {
        case N_program:
            execute (child1(node));
            execute (child2(node));
            break;
        case N_PU:
            pen = FALSE;

```

```

    break;
case N_PD:
    pen = TRUE;
    break;
case N_PW:
    line_width = eval (child1(node)); /* line width */
    break;
case N_FD:
    d = eval (child1(node)); /* distance */
    x = d * cos (angle*M_PI/180);
    y = d * sin (angle*M_PI/180);
    /* initialize the current point = start point of the line */
    cairo_move_to (cr, cur_x, cur_y);
    cur_x += x;
    cur_y += y;
    cairo_set_line_width (cr, line_width);
    cairo_set_source_rgb (cr, fc.red, fc.green, fc.blue);
    if (pen)
        cairo_line_to (cr, cur_x, cur_y);
    else
        cairo_move_to (cr, cur_x, cur_y);
    cairo_stroke (cr);
    break;
case N_TR:
    angle -= eval (child1(node));
    for (; angle < 0; angle += 360.0);
    for (; angle > 360; angle -= 360.0);
    break;
case N_BC:
    bc.red = eval (child1(node));
    bc.green = eval (child2(node));
    bc.blue = eval (child3(node));
#define fixcolor(c)  c = c < 0 ? 0 : (c > 1 ? 1 : c)
    fixcolor (bc.red);
    fixcolor (bc.green);
    fixcolor (bc.blue);
    /* clear the shapes and set the background color */
    cairo_set_source_rgb (cr, bc.red, bc.green, bc.blue);
    cairo_paint (cr);
    break;
case N_FC:
    fc.red = eval (child1(node));
    fc.green = eval (child2(node));
    fc.blue = eval (child3(node));
    fixcolor (fc.red);
    fixcolor (fc.green);
    fixcolor (fc.blue);
    break;
case N_ASSIGN:
    name = name(child1(node));
    d = eval (child2(node));
    if (! stack_replace (name, d)) /* First, tries to replace the value in the stack (parameter). */
        var_replace (name, d); /* If the above fails, tries to replace the value in the table. If th

```

```

    break;
case N_IF:
    if (eval (child1(node)))
        execute (child2(node));
    break;
case N_RT:
    ret_level--;
    break;
case N_RS:
    pen = TRUE;
    angle = 90.0;
    cur_x = 0.0;
    cur_y = 0.0;
    line_width = 2.0;
    fc.red = 0.0; fc.green = 0.0; fc.blue = 0.0;
    /* To change background color, use bc. */
    break;
case N_procedure_call:
    name = name(child1(node));
node_t *proc = proc_lookup (name);
    if (! proc)
        runtime_error ("Procedure %s not defined.\n", name);
    if (strcmp (name, name(child1(proc))) != 0)
        runtime_error ("Unexpected error. Procedure %s is called, but invoked procedure is %s.\n", name, name(child1(proc)));
    /* make tuples (parameter (name), argument (value)) and push them to the stack */
node_t *param_list;
node_t *arg_list;
    param_list = child2(proc);
    arg_list = child2(node);
    if (param_list == NULL) {
        if (arg_list == NULL) {
            stack_push (NULL, 0.0); /* number of argument == 0 */
        } else
            runtime_error ("Procedure %s has different number of argument and parameter.\n", name);
    } else {
        /* Don't change the stack until finish evaluating the arguments. */
#define TEMP_STACK_SIZE 20
        char *temp_param[TEMP_STACK_SIZE];
        double temp_arg[TEMP_STACK_SIZE];
        n = 0;
        for (; param_list->type == N_parameter_list; param_list = child1(param_list)) {
            if (arg_list->type != N_argument_list)
                runtime_error ("Procedure %s has different number of argument and parameter.\n", name);
            if (n >= TEMP_STACK_SIZE)
                runtime_error ("Too many parameters. the number must be %d or less.\n", TEMP_STACK_SIZE);
            temp_param[n] = name(child2(param_list));
            temp_arg[n] = eval (child2(arg_list));
            arg_list = child1(arg_list);
            ++n;
        }
        if (param_list->type == N_ID && arg_list -> type != N_argument_list) {
            temp_param[n] = name(param_list);
            temp_arg[n] = eval (arg_list);

```

```

        if (++n >= TEMP_STACK_SIZE)
            runtime_error ("Too many parameters. the number must be %d or less.\n", TEMP_STACK_SIZE);
        temp_param[n] = NULL;
        temp_arg[n] = (double) n;
        ++n;
    } else
        runtime_error ("Unexpected error.\n");
    for (i = 0; i < n; ++i)
        stack_push (temp_param[i], temp_arg[i]);
}
ret_level = ++proc_level;
execute (child3(proc));
ret_level = --proc_level;
stack_return ();
break;
case N_procedure_definition:
    name = name(child1(node));
    proc_install (name, node);
    break;
case N_primary_procedure_list:
    execute (child1(node));
    execute (child2(node));
    break;
default:
    runtime_error ("Unknown statement.\n");
}
}

```

A node `N_procedure_call` is created by the parser when it has found a user defined procedure call. The procedure has been defined in the prior statement. Suppose the parser reads the following example code.

```

dp drawline (angle, distance) {
    tr angle
    fd distance
}
drawline (90, 100)
drawline (90, 100)
drawline (90, 100)
drawline (90, 100)

```

This example draws a square.

When The parser reads the lines from one to four, it creates nodes like this:

Runtime routine just stores the procedure to the symbol table with its name and node.

When the parser reads the fifth line in the example, it creates nodes like this:

When the runtime routine meets `N_procedure_call` node, it behaves like this:

1. Searches the symbol table for the procedure with the name.
2. Gets pointers to the node to parameters and the node to the body.
3. Creates a temporary stack. Makes a tuple of each parameter name and argument value. Pushes the tuples into the stack, and (NULL, number of parameters) finally. If no error occurs, copies them from the temporary stack to the parameter stack.
4. Increases `prc_level` by one. Sets `ret_level` to the same value as `proc_level`. `proc_level` is zero when runtime routine runs on the main routine. If it goes into a procedure, `proc_level` increases

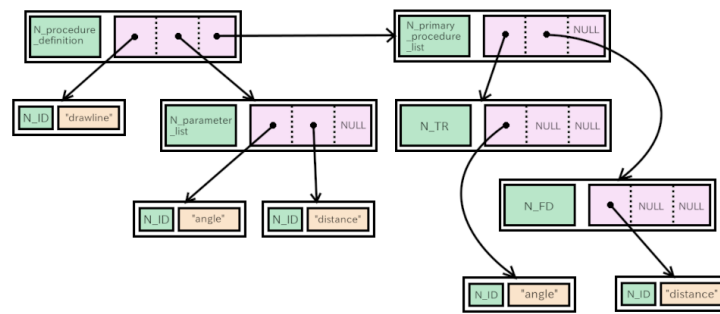


Figure 3.16: Nodes of drawline

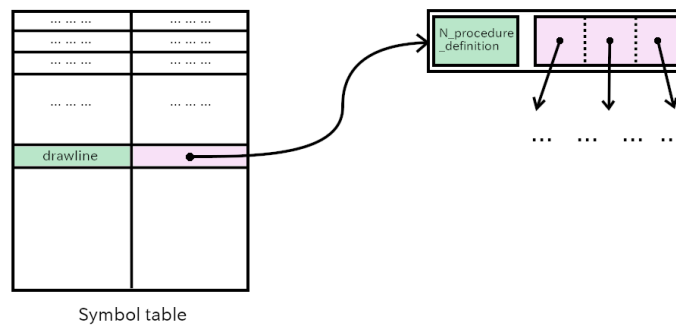


Figure 3.17: Symbol table

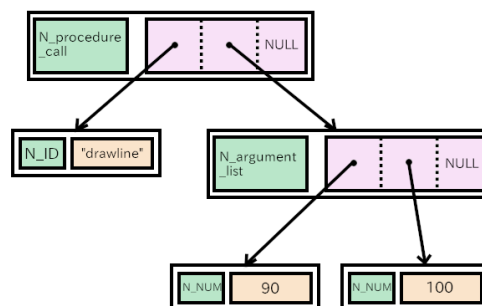


Figure 3.18: Nodes of procedure call

by one. Therefore, `proc_level` is the depth of the procedure call. `ret_level` is the level to return. If it is the same as `proc_level`, runtime routine executes commands in order of the commands in the procedure. If it is smaller than `proc_level`, runtime routine doesn't execute commands until it becomes the same level as `proc_level`. `ret_level` is used to return the procedure.

5. Executes the node of the body of the procedure.
6. Decreases `proc_level` by one. Sets `ret_level` to the same value as `proc_level`. Calls `stack_return`.

When the runtime routine meets `N_RT` node, it decreases `ret_level` by one so that the following commands in the procedure are ignored by the runtime routine.

3.4.6.5.7 Runtime Entry and Error Functions A function `run` is the entry of the runtime routine. A function `runtime_error` reports an error occurred during the runtime routine runs. (Errors which occur during the parsing are called syntax error and reported by `yyerror`.) After `runtime_error` reports an error, it stops the command execution and goes back to `run` to exit.

`Setjmp` and `longjmp` functions are used. They are declared in `<setjmp.h>`. `setjmp (buf)` saves state information in `buf` and returns zero. `longjmp(buf, 1)` restores the state information from `buf` and returns 1 (the second argument). Because the information is the status at the time `setjmp` is called, so `longjmp` resumes the execution at the next of `setjmp` function call. In the following program, `longjmp` resumes at the assignment to the variable `i`. When `setjmp` is called, 0 is assigned to `i` and `execute(node_top)` is called. On the other hand, when `longjmp` is called, 1 is assigned to `i` and `execute(node_top)` is not called..

`g_slist_free_full` frees all the allocated memories.

```
static jmp_buf buf;

void
run (void) {
    int i;

    if (! init_cairo()) {
        g_print ("Cairo not initialized.\n");
        return;
    }
    init_table();
    init_stack();
    ret_level = proc_level = 1;
    i = setjmp (buf);
    if (i == 0)
        execute(node_top);
    /* else ... get here by calling longjmp */
    destroy_cairo ();
    g_slist_free_full (g_steal_pointer (&list), g_free);
}

/* format supports only %s, %f and %d */
static void
runtime_error (char *format, ...) {
    va_list args;
    char *f;
    char b[3];
    char *s;
    double v;
```

```

int i;

va_start (args, format);
for (f = format; *f; f++) {
    if (*f != '%') {
        b[0] = *f;
        b[1] = '\\0';
        g_print ("%s", b);
        continue;
    }
    switch (*++f) {
        case 's':
            s = va_arg(args, char *);
            g_print ("%s", s);
            break;
        case 'f':
            v = va_arg(args, double);
            g_print ("%f", v);
            break;
        case 'd':
            i = va_arg(args, int);
            g_print ("%d", i);
            break;
        default:
            b[0] = '%';
            b[1] = *f;
            b[2] = '\\0';
            g_print ("%s", b);
            break;
    }
}
va_end (args);

longjmp (buf, 1);
}

```

A function `runtime_error` has a variable-length argument list.

```
void runtime_error (char *format, ...)
```

This is implemented with `<stdarg.h>` header file. The `va_list` type variable `args` will refer to each argument in turn. A function `va_start` initializes `args`. A function `va_arg` returns an argument and moves the reference of `args` to the next. A function `va_end` cleans up everything necessary at the end.

The function `runtime_error` has a similar format of `printf` standard function. But its format has only `%s`, `%f` and `%d`.

The functions declared in `<setjmp.h>` and `<stdarg.h>` are explained in the very famous book “The C programming language” written by Brian Kernighan and Dennis Ritchie. I referred to the book to write the program above.

The program `turtle` is unsophisticated and unpolished. If you want to make your own language, you need to know more and more. I don’t know any good textbook about compilers and interpreters. If you know a good book, please let me know.

However, the following information is very useful (but old).

- Bison documentation
- Flex documentation
- Software tools written by Brian W. Kernighan & P. J. Plauger (1976)
- Unix programming environment written by Brian W. Kernighan and Rob Pike (1984)
- Source code of a language, for example, ruby.

Lately, lots of source codes are on the internet. Maybe reading source codes is the most useful for programmers.

Chapter 4

Drag and Drop

4.1 Drag and Drop

4.1.1 What is Drag and Drop?

Drag and drop is also written as “Drag-and-Drop”, or “DND” in short. DND is like “copy and paste” or “cut and paste”. If a user drags a UI element, which is a widget, selected part or other elements, data is transferred from the source to the destination.

You probably have experience moving a file with DND.

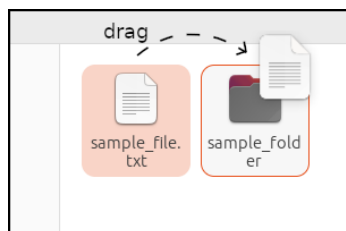


Figure 4.1: DND on the GUI file manager

When the DND starts, the file `sample_file.txt` is provided to the system. When it ends, the system provides it to the `sample_folder` in the file manager. Therefore, it is like “cut and paste”. The actual behavior may be different from the explanation here, but the concept is similar.

4.1.2 Example for DND

This tutorial provides a simple example in the `src/dnd` directory. It has three labels for the source and one label for the destination. The source labels have “red”, “green” or “blue” labels. If a user drags the label to the destination label, the font color will be changed.

4.1.3 UI File

The widgets are defined in the XML file `dnd.ui`.

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <interface>
3   <object class="GtkApplicationWindow" id="win">
4     <property name="default-width">800</property>
5     <property name="default-height">600</property>
6     <property name="resizable">FALSE</property>
7     <property name="title">Drag and Drop</property>
```

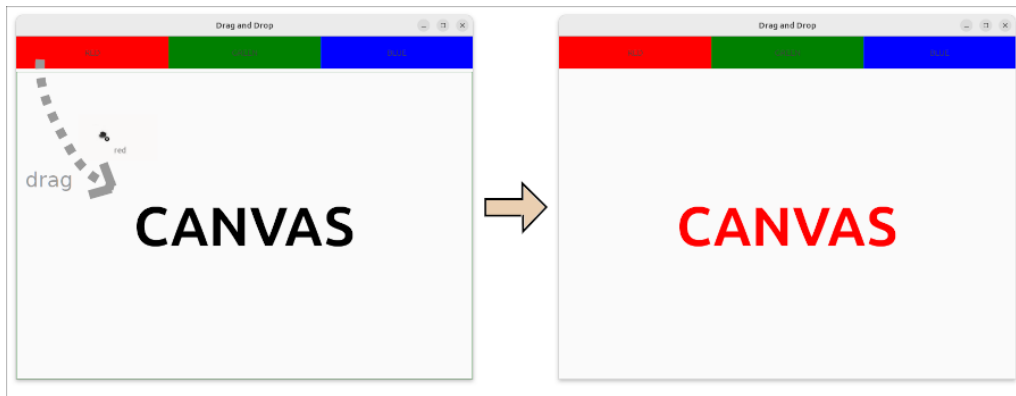


Figure 4.2: DND example

```

8 <child>
9   <object class="GtkBox">
10     <property name="hexpand">TRUE</property>
11     <property name="vexpand">TRUE</property>
12     <property name="orientation">GTK_ORIENTATION_VERTICAL</property>
13     <property name="spacing">5</property>
14     <child>
15       <object class="GtkBox">
16         <property name="orientation">GTK_ORIENTATION_HORIZONTAL</property>
17         <property name="homogeneous">TRUE</property>
18         <child>
19           <object class="GtkLabel" id="red">
20             <property name="label">RED</property>
21             <property name="justify">GTK_JUSTIFY_CENTER</property>
22             <property name="name">red</property>
23           </object>
24         </child>
25         <child>
26           <object class="GtkLabel" id="green">
27             <property name="label">GREEN</property>
28             <property name="justify">GTK_JUSTIFY_CENTER</property>
29             <property name="name">green</property>
30           </object>
31         </child>
32         <child>
33           <object class="GtkLabel" id="blue">
34             <property name="label">BLUE</property>
35             <property name="justify">GTK_JUSTIFY_CENTER</property>
36             <property name="name">blue</property>
37           </object>
38         </child>
39       </object>
40     </child>
41     <child>
42       <object class="GtkLabel" id="canvas">
43         <property name="label">CANVAS</property>
44         <property name="justify">GTK_JUSTIFY_CENTER</property>
45         <property name="name">canvas</property>

```

```

46         <property name="hexpand">TRUE</property>
47         <property name="vexpand">TRUE</property>
48     </object>
49 </child>
50 </object>
51 </child>
52 </object>
53 </interface>

```

It is converted to a resource file by `glib-compile-resources`. The compiler uses an XML file `dnd.gresource.xml`.

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <gresources>
3   <gresource prefix="/com/github/ToshioCP/dnd">
4     <file>dnd.ui</file>
5   </gresource>
6 </gresources>

```

4.1.4 C File `dnd.c`

The C file `dnd.c` isn't a big file. The number of the lines is less than a hundred. A `GtkApplication` object is created in the function `main`.

```

1 int
2 main (int argc, char **argv) {
3     GtkApplication *app;
4     int stat;
5
6     app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
7     g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
8     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
9     stat = g_application_run (G_APPLICATION (app), argc, argv);
10    g_object_unref (app);
11    return stat;
12 }

```

The application ID is defined as:

```
#define APPLICATION_ID "com.github.ToshioCP.dnd"
```

4.1.4.1 Startup Signal Handler

Most of the work is done in the “startup” signal handler.

Two objects `GtkDragSource` and `GtkDropTarget` are used for DND implementation.

- Drag source: A drag source (`GtkDragSource` instance) is an event controller. It initiates a DND operation when the user clicks and drags the widget. If data, in the form of `GdkContentProvider`, is set in advance, it gives the data to the system at the beginning of the drag.
- Drop target: A drop target (`GtkDropTarget`) is also an event controller. You can get the data in the `GtkDropTarget::drop` signal handler.

The example below uses these objects in a very simple way. You can use a number of features that the two objects have. See the following links for more information.

- [Drag-and-Drop in GTK](#)
- [GtkDragSource](#)
- [GtkDropTarget](#)

```

1  static void
2  app_startup (GApplication *application) {
3      GtkApplication *app = GTK_APPLICATION (application);
4      GtkBuilder *build;
5      GtkWidget *win;
6      GtkLabel *src_labels[3];
7      int i;
8      GtkLabel *canvas;
9      GtkDragSource *src;
10     GdkContentProvider *content;
11     GtkDropTarget *tgt;
12     GdkDisplay *display;
13     char *s;
14
15     build = gtk_builder_new_from_resource ("/com/github/ToshioCP/dnd/dnd.ui");
16     win = GTK_WINDOW (gtk_builder_get_object (build, "win"));
17     src_labels[0] = GTK_LABEL (gtk_builder_get_object (build, "red"));
18     src_labels[1] = GTK_LABEL (gtk_builder_get_object (build, "green"));
19     src_labels[2] = GTK_LABEL (gtk_builder_get_object (build, "blue"));
20     canvas = GTK_LABEL (gtk_builder_get_object (build, "canvas"));
21     gtk_window_set_application (win, app);
22     g_object_unref (build);
23
24     for (i=0; i<3; ++i) {
25         src = gtk_drag_source_new ();
26         content = gdk_content_provider_new_typed (G_TYPE_STRING, gtk_widget_get_name (GTK_WIDGET (src_labels[i])));
27         gtk_drag_source_set_content (src, content);
28         g_object_unref (content);
29         gtk_widget_add_controller (GTK_WIDGET (src_labels[i]), GTK_EVENT_CONTROLLER (src)); // The ownership of src is transferred to the widget.
30     }
31
32     tgt = gtk_drop_target_new (G_TYPE_STRING, GDK_ACTION_COPY);
33     g_signal_connect (tgt, "drop", G_CALLBACK (drop_cb), NULL);
34     gtk_widget_add_controller (GTK_WIDGET (canvas), GTK_EVENT_CONTROLLER (tgt)); // The ownership of tgt is transferred to the widget.
35
36     provider = gtk_css_provider_new ();
37     s = g_strdup_printf (format, "black");
38     gtk_css_provider_load_from_string (provider, s);
39     g_free (s);
40     display = gdk_display_get_default ();
41     gtk_style_context_add_provider_for_display (display, GTK_STYLE_PROVIDER (provider),
42                                                GTK_STYLE_PROVIDER_PRIORITY_APPLICATION);
43     g_object_unref (provider); // The provider is still alive because the display owns it.
44 }

```

- 15-22: Builds the widgets. The array `src_labels[]` points to the source labels red, green and blue in the UI file. The variable `canvas` points to the destination label.
- 24-30: Sets the DND source widgets. The for-loop iterates through the array `src_labels[]` each of which points to the source widget, red, green or blue label.
- 25: Creates a new `GtkDragSource` instance.
- 26: Creates a new `GdkContentProvider` instance with the string “red”, “green” or “blue”. They are the name of the widgets. These strings are the data to transfer through the DND operation.
- 27: Sets the content of the drag source to the `GdkContentProvider` instance above.
- 28: Content is useless so it is destroyed.

- 29: Add the event controller, which is actually the drag source, to the widget. If a DND operation starts on the widget, the corresponding drag source works and the data is given to the system.
- 32-34: Sets the DND drop target.
- 32: Creates a new `GtkDropTarget` instance. The first parameter is the `GType` of the data. The second parameter is a `GdkDragAction` enumeration constant. The arguments here are string type and the constant for copy.
- 33: Connects the “drop” signal and the handler `drop_cb`.
- 34: Add the event controller, which is actually the drop target, to the widget.
- 36-43: Sets CSS.
- 37: A variable `format` is static and defined at the top of the program. Static variables are shown below.

```
static GtkCssProvider *provider = NULL;
static const char *format = "label {padding: 20px;} label#red {background: red;} "
    "label#green {background: green;} label#blue {background: blue;} "
    "label#canvas {color: %s; font-weight: bold; font-size: 72pt;}";
```

4.1.4.2 Activate Signal Handler

```
1 static void
2 app_activate (GApplication *application) {
3     GtkApplication *app = GTK_APPLICATION (application);
4     GtkWindow *win;
5
6     win = gtk_application_get_active_window (app);
7     gtk_window_present (win);
8 }
```

This handler just shows the window.

4.1.4.3 Drop Signal Handler

```
1 static gboolean
2 drop_cb (GtkDropTarget *self, const GValue *value, gdouble x, gdouble y, gpointer user_data) {
3     char *s;
4
5     s = g_strdup_printf (format, g_value_get_string (value));
6     gtk_css_provider_load_from_string (provider, s);
7     g_free (s);
8     return TRUE;
9 }
```

The “drop” signal handler has five parameters.

- `GtkDropTarget` instance on which the signal has been emitted.
- `GValue` that holds the data from the source.
- The arguments `x` and `y` are the coordinate of the mouse when released.
- User data was set when the signal and the handler were connected.

The string from the `GValue` is “red”, “green” or “blue”. It replaces “%s” in the variable `format`. That means the font color of the label `canvas` will be updated to the color.

4.1.5 Meson.build

The file `meson.build` controls the building process.

```
1 project('dnd', 'c')
2
```



```
3 gtkdep = dependency('gtk4')
4
5 gnome = import('gnome')
6 resources = gnome.compile_resources('resources', 'dnd.gresource.xml')
7
8 executable(meson.project_name(), 'dnd.c', resources, dependencies: gtkdep, export_dynamic: true, ins
```

You can build it from the command line.

```
$ cd src/dnd
$ meson setup _build
$ ninja -C _build
$ _build/dnd
```

The source files are under the directory `src/dnd` of the [repository](#). Download it and see the directory.

Chapter 5

List Widgets

5.1 GtkListView

GTK 4 has added new list objects `GtkListView`, `GtkGridView` and `GtkColumnView`. The new feature is described in [Gtk API Reference – List Widget Overview](#).

GTK 4 has other means to implement lists. They are `GtkListBox` and `GtkTreeView` which are took over from GTK 3. There's an article in [Gtk Development blog](#) about list widgets by Matthias Clasen. He described why `GtkListView` are developed to replace `GtkTreeView`. `GtkTreeView` is deprecated since version 4.10.

`GtkListView`, `GtkGridView`, `GtkColumnView` and related objects are described in Section 29 to 33.

5.1.1 Outline

A list is a sequential data structure. For example, an ordered string sequence “one”, “two”, “three”, and “four” is a list. Each element is called an item. A list is like an array, but in many cases it is implemented with pointers which point to the next items of the list. And it has a start point. So, each item can be referred to by the index of the item (first item, second item, ..., nth item, ...).

Gio provides `GListModel` interface. It is a zero-based list and its items are the same type of `GObject` descendants, or objects that implement the same interface. An object that implements `GListModel` is not a widget. So, the list is not displayed on the screen directly. There's another object `GtkListView` which is a widget to display the list. The items in the list need to be connected to the items in `GtkListView`. `GtkListItemFactory` instance maps items in the list to `GtkListView`.

5.1.2 GListModel and GtkStringList

If you want to make a list of strings with `GListModel`, for example, “one”, “two”, “three”, “four”, note that strings can't be items of the list. Because `GListModel` is a list of `GObject` objects and strings aren't `GObject` objects. The word “`GObject`” here means `GObject` class or its derived class. So, you need a wrapper which is a `GObject` and contains a string. `GtkStringObject` is the wrapper object and `GtkStringList`, which implements `GListModel`, is a list of `GtkStringObject`.

```
char *array[] = {"one", "two", "three", "four", NULL};
GtkStringList *stringlist = gtk_string_list_new ((const char * const *) array);
```

The function `gtk_string_list_new` creates a `GtkStringList` object. Its items are `GtkStringObject` objects which contain the strings “one”, “two”, “three” and “four”. There are functions to add items to the list or remove items from the list.

- `gtk_string_list_append` appends an item to the list

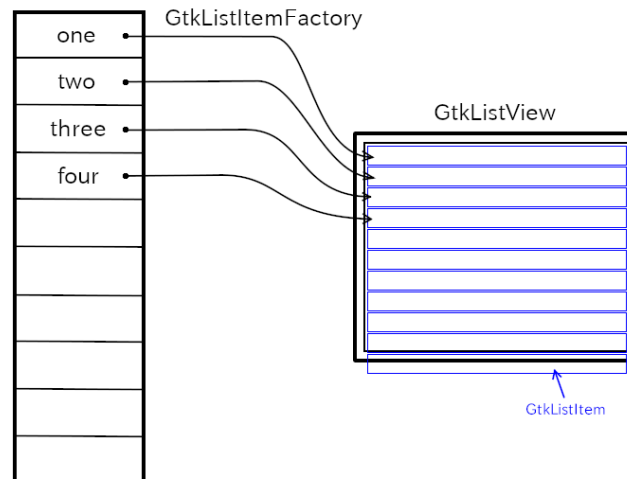


Figure 5.1: List

- `gtk_string_list_remove` removes an item from the list
- `gtk_string_list_get_string` gets a string in the list

See [GTK 4 API Reference – GtkStringList](#) for further information.

Other list objects will be explained later.

5.1.3 GtkSelectionMode

`GtkSelectionMode` is an interface to support for selection. Thanks to this model, a user can select items by clicking on them. It is implemented by `GtkMultiSelection`, `GtkNoSelection` and `GtkSingleSelection` objects. These three objects are usually enough to build an application. They are created with another `GListModel`. You can also create them alone and add a `GListModel` later.

- `GtkMultiSelection` supports multiple selection.
- `GtkNoSelection` supports no selection. This is a wrapper to `GListModel` when `GtkSelectionMode` is needed.
- `GtkSingleSelection` supports single selection.

5.1.4 GtkListView

`GtkListView` is a widget to show `GListModel` items. `GtkListItem` is used by `GtkListView` to represent items of a list model. But, `GtkListItem` is not a widget (not a descendant of `GtkWidget`), so a user needs to set a widget, for example `GtkLabel`, as a child of `GtkListItem` to display an item of the list model. “item” property of `GtkListItem` points to an object that belongs to the list model.

In case the number of items is very big, for example more than a thousand, `GtkListItem` is recycled and connected to another item which is newly displayed. This recycle makes the number of `GtkListItem` objects fairly small, less than 200. This is very effective to restrain the growth of memory consumption so that `GListModel` can contain lots of items, for example, more than a million items.

5.1.5 GtkListItemFactory

`GtkListItemFactory` creates or recycles `GtkListItem` and connects it with an item of the list model. There are two child classes of this factory, `GtkSignalListItemFactory` and `GtkBuilderListItemFactory`.

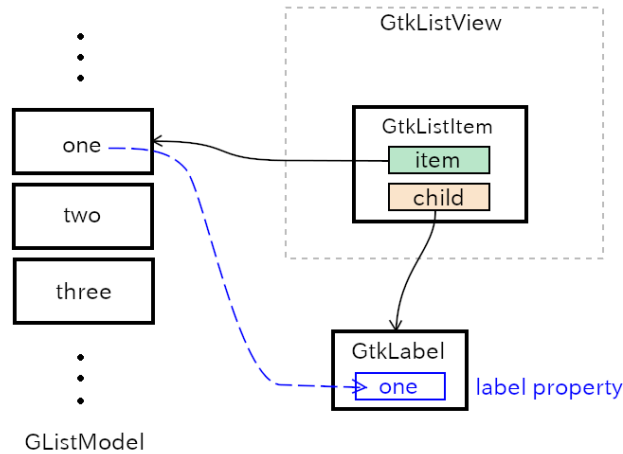


Figure 5.2: GtkListItem

5.1.5.1 GtkSignalListItemFactory

`GtkSignalListItemFactory` provides signals for users to configure a `GtkListItem` object. There are four signals.

1. “setup” is emitted to set up `GtkListItem` object. A user sets its child widget in the handler. For example, creates a `GtkLabel` widget and sets the child property of the `GtkListItem` to it. This setting is kept even when the `GtkListItem` instance is recycled (to bind to another item of `GListModel`).
2. “bind” is emitted to bind an item in the list model to the widget. For example, a user gets the item from “item” property of the `GtkListItem` instance. Then gets the string of the item and sets the label property of the `GtkLabel` instance with the string. This signal is emitted when the `GtkListItem` is newly created, recycled or some changes have happened to the item of the list.
3. “unbind” is emitted to unbind an item. A user undoes everything done in step 2 in the signal handler. If some object are created in step 2, they must be destroyed.
4. “teardown” is emitted to undo everything done in step 1. So, the widget created in step 1 must be destroyed. After this signal, the list item will be destroyed.

The following program `list1.c` shows a list of strings “one”, “two”, “three” and “four”. `GtkNoSelection` is used, so user can’t select any item.

```

1  #include <gtk/gtk.h>
2
3  static void
4  setup_cb (GtkSignalListItemFactory *self, GtkListItem *listitem, gpointer user_data) {
5      GtkWidget *lb = gtk_label_new (NULL);
6      gtk_list_item_set_child (listitem, lb);
7      /* Because gtk_list_item_set_child sunk the floating reference of lb, releasing (unref) isn't nece.
8  }
9
10 static void
11 bind_cb (GtkSignalListItemFactory *self, GtkListItem *listitem, gpointer user_data) {
12     GtkWidget *lb = gtk_list_item_get_child (listitem);
13     /* Strobj is owned by the instance. Caller mustn't change or destroy it. */
14     GtkStringObject *strobj = gtk_list_item_get_item (listitem);
15     /* The string returned by gtk_string_object_get_string is owned by the instance. */
16     gtk_label_set_text (GTK_LABEL (lb), gtk_string_object_get_string (strobj));

```

```

17 }
18
19 static void
20 unbind_cb (GtkSignalListItemFactory *self, GtkListItem *listitem, gpointer user_data) {
21     /* There's nothing to do here. */
22 }
23
24 static void
25 teardown_cb (GtkSignalListItemFactory *self, GtkListItem *listitem, gpointer user_data) {
26     /* There's nothing to do here. */
27     /* GtkListItem instance will be destroyed soon. You don't need to set the child to NULL. */
28 }
29
30 static void
31 app_activate (GApplication *application) {
32     GtkApplication *app = GTK_APPLICATION (application);
33     GtkWidget *win = gtk_application_window_new (app);
34     gtk_window_set_default_size (GTK_WINDOW (win), 600, 400);
35     GtkWidget *scr = gtk_scrolled_window_new ();
36     gtk_window_set_child (GTK_WINDOW (win), scr);
37
38     char *array[] = {
39         "one", "two", "three", "four", NULL
40     };
41     /* sl is owned by ns */
42     /* ns and factory are owned by lv. */
43     /* Therefore, you don't need to care about their destruction. */
44     GtkStringList *sl = gtk_string_list_new ((const char * const *) array);
45     GtkNoSelection *ns = gtk_no_selection_new (G_LIST_MODEL (sl));
46
47     GtkListItemFactory *factory = gtk_signal_list_item_factory_new ();
48     g_signal_connect (factory, "setup", G_CALLBACK (setup_cb), NULL);
49     g_signal_connect (factory, "bind", G_CALLBACK (bind_cb), NULL);
50     /* The following two lines can be left out. The handlers do nothing. */
51     g_signal_connect (factory, "unbind", G_CALLBACK (unbind_cb), NULL);
52     g_signal_connect (factory, "teardown", G_CALLBACK (teardown_cb), NULL);
53
54     GtkWidget *lv = gtk_list_view_new (GTK_SELECTION_MODEL (ns), factory);
55     gtk_scrolled_window_set_child (GTK_SCROLLED_WINDOW (scr), lv);
56     gtk_window_present (GTK_WINDOW (win));
57 }
58
59 /* ----- main ----- */
60 #define APPLICATION_ID "com.github.ToshioCP.list1"
61
62 int
63 main (int argc, char **argv) {
64     GtkApplication *app;
65     int stat;
66
67     app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
68
69     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);

```

```

70
71     stat = g_application_run (G_APPLICATION (app), argc, argv);
72     g_object_unref (app);
73     return stat;
74 }

```

The file `list1.c` is located under the directory `/src/misc`. Make a shell script below and save it to your bin directory, for example `$HOME/bin`.

```
gcc `pkg-config --cflags gtk4` $1.c `pkg-config --libs gtk4`
```

Change the current directory to the directory includes `list1.c` and type as follows.

```

$ chmod 755 $HOME/bin/comp # or chmod 755 (your bin directory)/comp
$ comp list1
$ ./a.out

```

Then, the following window appears.



Figure 5.3: list1

The program is not so difficult. If you feel some difficulty, read this section again, especially `GtkSignalListItemFactory` subsection.

5.1.5.2 GtkBuilderListItemFactory

`GtkBuilderListItemFactory` is another `GtkListItemFactory`. Its behavior is defined in a UI file.

```

<interface>
  <template class="GtkListItem">
    <property name="child">
      <object class="GtkLabel">
        <binding name="label">
          <lookup name="string" type="GtkStringObject">
            <lookup name="item">GtkListItem</lookup>
          </lookup>
        </binding>
      </object>
    </property>
  </template>
</interface>

```

Template tag is used to define `GtkListItem`. And its child property is `GtkLabel` object. The factory sees this template and creates `GtkLabel` and sets the child property of `GtkListItem`. This is the same as what setup handler of `GtkSignalListItemFactory` did.

Then, bind the label property of the `GtkLabel` to the string property of a `GtkStringObject`. The string object refers to the item property of the `GtkListItem`. So, the lookup tag is like this:

```
label <- string <- GtkStringObject <- item <- GtkListItem
```

The last lookup tag has a content `GtkListItem`. Usually, C type like `GtkListItem` doesn't appear in the content of tags. This is a special case. There is an explanation in the [GTK Development Blog](#) by Matthias Clasen.

Remember that the classname (`GtkListItem`) in a ui template is used as the “this” pointer referring to the object that is being instantiated.

Therefore, `GtkListItem` instance is used as the `this` object of the lookup tag when it is evaluated. `this` object will be explained in [Section 31](#).

The C source code is as follows. Its name is `list2.c` and located under `/src/misc` directory.

```
1  #include <gtk/gtk.h>
2
3  static void
4  app_activate (GApplication *application) {
5      GtkApplication *app = GTK_APPLICATION (application);
6      gtk_window_present (gtk_application_get_active_window(app));
7  }
8
9  static void
10 app_startup (GApplication *application) {
11     GtkApplication *app = GTK_APPLICATION (application);
12     GtkWidget *win = gtk_application_window_new (app);
13     gtk_window_set_default_size (GTK_WINDOW (win), 600, 400);
14     GtkWidget *scr = gtk_scrolled_window_new ();
15     gtk_window_set_child (GTK_WINDOW (win), scr);
16
17     char *array[] = {
18         "one", "two", "three", "four", NULL
19     };
20     GtkStringList *sl = gtk_string_list_new ((const char * const *) array);
21     GtkSingleSelection *ss = gtk_single_selection_new (G_LIST_MODEL (sl));
22
23     const char *ui_string =
24     "<interface>"
25     "<template class=\"GtkListItem\">"
26     "  <property name=\"child\">"
27     "    <object class=\"GtkLabel\">"
28     "      <binding name=\"label\">"
29     "        <lookup name=\"string\" type=\"GtkStringObject\">"
30     "          <lookup name=\"item\">GtkListItem</lookup>"
31     "        </lookup>"
32     "      </binding>"
33     "    </object>"
34     "  </property>"
35     "</template>"
36     "</interface>"
37 ;
38     GBytes *gbytes = g_bytes_new_static (ui_string, strlen (ui_string));
39     GtkListItemFactory *factory = gtk_builder_list_item_factory_new_from_bytes (NULL, gbytes);
40
41     GtkWidget *lv = gtk_list_view_new (GTK_SELECTION_MODEL (ss), factory);
42     gtk_scrolled_window_set_child (GTK_SCROLLED_WINDOW (scr), lv);
```

```

43 }
44
45 /* ----- main ----- */
46 #define APPLICATION_ID "com.github.ToshioCP.list2"
47
48 int
49 main (int argc, char **argv) {
50     GtkApplication *app;
51     int stat;
52
53     app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
54
55     g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
56     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
57
58     stat = g_application_run (G_APPLICATION (app), argc, argv);
59     g_object_unref (app);
60     return stat;
61 }

```

No signal handler is needed for `GtkBuilderListItemFactory`. `GtkSingleSelection` is used, so user can select one item at a time.

Because this is a small program, the UI data is given as a string.

5.1.6 GtkDirectoryList

`GtkDirectoryList` is a list model containing `GFileInfo` objects which are information of files under a certain directory. It uses `g_file_enumerate_children_async()` to get the `GFileInfo` objects. The list model is created by `gtk_directory_list_new` function.

```
GtkDirectoryList *gtk_directory_list_new (const char *attributes, GFile *file);
```

`attributes` is a comma separated list of file attributes. File attributes are key-value pairs. A key consists of a namespace and a name. For example, “standard::name” key is the name of a file. “standard” means general file information. “name” means filename. The following table shows some example.

key	meaning
standard::type	file type. for example, regular file, directory, symbolic link, etc.
standard::name	filename
standard::size	file size in bytes
access::can-read	read privilege if the user is able to read the file
time::modified	the time the file was last modified in seconds since the UNIX epoch

The current directory is “.”. The following program makes `GtkDirectoryList` `dl` and its contents are `GFileInfo` objects under the current directory.

```

GFile *file = g_file_new_for_path (".");
GtkDirectoryList *dl = gtk_directory_list_new ("standard::name", file);
g_object_unref (file);

```

It is not so difficult to make a file listing program by changing `list2.c` in the previous subsection. One problem is that `GFileInfo` doesn’t have properties. Lookup tag look for a property, so it is useless for looking for a filename from a `GFileInfo` object. Instead, a closure tag is appropriate in this case. A closure tag specifies a function and the type of the return value of the function.


```

char *
get_file_name (GtkListItem *item, GFileInfo *info) {
    return G_IS_FILE_INFO (info) ? g_strdup (g_file_info_get_name (info)) : NULL;
}
...
...

"<interface>"
"<template class=\"GtkListItem\">"
  "<property name=\"child\">"
    "<object class=\"GtkLabel\">"
      "<binding name=\"label\">"
        "<closure type=\"gchararray\" function=\"get_file_name\">"
          "<lookup name=\"item\">GtkListItem</lookup>"
        "</closure>"
      "</binding>"
    "</object>"
  "</property>"
"</template>"
"</interface>"

```

- The string “gchararray” is a type name registered in the GObject type system. While the type “gchar” is the same as the C “char” type, “gchararray” specifically represents a NULL-terminated string (char *). In GtkBuilder XML files, you must use these registered type names instead of native C types to let the GValue system handle the data correctly. For example, common type names include gboolean, gchar, gint, gfloat, gdouble, and gchararray. See [GObject tutorial](#) for further information.
- Closure tag has type attribute and function attribute. Function attribute specifies the function name and type attribute specifies the type of the return value of the function. The content of closure tag (it is between <closure...> and </closure>) consists of the parameters for the function. <lookup name="item">GtkListItem</lookup> gives the value of the item property of the GtkListItem. This will be the second argument of the function. The first parameter is always the GListItem instance, which is a ‘this’ object. The ‘this’ object is explained in Section 31.
- get_file_name function is the callback function for the closure tag. It first checks the info parameter. Because it can be NULL when GListItem item is unbounded. If it’s GFileInfo, it returns the copied filename. Because the return value (filename) of g_file_info_get_name is owned by GFileInfo object. So, the the string needs to be duplicated to give the ownership to the caller. Binding tag binds the “label” property of the GtkLabel to the closure tag.

The whole program (list3.c) is as follows. The program is located in [/src/misc](#) directory.

```

1  #include <gtk/gtk.h>
2
3  char *
4  get_file_name (GtkListItem *item, GFileInfo *info) {
5      return G_IS_FILE_INFO (info) ? g_strdup (g_file_info_get_name (info)) : NULL;
6  }
7
8  static void
9  app_activate (GApplication *application) {
10     GtkApplication *app = GTK_APPLICATION (application);
11     gtk_window_present (gtk_application_get_active_window(app));
12 }
13
14 static void

```

```

15 app_startup (GApplication *application) {
16     GtkApplication *app = GTK_APPLICATION (application);
17     GtkWidget *win = gtk_application_window_new (app);
18     gtk_window_set_default_size (GTK_WINDOW (win), 600, 400);
19     GtkWidget *scr = gtk_scrolled_window_new ();
20     gtk_window_set_child (GTK_WINDOW (win), scr);
21
22     GFile *file = g_file_new_for_path (".");
23     GtkDirectoryList *dl = gtk_directory_list_new ("standard::name", file);
24     g_object_unref (file);
25     GtkNoSelection *ns = gtk_no_selection_new (G_LIST_MODEL (dl));
26
27     const char *ui_string =
28 "<interface>"
29 "<template class=\"GtkListItem\">"
30     "<property name=\"child\">"
31         "<object class=\"GtkLabel\">"
32             "<binding name=\"label\">"
33                 "<closure type=\"gchararray\" function=\"get_file_name\">"
34                     "<lookup name=\"item\">GtkListItem</lookup>"
35                 "</closure>"
36             "</binding>"
37         "</object>"
38     "</property>"
39 "</template>"
40 "</interface>"
41 ;
42     GBytes *gbytes = g_bytes_new_static (ui_string, strlen (ui_string));
43     GtkListItemFactory *factory = gtk_builder_list_item_factory_new_from_bytes (NULL, gbytes);
44
45     GtkWidget *lv = gtk_list_view_new (GTK_SELECTION_MODEL (ns), factory);
46     gtk_scrolled_window_set_child (GTK_SCROLLED_WINDOW (scr), lv);
47 }
48
49 /* ----- main ----- */
50 #define APPLICATION_ID "com.github.ToshioCP.list3"
51
52 int
53 main (int argc, char **argv) {
54     GtkApplication *app;
55     int stat;
56
57     app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
58
59     g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
60     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
61
62     stat = g_application_run (G_APPLICATION (app), argc, argv);
63     g_object_unref (app);
64     return stat;
65 }

```

The ui data (xml data above) is used to build the `GListItem` template at runtime. `GtkBuilder` refers to the symbol table to find the function `get_file_name`.

Generally, a symbol table is used by a linker to link objects to an executable file. It includes function names and their location. A linker usually doesn't put a symbol table into the created executable file. But if `--export-dynamic` option is given, the linker adds the symbol table to the executable file.

To accomplish it, an option `-Wl,--export-dynamic` is given to the C compiler.

- `-Wl` is a C compiler option that passes the following option to the linker.
- `--export-dynamic` is a linker option. The following is cited from the linker document. "When creating a dynamically linked executable, add all symbols to the dynamic symbol table. The dynamic symbol table is the set of symbols which are visible from dynamic objects at run time."

Compile and execute it.

```
$ gcc -Wl,--export-dynamic `pkg-config --cflags gtk4` list3.c `pkg-config --libs gtk4`
```

You can also make a shell script to compile `list3.c`

```
gcc -Wl,--export-dynamic `pkg-config --cflags gtk4` $1.c `pkg-config --libs gtk4`
```

Save this one liner to a file `comp`. Then, copy it to `$HOME/bin` and give it executable permission.

```
$ cp comp $HOME/bin/comp
$ chmod +x $HOME/bin/comp
```

You can compile `list3.c` and execute it, like this:

```
$ comp list3
$ ./a.out
```

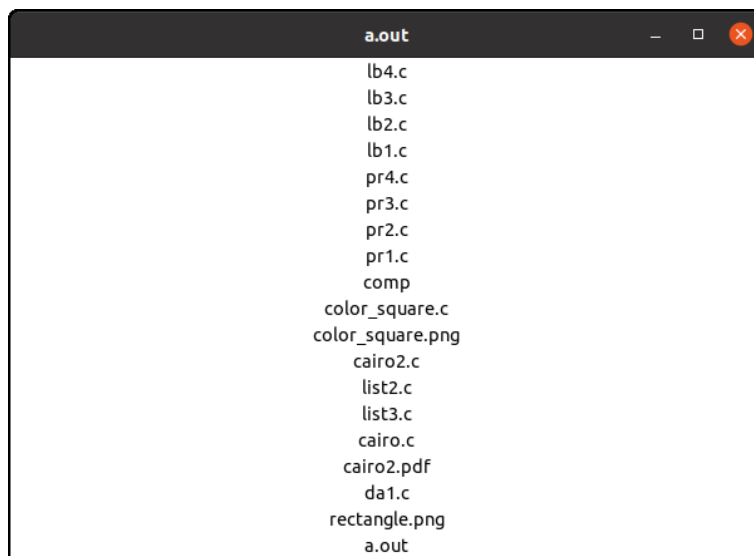


Figure 5.4: screenshot list3

5.2 GtkGridView and Activate Signal

GtkGridView is similar to GtkListView. It displays a `GListModel` as a grid, which is like a square tessellation.

This is often seen when you use a file browser like GNOME Files (Nautilus).

In this section, let's make a very simple file browser `list4`. It just shows the files in the current directory. And a user can choose either list or grid layout by clicking on buttons in the tool bar. Each item in the list or grid has an icon and a filename. In addition, `list4` provides the way to open the `tfe` text editor to show a text file. A user can do that by double clicking on an item or pressing enter key when an item is selected.

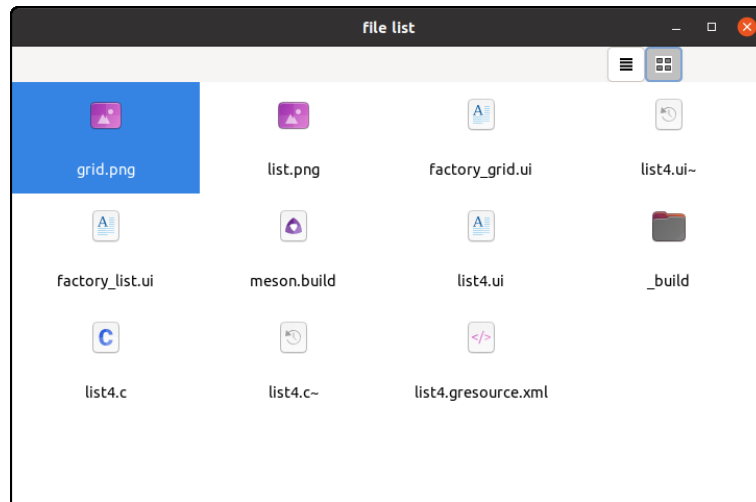


Figure 5.5: Grid

5.2.1 GtkDirectoryList

GtkDirectoryList implements GListModel and it contains information of files in a certain directory. The items of the list are GFileInfo objects.

In the `list4` source files, GtkDirectoryList is described in a UI file and built by GtkBuilder. The GtkDirectoryList instance is assigned to the “model” property of a GtkSingleSelection instance. And the GtkSingleSelection instance is assigned to the “model” property of a GListView or GGridView instance.

GtkListView (model property) => GtkSingleSelection (model property) => GtkDirectoryList
 GtkGridView (model property) => GtkSingleSelection (model property) => GtkDirectoryList

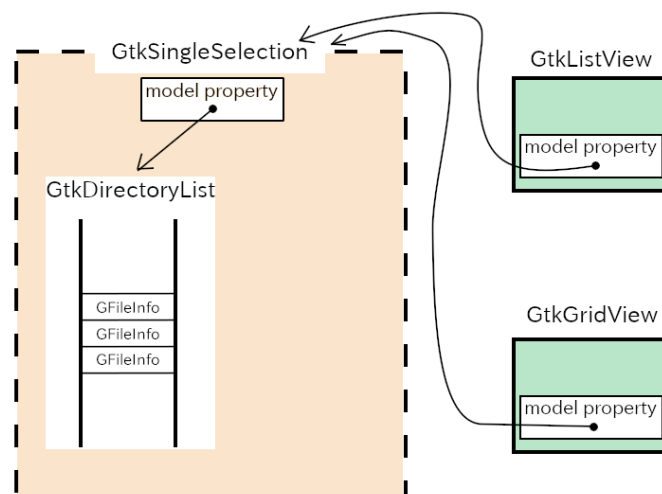


Figure 5.6: DirectoryList

The following is a part of the UI file `list4.ui`.

```
<object class="GtkListView" id="list">
  <property name="model">
    <object class="GtkSingleSelection" id="singleselection">
      <property name="model">
        <object class="GtkDirectoryList" id="directory_list">
```

```

        <property name="attributes">standard::name,standard::icon,standard::content-type</property>
    </object>
</property>
</object>
</property>
</object>
<object class="GtkGridView" id="grid">
    <property name="model">singlselection</property>
</object>

```

GtkDirectoryList has an “attributes” property. They are attributes of GFileInfo such as “standard::name”, “standard::icon” and “standard::content-type”.

- standard::name is a filename.
- standard::icon is an icon of the file. It is a GIcon object.
- standard::content-type is a content-type. Content-type is the same as MIME type. For example, “text/plain” is a text file, “text/x-csrc” is a C source code and so on. (“text/x-csrc” is not registered to IANA media types. Such “x-” subtype is not a standard mime type.) Content type is also used by desktop systems.

GtkGridView uses the same GtkSingleSelection instance (`singlselection`). So, its model property is set to it.

5.2.2 Ui File of the Window

The window is built with the following UI file. (See the screenshot at the beginning of this section).

```

1  <?xml version="1.0" encoding="UTF-8"?>
2  <interface>
3      <object class="GtkApplicationWindow" id="win">
4          <property name="title">file list</property>
5          <property name="default-width">600</property>
6          <property name="default-height">400</property>
7          <child>
8              <object class="GtkBox" id="boxv">
9                  <property name="orientation">GTK_ORIENTATION_VERTICAL</property>
10                 <child>
11                     <object class="GtkBox" id="boxh">
12                         <property name="orientation">GTK_ORIENTATION_HORIZONTAL</property>
13                         <child>
14                             <object class="GtkLabel" id="dmy1">
15                                 <property name="hexpand">TRUE</property>
16                             </object>
17                         </child>
18                         <child>
19                             <object class="GtkButton" id="btnlist">
20                                 <property name="name">btnlist</property>
21                                 <property name="action-name">win.view</property>
22                                 <property name="action-target">&apos;list&apos;</property>
23                             <child>
24                                 <object class="GtkImage">
25                                     <property name="resource">/com/github/ToshioCP/list4/list.png</property>
26                                 </object>
27                             </child>
28                         </object>
29                     </child>

```

```

30         <child>
31             <object class="GtkButton" id="btngrid">
32                 <property name="name">btngrid</property>
33                 <property name="action-name">win.view</property>
34                 <property name="action-target">&apos;grid&apos;</property>
35                 <child>
36                     <object class="GtkImage">
37                         <property name="resource">/com/github/ToshioCP/list4/grid.png</property>
38                     </object>
39                 </child>
40             </object>
41         </child>
42         <child>
43             <object class="GtkLabel" id="dmy2">
44                 <property name="width-chars">10</property>
45             </object>
46         </child>
47     </object>
48 </child>
49 <child>
50     <object class="GtkScrolledWindow" id="scr">
51         <property name="hexpand">TRUE</property>
52         <property name="vexpand">TRUE</property>
53     </object>
54 </child>
55 </object>
56 </child>
57 </object>
58 <object class="GtkListView" id="list">
59     <property name="model">
60         <object class="GtkSingleSelection" id="singleselection">
61             <property name="model">
62                 <object class="GtkDirectoryList" id="directory_list">
63                     <property name="attributes">standard::name,standard::icon,standard::content-type</property>
64                 </object>
65             </property>
66         </object>
67     </property>
68 </object>
69 <object class="GtkGridView" id="grid">
70     <property name="model">singleselection</property>
71 </object>
72 </interface>

```

The file consists of two parts. The first part begins at the line 3 and ends at line 57. This part is the widgets from the top level window to the scrolled window. It also includes two buttons. The second part begins at line 58 and ends at line 71. This is the part of `GtkListView` and `GtkGridView`.

- 13-17, 42-46: Two labels are dummy labels. They just work as a space to put the two buttons at the appropriate position.
- 18-41: `GtkButton` `btnlist` and `btngrid`. These two buttons work as selection buttons to switch from list to grid and vice versa. These two buttons are connected to a stateful action `win.view`. This action has a parameter. Such action consists of prefix, action name and parameter. The prefix of the action is `win`, which means the action belongs to the top level window (`GtkApplicationWindow`).

The prefix defines the scope of the action. The action name is **view**. The parameters are **list** or **grid**, which show the state of the action. A parameter is also called a target, because it is a target to which the action changes its state. We often write the detailed action like “win.view::list” or “win.view::grid”.

- 21-22: The properties “action-name” and “action-target” belong to GtkActionable interface. The GtkButton class implements the GtkActionable interface. The action name is “win.view” and the target is “list”. Generally, a target is GVariant, which can be string, integer, float and so on. You need to use GVariant text format to write GVariant value in UI files. If the type of the GVariant value is string, then the value with GVariant text format is enclosed in single quotes or double quotes. Because UI files are xml format text, single quote cannot be written without escaping it. Its escape sequence is `'`. Therefore, the target ‘list’ is written as `'list'`. Because the button is connected to the action, “clicked” signal handler isn’t needed.
- 23-27: The child widget of the button is GtkImage. GtkImage has a “resource” property. It is a GResource and GtkImage reads an image data from the resource and sets the image. This resource is built from 24x24-sized png image data, which is an original icon.
- 50-53: GtkScrolledWindow. Its child widget will be set with GtkListView or GtkGridView.

The action **view** is created, connected to the “activate” signal handler and inserted to the window (action map) as follows.

```
act_view = g_simple_action_new_stateful ("view", g_variant_type_new("s"), g_variant_new_string ("list"));
g_signal_connect (act_view, "activate", G_CALLBACK (view_activated), NULL);
g_action_map_add_action (G_ACTION_MAP (win), G_ACTION (act_view));
```

The signal handler `view_activated` will be explained later.

5.2.3 Factories

Each view (GtkListView and GtkGridView) has its own factory because its items have different structure of widgets. The factories are GtkBuilderListItemFactory objects. Their UI files are as follows.

factory_list.ui

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <interface>
3   <template class="GtkListItem">
4     <property name="child">
5       <object class="GtkBox">
6         <property name="orientation">GTK_ORIENTATION_HORIZONTAL</property>
7         <property name="spacing">20</property>
8         <child>
9           <object class="GtkImage">
10            <binding name="gicon">
11              <closure type="GIcon" function="get_icon">
12                <lookup name="item">GtkListItem</lookup>
13              </closure>
14            </binding>
15          </object>
16        </child>
17      </child>
18      <object class="GtkLabel">
19        <property name="hexpand">TRUE</property>
20        <property name="xalign">0</property>
21        <binding name="label">
22          <closure type="gchararray" function="get_file_name">
23            <lookup name="item">GtkListItem</lookup>
```

```

24         </closure>
25     </binding>
26 </object>
27 </child>
28 </object>
29 </property>
30 </template>
31 </interface>

factory__grid.ui

1 <?xml version="1.0" encoding="UTF-8"?>
2 <interface>
3     <template class="GtkListItem">
4         <property name="child">
5             <object class="GtkBox">
6                 <property name="orientation">GTK_ORIENTATION_VERTICAL</property>
7                 <property name="spacing">20</property>
8                 <child>
9                     <object class="GtkImage">
10                        <property name="icon-size">GTK_ICON_SIZE_LARGE</property>
11                        <binding name="gicon">
12                            <closure type="GIcon" function="get_icon">
13                                <lookup name="item">GtkListItem</lookup>
14                            </closure>
15                        </binding>
16                    </object>
17                </child>
18                <child>
19                    <object class="GtkLabel">
20                        <property name="hexpand">TRUE</property>
21                        <property name="xalign">0.5</property>
22                        <binding name="label">
23                            <closure type="gchararray" function="get_file_name">
24                                <lookup name="item">GtkListItem</lookup>
25                            </closure>
26                        </binding>
27                    </object>
28                </child>
29            </object>
30        </property>
31    </template>
32 </interface>

```

The two files above are almost same. The difference is:

- The orientation of the box
- The icon size
- The position of the text of the label

```
$ cd list4; diff factory_list.ui factory_grid.ui
```

```
6c6
```

```
<         <property name="orientation">GTK_ORIENTATION_HORIZONTAL</property>
```

```
---
```

```
>         <property name="orientation">GTK_ORIENTATION_VERTICAL</property>
```

```
9a10
```



```

>          <property name="icon-size">GTK_ICON_SIZE_LARGE</property>
20c21
<          <property name="xalign">0</property>
---
>          <property name="xalign">0.5</property>
$

```

Two properties “gicon” (property of GtkImage) and “label” (property of GtkLabel) are in the UI files above. Because GFileInfo doesn’t have properties correspond to icon or filename, the factory uses closure tag to bind “gicon” and “label” properties to GFileInfo information. A function `get_icon` gets GIcon from the GFileInfo object. And a function `get_file_name` gets a filename from the GFileInfo object.

```

1  GIcon *
2  get_icon (GtkListItem *item, GFileInfo *info) {
3      GIcon *icon;
4
5      /* g_file_info_get_icon can return NULL */
6      icon = G_IS_FILE_INFO (info) ? g_file_info_get_icon (info) : NULL;
7      return icon ? g_object_ref (icon) : NULL;
8  }
9
10 char *
11 get_file_name (GtkListItem *item, GFileInfo *info) {
12     return G_IS_FILE_INFO (info) ? g_strdup (g_file_info_get_name (info)) : NULL;
13 }

```

One important thing is the ownership of the return values. The return value is owned by the caller. So, `g_object_ref` or `g_strdup` is necessary.

5.2.4 An Activate Signal Handler of the Button Action

An activate signal handler `view_activate` switches the view. It does two things.

- Changes the child widget of GtkScrolledWindow.
- Changes the CSS of buttons to show the current state.

```

1  static void
2  view_activated(GSimpleAction *action, GVariant *parameter) {
3      const char *view = g_variant_get_string (parameter, NULL);
4      const char *other;
5      char *css;
6
7      if (strcmp (view, "list") == 0) {
8          other = "grid";
9          gtk_scrolled_window_set_child (scr, GTK_WIDGET (list));
10     }else {
11         other = "list";
12         gtk_scrolled_window_set_child (scr, GTK_WIDGET (grid));
13     }
14     css = g_strdup_printf ("button#btn%s {background: silver;} button#btn%s {background: white;}", view, other);
15     gtk_css_provider_load_from_string (provider, css);
16     g_free (css);
17     g_action_change_state (G_ACTION (action), parameter);
18 }

```

The second parameter of this handler is the target value associated with the clicked button. Its type is GVariant.

- If `btnlist` has been clicked, then `parameter` is a `GVariant` of the string “list”.
- If `btngrid` has been clicked, then `parameter` is a `GVariant` of the string “grid”.

The third parameter `user_data` points to `NULL` and it is ignored here.

- 3: `g_variant_get_string` gets the string from the `GVariant` variable.
- 7-13: Sets the child of `scr`. The function `gtk_scrolled_window_set_child` decreases the reference count of the old child by one. And it increases the reference count of the new child by one.
- 14-16: Sets the CSS for the buttons. The background of the clicked button will be silver color and the other button will be white.
- 17: Changes the state of the action.

5.2.5 Activate Signal on GtkListView and GtkGridView

Views (`GtkListView` and `GtkGridView`) have an “activate” signal. It is emitted when an item in the view is double clicked or the enter key is pressed. You can do anything you like by connecting the “activate” signal to the handler.

The example `list4` launches `tfe` text file editor if the item of the list is a text file.

```
static void
list_activate (GtkListView *list, int position, gpointer user_data) {
    GFileInfo *info = G_FILE_INFO (g_list_model_get_item (G_LIST_MODEL (gtk_list_view_get_model (list)),
        position));
    launch_tfe_with_file (info);
}

static void
grid_activate (GtkGridView *grid, int position, gpointer user_data) {
    GFileInfo *info = G_FILE_INFO (g_list_model_get_item (G_LIST_MODEL (gtk_grid_view_get_model (grid)),
        position));
    launch_tfe_with_file (info);
}

... ..
... ..

g_signal_connect (GTK_LIST_VIEW (list), "activate", G_CALLBACK (list_activate), NULL);
g_signal_connect (GTK_GRID_VIEW (grid), "activate", G_CALLBACK (grid_activate), NULL);
```

The second parameter of each handler is the position of the item (`GFileInfo`) of the `GListModel`. So you can get the item with `g_list_model_get_item` function.

5.2.5.1 Content Type and Application Launch

The function `launch_tfe_with_file` gets a file from the `GFileInfo` instance. If the file is a text file, it launches `tfe` with the file.

`GFileInfo` has information about file type. The file type is like “text/plain”, “text/x-csrc” and so on. This is referred to as the content type. Content type can be got with `g_file_info_get_content_type` function.

```
1 static void
2 launch_tfe_with_file (GFileInfo *info) {
3     GError *err = NULL;
4     GFile *file;
5     GList *files = NULL;
6     const char *content_type;
7     const char *text_type = "text/";
8     GAppInfo *appinfo;
```

```

9   int i;
10
11   if (! info)
12       return;
13   content_type = g_file_info_get_content_type (info);
14   #ifdef debug
15       g_print ("%s\n", content_type);
16   #endif
17   if (! content_type)
18       return;
19   for (i=0;i<5;++i) { /* compare the first 5 characters */
20       if (content_type[i] != text_type[i])
21           return;
22   }
23   appinfo = g_app_info_create_from_commandline ("tfe", "tfe", G_APP_INFO_CREATE_NONE, &err);
24   if (err) {
25       g_printerr ("%s\n", err->message);
26       g_error_free (err);
27       return;
28   }
29   err = NULL;
30   file = g_file_new_for_path (g_file_info_get_name (info));
31   files = g_list_append (files, file);
32   if (! (g_app_info_launch (appinfo, files, NULL, &err))) {
33       g_printerr ("%s\n", err->message);
34       g_error_free (err);
35   }
36   g_list_free_full (files, g_object_unref);
37   g_object_unref (appinfo);
38 }

```

- 13: Gets the content type of the file from GFileInfo.
- 14-16: Prints the content type if “debug” is defined. This is only useful to know a content type of a file. If you don’t want this, delete or uncomment the definition `#define debug 1` at line 6 in the source file.
- 17-22: If there is no content type, or if it doesn’t begin with “text/”, the function returns.
- 23: Creates GAppInfo object of `tfe` application. GAppInfo is an interface and the variable `appinfo` points to a GDesktopAppInfo instance. GAppInfo is a collection of information of applications.
- 32: Launches the application (`tfe`) with an argument `file`. `g_app_info_launch` has four parameters. The first parameter is GAppInfo object. The second parameter is a list of GFile objects. In this function, only one GFile instance is given to `tfe`, but you can give more arguments. The third parameter is GAppLaunchContext, but this program gives NULL instead. The last parameter is the pointer to the pointer to a GError.
- 36: `g_list_free_full` frees the memories used by the list and items.

If your distribution supports GTK 4, using `g_app_info_launch_default_for_uri` is convenient. The function automatically determines the default application from the file and launches it. For example, if the file is text, then it launches GNOME text editor with the file. Such feature comes from desktop.

5.2.6 Compilation and Execution

The source files are located in `/src/list4` directory. To compile and execute list4, type as follows.

```

$ cd list4 # or cd src/list4. It depends your current directory.
$ meson setup _build

```

```
$ ninja -C _build
$ _build/list4
```

Then a file list appears as a list style. Click on a button on the tool bar so that you can change the style to grid or back to list. Double click “list4.c” item, then `tfe` text editor runs with the argument “list4.c”. The following is the screenshot.

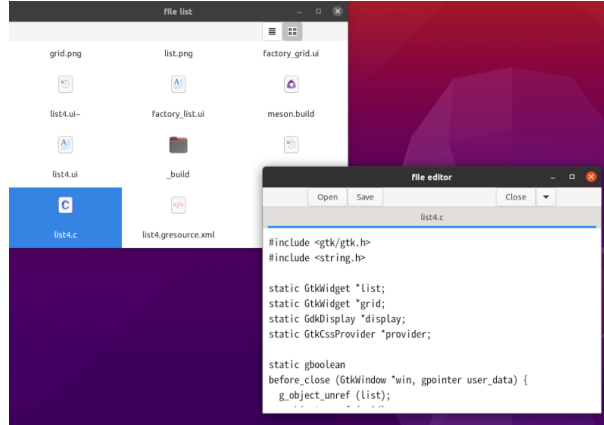


Figure 5.7: Screenshot

5.2.7 “gbytes” Property of GtkBuilderListItemFactory

GtkBuilderListItemFactory has “gbytes” property. The property contains a byte sequence of UI data. If you use this property, you can put the contents of `factory_list.ui` and `factory_grid.ui` into `list4.ui`. The following shows a part of the new UI file (`list5.ui`).

```
<object class="GtkListView" id="list">
  <property name="model">
    <object class="GtkSingleSelection" id="singleselection">
      <property name="model">
        <object class="GtkDirectoryList" id="directory_list">
          <property name="attributes">standard::name,standard::icon,standard::content-type</property>
        </object>
      </property>
    </object>
  </property>
  <property name="factory">
    <object class="GtkBuilderListItemFactory">
      <property name="bytes"><![CDATA[
<?xml version="1.0" encoding="UTF-8"?>
<interface>
  <template class="GtkListItem">
    <property name="child">
      <object class="GtkBox">
        <property name="orientation">GTK_ORIENTATION_HORIZONTAL</property>
        <property name="spacing">20</property>
        <child>
          <object class="GtkImage">
            <binding name="gicon">
              <closure type="GIcon" function="get_icon">
                <lookup name="item">GtkListItem</lookup>
              </closure>
            </binding>
          </object>
        </child>
      </object>
    </property>
  </template>
</interface>
</property>
</object>
```

```

        </binding>
    </object>
</child>
<child>
    <object class="GtkLabel">
        <property name="hexpand">TRUE</property>
        <property name="xalign">0</property>
        <binding name="label">
            <closure type="gchararray" function="get_file_name">
                <lookup name="item">GtkListItem</lookup>
            </closure>
        </binding>
    </object>
</child>
</object>
</property>
</template>
</interface>
]]></property>
</object>
</property>
</object>

```

CDATA section begins with “<[CDATA[” and ends with ”]>”. The contents of CDATA section is recognized as a string. Any character, even if it is a key syntax marker such as ‘<’ or ‘>’, is recognized literally. Therefore, the text between “<[CDATA[” and ”]>” is inserted to “bytes” property as it is.

This approach reduces the number of UI files. But, the new UI file is a bit complicated especially for the beginners. If you feel some difficulty, it is better for you to separate the UI file.

A directory `/src/list5` includes the UI file above.

5.3 GtkExpression

The GtkExpression is a fundamental type. It is not a descendant of GObject. GtkExpression provides a way to describe references to values. GtkExpression needs to be evaluated to obtain a value.

It is similar to arithmetic calculation.

`1 + 2 = 3`

`1+2` is an expression. It shows the way how to calculate. `3` is the value that comes from the expression. Evaluation is to calculate the expression and get the value.

GtkExpression is a way to get a value. Evaluation is like a calculation. A value is obtained by evaluating the expression.

5.3.1 Constant Expression

A constant expression (GtkConstantExpression) provides a constant value or an instance when it is evaluated.

```

GValue value = G_VALUE_INIT;
expression = gtk_constant_expression_new (G_TYPE_INT, 100);
gtk_expression_evaluate (expression, NULL, &value);

```

- GtkExpression uses GValue to hold a value. GValue is a structure and container to hold a type and value. It must be initialized with `G_VALUE_INIT`, first. Be careful that `value` is a structure, not a

pointer to a structure.

- Constant expression is created with `gtk_constant_expression_new` function. The parameter of the function is a type (GType) and a value (or instance). This expression holds a constant value. `G_TYPE_INT` is a type that is registered to the type system. It is integer type. Some types are shown in the following table.
- `gtk_expression_evaluate` evaluates the expression. It has three parameters, the expression to evaluate, `this` instance and a pointer to a GValue for being set with the value. `this` instance isn't necessary for constant expressions. Therefore, the second argument is NULL. `gtk_expression_evaluate` returns TRUE if it successfully evaluates the expression. Otherwise it returns FALSE.
- If it returns TRUE, the GValue `value` is set with the value of the expression. The type of the value is int.

GType	C type	type name	notes
<code>G_TYPE_CHAR</code>	char	gchar	
<code>G_TYPE_BOOLEAN</code>	int	gboolean	
<code>G_TYPE_INT</code>	int	gint	
<code>G_TYPE_FLOAT</code>	float	gfloat	
<code>G_TYPE_DOUBLE</code>	double	gdouble	
<code>G_TYPE_POINTER</code>	void *	gpointer	general pointer
<code>G_TYPE_STRING</code>	char *	gchararray	null-terminated Cstring
<code>G_TYPE_OBJECT</code>		GObject	
<code>GTK_TYPE_WINDOW</code>		GtkWindow	

A sample program `exp_constant_simple.c` is in `/src/expression` directory.

```

1  #include <gtk/gtk.h>
2
3  int
4  main (int argc, char **argv) {
5      GtkExpression *expression;
6      GValue value = G_VALUE_INIT;
7
8      /* Create an expression */
9      expression = gtk_constant_expression_new (G_TYPE_INT, 100);
10     /* Evaluate the expression */
11     if (gtk_expression_evaluate (expression, NULL, &value))
12         g_print ("The value is %d.\n", g_value_get_int (&value));
13     else
14         g_print ("The constant expression wasn't evaluated correctly.\n");
15     gtk_expression_unref (expression);
16     g_value_unset (&value);
17
18     return 0;
19 }
```

- 9: A constant expression is created. It holds an int value 100. The variable `expression` points to the expression.
- 11-14: Evaluates the expression. If it succeeds, show the value to the stdout. Otherwise show an error message.
- 15-16: Releases the expression and unsets the GValue.

Constant expression is usually used to give a constant value or instance to another expression.

5.3.2 Property Expression

A property expression (`GtkPropertyExpression`) looks up a property in a `GObject` instance. For example, a property expression that refers “label” property in a `GtkLabel` object is created like this.

```
expression = gtk_property_expression_new (GTK_TYPE_LABEL, another_expression, "label");
```

The second parameter `another_expression` is one of:

- An expression that gives a `GtkLabel` instance when it is evaluated.
- `NULL`. When `NULL` is given, a `GtkLabel` instance will be given when it is evaluated. The instance is called `this` object.

For example,

```
label = gtk_label_new ("Hello");
another_expression = gtk_constant_expression_new (GTK_TYPE_LABEL, label);
expression = gtk_property_expression_new (GTK_TYPE_LABEL, another_expression, "label");
```

If `expression` is evaluated, its second parameter `another_expression` is evaluated first in the evaluation process. The value of `another_expression` is the `label` (`GtkLabel` instance). Then, `expression` looks up `label` property of the `label` and the evaluation results in “Hello”.

In the example above, the second argument of `gtk_property_expression_new` is `another_expression`. But the second argument can be `NULL`. If it is `NULL`, `this` instance is used instead. `this` is given by `gtk_expression_evaluate` function.

There’s a simple program `exp_property_simple.c` in `/src/expression` directory.

```
1  #include <gtk/gtk.h>
2
3  int
4  main (int argc, char **argv) {
5      GtkWidget *label;
6      GtkExpression *expression;
7      GValue value = G_VALUE_INIT;
8
9      gtk_init ();
10     label = gtk_label_new ("Hello world.");
11     /* Create an expression */
12     expression = gtk_property_expression_new (GTK_TYPE_LABEL, NULL, "label");
13     /* Evaluate the expression */
14     if (gtk_expression_evaluate (expression, label, &value))
15         g_print ("The value is %s.\n", g_value_get_string (&value));
16     else
17         g_print ("The property expression wasn't evaluated correctly.\n");
18     gtk_expression_unref (expression);
19     g_value_unset (&value);
20
21     return 0;
22 }
```

- 9-10: `gtk_init` initializes GTK GUI toolkit. It isn’t usually necessary because the `GtkApplication` default startup handler does the initialization. A `GtkLabel` instance is created with the text “Hello world.”.
- 12: A property expression is created. It looks up a “label” property of a `GtkLabel` instance. But at the creation, no instance is given because the second argument is `NULL`. The expression just knows how to take the property from a future-given `GtkLabel` instance.

- 14-17: The function `gtk_expression_evaluate` evaluates the expression with a ‘this’ instance `label`. The result is stored in the `GValue` `value`. The function `g_value_get_string` gets a string from the `GValue`. But the string is owned by the `GValue` so you must not free the string.
- 18-19: Releases the expression and unset the `GValue`. At the same time the string in the `GValue` is freed.

If the second argument of `gtk_property_expression_new` isn’t `NULL`, it is another expression. The expression is owned by a newly created property expression. So, when the expressions are useless, you just release the last expression. Then it releases another expression it has.

5.3.3 Closure Expression

A closure expression calls a closure when it is evaluated. A closure is a generic representation of a callback (a pointer to a function). For information about closure, see [GObject API Reference – The GObject messaging system](#). There are simple closure example files `closure.c` and `closure_each.c` in the `/src/expression` directory.

There are two types of closure expressions, `GtkCClosureExpression` and `GtkClosureExpression`. They corresponds to `GCClosure` and `GClosure` respectively. When you program in C language, `GtkCClosureExpression` and `GCClosure` are appropriate.

A closure expression is created with `gtk_cclosure_expression_new` function.

`GtkExpression *`

```
gtk_cclosure_expression_new (GType value_type,
                             GClosureMarshal marshal,
                             guint n_params,
                             GtkExpression **params,
                             GCallback callback_func,
                             gpointer user_data,
                             GClosureNotify user_destroy);
```

- `value_type` is the type of the value when it is evaluated.
- `marshal` is a marshaller. You can assign `NULL`. If it is `NULL`, then `g_cclosure_marshal_generic()` is used as a marshaller. It is a generic marshaller function implemented via `libffi`.
- `n_params` is the number of parameters.
- `params` is an array of pointers to expressions. When the C closure expression is evaluated, each of these expressions is evaluated first, and their resulting values are passed to the callback function as arguments.
- `callback_func` is a callback function. It receives the `this` object followed by the evaluated params above. So, if `n_params` is 3, the number of arguments for the callback function is 4. (`this` and `params`. See below.)
- `user_data` is user data. You can add it for the closure. It is like `user_data` in `g_signal_connect`. If it is not necessary, assign `NULL`.
- `user_destroy` is a destroy notify for `user_data`. It is called to destroy `user_data` when it is no longer needed. If `NULL` is assigned to `user_data`, assign `NULL` to `user_destroy`, too.

Call back functions have the following format.

C-type

```
callback (this, param1, param2, ...)
```

For example,

```
int
callback (GObject *object, int x, const char *s)
```

The following is `exp_closure_simple.c` in `/src/expression`.


```

1  #include <gtk/gtk.h>
2
3  static int
4  calc (GtkLabel *label) { /* this object */
5      const char * s;
6      int i, j;
7
8      s = gtk_label_get_text (label); /* s is owned by the label. */
9      sscanf (s, "%d+%d", &i, &j);
10     return i+j;
11 }
12
13 int
14 main (int argc, char **argv) {
15     GtkExpression *expression;
16     GValue value = G_VALUE_INIT;
17     GtkLabel *label;
18
19     gtk_init ();
20     label = GTK_LABEL (gtk_label_new ("123+456"));
21     g_object_ref_sink (label);
22     expression = gtk_cclosure_expression_new (G_TYPE_INT, NULL, 0, NULL,
23                                              G_CALLBACK (calc), NULL, NULL);
24     if (gtk_expression_evaluate (expression, label, &value)) /* 'this' object is the label. */
25         g_print ("%d\n", g_value_get_int (&value));
26     else
27         g_print ("The closure expression wasn't evaluated correctly.\n");
28     gtk_expression_unref (expression);
29     g_value_unset (&value);
30     g_object_unref (label);
31
32     return 0;
33 }

```

- 3-11: A call back function. The parameter is only one and it is a ‘this’ object. It is a GtkLabel and its label is assumed to be “(number)+(number)”.
- 8-10: Retrieves two integers from the label and returns the sum of them. This function has no error report. If you want to return error report, change the return value type to be a pointer to a structure of gboolean and integer. One is for errors and the other is for the sum. The first argument of `gtk_cclosure_expression_new` is `G_TYPE_POINTER`. There is a sample program `exp_closure_with_error_report` in `/src/expression` directory.
- 19: The function ‘`gtk_init`’ initializes GTK. It is necessary for GtkLabel.
- 20: A GtkLabel instance is created with “123+456”.
- 21: The instance has floating reference. It is changed to an ordinary reference count.
- 22-23: Creates a closure expression. The value type is `G_TYPE_INT` when it is evaluated. This expression has no parameters. The callback function `calc` receives only `this` object (GtkLabel) as an argument.
- 24: Evaluates the expression with the label as a ‘this’ object.
- 25: If the evaluation succeeds, the sum of “123+456”, which is 579, is shown.
- 27: If it fails, an error message appears.
- 28-30: Releases the expression and the label. Unsets the value.

A closure expression is more flexible than other type of expressions because you can specify your own callback function.

5.3.4 GtkExpressionWatch

GtkExpressionWatch is a structure, not an object. It represents a watched GtkExpression. Two functions create GtkExpressionWatch structure. They are `gtk_expression_bind` and `gtk_expression_watch`.

5.3.4.1 gtk_expression_bind Function

This function binds the target object's property to the expression. If the value of the expression changes, the property reflects the value immediately.

```
GtkExpressionWatch*
gtk_expression_bind (
    GtkExpression* self,
    GObject* target,
    const char* property,
    GObject* this_
)
```

This function takes the ownership of the expression. So, if you want to own the expression, call `gtk_expression_ref ()` to increase the reference count of the expression. And you should unref it when it becomes no longer needed. If you don't own the expression, you don't care about releasing the expression.

An example `exp_bind.c` and `exp_bind.ui` is in `/src/expression` directory.



Figure 5.8: exp_bind

It includes a label and a scale. If you move the slider to the right, the scale value increases and the number on the label also increases. In the same way, if you move it to the left, the number on the label decreases. The label is bound to the scale value via an adjustment.

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <interface>
3   <object class='GtkApplicationWindow' id='win'>
4     <property name='default-width'>600</property>
5     <child>
6       <object class='GtkBox'>
7         <property name='orientation'>GTK_ORIENTATION_VERTICAL</property>
8         <child>
9           <object class='GtkLabel' id='label'>
10            <property name="label">10</property>
11          </object>
12        </child>
13      <child>
14        <object class='GtkScale'>
15          <property name='adjustment'>
16            <object class='GtkAdjustment' id='adjustment'>
17              <property name='upper'>20.0</property>
18              <property name='lower'>0.0</property>
19              <property name='value'>10.0</property>
20              <property name='step-increment'>1.0</property>
21              <property name='page-increment'>5.0</property>
```

```

22         <property name='page-size'>0.0</property>
23     </object>
24 </property>
25     <property name='digits'>0</property>
26     <property name='draw-value'>true</property>
27     <property name='has-origin'>true</property>
28     <property name='round-digits'>0</property>
29 </object>
30 </child>
31 </object>
32 </child>
33 </object>
34 </interface>

```

The UI file describes the following parent-child relationship.

```

GtkApplicationWindow (win) -- GtkBox +-+ GtkLabel (label)
                                +-+ GtkScale

```

Four GtkScale properties are defined.

- adjustment. GtkAdjustment provides the followings.
 - upper and lower: the range of the scale.
 - value: current value of the scale. It reflects the value of the scale.
 - step increment and page increment: When a user presses an arrow key or page up/down key, the scale moves by the step increment or page increment respectively.
 - page-size: When an adjustment is used with a scale, page-size is zero.
- digits: The number of decimal places that are displayed in the value.
- draw-value: Whether the value is displayed.
- has-origin: Whether the scale has the origin. If it's true, an orange bar appears between the origin and the current point.
- round-digits: The number of digits to round the value to when it changes. For example, if it is zero, the slider moves to an integer point.

```

1  #include <gtk/gtk.h>
2
3  GtkExpressionWatch *watch;
4
5  static int
6  f2i (GObject *object, double d) {
7      return (int) d;
8  }
9
10 static int
11 close_request_cb (GtkWindow *win) {
12     gtk_expression_watch_unwatch (watch);
13     return false;
14 }
15
16 static void
17 app_activate (GApplication *application) {
18     GtkApplication *app = GTK_APPLICATION (application);
19     gtk_window_present (gtk_application_get_active_window(app));
20 }
21
22 static void

```

```

23 app_startup (GApplication *application) {
24     GtkApplication *app = GTK_APPLICATION (application);
25     GtkBuilder *build;
26     GtkWidget *win, *label;
27     GtkAdjustment *adjustment;
28     GtkExpression *expression, *params[1];
29
30     /* Builds a window with exp.ui resource */
31     build = gtk_builder_new_from_resource ("/com/github/ToshioCP/exp/exp_bind.ui");
32     win = GTK_WIDGET (gtk_builder_get_object (build, "win"));
33     label = GTK_WIDGET (gtk_builder_get_object (build, "label"));
34     // scale = GTK_WIDGET (gtk_builder_get_object (build, "scale"));
35     adjustment = GTK_ADJUSTMENT (gtk_builder_get_object (build, "adjustment"));
36     gtk_window_set_application (GTK_WINDOW (win), app);
37     g_signal_connect (win, "close-request", G_CALLBACK (close_request_cb), NULL);
38     g_object_unref (build);
39
40     /* GtkExpressionWatch */
41     params[0] = gtk_property_expression_new (GTK_TYPE_ADJUSTMENT, NULL, "value");
42     expression = gtk_cclosure_expression_new (G_TYPE_INT, NULL, 1, params, G_CALLBACK (f2i), NULL, NULL);
43     watch = gtk_expression_bind (expression, label, "label", adjustment); /* watch takes the ownership */
44 }
45
46 #define APPLICATION_ID "com.github.ToshioCP.exp_watch"
47
48 int
49 main (int argc, char **argv) {
50     GtkApplication *app;
51     int stat;
52
53     app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
54
55     g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
56     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
57
58     stat = g_application_run (G_APPLICATION (app), argc, argv);
59     g_object_unref (app);
60     return stat;
61 }

```

The point of the program is:

- 41-42: Two expressions are defined. One is a property expression and the other is a closure expression. The property expression looks up the “value” property of the adjustment instance. The closure expression just converts the double into an integer.
- 43: `gtk_expression_bind` binds the label property of the `GtkLabel` instance to the integer returned by the closure expression. It creates a `GtkExpressionWatch` structure. The binding remains active while the watch exists. When the window is destroyed, the scale and adjustment are also destroyed. The watch recognizes a change in the value of the expression and tries to change the property of the label. Obviously, it is not a correct behavior. The watch should be unwatched before the window is destroyed.
- 37: Connects the “close-request” signal on the window to a handler `close_request_cb`. This signal is emitted when the close button is clicked. The “close-request” handler is called just before the window closes. It is the right moment to make the `GtkExpressionWatch` unwatched.

- 10-14: “close-request” signal handler. The function `gtk_expression_watch_unwatch (watch)` makes the watch stop watching the expression. It also releases the expression.

If you want to bind a property to an expression, `gtk_expression_bind` is the best choice. You can do it with `gtk_expression_watch` function, but it is less suitable.

5.3.4.2 `gtk_expression_watch` Function

```
GtkExpressionWatch*
gtk_expression_watch (
    GtkExpression* self,
    GObject* this_,
    GtkExpressionNotify notify,
    gpointer user_data,
    GDestroyNotify user_destroy
)
```

The function doesn’t take the ownership of the expression. It differs from `gtk_expression_bind`. So, you need to release the expression when it is useless. It creates a `GtkExpressionWatch` structure. The third parameter `notify` is a callback to invoke when the expression changes. You can set `user_data` to give it to the callback. The last parameter is a function to destroy the `user_data` when the watch is unwatched. Put `NULL` if you don’t need them.

Notify callback has the following format.

```
void
notify (
    gpointer user_data
)
```

This function is used to do something when the value of the expression changes. But if you want to bind a property to the value, use `gtk_expression_bind` instead.

There’s a sample program `exp_watch.c` in `/src/expression` directory. It outputs the width of the window to the standard output.

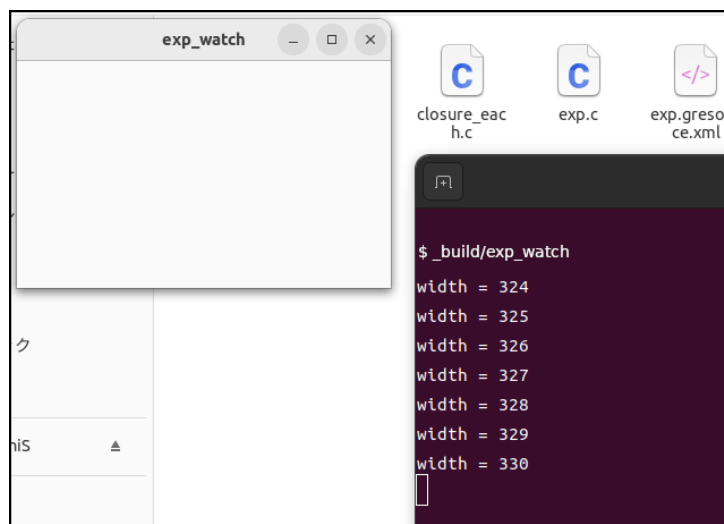


Figure 5.9: `exp_watch`

When you resize the window, the width is displayed in the terminal.

```
1 #include <gtk/gtk.h>
2
```

```

3  GtkExpression *expression;
4  GtkExpressionWatch *watch;
5
6  static void
7  notify (gpointer user_data) {
8      GValue value = G_VALUE_INIT;
9
10     if (gtk_expression_watch_evaluate (watch, &value))
11         g_print ("width = %d\n", g_value_get_int (&value));
12     else
13         g_print ("evaluation failed.\n");
14 }
15
16 static int
17 close_request_cb (GtkWindow *win) {
18     gtk_expression_watch_unwatch (watch);
19     gtk_expression_unref (expression);
20     return false;
21 }
22
23 static void
24 app_activate (GApplication *application) {
25     GtkApplication *app = GTK_APPLICATION (application);
26     gtk_window_present (gtk_application_get_active_window(app));
27 }
28
29 static void
30 app_startup (GApplication *application) {
31     GtkApplication *app = GTK_APPLICATION (application);
32     GtkWidget *win;
33
34     win = GTK_WIDGET (gtk_application_window_new (app));
35     g_signal_connect (win, "close-request", G_CALLBACK (close_request_cb), NULL);
36
37     expression = gtk_property_expression_new (GTK_TYPE_WINDOW, NULL, "default-width");
38     watch = gtk_expression_watch (expression, win, notify, NULL, NULL);
39 }
40
41 #define APPLICATION_ID "com.github.ToshioCP.exp_watch"
42
43 int
44 main (int argc, char **argv) {
45     GtkApplication *app;
46     int stat;
47
48     app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
49
50     g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
51     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
52
53     stat = g_application_run (G_APPLICATION (app), argc, argv);
54     g_object_unref (app);
55     return stat;

```

56 }

- 37: A property expression looks up the “default-width” property of the window.
- 38: Create a watch structure for the expression. The callback `notify` is called every time the value of the expression changes. The ‘this’ object is `win`, so the expression returns the default width of the window.
- 6-14: The callback function `notify`. It uses `gtk_expression_watch_evaluate` to get the value of the expression. The ‘this’ object is given in advance (when the watch is created). It outputs the window width to the standard output.
- 16-21: A handler for the “close-request” signal on the window. It stops the watch. In addition, it releases the reference to the expression. Because `gtk_expression_watch` doesn’t take the ownership of the expression, you own it. So, the release is necessary.

5.3.5 Gtkexpression in Ui Files

GtkBuilder supports GtkExpressions. There are four tags.

- constant tag to create constant expression. Type attribute specifies the type name of the value. If no type is specified, the type is assumed to be an object. The content is the value of the expression.
- lookup tag to create property expression. Type attribute specifies the type of the object. Name attribute specifies the property name. The content is an expression or object which has the property to look up. If there’s no content, ‘this’ object is used.
- closure tag to create closure expression. Type attribute specifies the type of the returned value. Function attribute specifies the callback function. The contents of the tag are arguments that are expressions.
- binding tag to bind a property to an expression. It is put in the content of an object tag. Name attribute specifies the property name of the object. The content is an expression.

```
<constant type="gchararray">Hello world</constant>
<lookup name="label" type="GtkLabel">label</lookup>
<closure type="gint" function="callback_function"></closure>
<bind name="label">
  <lookup name="default-width">win</lookup>
</bind>
```

These tags are usually used for `GtkBuilderListItemFactory`.

```
<interface>
  <template class="GtkListItem">
    <property name="child">
      <object class="GtkLabel">
        <binding name="label">
          <lookup name="string" type="GtkStringObject">
            <lookup name="item">GtkListItem</lookup>
          </lookup>
        </binding>
      </object>
    </property>
  </template>
</interface>
```

`GtkBuilderListItemFactory` uses `GtkBuilder` to extends the `GtkListItem` with the XML data.

In the xml file above, “`GtkListItem`” is an instance of the `GtkListItem` template. It is the ‘this’ object given to the expressions. Please refer to the [GTK Development Blog](#) for further information about “this” objects.

GtkBuilder calls `gtk_expression_bind` function when it finds a binding tag. The function sets the ‘this’ object like this:

1. If the binding tag has object attribute, the object will be the ‘this’ object.
2. If the current object of the GtkBuilder exists, it will be the ‘this’ object. That’s why a `GtkListItem` instance is the ‘this’ object of the XML data for a `GtkBuilderListItemFactory`.
3. Otherwise, the target object of the binding tag will be the ‘this’ object.

GTK 4 document doesn’t describe information about “this” object when expressions are defined in a UI file. The information above is found from the GTK 4 source files and it is possible to include mistakes. If you have accurate information, please let me know.

A sample program `exp.c` and a UI file `exp.ui` is in `/src/expression` directory. The UI file includes lookup, closure and bind tags. No constant tag is included. However, constant tags are not used so often.

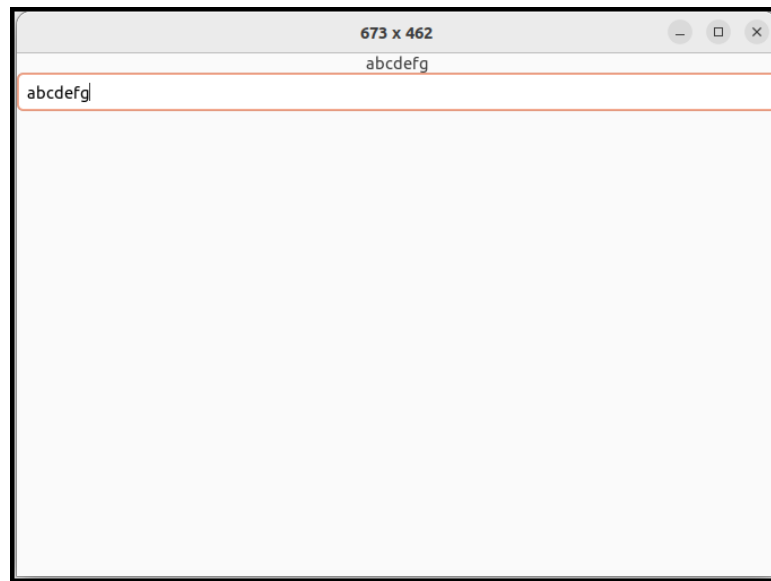


Figure 5.10: `exp.c`

If you resize the window, the size is shown at the title of the window. If you type characters in the entry, the same characters appear on the label.

The UI file is as follows.

```

1  <?xml version="1.0" encoding="UTF-8"?>
2  <interface>
3    <object class="GtkApplicationWindow" id="win">
4      <binding name="title">
5        <closure type="gchararray" function="set_title">
6          <lookup name="default-width" type="GtkWindow"></lookup>
7          <lookup name="default-height" type="GtkWindow"></lookup>
8        </closure>
9      </binding>
10     <property name="default-width">600</property>
11     <property name="default-height">400</property>
12     <child>
13       <object class="GtkBox">
14         <property name="orientation">GTK_ORIENTATION_VERTICAL</property>
15         <child>
16           <object class="GtkLabel">
17             <binding name="label">
```



```

18         <lookup name="text">
19             buffer
20         </lookup>
21     </binding>
22 </object>
23 </child>
24 <child>
25     <object class="GtkEntry">
26         <property name="buffer">
27             <object class="GtkEntryBuffer" id="buffer"></object>
28         </property>
29     </object>
30 </child>
31 </object>
32 </child>
33 </object>
34 </interface>

```

- 4-9: The title property of the main window is bound to a closure expression. Its callback function `set_title` is defined in the C source file. It returns a string because the type attribute of the tag is “gchararray”. Two parameters are given to the function. They are width and height of the window. Lookup tags don’t have contents, so ‘this’ object is used to look up the properties. The ‘this’ object is `win`, which is the target of the binding (`win` includes the binding tag).
- 17-21: The “label” property of the `GtkLabel` instance is bound to the “text” property of `buffer`, which is the buffer of `GtkEntry` defined in line 25. If a user types characters in the entry, the same characters appear on the label.

The C source file is as follows.

```

1  #include <gtk/gtk.h>
2
3  char *
4  set_title (GtkWidget *win, int width, int height) {
5      return g_strdup_printf ("%d x %d", width, height);
6  }
7
8  static void
9  app_activate (GApplication *application) {
10     GtkApplication *app = GTK_APPLICATION (application);
11     gtk_window_present (gtk_application_get_active_window(app));
12 }
13
14 static void
15 app_startup (GApplication *application) {
16     GtkApplication *app = GTK_APPLICATION (application);
17     GtkBuilder *build;
18     GtkWidget *win;
19
20     build = gtk_builder_new_from_resource ("/com/github/ToshioCP/exp/exp.ui");
21     win = GTK_WIDGET (gtk_builder_get_object (build, "win"));
22     gtk_window_set_application (GTK_WINDOW (win), app);
23     g_object_unref (build);
24 }
25
26 #define APPLICATION_ID "com.github.ToshioCP.exp"

```

```

27
28 int
29 main (int argc, char **argv) {
30     GtkApplication *app;
31     int stat;
32
33     app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
34
35     g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
36     g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
37
38     stat = g_application_run (G_APPLICATION (app), argc, argv);
39     g_object_unref (app);
40     return stat;
41 }

```

- 4-6: The callback function. It returns a string (w)x(h), where the w and h are the width and height of the window. String duplication is necessary.

The C source file is very simple because almost everything is done in the UI file.

5.3.6 Conversion between GValues

If you bind different type properties, type conversion is automatically done. Suppose the `label` property (string) is bound to the `default-width` property (int).

```

<object class="GtkLabel">
  <binding name="label">
    <lookup name="default-width">
      win
    </lookup>
  </binding>
</object>

```

The expression created by the lookup tag returns a int type GValue. On the other hand “label” property holds a string type GValue. When a GValue is copied to another GValue, the type is automatically converted if possible. If the current width is 100, an int 100 is converted to a string "100".

If you use `g_object_get` and `g_object_set` to copy properties, the value is automatically converted.

5.3.7 Meson.build

The source files are in `/src/expression` directory. You can build all the files at once.

```

$ cd src/expression
$ meson setup _build
$ ninja -C _build

```

For example, if you want to run “exp”, which is the executable file from “exp.c”, type `_build/exp`. You can run other programs as well.

The file `meson.build` is as follows.

```

1 project('exp', 'c')
2
3 gtkdep = dependency('gtk4')
4
5 gnome=import('gnome')
6 resources = gnome.compile_resources('resources','exp.gresource.xml')

```

7

8 `sourcefiles=files('exp.c')`

9

10 `executable('exp', sourcefiles, resources, dependencies: gtkdep, export_dynamic: true, install: false)`11 `executable('exp_constant', 'exp_constant.c', dependencies: gtkdep, export_dynamic: true, install: false)`12 `executable('exp_constant_simple', 'exp_constant_simple.c', dependencies: gtkdep, export_dynamic: true, install: false)`13 `executable('exp_property_simple', 'exp_property_simple.c', dependencies: gtkdep, export_dynamic: true, install: false)`14 `executable('closure', 'closure.c', dependencies: gtkdep, export_dynamic: true, install: false)`15 `executable('closure_each', 'closure_each.c', dependencies: gtkdep, export_dynamic: true, install: false)`16 `executable('exp_closure_simple', 'exp_closure_simple.c', dependencies: gtkdep, export_dynamic: true, install: false)`17 `executable('exp_closure_with_error_report', 'exp_closure_with_error_report.c', dependencies: gtkdep, export_dynamic: true, install: false)`18 `executable('exp_bind', 'exp_bind.c', resources, dependencies: gtkdep, export_dynamic: true, install: false)`19 `executable('exp_watch', 'exp_watch.c', dependencies: gtkdep, export_dynamic: true, install: false)`20 `executable('exp_test', 'exp_test.c', resources, dependencies: gtkdep, export_dynamic: true, install: false)`

5.4 GtkColumnView

5.4.1 GtkColumnView

GtkColumnView is like GtkListView, but it has multiple columns. Each column is GtkColumnViewColumn.



Figure 5.11: Column View

- GtkColumnView has “model” property. The property points to a GtkSelectionModel object.
- Each GtkColumnViewColumn has the “factory” property. The property points to a GtkListItemFactory (GtkSignalListItemFactory or GtkBuilderListItemFactory).
- The factory connects GtkListItem and items of GtkSelectionModel. And the factory builds the descendant widgets of GtkColumnView to display the items on the screen. This process is the same as the one in GtkListView.

The following diagram shows how it works.

The example in this section is a window that displays information of files in the current directory. The information is the name, size and last modified datetime of files. So, there are three columns.

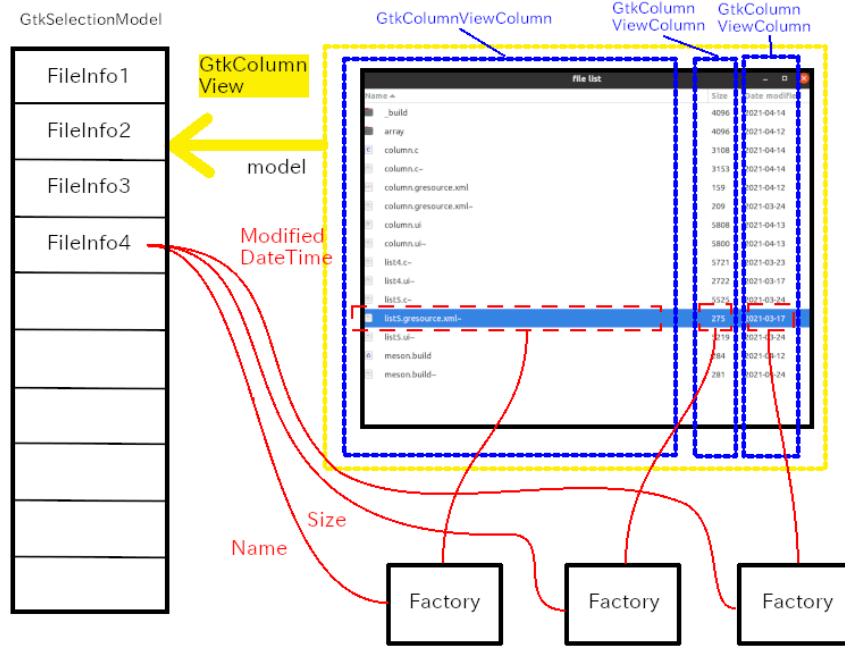


Figure 5.12: ColumnView

In addition, the example uses `GtkSortListModel` and `GtkSorter` to sort the information.

5.4.2 column.ui

UI file specifies widgets and list item templates.

```

1  <?xml version="1.0" encoding="UTF-8"?>
2  <interface>
3      <object class="GtkApplicationWindow" id="win">
4          <property name="title">file list</property>
5          <property name="default-width">800</property>
6          <property name="default-height">600</property>
7          <child>
8              <object class="GtkScrolledWindow">
9                  <property name="hexpand">TRUE</property>
10                 <property name="vexpand">TRUE</property>
11                 <child>
12                     <object class="GtkColumnView" id="columnview">
13                         <property name="model">
14                             <object class="GtkNoSelection">
15                                 <property name="model">
16                                     <object class="GtkSortListModel">
17                                         <property name="model">
18                                             <object class="GtkDirectoryList" id="directorylist">
19                                                 <property name="attributes">standard::name,standard::icon,standard::size,tim
20                                             </object>
21                                         </property>
22                                         <binding name="sorter">
23                                             <lookup name="sorter">columnview</lookup>
24                                         </binding>
25                                     </object>

```

```

26         </property>
27     </object>
28 </property>
29 <child>
30     <object class="GtkColumnViewColumn">
31         <property name="title">Name</property>
32         <property name="expand">TRUE</property>
33         <property name="factory">
34             <object class="GtkBuilderListItemFactory">
35                 <property name="bytes"><![CDATA[
36 <?xml version="1.0" encoding="UTF-8"?>
37 <interface>
38     <template class="GtkListItem">
39         <property name="child">
40             <object class="GtkBox">
41                 <property name="orientation">GTK_ORIENTATION_HORIZONTAL</property>
42                 <property name="spacing">20</property>
43                 <child>
44                     <object class="GtkImage">
45                         <binding name="gicon">
46                             <closure type="GIcon" function="get_icon_factory">
47                                 <lookup name="item">GtkListItem</lookup>
48                             </closure>
49                         </binding>
50                     </object>
51                 </child>
52                 <child>
53                     <object class="GtkLabel">
54                         <property name="hexpand">TRUE</property>
55                         <property name="xalign">0</property>
56                         <binding name="label">
57                             <closure type="gchararray" function="get_file_name_factory">
58                                 <lookup name="item">GtkListItem</lookup>
59                             </closure>
60                         </binding>
61                     </object>
62                 </child>
63             </object>
64         </property>
65     </template>
66 </interface>
67         ]]></property>
68     </object>
69 </property>
70 <property name="sorter">
71     <object class="GtkStringSorter">
72         <property name="expression">
73             <closure type="gchararray" function="get_file_name">
74                 </closure>
75             </property>
76         </object>
77     </property>
78 </object>

```

```

79         </child>
80     <child>
81         <object class="GtkColumnViewColumn">
82             <property name="title">Size</property>
83             <property name="factory">
84                 <object class="GtkBuilderListItemFactory">
85                     <property name="bytes"><![CDATA[
86 <?xml version="1.0" encoding="UTF-8"?>
87 <interface>
88     <template class="GtkListItem">
89         <property name="child">
90             <object class="GtkLabel">
91                 <property name="hexpand">TRUE</property>
92                 <property name="xalign">0</property>
93                 <binding name="label">
94                     <closure type="gint64" function="get_file_size_factory">
95                         <lookup name="item">GtkListItem</lookup>
96                     </closure>
97                 </binding>
98             </object>
99         </property>
100     </template>
101 </interface>
102         ]]></property>
103     </object>
104 </property>
105     <property name="sorter">
106         <object class="GtkNumericSorter">
107             <property name="expression">
108                 <closure type="gint64" function="get_file_size">
109                     </closure>
110                 </property>
111                 <property name="sort-order">GTK_SORT_ASCENDING</property>
112             </object>
113         </property>
114     </object>
115 </child>
116 <child>
117     <object class="GtkColumnViewColumn">
118         <property name="title">Date modified</property>
119         <property name="factory">
120             <object class="GtkBuilderListItemFactory">
121                 <property name="bytes"><![CDATA[
122 <?xml version="1.0" encoding="UTF-8"?>
123 <interface>
124     <template class="GtkListItem">
125         <property name="child">
126             <object class="GtkLabel">
127                 <property name="hexpand">TRUE</property>
128                 <property name="xalign">0</property>
129                 <binding name="label">
130                     <closure type="gchararray" function="get_file_time_modified_factory">
131                         <lookup name="item">GtkListItem</lookup>

```

```

132         </closure>
133     </binding>
134 </object>
135 </property>
136 </template>
137 </interface>
138
139         <property name="model">
140             <object class="GtkNoSelection">
141                 <property name="expression">
142                     <closure type="gint64" function="get_file_unixtime_modified">
143                         </closure>
144                     </property>
145                     <property name="sort-order">GTK_SORT_ASCENDING</property>
146                 </object>
147             </property>
148         </object>
149     </child>
150 </object>
151 </child>
152 </object>
153 </child>
154 </object>
155 </child>
156 </object>
157 </interface>

```

- 3-12: GtkApplicationWindow has a child widget GtkScrolledWindow. GtkScrolledWindow has a child widget GtkColumnView.
- 12-18: GtkColumnView has “model” property. It points to GtkSelectionModel interface. GtkNoSelection class is used as GtkSelectionModel. Similarly, it also has a “model” property. It points to GtkSortListModel. This list model supports sorting the list. It will be explained in the later subsection. And it also has “model” property. It points to GtkDirectoryList. Therefore, the chain is: GtkColumnView => GtkNoSelection => GtkSortListModel => GtkDirectoryList.
- 18-20: GtkDirectoryList. It is a list of GFileInfo, which holds information of files under a directory. It has “attributes” property. It specifies which attributes are kept in each GFileInfo.
 - “standard::name” is the name of the file.
 - “standard::icon” is the GIcon object of the file
 - “standard::size” is the file size.
 - “time::modified” is the date and time the file was last modified.
- 29-79: The first GtkColumnViewColumn object. There are four properties, “title”, “expand”, “factory” and “sorter”.
- 31: Sets the “title” property to “Name”. This is the title on the header of the column.
- 32: Sets the “expand” property to TRUE to allow the column to expand as much as possible. (See the image above).
- 33- 69: Sets the “factory” property to GtkBuilderListItemFactory. The factory has “bytes” property which holds a UI string to define a template to extend GtkListItem class. The CDATA section (line 36-66) is the UI string to put into the “bytes” property. The contents are the same as the UI file `factory_list.ui` in Section 30.
- 70-77: Sets the “sorter” property to GtkStringSorter object. This object provides a sorter that compares strings. It has “expression” property. A closure tag with a string type function `get_file_name` is used here. The function will be explained later.
- 80-115: The second GtkColumnViewColumn object. Its sorter property is set to GtkNumericSorter.
- 116-151: The third GtkColumnViewColumn object. Its sorter property is set to GtkNumericSorter.

5.4.3 GtkSortListModel and GtkSorter

GtkSortListModel is a list model that sorts its elements according to a GtkSorter instance assigned to the “sorter” property. The property is bound to “sorter” property of GtkColumnView in line 22 to 24.

```
<object class="GtkSortListModel" id="sortlist">
... ..
  <binding name="sorter">
    <lookup name="sorter">columnview</lookup>
  </binding>
```

Therefore, `columnview` determines how to sort the list model. The “sorter” property of GtkColumnView is a read-only GtkSorter and it always refers to the current column’s “sorter” property. When a user clicks a column header, GtkColumnView updates its own “sorter” property to point to that column’s sorter.

The binding above makes an indirect connection between the “sorter” property of GtkSortListModel and the “sorter” property of each column.

GtkSorter compares two items (GObject or its descendant). GtkSorter has several child objects.

- GtkStringSorter compares strings taken from the items.
- GtkNumericSorter compares numbers taken from the items.
- GtkCustomSorter uses a callback to compare.
- GtkMultiSorter combines multiple sorters.

The example uses GtkStringSorter and GtkNumericSorter.

GtkStringSorter uses GtkExpression to get the strings from the items (objects). The GtkExpression is stored in the “expression” property of the GtkStringSorter. When GtkStringSorter compares two items, it evaluates the expression by calling `gtk_expression_evaluate` function. It assigns each item to the second argument (‘this’ object) of the function.

In the UI file above, the GtkExpression is in the line 71 to 76.

```
<object class="GtkStringSorter">
  <property name="expression">
    <closure type="gchararray" function="get_file_name">
    </closure>
  </property>
</object>
```

The GtkExpression calls `get_file_name` function when it is evaluated.

```
1 char *
2 get_file_name (GFileInfo *info) {
3   return G_IS_FILE_INFO (info) ? g_strdup (g_file_info_get_name (info)) : NULL;
4 }
```

The function is given the item (GFileInfo) of the GtkSortListModel as an argument (this object). But you need to be careful that it can be NULL while the list item is being recycled. So, `G_IS_FILE_INFO (info)` is always necessary in callback functions. The function retrieves a filename from `info`. The string is owned by `info` so it is necessary to duplicate. And it returns the copied string.

GtkNumericSorter compares numbers. It is used in the line 106 to 112 and line 142 to 148. The lines from 106 to 112 is:

```
<object class="GtkNumericSorter">
  <property name="expression">
    <closure type="gint64" function="get_file_size">
    </closure>
  </property>
```



```
<property name="sort-order">GTK_SORT_ASCENDING</property>
</object>
```

The closure tag specifies a callback function `get_file_size`.

```
1 goffset
2 get_file_size (GFileInfo *info) {
3     return G_IS_FILE_INFO (info) ? g_file_info_get_size (info) : -1;
4 }
```

It just returns the size of `info`. The type of the size is `goffset`. The type `goffset` is the same as `gint64`.

The lines from 142 to 148 is:

```
<object class="GtkNumericSorter" id="sorter_datetime_modified">
  <property name="expression">
    <closure type="gint64" function="get_file_unixtime_modified">
    </closure>
  </property>
  <property name="sort-order">GTK_SORT_ASCENDING</property>
</object>
```

The closure tag specifies a callback function `get_file_unixtime_modified`.

```
1 gint64
2 get_file_unixtime_modified (GFileInfo *info) {
3     GDateTime *dt;
4
5     dt = G_IS_FILE_INFO (info) ? g_file_info_get_modification_date_time (info) : NULL;
6     return dt ? g_date_time_to_unix (dt) : -1;
7 }
```

It gets the modification date and time (`GDateTime` type) of `info`. Then it gets a Unix time from `dt`. Unix time, sometimes called Unix epoch, is the number of seconds that have elapsed since 00:00:00 UTC on 1 January 1970. It returns the Unix time as a `gint64`.

5.4.4 column.c

`column.c` is as follows. The code is simple and short thanks to `column.ui`.

```
1 #include <gtk/gtk.h>
2
3 /* functions (closures) for GtkBuilderListItemFactory */
4 GIcon *
5 get_icon_factory (GtkListItem *item, GFileInfo *info) {
6     GIcon *icon;
7
8     /* g_file_info_get_icon can return NULL */
9     icon = G_IS_FILE_INFO (info) ? g_file_info_get_icon (info) : NULL;
10    return icon ? g_object_ref (icon) : NULL;
11 }
12
13 char *
14 get_file_name_factory (GtkListItem *item, GFileInfo *info) {
15     return G_IS_FILE_INFO (info) ? g_strdup (g_file_info_get_name (info)) : NULL;
16 }
17
18 /* goffset is defined as gint64 */
```

```

19  /* It is used for file offsets. */
20  goffset
21  get_file_size_factory (GtkListItem *item, GFileInfo *info) {
22      return G_IS_FILE_INFO (info) ? g_file_info_get_size (info) : -1;
23  }
24
25  char *
26  get_file_time_modified_factory (GtkListItem *item, GFileInfo *info) {
27      GDateTime *dt;
28
29      /* g_file_info_get_modification_date_time can return NULL */
30      dt = G_IS_FILE_INFO (info) ? g_file_info_get_modification_date_time (info) : NULL;
31      return dt ? g_date_time_format (dt, "%F") : NULL;
32  }
33
34  /* Functions (closures) for GtkSorter */
35  char *
36  get_file_name (GFileInfo *info) {
37      return G_IS_FILE_INFO (info) ? g_strdup (g_file_info_get_name (info)) : NULL;
38  }
39
40  goffset
41  get_file_size (GFileInfo *info) {
42      return G_IS_FILE_INFO (info) ? g_file_info_get_size (info) : -1;
43  }
44
45  gint64
46  get_file_unixtime_modified (GFileInfo *info) {
47      GDateTime *dt;
48
49      dt = G_IS_FILE_INFO (info) ? g_file_info_get_modification_date_time (info) : NULL;
50      return dt ? g_date_time_to_unix (dt) : -1;
51  }
52
53  static void
54  app_activate (GApplication *application) {
55      GtkApplication *app = GTK_APPLICATION (application);
56      gtk_window_present (gtk_application_get_active_window(app));
57  }
58
59  static void
60  app_startup (GApplication *application) {
61      GtkApplication *app = GTK_APPLICATION (application);
62      GFile *file;
63      GtkBuilder *build = gtk_builder_new_from_resource ("/com/github/ToshioCP/column/column.ui");
64      GtkWidget *win = GTK_WIDGET (gtk_builder_get_object (build, "win"));
65      GtkDirectoryList *directorylist = GTK_DIRECTORY_LIST (gtk_builder_get_object (build, "directorylist"));
66      g_object_unref (build);
67
68      gtk_window_set_application (GTK_WINDOW (win), app);
69
70      file = g_file_new_for_path (".");
71      gtk_directory_list_set_file (directorylist, file);

```

```

72  g_object_unref (file);
73  }
74
75  #define APPLICATION_ID "com.github.ToshioCP.columnview"
76
77  int
78  main (int argc, char **argv) {
79      GtkApplication *app;
80      int stat;
81
82      app = gtk_application_new (APPLICATION_ID, G_APPLICATION_DEFAULT_FLAGS);
83
84      g_signal_connect (app, "startup", G_CALLBACK (app_startup), NULL);
85      g_signal_connect (app, "activate", G_CALLBACK (app_activate), NULL);
86
87      stat = g_application_run (G_APPLICATION (app), argc, argv);
88      g_object_unref (app);
89      return stat;
90  }

```

5.4.5 Compilation and Execution.

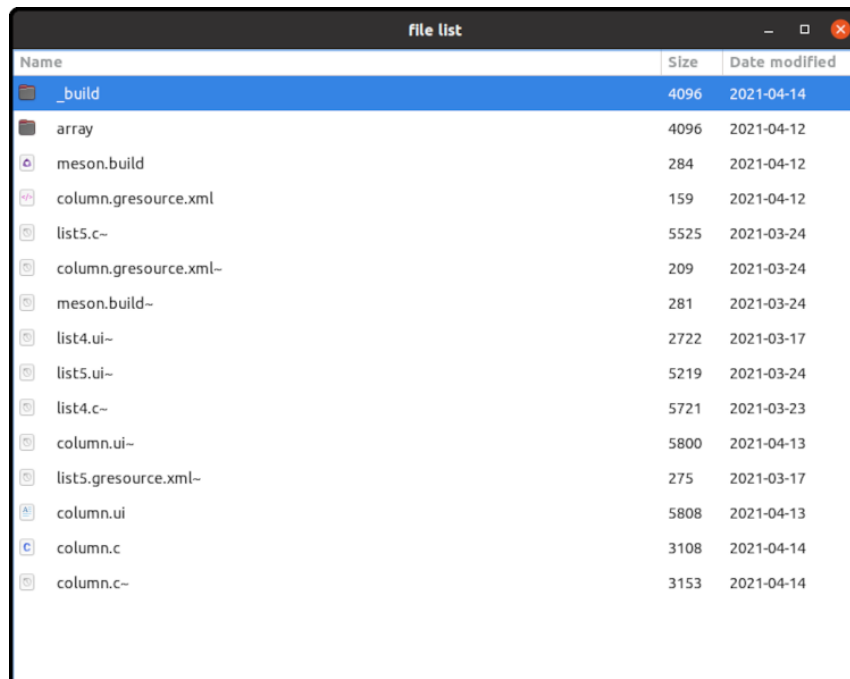
All the source files are in `src/column` directory. Change your current directory to the directory and type the following.

```

$ cd src/column
$ meson setup _build
$ ninja -C _build
$ _build/column

```

Then, a window appears.



Name	Size	Date modified
_build	4096	2021-04-14
array	4096	2021-04-12
meson.build	284	2021-04-12
column.gresource.xml	159	2021-04-12
list5.c~	5525	2021-03-24
column.gresource.xml~	209	2021-03-24
meson.build~	281	2021-03-24
list4.ui~	2722	2021-03-17
list5.ui~	5219	2021-03-24
list4.c~	5721	2021-03-23
column.ui~	5800	2021-04-13
list5.gresource.xml~	275	2021-03-17
column.ui	5808	2021-04-13
column.c	3108	2021-04-14
column.c~	3153	2021-04-14

Figure 5.13: Column View

If you click the header of a column, then the entire list is sorted by the column. If you click the header of

another column, then the list is sorted by the newly selected column.

GtkColumnView is very useful and it can manage very big GListModel. It is possible to use it for file list, application list, database frontend and so on.

5.5 GtkSignalListItemFactory

5.5.1 GtkSignalListItemFactory vs GtkBuilderListItemFactory

GtkBuilderListItemFactory is convenient when GtkListView just shows the contents of a list model. Its binding direction is always from an item of the list to a child of GtkListItem.

However, it is insufficient for dynamic connections. For example, suppose you want to edit the contents of a list. You can set a GtkText widget as the child of a GtkListItem to allow users to edit text. You need to update the list item when the text in the GtkText is edited. The data flow is in the opposite direction compared to GtkBuilderListItemFactory—it goes from the GtkText widget back to the list item. You can achieve this using GtkSignalListItemFactory, which provides more flexibility.

This section shows just some parts of the source file [listeditor.c](#). If you want to see the whole codes, see [src/listeditor](#) directory of the Gtk4 tutorial repository.

5.5.2 List Editor

The sample program is a list editor and the data in the list are strings. It's similar to a line editor. It reads a text file line by line. Each line is an item of the list. The list is displayed with a GtkListView. The list view contains strings which are the contents of the corresponding items in the list model.

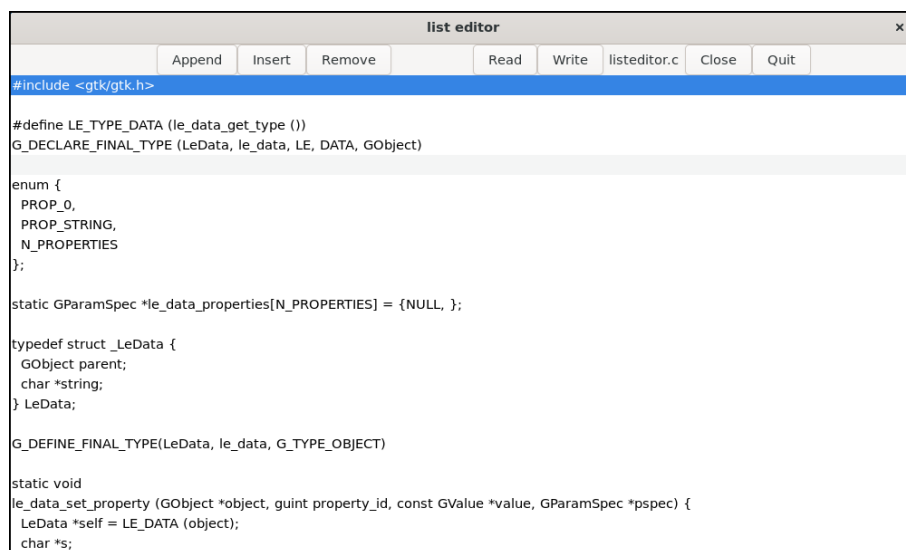


Figure 5.14: List editor

You can compile the source file and execute it as follows.

- Download the program from the [repository](#).
- Change your current directory to `src/listeditor`.
- Type the following on your commandline.

```

$ meson setup _build
$ ninja -C _build
$ _build/listeditor

```

The window has buttons:

- **Append button:** appends a data after the current item, or at the end of the model if no current item exists.
- **Insert button:** inserts a row before the current item, or at the top of the model if no current item exists.
- **Remove button:** removes the current item.
- **Read button:** reads a file.
- **Write button:** Writes the contents to the file shown on the toolbar. If no filename is set, a file dialog opens to let you choose one.
- **Close button:** remove all the items from the model.
- **Quit button:** quits the application.

You can click on the content area and select a row. The corresponding item to the row will be the current item. The current position (zero-based) is shown at the left of the tool bar.

5.5.3 Connect a GtkText Instance and an Item in the Model

The ListView sets its factory property to GtkSignalListItemFactory. Its two signals “setup” and “bind” are connected to the handlers `setup_cb` and `bind_cb` respectively. The remaining two signals “unbind” and “teardown” aren’t used in this program. The following shows the signal handlers.

```

1  static void
2  setup_cb (GtkListItemFactory *factory, GtkListItem *listitem) {
3      GtkWidget *text = gtk_text_new ();
4
5      gtk_list_item_set_child (listitem, GTK_WIDGET (text));
6      gtk_editable_set_alignment (GTK_EDITABLE (text), 0.0);
7      gtk_widget_set_can_target (GTK_WIDGET (text), FALSE);
8
9      g_signal_connect (GTK_EDITABLE (text), "changed", G_CALLBACK (text_changed_cb), listitem);
10     g_signal_connect (listitem, "notify::selected", G_CALLBACK (item_selected_cb), text);
11 }
12
13 static void
14 bind_cb (GtkListItemFactory *factory, GtkListItem *listitem) {
15     GtkWidget *text = gtk_list_item_get_child (listitem);
16     LeData *data = LE_DATA (gtk_list_item_get_item(listitem));
17
18     if (data) {
19         g_signal_handlers_block_by_func (text, (gpointer) text_changed_cb, listitem);
20         gtk_editable_set_text (GTK_EDITABLE (text), le_data_look_string (data));
21         gtk_editable_set_position (GTK_EDITABLE (text), -1);
22         g_signal_handlers_unblock_by_func (text, (gpointer) text_changed_cb, listitem);
23     }
24 }
```

- 5-7: `setup_cb` creates the child widget of the list item. This widget will be a row of the GtkListView instance.
- 9-10: `setup_cb` connects the “changed” signal on the GtkText instance to the handler `text_changed_cb`. This handler copies the text in the GtkText to the item when the GtkText is changed. `setup_cb` also connects the “notify::selected” signal on the GtkListItem to the handler `item_selected_cb`. The handler make the GtkText grab the keyboard focus to prepare keyboard input.
- These instance and signals created by `setup_cb` will be automatically destroyed when the list item disposes itself. So, teardown signal handler is not necessary.

- 83-86: `bind_cb` just copy the string in the item in the model to the `GtkText`. The macro function `g_signal_block_by_func` avoids unnecessary copy from `GtkText` to item at the update of the `GtkText`. No instances are created in this process, so “unbind” signal handler is not necessary.
- 5-7: `setup_cb` creates the child widget for the list item, which serves as a row in the `GtkListView`.
- 9-10: `setup_cb` connects the `GtkText`’s “changed” signal to the `text_changed_cb` handler. This handler updates the item in the model with the new text whenever the `GtkText` is modified. Additionally, it connects the “notify::selected” signal of the `GtkListItem` to `item_selected_cb`, which makes the `GtkText` grab keyboard focus to prepare for immediate input. The “notify” signal is inherited from `GObject` class.
- The instances and signal connections created in `setup_cb` are automatically cleaned up when the list item is disposed. Therefore, a “teardown” signal handler is unnecessary.
- 19-22: `bind_cb` simply copies the string from the model’s item to the `GtkText`. The `g_signal_handlers_block_by_func` macro prevents an unnecessary write-back from the `GtkText` to the item during this update. Since no new resources are allocated in this step, an “unbind” signal handler is not needed.

The two signal handlers `text_changed_cb` and `item_selected_cb` are shown below.

```

1  static void
2  text_changed_cb (GtkEditable *self, gpointer user_data) {
3      GtkListItem *listitem = GTK_LIST_ITEM (user_data);
4      LeData *data = LE_DATA (gtk_list_item_get_item(listitem));
5      const char *s;
6
7      if (data) {
8          s = gtk_editable_get_text (self);
9          le_data_set_string (data, s);
10     }
11 }
12
13 static void
14 item_selected_cb (GtkListItem *listitem, GParamSpec *pspec, gpointer user_data) {
15     GtkWidget *text = GTK_WIDGET (user_data);
16
17     if (gtk_list_item_get_selected (listitem)) {
18         gtk_widget_grab_focus (text);
19     }
20 }
```

5.5.4 GtkSingleSelection

In the `liststore.ui` file, the `GtkListView`’s model property is set to a `GtkSingleSelection`, which in turn wraps the underlying `GListModel`.

Clicking a row in the list view selects and highlights it, automatically focusing the `GtkText` widget for immediate editing. Since `GtkSingleSelection` handles all the selection logic internally, no additional code is required for this behavior.

5.5.5 Another Possible Implementation

Alternatively, you can use a `GBinding` object to bind the `GtkText`’s “text” property to the `LeData`’s “string” property. To do this, you first need to implement the “string” property in the `LeData` class.

The code for this version is available in the [src/listeditor_binding/](#) directory. The GBinding instance is created in the “bind” handler and unbound in the “unbind” handler. We use the `g_object_set_data` function to store the binding object within the list item.

This is just one example, and other implementations are possible. For instance, you could subclass GtkText and store the binding instance directly inside the subclass.

However, these alternatives require more code. Ultimately, using the “changed” signal remains the simplest and most straightforward approach.

Appendix A

Appendix

A.1 Author Guide

A.1.1 Quick Start Guide

This section describes the steps to view the GTK 4 Tutorial on your local machine.

1. You will need a Linux operating system (or a Linux environment like WSL on Windows), Ruby (ver 4.0.1), Rake, Pandoc, and a LaTeX system.
2. Download this repository and extract its contents. See the next section for instructions on how to download the repository.
3. Change your current directory to the root directory of the source files.
4. Run `bundle install` to install the required Gems. `bundle` is a command from Bundler, a Ruby library management tool included with Ruby.
5. To view the tutorial in a web browser on your local PC:
 - Run `bundle exec rake html` to generate HTML files for local use.
 - Run `bundle exec rake serve` and access `http://localhost:8000` in your browser to view the HTML files.
6. To create a PDF version of the tutorial:
 - Run `bundle exec rake pdf` to generate the PDF file.
 - The PDF file will be generated in the `pdf` directory.

A.1.2 Recommended Environment and Software Setup

To generate the tutorial documents in your local repository, the following environment and software are required. I have also included the versions used in my own environment for reference.

- Linux Operating System: The programs in this repository were tested on Ubuntu 24.04.3 LTS running via WSL (Windows Subsystem for Linux).
- Ruby, Rake, Pandoc, and LaTeX: These are used to generate the documentation.
 - Ruby: Version 4.0.1 is used. Ruby version 3 or later may work, but you may need to modify the Gemfile, `.ruby-version`, and `mise.toml`.
 - Pandoc: Version 3.1.3 was used, but the version provided by your Linux distribution's package manager is likely sufficient. Pandoc converts Markdown (Pandoc's Markdown) into HTML or PDF.
 - LaTeX: When Pandoc converts Markdown to PDF, it generates LaTeX as an intermediate step. A LaTeX system then converts that into a PDF. For this tutorial, TeXLive 2025 with LuaLaTeX (LuaHBTeX 1.22.0 engine) was used.

There are two ways to download the repository to your local PC: - Using Git: Run `git clone https://github.com/ToshioCP/Gtk4-tutorial.git` in your terminal. - Downloading as a Zip: Click the green Code button on the repository's main page and select "Download ZIP." Extract the ZIP file after downloading.

A.1.3 How to Write .src.md Files

The source files for this tutorial are located in the `src` directory and have the `.src.md` extension. From here on, these will be referred to as ".src.md files."

A.1.3.1 The Document Generation Process

Three types of documents are generated and stored in the `gfm`, `docs`, and `pdf` directories, corresponding to GFM (GitHub Flavored Markdown), HTML, and PDF formats.

These three output formats—`gfm`, `html`, and `pdf`—are referred to as targets. Note that for the `html` target, the output directory is named `docs` instead of `html`. This is to align with the directory name used for hosting files on GitHub Pages.

Conversion to a target can happen in two ways: direct generation (no intermediate steps) or indirect generation (using intermediate steps).

Direct Generation

- `gfm` target: `.src.md => GFM`

Indirect Generation

- `html` target: `.src.md => Pandoc's Markdown => HTML`
- `pdf` target: `.src.md => Pandoc's Markdown => LaTeX => PDF`

As seen in this process, the `.src.md` format is converted into either GFM or Pandoc's Markdown. This conversion is handled by Ruby scripts.

GFM and Pandoc's Markdown are both extensions of the original Markdown. Since Markdown has several such "dialects," they share common features but also have distinct differences. While GFM and Pandoc's Markdown are identical in their basic syntax, they differ in areas such as how info strings and attributes are handled in fenced code blocks.

A.1.3.2 The Two Components of .src.md

The content of a `.src.md` file is divided into two parts:

1. `@@@` commands
2. Markdown: specifically, GFM or Pandoc's Markdown

Let's look at a concrete example:

This line belongs to "2. Markdown."

```
@@@shell
echo Hello World.
@@@
```

This line also belongs to "2. Markdown."

The three lines starting from `@@@shell` to `@@@` belong to "1. `@@@` commands."

A.1.3.3 @@@ Commands

`@@@` commands are defined as blocks with the following structures:

- `@@@if` command: Start with `@@@if` and end with `@@@end`.
- `@@@include` command: Start with `@@@include` and end with `@@@`.
- `@@@shell` command: Start with `@@@shell` and end with `@@@`.
- `@@@table` command: Start with `@@@table` and end with `@@@`.

The table command (`@@@table`) isn't used in the conversion and will be explained in a later section.

A.1.3.4 The `@@@if` Command

This command begins with `@@@if`, optionally followed by `@@@elif` or `@@@else`, and ends with `@@@end`. These work similarly to the C preprocessor directives `#if`, `#elif`, `#else`, and `#endif`.

`@@@if` and `@@@elif` are followed by a condition. Conditions can be:

- One of the generation targets: `gfm`, `html`, or `pdf`.
- Two targets joined by the `|` (OR) operator. The condition is true if the generation target matches either one.

Example:

```
@@@if gfm
```ruby
@@@else
```{.ruby .numberLines}
@@@end
def hello
  print("Hello World.\n")
end
```
```

This example resolves the difference in how fenced code blocks handle metadata between GFM and Pandoc's Markdown:

- In GFM, you can append an info string (usually the language name) after the backticks.
- In Pandoc's Markdown, you can append attributes enclosed in curly braces after the backticks. This allows for language specification as well as additional features like line numbering.

#### A.1.3.5 `@@@include`

The `@@@include` command imports external files.

```
@@@include
sample.c
@@@
```

In this example, the C source file `sample.c` specified between `@@@include` and the `@@@` marker is imported. The source file path must be relative from the `src` directory. If the content of `sample.c` is as shown below, the `@@@include` command is replaced by that C code. To ensure it fits within the Markdown text, it is wrapped in a fenced code block.

```
#include <stdio.h>

void
hello() {
 printf("Hello World.\n");
}

int
main() {
```

```

 hello();
 return 0;
}

```

You can extract a specific function by adding the function name after the filename.

```

@@@include
sample.c hello
@@@

```

This will only include the `hello` function from `sample.c`. The resulting replacement will look like this (wrapped in a fenced code block, which is omitted here for clarity):

```

void
hello() {
 printf("Hello World.\n");
}

```

While you can import source files from other languages, function-level extraction is only supported for C source files. Additionally, the C function extraction feature assumes the following formatting:

```

Return_Type
Function_Name(Arguments) {
 Function body
}

```

Specifically, the return type must be on the line immediately preceding the function name, and the function must end with a closing brace `}` at the beginning of a line. If these conditions are not met, the function cannot be extracted correctly.

Inserted text is converted into a fenced code block, surrounded by lines starting with three backticks.

If the output target is `gfm`, an info string (language name) is added to the starting fence. Using the previous C code example:

```

@@@include
sample.c hello
@@@

```

Since the file extension `.c` indicates C, the info string will be `c`. The resulting output becomes:

```

```c
void
hello() {
    printf("Hello World.\n");
}
```

```

Info strings like `c`, `ruby`, or `xml` are determined by the file extension:

- `.c => c`
- `.rb => ruby`
- `.xml => xml`

A list of supported languages is defined in </data/ext.yml>. Note that this path starts with a slash to represent the repository root, not the operating system root.

If the output target is `html` or `pdf`, attributes of Pandoc's Markdown are added after the starting fence.

```

@@@include
sample.c hello
@@@

```

This `@@@` command is converted as follows:

```
```{.c .numberLines}
void
hello() {
    printf("Hello World.\n");
}
```
```

These Pandoc's Markdown attributes provide both syntax highlighting for the C code and line numbers. Line numbering is only available for `html` and `pdf` targets that use Pandoc's Markdown. Unfortunately, GFM does not have a Markdown extension for line numbers, so only the code itself is output for that target.

The `@@@include` command must be written in exactly three lines (start marker, pathname with optional function names, end marker). Consequently, you cannot include multiple files in a single command block.

#### A.1.3.6 `@@@shell`

This command consists of the following three lines:

```
@@@shell
Shell command to execute
@@@
```

Make sure your current directory is the `src` directory of the local repository. All paths in the shell commands must be within `src`; any attempt to access files outside this directory will cause an error.

The three lines of the `@@@shell` command are replaced by the command line, its standard output, and the subsequent prompt.

For example:

```
@@@shell
wc author_guide.src.md
@@@
```

This is converted into:

```
```
$ wc author_guide.src.md
  509  2850 18698 author_guide.src.md
$
```
```

No info strings or attributes are added to this fenced code block.

#### A.1.3.7 How to Write Links

Links use URLs to direct the browser to a destination. There are three ways to write a URL:

- Absolute URL: A URL starting with `http` or `https`. (e.g., `https://example.com/a/b/c.html`)
- Root-relative URL: A path starting with a slash, representing the root of the site. In the example above, the root-relative version would be `/a/b/c.html`.
- Relative URL: A path relative to the current URL. If the current page is `https://example.com/a/index.html`, the relative URL to `https://example.com/a/b/c.html` would be `b/c.html`. Note that a relative URL does not start with a slash.

**A.1.3.7.1 External links** Use absolute URLs for links to external sites. For example, the following sentence includes an external link to the GFM specification page:

The GFM documentation can be found in the [GitHub Flavored Markdown Spec](https://github.github.com/

**A.1.3.7.2 Internal links** For internal links, use root-relative or relative URLs to ensure portability. When writing root-relative URLs in `.src.md` files, treat the `src` directory as the site root. The Ruby script automatically adjusts these paths to refer to the appropriate locations during the conversion.

We recommend using root-relative links for image files, though relative links also work.

Examples:

Root-relative URL:

- [Section 1](/sec1.src.md)
- [Image file](/images/sample.png)
- [C file pr1.c](/misc/pr1.c)
- [Directory misc](/misc/)

Relative URL:

- [Section 1](sec1.src.md)
- [Image file](images/sample.png)
- [C file pr1.c](misc/pr1.c)
- [Directory misc](misc/)

The requirements for the pdf target differ from the gfm or html targets:

- Section links: Must use fragment identifiers (e.g., `#link-to-files`). They link to ATX header IDs created by Pandoc. For further information, see [Pandoc User's Manual - Extension: auto\\_identifiers](#).
- Other files: Both root-relative and relative URLs are supported. The links usually refer to C source files or directories. The Ruby script converts these links to point to the corresponding blob or tree pages in the GitHub Gtk4-tutorial repository.

Example:

- [Section 1](#prerequisite-and-license)
- [Image file](/images/image.png)
- [C source file](misc/pr1.c)
- [Directory](/misc/)

#### Path Validation:

Ensure that all internal links remain within the `src` directory. The Ruby script will detect and warn you if a path (e.g., `../../etc/passwd`) attempts to escape the directory.

Note: This check prevents broken links and helps avoid “path traversal” issues, ensuring all referenced resources are correctly included in the final output.

For the pdf target, you can specify image sizes in units like cm.

```
! [Image a](/images/a.png){width=6cm height=5cm}
```

This attribute is a Pandoc's Markdown extension and is not available in GFM. The units (cm, % etc.) depend on the target (HTML or PDF). These attributes are preserved only when the target is html or pdf. If the target is gfm, the Ruby script removes the attribute.

#### A.1.3.8 Other Notes on Writing `.src.md`

The conversion from `.src.md` to GFM or Pandoc's Markdown by the Ruby script follows this order:

1. Pass 1: Replacement of `@@@if` commands
2. Pass 2: Replacement of `@@@include` and `@@@shell` commands
3. Pass 3: Separation into code and non-code sections; code is left as-is, while links in non-code sections are converted.

Failure to account for this order can lead to unexpected results.

```

...
@@@if gfm
Some statement for the target gfm
@@@elif html
Some statement for the target html
@@@else
Some statement for the target pdf
@@@end
...

```

Even if you intend to display the @@@if command itself, it will not be preserved as-is. This is because @@@if replacement (Pass 1) occurs before code block protection (Pass 3). If the target is gfm, the resulting output will be:

```

...
Some statement for the target gfm
...

```

To achieve the intended output, use indented code blocks instead of fenced code blocks. @@@ commands must start at the beginning of a line. If they are indented, they are not recognized as commands, allowing them to pass through Pass 1 unchanged and be treated as an (indented) code block in Pass 3.

As explained later, section files (`sec<num>.src.md`) must be placed directly under the root of the source directory. However, other `.src.md` files can be organized in subdirectories within the source directory. This directory structure is mirrored in the target directory when the target is gfm or html. If the target is pdf, these `.src.md` files are included in the appendices. Note that only directories containing `.src.md` files are copied. C programs are not copied, and image files are copied to the `docs` directory only when generating HTML.

Finally, since `.src.md` uses GFM or Pandoc's Markdown, please refer to their respective manuals for proper syntax. If the dialects differ, use @@@if to write the appropriate version for each. Links to the manuals are below:

- [GitHub Flavored Markdown Spec](#)
- [Pandoc User's Guide](#)

#### A.1.3.9 Table Formatting

Formatting tables is not strictly required, but it can improve the readability of `.src.md` files.

For example, suppose you have the following table in a text file named `sample.src.md`:

Price List

```

@@@table
item	price
mouse	$10
PC	$500
@@@

```

The @@@table command formats it as follows, aligning items vertically for better aesthetics:

Price List

```

item	price

```

```
|mouse| $10 |
| PC |$500 |
```

This command only affects the visual alignment within the Markdown file and does not change the final HTML or PDF output. Note that this command only supports the specific table syntax shown above.

The `@@@table` command is not supported during Rakefile-based conversion. You must use the standalone script `bin/table_formatter.rb`.

```
$ bundle exec ruby bin/table_formatter.rb sample.src.md
```

Whether to format tables is a matter of personal preference. If you don't find it necessary, you don't need to use the `@@@table` command.

### A.1.4 Directory Structure

This section covers the primary directory structure. This structure can be modified, but if you do so, you must also update `/path_manager.rb`, which manages these directories.

- `/src`: Contains all source files (`.src.md` files, C source code, etc.).
- `/src/images`: Contains image files like PNG and JPG.
- `/gfm`: Contains generated GFM files.
- `/docs`: Contains generated HTML files and copies of image files.
- `/pdf`: Contains generated PDF files.
- `/lib`: Contains Ruby library files.
- `/test`: Contains test files. Run all tests by executing `bundle exec rake -f test/Rakefile` from the terminal.
- `/data`: Contains `chapters.yml` (defining chapters for PDF generation), other definition files, and templates.
- `/bin`: Contains standalone Ruby scripts.

#### A.1.4.1 Source Directory Structure

**A.1.4.1.1 `index.src.md`** `index.src.md` is placed directly in the source directory. In GFM generation, it is converted to `README.md` at the repository root. In HTML generation, it becomes `/docs/index.html`, serving as the website's homepage. In PDF generation, it forms the initial introductory section.

**A.1.4.1.2 Section Files** Section files are named with the prefix “sec” followed by the section number and the `.src.md` extension (e.g., `sec1.src.md`, `sec5.src.md`, `sec12.src.md`). These correspond to Section 1, Section 5, and Section 12, respectively.

Section numbers must be consecutive natural numbers starting from 1. However, if you want to insert a new section in the middle, renaming many files manually is tedious. For example, if you want to insert a new section between `sec8.src.md` and `sec9.src.md`, first create a temporary section using a decimal, like `sec8.5.src.md`. Then, use the `/bin/renumber.rb` script to automatically renumber the sections:

- `sec8.src.md` remains `sec8.src.md`
- `sec8.5.src.md` becomes `sec9.src.md`
- `sec9.src.md` becomes `sec10.src.md`

All subsequent sections will also be renumbered accordingly.

### A.1.5 Rake

This repository's conversion system uses Rake, a Ruby build tool. Execute Rake by passing a task as an argument:

```
$ bundle exec rake <task name>
```

Available tasks:

- **gfm**: Generates GFM files. (Default task; executed if the task name is omitted).
- **html**: Generates HTML files.
- **pdf**: Generates PDF files.
- **all**: Generates GFM, HTML, and PDF files.
- **serve**: Starts a local server with **docs** as the site root (Port 8000).

#### A.1.5.1 Modes

The environment variable `G4T_ENV` determines the mode used when generating HTML files.

- **development**: Development mode. Generates files for local serving.
- **production**: Production mode. Generates files for GitHub Pages.

The primary difference is the base URL, which affects root-relative link URLs. GitHub Pages uses the repository name as the base URL, while the local environment has no base URL. The Ruby script determines the mode by reading `G4T_ENV`.

```
$ G4T_ENV=development bundle exec rake html # Development mode
$ G4T_ENV=production bundle exec rake html # Production mode
```

You must generate the HTML files in production mode before pushing to GitHub. If the environment variable assignment is omitted, the files will be generated in development mode.

## A.2 TfeTextView API reference

### A.2.1 Description

`TfeTextView` is a child object of `GtkTextView`. If its contents comes from a file, it holds the pointer to the `GFile`. Otherwise, the pointer is `NULL`.

### A.2.2 Hierarchy

```
GObject -- GInitiallyUnowned -- GtkWidget -- GtkTextView -- TfeTextView
```

### A.2.3 Ancestors

- `GtkTextView`
- `GtkWidget`
- `GInitiallyUnowned`
- `GObject`

### A.2.4 Constructors

- [`tfe\_text\_view\_new \(\)`](#)
- [`tfe\_text\_view\_new\_with\_file \(\)`](#)

### A.2.5 Instance methods

- [`tfe\_text\_view\_get\_file \(\)`](#)
- [`tfe\_text\_view\_open \(\)`](#)
- [`tfe\_text\_view\_save \(\)`](#)
- [`tfe\_text\_view\_saveas \(\)`](#)

### A.2.6 Signals

- [`Tfe.TextView::change-file`](#)
- [`Tfe.TextView::open-response`](#)



## A.2.7 API for constructors, instance methods and signals

### constructors

#### A.2.7.1 `tfe_text_view_new()`

```
GtkWidget *
tfe_text_view_new (void);
```

Creates a new `TfeTextView` instance and returns the pointer to it as `GtkWidget`. If an error happens, it returns `NULL`.

Return value

- a new `TfeTextView`.

#### A.2.7.2 `tfe_text_view_new_with_file()`

```
GtkWidget *
tfe_text_view_new_with_file (GFile *file);
```

Creates a new `TfeTextView`, reads the contents of the `file` and set it to the `GtkTextBuffer` corresponds to the newly created `TfeTextView`. Then returns the pointer to the `TfeTextView` as `GtkWidget`. If an error happens, it returns `NULL`.

Parameters

- `file`: a `GFile`

Return value

- a new `TfeTextView`.

### Instance methods

#### A.2.7.3 `tfe_text_view_get_file()`

```
GFile *
tfe_text_view_get_file (TfeTextView *tv);
```

Returns the copy of the `GFile` in the `TfeTextView`.

Parameters

- `tv`: a `TfeTextView`

Return value

- the pointer to the `GFile`

#### A.2.7.4 `tfe_text_view_open()`

```
void
tfe_text_view_open (TfeTextView *tv, GtkWidget *win);
```

Shows a file chooser dialog so that a user can choose a file to read. Then, read the file and set the buffer with the contents. This function doesn't return the I/O status. Instead, the status is informed by `open-response` signal. The caller needs to set a handler to this signal in advance.

parameters

- `tv`: a `TfeTextView`
- `win`: the top level window

**A.2.7.5 tfe\_text\_view\_save()**

```
void
tfe_text_view_save (TfeTextView *tv);
```

Saves the contents of the buffer to the file. If `tv` holds a `GFile`, it is used. Otherwise, this function shows a file chooser dialog so that the user can choose a file to save.

Parameters

- `tv`: a `TfeTextView`

**A.2.7.6 tfe\_text\_view\_saveas()**

```
void
tfe_text_view_saveas (TfeTextView *tv);
```

Saves the contents of the buffer to a file. This function shows file chooser dialog so that a user can choose a file to save.

Parameters

- `tv`: a `TfeTextView`

**Signals****A.2.7.7 change-file**

```
void
user_function (TfeTextView *tv,
 gpointer user_data)
```

Emitted when the `GFile` in the `TfeTextView` object is changed. The signal is emitted when:

- a new file is opened and read
- a user chooses a file with the file chooser dialog and save the contents.

**A.2.7.8 open-response**

```
void
user_function (TfeTextView *tv,
 TfeTextViewOpenResponseType response-id,
 gpointer user_data)
```

Emitted after the user calls `tfe_text_view_open`. This signal informs the status of file I/O operation.

**Enumerations****A.2.7.9 TfeTextViewOpenResponseType**

Predefined values for the response id given by `open-response` signal.

Members:

- `TFE_OPEN_RESPONSE_SUCCESS`: The file is successfully opened.
- `TFE_OPEN_RESPONSE_CANCEL`: Reading file is canceled by the user.
- `TFE_OPEN_RESPONSE_ERROR`: An error happened during the opening or reading process.

**A.3 Turtle manual**

Turtle is a simple interpreter for turtle graphics.

### A.3.1 Prerequisite and compiling

Turtle is written in C language. You need:

- Linux. Turtle is tested on ubuntu 22.10
- gcc, meson and ninja
- gtk4

It is easy to compile the source file of turtle. If you want to install it in your local area, put an option `--prefix=$HOME/.local` to your meson command line. Then, it will be installed under `$HOME/.local/bin`. The instruction is:

```
$ meson setup --prefix=$HOME/.local _build
$ ninja -C _build
$ ninja -C _build install
```

If you want to install it in the system area, no option is necessary. It will be installed under `/usr/local/bin`.

```
$ meson setup _build
$ ninja -C _build
$ sudo ninja -C _build install
```

Type the following command then turtle shows the following window.

```
$ turtle
```

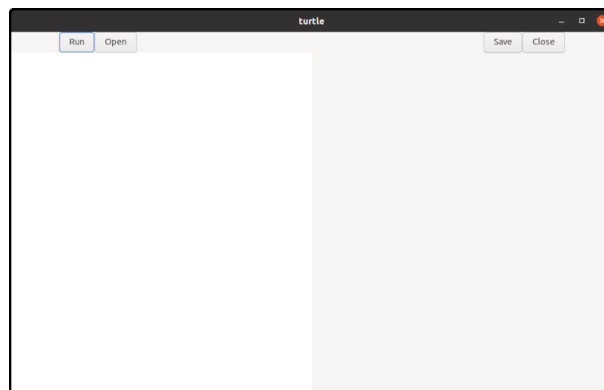


Figure A.1: Screenshot just after it's executed

The left half is a text editor and the right half is a surface. Surface is like a canvas to draw shapes.

Write turtle language in the text editor and click on run button, then the program will be executed and it draws shapes on the surface.

If you uncomment the following line in `turtle.y`, then codes for debug will also be compiled. Turtle shows the status to the standard output, but the speed is quite slow. It is not recommended except you are developing the program.

```
/* # define debug 1 */
```

### A.3.2 Example

Imagine a turtle. The turtle has a pen and initially it is at the center of the screen, facing to the north (to the north means up on the screen). You can let the turtle down the pen or up the pen. You can order the turtle to move forward.

```
pd
fd 100
```

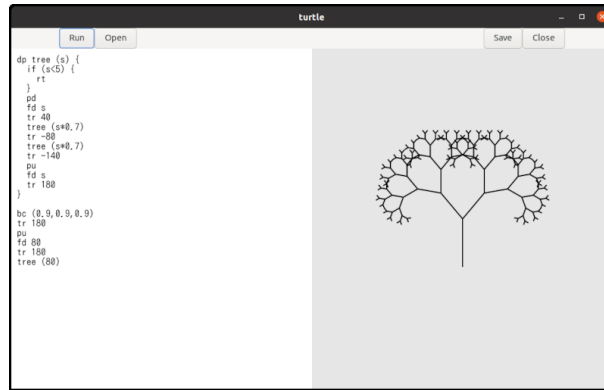


Figure A.2: Tree

- `pd`: Pen Down. The turtle put the pen down so that the turtle will draw a line if it moves.
- `fd 100`: move Forward 100. The turtle goes forward 100 pixels.

If you click on **run** button, then a line segment appears on the screen. One of the endpoints of the line segment is at the center of the surface and the other is at 100 pixels up from the center. The point at the center is the start point of the turtle and the other endpoint is the end point of the movement.

If the turtle picks the pen up, then no line segment appears.

```
pu
fd 100
```

The command `pu` means “Pen Up”.

The turtle can change the direction.

```
pd
fd 100
tr 90
fd 100
```

The command `tr` is “Turn Right”. The argument is angle with degrees. Therefore, `tr 90` means “Turn right by 90 degrees”. If you click on the **run** button, then two line segments appears. One is vertical and the other is horizontal.

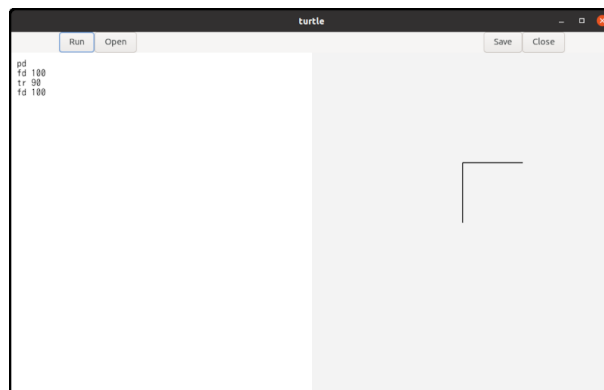


Figure A.3: Two line segments on the surface

You can use `tl` (Turn Left) as well.

### A.3.3 Background and foreground color

Colors are specified with RGB. A vector (r, g, b) denotes RGB color. Each of the elements is a real number between 0 and 1.

- Red is (1.0, 0.0, 0.0). You can write (1, 0, 0) instead.
- Green is (0.0, 1.0, 0.0)
- Blue is (0.0, 0.0, 1.0)
- Black is (0.0, 0.0, 0.0)
- White is (1.0, 1.0, 1.0)

You can express a variety of colors by changing each element.

There are two commands to change colors.

- bc: Background Color. `bc (1,0,0)` changes the background color to red. This command clear the surface and change the background color. So, the shapes on the surface disappears.
- fc: Foreground Color. `fc (0,1,0)` changes the foreground color to green. This command changes the pen color. The prior shapes on the surface aren't affected. After this command, the turtle draws lines with the new color.

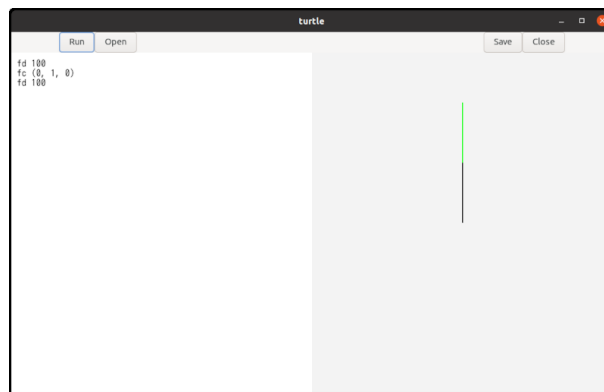


Figure A.4: Change the foreground color

### A.3.4 Other simple commands

- pw: Pen Width. This is the same as pen size or line width. For example, `pw 5` makes lines thick and `pw 1` makes it thin.
- rs: ReSet. The turtle moves back to the initial position and direction. In addition, The command initialize the pen, line width (pen size), and foreground color. The pen is down, the line width is 2 and the foreground color is black.

An order such as `fd 100`, `pd` and so on is a statement. Statements are executed in the order from the top to the end in the program.

### A.3.5 Comment and spaces

Characters between # (hash mark) and \n (new line) are comment. If the comment is at the end of the file, the trailing new line can be left out. Comments are ignored.

```
draw a triangle<NEW LINE>
fd 100 # forward 100 pixels<NEW LINE>
tr 120 # turn right by 90 degrees<NEW LINE>
fd 100<NEW LINE>
tr 120<NEW LINE>
fd 100 # Now a triangle appears.<EOF>
```

<NEW LINE> and <EOF> indicate newline code and end of file respectively. The comments in the line 1, 2, 3 and 6 are correct syntactically.

Spaces (white space, tab and new line) are ignored. They are used only as delimiters. Tabs are recognized as eight spaces to calculate the column number.

### A.3.6 Variables and expressions

Variables begin alphabet followed by alphabet or digit. Key words like **fd** or **tr** can't be variables. **Distance** and **angle5** are variables, but **1step** isn't a variable because the first character isn't alphabet. Variable names are case sensitive. Variables keep real numbers. Their type is the same as **double** in C language. Integers are casted to real numbers automatically. So 1 and 1.0 are the same value. Numbers begin digits, not signs (+ or -).

- 100, 20.34 and 0.01 are numbers
- +100 isn't a number. It causes syntax error. Use 100 instead.
- -100 isn't a number. But turtle recognizes it unary minus and a number 100. So turtle calculate it and the result is -100.
- 100 + -20: This is recognized 100 + (- 20). However, using bracket, 100 + (-20), is better for easy reading.

```
distance = 100
fd distance
```

A value 100 is assigned to the variable **distance** in the first line. Assignment is a statement. Most of statements begin with commands like **fd**. Assignment is the only exception.

The example above draws a line segment of 100 pixels long.

You can use variables in expressions. There are 8 kinds of calculations available.

- addition:  $x + y$
- subtraction:  $x - y$
- multiplication:  $x * y$
- division:  $x / y$
- unary minus:  $-x$
- logical equal:  $x = y$ . This symbol = works as == in C language.
- greater than:  $x > y$
- less than:  $x < y$

The last three symbols are mainly used in the condition of if statement.

Variables are registered to a symbol table when it is assigned a value for the first time. Evaluating a variable before the registration isn't allowed and occurs an error.

### A.3.7 If statement

Turtle language has very simple if statement.

```
if (x > 50) {
 fd x
}
```

There is no else part.

### A.3.8 Loop

Turtle has very simple loop statement. It is **rp** (RePeat) statement.

```
rp (4) {
```

```

 fd 100
 tr 90
}

```

The program repeats the statements in the brace four times.

### A.3.9 Procedures

Procedures are similar to functions in C language. The difference is that procedures don't have return values.

```

dp triangle (side) {
 fd side
 tr 120
 fd side
 tr 120
 fd side
}

```

```
triangle (100)
```

dp (Define Procedure) is a key word followed by procedure name, parameters, and body. Procedure names start alphabet followed by alphabet or digit. Parameters are a list of variables. For example

```

dp abc (a) { ... }
dp abc (a, b) { ... }
dp abc (a, b, c) { ... }

```

Body is a sequence of statements. The statements aren't executed when the procedure is defined. They will be executed when the procedure is called later.

Procedures are called by the name followed by arguments.

```
dp proc (a, b, c) { ... }
```

```
proc (100, 0, -20*3)
```

The number of parameters and arguments must be the same. Arguments can be any expressions. When you call a procedure, brackets following the procedure name must exist even if the procedure has no argument.

Procedure names and variable names don't conflict.

```

dp a () {fd a}
a=100
a ()

```

This is a correct program.

- 1: Defines a procedure **a**. A variable **a** is in its body.
- 2: Assigns 100 to a variable **a**.
- 3: Procedure **a** is called.

However, using the same name to a procedure and variable makes confusion. You should avoid that.

### A.3.10 Recursive call

Procedures can be called recursively.

```

dp repeat (n) {
 n = n - 1
}

```

```

 if (n < 0) {
 rt
 }
 fd 100
 tr 90
 repeat (n)
}

```

```
repeat (4)
```

Repeat is called in the body of repeat. The call to itself is a recursive call. Parameters are created and set each time the procedure is called. So, parameter *n* is 4 at the first call but it is 3 at the second call. Every time the procedure is called, the parameter *n* decreases by one. Finally, it becomes less than zero, then the procedures return.

The program above draws a square.

It shows that we can program loop with a recursive call.

### A.3.11 Fractal curves

Recursive call can be applied to draw fractal curves. Fractal curves appear when a procedure is applied to it repeatedly. The procedure replaces a part of the curve with the contracted curve.

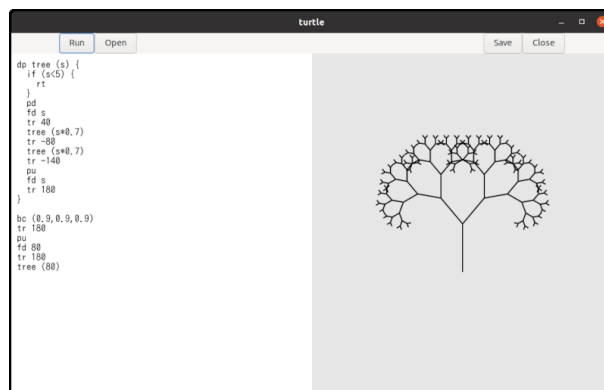


Figure A.5: Tree

This shape is called tree. The basic pattern of this shape is a line segment. It is the first stage. The second stage adds two shorter line segments at the endpoint of the original segment. The new segment has 70 percent length to the original segment and the orientation is +30 or -30 degrees different. The third stage adds two shorter line segments to the second stage line segments. And repeats it several times.

This repeating is programmed by recursive call. Two more examples are shown here. They are Koch curve and Square Koch curve.

### A.3.12 Tokens and punctuations

The following is the list of tokens.

Keywords:

- pu: pen up
- pd: pen down
- pw: pen width = line width
- fd: forward
- tr: turn right



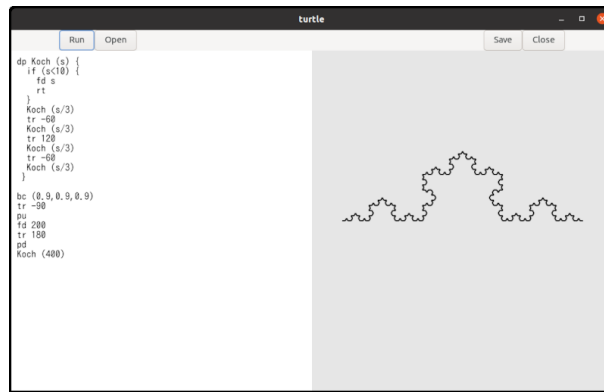


Figure A.6: Koch curve

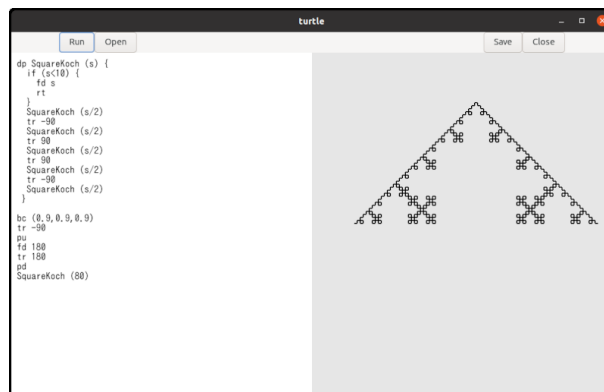


Figure A.7: Square Koch curve

- tl: turn left
- bc: background color
- fc: foreground color
- if: if statement
- rt: return
- rs: reset
- rp: repeat
- dp: define procedure

identifiers and numbers:

- identifier: This is used for the name of variables, parameters and procedures. It is expressed `[a-zA-Z][a-zA-Z0-9]*` in regular expression.
- number: This is expressed `(0|[1-9][0-9]*)(\.[0-9]+)?` in regular expression. It doesn't have `+` or `-` sign because they bring some syntactic confusion. However negative number such as `-10` can be recognized as unary minus and a number.

Symbols for expression

- =
- >
- <
- +
- -
- \*
- /
- (

- )

Delimiters

- (
- )
- {
- }
- ,

Comments and spaces:

- comment: This is characters between # and new line. If a comment is at the end of the file, the trailing new line can be left out.
- white space:
- horizontal tab: tab is recognized as eight spaces.
- new line: This is the end of a line.

These characters are used to separate tokens explicitly. They doesn't have any syntactic meaning and are ignored by the parser.

### A.3.13 Grammar

program:

```
 statement
| program statement
;
```

statement:

```
 primary_procedure
| procedure_definition
;
```

primary\_procedure:

```
 pu
| pd
| pw expression
| fd expression
| tr expression
| tl expression
| bc '(' expression ',' expression ',' expression ')'
| fc '(' expression ',' expression ',' expression ')'
| ID '=' expression /* ID is an identifier which is a name of variable */
| if '(' expression ')' '{' primary_procedure_list '}'
| rt
| rs
| rp '(' expression ')' '{' primary_procedure_list '}'
| ID '(' ')' /* ID is an identifier which is a name of procedure */
| ID '(' argument_list ')' /* the same as above */
;
```

procedure\_definition:

```
 dp ID '(' ')' '{' primary_procedure_list '}'
| dp ID '(' parameter_list ')' '{' primary_procedure_list '}'
;
```

```
parameter_list:
 ID
| parameter_list ',' ID
;

argument_list:
 expression
| argument_list ',' expression
;

primary_procedure_list:
 primary_procedure
| primary_procedure_list primary_procedure
;

expression:
 expression '=' expression
| expression '>' expression
| expression '<' expression
| expression '+' expression
| expression '-' expression
| expression '*' expression
| expression '/' expression
| '-' expression %prec UMINUS
| '(' expression ')'
| ID
| NUM /* NUM is a number */
;
```